Jet Ranger Cockpit User Guide

Setting up, running, wiring, flashing and troubleshooting the Jet Ranger panel

Regenerated 31 October 2026

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1 Overview

The Jet Ranger cockpit drives real gauge needles, warning lights, radio displays and switches from the simulator. A Windows PC exchanges UDP packets over Ethernet with Arduino Mega 2560 panel boards, and a small USB board presents the panel switches to Windows as joysticks.

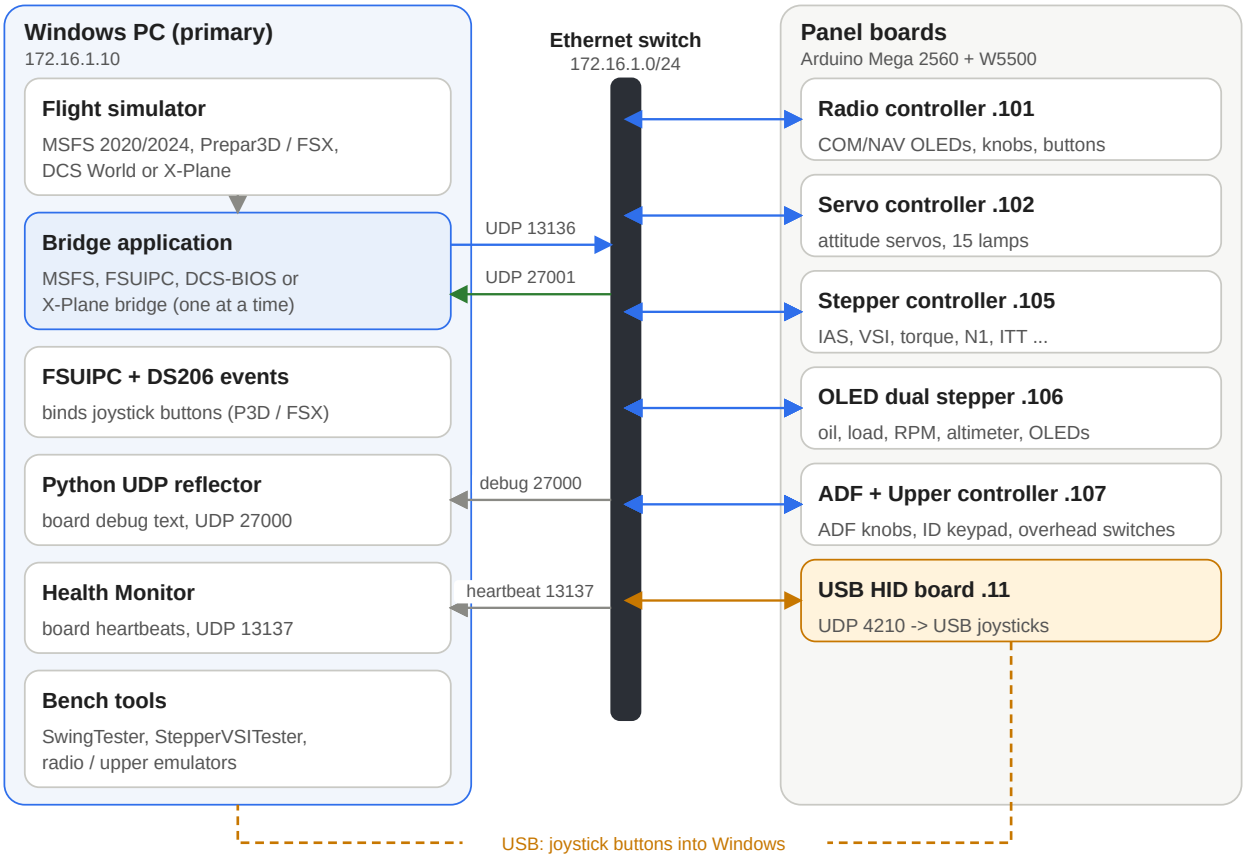


Figure 1 — The whole system

Data flows from the PC to the boards as short text packets. On the primary PC a Python listener shows the boards' debug messages and a separate Windows app, the Health Monitor, tracks their keep-alives. Cockpit switches reach the simulator separately, as joystick buttons (and, in Prepar3D and FSX, FSUIPC events).

Part	Hardware and software	Address	Section
Primary PC	Windows, MSFS 2020 or 2024, Prepar3D / FSX with DodoSim 206 V2, DCS World or X-Plane, one bridge program, debug listener, Health Monitor	172.16.1.10	Windows and simulator
Radio controller	Mega 2560 with the Radio Auto Pilot PCB and the COM and NAV panel	172.16.1.101	Radio controller
Servo board	Mega 2560 and Ethernet module on the Front Panel Servo PCB	172.16.1.102	Servo controller
ADF and Upper controller	Mega 2560 on the combined Upper ADF Controller PCB: the ADF knobs, aircraft ID keypad, gear, flaps, speed brake, brakes, anti-ice, hyd boost and caution dim switches, landing gear lamps, and the overhead switch matrix and keypad of the old Upper controller (172.16.1.103, retired)	172.16.1.107	ADF and Upper controller, Joystick (HID) interface
Stepper board	Mega 2560 on the Jet Ranger 13 Stepper PCB	172.16.1.105	Stepper controller
Dual stepper board	Mega 2560 on the OLED Dual Stepper PCB	172.16.1.106	Dual stepper controller

Part	Hardware and software	Address	Section
USB HID board	Leonardo-class board with native USB	172.16.1.11	Joystick (HID) interface

2 Windows and simulator

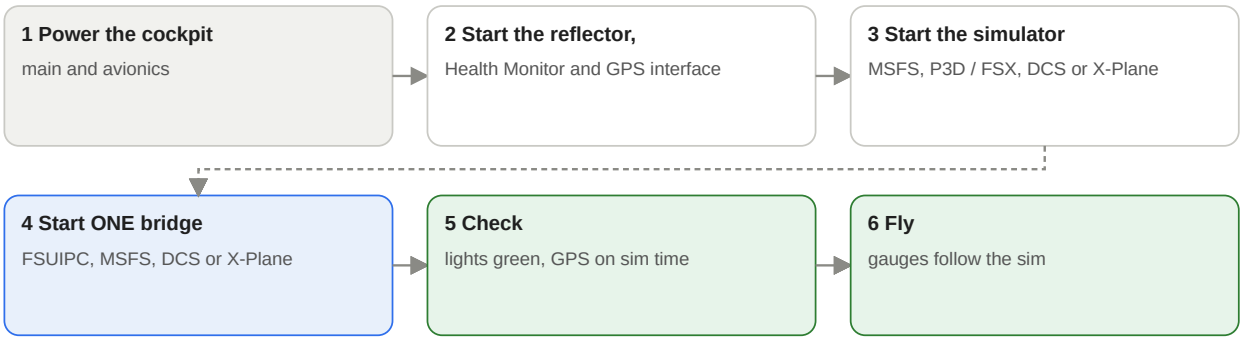
The Windows PC runs the simulator and a **bridge program** that turns simulator variables into UDP text packets for the Arduino boards. Switch inputs travel the other way as joystick buttons.

The bridge sends the Servo, Stepper and Dual Stepper boards each their own packet in **real units** (degrees, knots, feet per minute, psi, percent, degrees C), and the boards convert them with their own calibration. It sends about 10 packets per second to each board; the radio gets a packet whenever a frequency changes and at least every 5 seconds. The Dual Stepper's temperature OLED shows the outside air temperature from the simulator.

Component	Role	Notes
Simulator	Flies the DodoSim 206 or another Bell 206 Jet Ranger, or (with the DCS and X-Plane bridges) the Huey or any other helicopter	MSFS 2020 or 2024, Prepar3D, FSX, DCS World or X-Plane
FSUIPCWinformsAutoCS	Bridge for Prepar3D and FSX, reads FSUIPC offsets	Needs FSUIPC
MSFSWinformsAutoCS	Bridge for MSFS, reads SimConnect	Has estimates for the FlyInside B206 and CowaSim Jet Ranger
DCSBiosBridge	Bridge for DCS World, reads the DCS-BIOS export stream (output only; first the UH-1H Huey)	Needs DCS-BIOS installed in DCS
XPlaneBridge	Bridge for X-Plane 11 and 12, reads datarefs over UDP (output only, no plugin)	Not yet tried against a real X-Plane
FSUIPC and DODSIM206V2.EVT	Exposes 234 DodoSim custom events so switches can be bound	Prepar3D and FSX only
Python debug listener	Shows debug messages from the boards	Listens on UDP 27000
Health Monitor	Tracks each board's keep-alive and flags a board that goes quiet	Listens on UDP 13137
LowranceAirMapGps	Feeds a Lowrance AirMap 2000C GPS from the simulator over a serial port	Listens on UDP 27002; see Lowrance AirMap GPS

Run **only one bridge at a time**: the FSUIPC and MSFS bridges both listen for the radio board's commands on UDP 27001, and all four bridges send to the same boards. Both bridges also send the aircraft position, altitude, airspeed, heading and simulator date and time to the GPS interface on UDP 27002 (on the same PC).

2.1 Starting up



Order matters little: every board waits for data, and the bridge reconnects to the simulator on its own every second.

If a gauge stays at zero for 30 s without data, the board's watchdog parked it on purpose.

Figure 2 — Start-up order

1. Power the cockpit (main and avionics).
2. Start the debug listener (UDP 27000), the Health Monitor and the Lowrance GPS interface.
3. Start the simulator.

4. Start **one** bridge.
5. Check: every Health Monitor light should turn green within about 10 seconds, and the GPS window should show the simulator as its time source.
6. Fly. The gauges follow the simulator.

The procedure for each simulator (MSFS, Prepar3D / FSX, DCS World and X-Plane), and the `Start Jet Ranger.ps1` script that does the PC side of it, are in the next chapters (Starting each simulator).

The order matters little: every board waits for data and the bridge reconnects to the simulator on its own every second. A gauge that stays at zero for 30 seconds without data was parked on purpose by the board's watchdog.

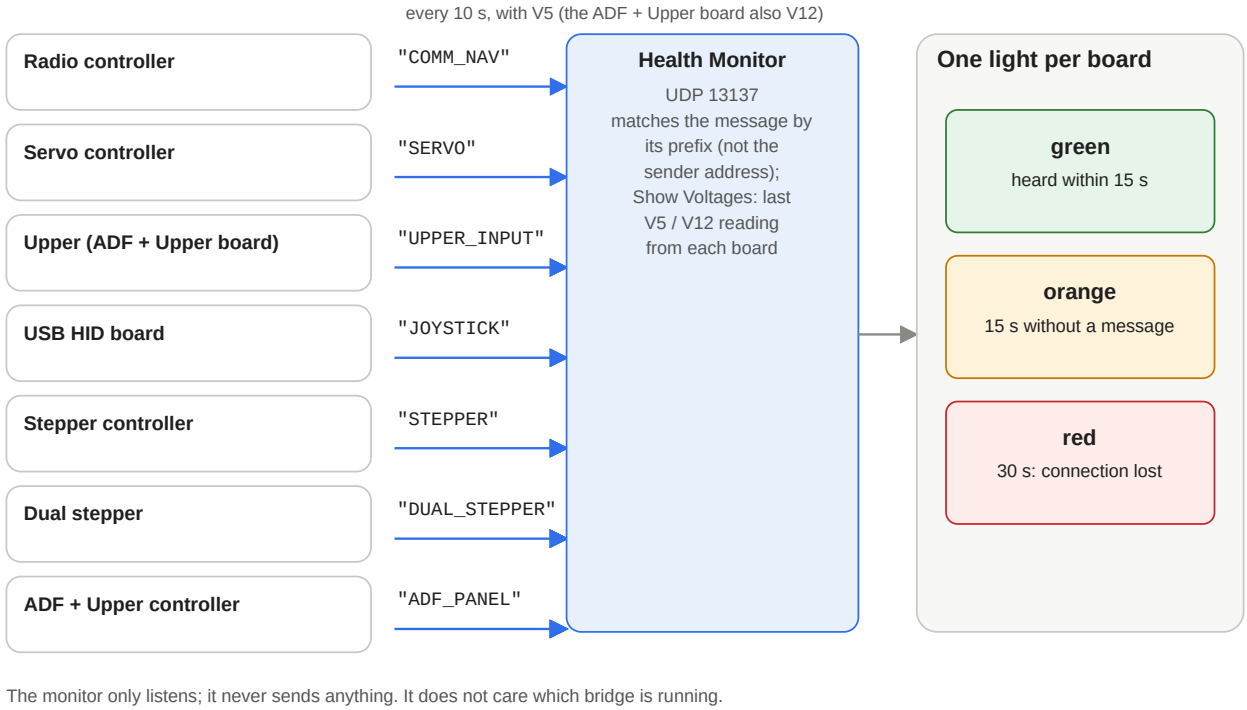
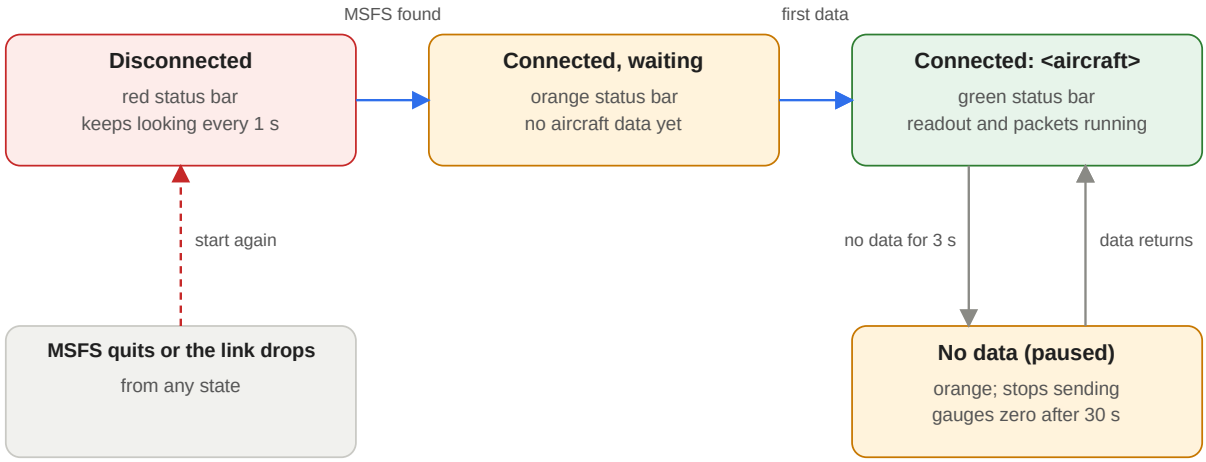


Figure 3 — Health Monitor

2.2 Running the MSFS bridge

1. Build the project `MSFSWinformsAutoCS` (in the repository's `C# Code` folder) and run `MSFSWinformsAutoCS.exe` from its `bin\Debug` folder. Build output is not stored in git, so a fresh copy of the repository has to be built first.
2. Close the FSUIPC bridge if it is running, then start MSFS and load a Jet Ranger. The status bar changes from *Disconnected* (red) to *Connected to MSFS* with the aircraft name (green) once data arrives.
3. The window lists the values being sent and the warning lamps. The Avionics Master box follows the simulator and can also be used to switch it.
4. If the status turns orange and says no data, the simulator is paused or loading and the bridge has stopped sending; the boards return their gauges to zero after 30 seconds and carry on when data returns.
5. Five lamps are driven: engine out, rotor low, low fuel, generator fail (engine speed below 10 percent) and transmission oil pressure (below 40 psi, using the estimated pressure on the FlyInside and CowaSim). The other lamps need aircraft-specific variables. Some MSFS Jet Rangers do not report every value (for example transmission pressure and temperature), so those gauges may stay at zero. With the FlyInside B206 the estimated transmission pressure (35 psi) is below the 40 psi threshold, so that lamp stays on.
6. The status bar shows the aircraft name the simulator reports and follows it when you load a different one.

7. The electrical load follows the size of the total load whichever way the current flows, so with the engine stopped it still shows a load.



A new aircraft is picked up while connected: the name in the status bar changes and the estimate models restart.

SimConnect errors are caught on every path, so a closed simulator never stops the bridge. FSUIPCWinformsAutoCS behaves the same way with FSUIPC.

Figure 4 — Bridge status

2.2.1 Aircraft with estimated dials

Value	Standard SimVars	FlyInside B206	CowaSim Jet Ranger
Rotor, engine speed	**ROTOR RPM PCT,** GENERAL ENG PCT MAX RPM	**L:Aircraft.Rotor.RPM.Percent,** L:...Turbine.N2.Smooth	**standard**
Gas producer (N1)	**TURB ENG CORRECTED N1**	**0.92 x turbine N2**	**standard**
Torque (TQ)	**ENG TORQUE PERCENT**	**collective lever curve,** up to 120 %	**0.42 x SimVar,** smoothed 2.5 s
Turbine out temp (ITT)	**TURB ENG ITT**	**6 x N2 + 1.05 x** (torque - 21)	**EGT - 200 C**
Fuel load (psi)	**ENG FUEL PRESSURE**	**10 x SimVar**	**standard**
Electrical load (%)	**|total amps| / 300 A**	**|total amps| / 300 A**	**|total amps| / 139 A**
Transmission pressure	**SimVar (usually 0)**	**35 psi x rotor fraction**	**48 psi x rotor fraction**
Transmission temp	**SimVar (usually 0)**	**slow warm-up from air temp**	**0.92 x engine oil temp**
Recognised by	**otherwise**	**turbine L:var, or title** contains FlyInside	**title contains 206B3**

Green = the standard SimVar is used unchanged. Estimates come from a few dial readings and may be a few percent off.

Figure 5 — Which values come from where

FlyInside B206. Its own gauges show values that are not in the standard simulator variables: rotor speed, engine speed, gas producer, torque, turbine out temperature, fuel pressure and transmission oil. The bridge reads the aircraft's own rotor and engine speed variables and estimates the rest from cockpit dial readings: torque follows the collective lever, turbine out temperature rises with torque, and the transmission oil pressure and temperature are modelled (the temperature warms up slowly from the outside air temperature and starts again from cold each time the bridge restarts). In this aircraft holding full collective for more than about 2 seconds over-torques the engine and breaks it. Its cockpit load dial stays at about 50 percent whatever the load.



Both curves were fitted to dial readings from screenshots; check them against the cockpit at other power settings.

FlyInside above lever 73.6 climbs at about 1.74 points per percent; the engine over-torques if full collective is held for 2 s.

Figure 6 — Torque estimates

CowaSim Jet Ranger (shown as `206B3` with its registration). Most gauges match the standard variables. Its torque, turbine out temperature and electrical load dials do not, and it has no transmission oil values: the bridge smooths the torque, takes the turbine out temperature from the exhaust gas temperature, rescales the load and estimates the transmission oil from rotor speed and engine oil temperature.

These estimates were fitted to a few dial readings and may be a few percent off at other power settings, so compare the panel gauges with the cockpit dials. Other aircraft use the standard variables.

2.3 Running the FSUIPC bridge (Prepar3D and FSX)

Start FSUIPC with the simulator, then run `FSUIPCWinformsAutoCS.exe`. The status bar shows when it is connected to Prepar3D. The window lists the values being sent and the 21 annunciator bits. The bridge sends about 10 packets per second to each board and listens on UDP 27001 for the radio board's commands (frequency swaps, standby tuning, battery, alternator, avionics master).

2.4 Running the DCS bridge (DCS World)

The DCS bridge reads **DCS-BIOS**, a free add-on that makes DCS broadcast its cockpit state on the PC. It only sends data to the panel: the panel's switches reach DCS as joystick buttons (bind them in DCS as usual), and the radio knobs do not tune DCS radios yet.

1. Install DCS-BIOS once (it is already installed on the cockpit PC, version 0.11.8, in `Saved Games\DCS\Scripts`, with a one line `Export.lua`). Start DCS and load a mission with an aircraft; DCS-BIOS writes its aircraft files into `Saved Games\DCS\Scripts\DCS-BIOS\doc\json`.
2. Build `DCSBiosBridge` (in the repository's `C# Code` folder) and run `DCSBiosBridge.exe` from its `bin` folder. Close the other bridges first.
3. The window shows the stream status, the aircraft name DCS-BIOS reports and the profile it chose. With **Send to the boards** ticked the values go to the boards; the token table shows each value and where it came from.
4. **UH-1H Huey.** The 20 caution lights drive the Huey Warning Panel LEDs on the Radio controller; they light only while the Radio is powered, that is while the aircraft has power. The radar altitude goes to the radar altitude gauge. For every aircraft the airspeed, altitude, heading and position are sent (the GPS interface follows the aircraft).
5. **GPS time.** The Lowrance GPS shows the mission clock. DCS gives the mission's local time, so set **Mission time zone (hours from UTC)** to the theatre's offset (for example 4 for the Caucasus and

the Persian Gulf, 3 for Syria, -8 for Nevada); the box starts at 0 each run, or start the bridge with `--utc-offset 4`.

6. **Gauges that need calibrating.** The other Huey gauges (torque, gas producer, turbine temperature, fuel, oil, rotor and engine speed, electrical load, vertical speed, pitch and bank) are read as needle positions and shown with a first-guess scale, marked *uncalibrated*. To correct one: fly so the cockpit gauge reads a known value, select its row, press **Calibrate selected...** and type the reading; repeat at two or more readings. The points are kept in `profiles\UH-1H.calibration.json`. **Clear its calibration** removes them.
7. For another aircraft the bridge uses the common values until a profile file for it is added.

2.5 Running the X-Plane bridge

The X-Plane bridge needs no plugin: it asks X-Plane for datarefs over UDP, and it only sends data to the panel (the switches reach X-Plane as joystick buttons). It has not been tried against a real X-Plane yet, so expect to check the values.

1. In X-Plane open Settings, Network and allow incoming UDP connections (the default receive port is 49000).
2. Build `XPlaneBridge` and run `XPlaneBridge.exe` from its `bin` folder. Close the other bridges first. With no options it finds X-Plane from its beacon, or tries `127.0.0.1:49000`; use `--xplane-host` and `--xplane-port` if X-Plane is on another PC.
3. Choose the profile in the Profile box: *Generic (common values, any aircraft)* works with any aircraft that fills in the standard datarefs, *Robinson R22* and *Sikorsky S-76* add their rotor speed and bus voltage. X-Plane does not tell the bridge which aircraft is loaded, so choose it yourself.
4. The window shows whether data is arriving. A gauge that stays at zero with the generic profile means the aircraft does not fill in that dataref: press **Search datarefs...**, type words such as `oil pressure`, and add the right dataref to a profile file in the `profiles` folder.
5. The warning lamps and the Huey LEDs follow X-Plane's generic annunciators, so they are approximate.
6. The Lowrance GPS follows X-Plane's zulu clock and date (the PC's year is used).

2.6 Installing the DodoSim event file (FSUIPC)

Copy `DODSIM206V2.EVT` into the `{P3D or FSX}/Modules` folder, next to `FSUIPC.ini`. Add this to `FSUIPC.ini` if FSUIPC has not added it itself:

```
[EventFiles]
0=DodoSim206
```

Restart the simulator. The `DS206.*` events now appear in FSUIPC's button and key assignment lists. Bind each physical switch to its event. After an EVT update FSUIPC re-indexes events alphabetically, so check and re-map bindings.

2.6.1 Event naming

Suffix	Parameter	Behaviour	Example
`_SET`	Integer (usually 1 = on, 0 = off)	Sets a state directly; best for latching toggle switches	`DS206.BATTERY_SWITCH_SET`
`_TOGGLE`	None	Flips the state each press	`DS206.FUEL_VALVE_SWITCH_TOGGLE`
`_TRIGGER`	None (some take 1 to 10 as a multiplier)	Momentary; tick Control to repeat while held	`DS206.STARTER_BUTTON_TRIGGER`
none	None	One-shot action	`DS206.CLOCK_MODE_BUTTON`

For physical latching switches prefer the `_SET` events: send 1 on switch-on and 0 on switch-off so the simulator never drifts out of step with the panel.

2.6.2 Event groups at a glance

Group	Event numbers	Examples
Circuit breakers	36 to 95, 107 to 110	Hoist, SAS, fuel pumps, governor, ignition, starter
Lights and master switches	96 to 101, 111 to 122, 209 to 212, 224 to 227	Battery, generator, avionics, pitot heat, landing lights
Engine and flight controls	1 to 2, 207 to 208, 213 to 223, 228 to 232	Rotor brake, hydraulics, force trim, starter, governor beep
Instruments	3 to 33	Annunciator test, AI cage, radar altimeter, HSI and OBS knobs
Radios and audio	129 to 192	COM and NAV swap and tuning, ADF, DME, audio panel
Transponder	193 to 206	Mode knob, digit buttons, IDT, VFR
SAS modes	123 to 128	Master, HDG, NAV, SPD, ALT, VRT
Other	102 to 106, 233 to 234	Difficulty rotary, heat and vent, external power

3 Colour codes and cabling

This chapter explains the colours used in the diagrams and programs, and the colour codes used for cables and connectors.

3.1 Reading the diagrams

Each diagram box is shaded to say what kind of item it is. The same shades are used in every diagram.

Colour	What it means in the diagrams
White	An ordinary item: a board, program, step or detail
Blue	The subject of the figure or the main step: the bridge, the Health Monitor, the code that decides
Green	Running or healthy, power on, or an output: connected, the gauge or backlight being driven
Amber	Waiting, a warning or a special path: no data, the default position, the USB HID path
Red	Lost, off, failed or a reset
Gray	Supporting or inactive: notes, items that stay off
Purple	Tells boards or groups apart where the figure says so (for example the Dual Stepper in the Swing ALL sequence, the Upper controller in the PCB map)
Teal	The same, for the ADF controller in the PCB map
Dark	Column headers and the Ethernet switch

Arrows follow the same idea: blue is data going to a board over UDP, green is commands or events coming back, amber is the USB HID path, gray is an ordinary link, and a dashed line is an optional or recovery path.

3.2 Colours in the programs

Where	Colour	Meaning
Health Monitor light	Green	A heartbeat was heard within the last 15 seconds
Health Monitor light	Orange	15 seconds without a heartbeat (connection warning)
Health Monitor light	Red	30 seconds without a heartbeat (connection lost)
Bridge status bar	Red	Disconnected from the simulator
Bridge status bar	Orange	Connected but no data (the simulator is paused or loading)
Bridge status bar	Green	Connected, with the aircraft name
Board status LEDs (pins 14 green, 15 red)	Green and red	Alternate every 200 ms while the sketch runs; red flashes fast while Ethernet starts
SwingTester	Gold	The gauge row being swung, or a warning lamp that is on
GPS bench tester	Pink	A box whose value could not be read (it is not used)
Upper controller emulator	Green	A switch that is on
GPS interface	Green	The position comes from aircraft data over UDP
GPS interface	Orange	No data for 30 seconds: the Brisbane Airport default is being sent

3.3 Cabling colour codes

3.3.1 Round power connectors

Colour	Supply
Red	12 V
Gray	5 V

3.3.2 PCB connectors

Colour	Use	Example
Uncoloured	Digital input	
Blue	Analog	
Red	12 V	Magnetic coils
Green	12 V backlighting	
Yellow	OLED	
Orange	LEDs	

The two 12 V colours are different on purpose: red is 12 V for coils, green is 12 V for backlighting.

3.4 OLED connectors

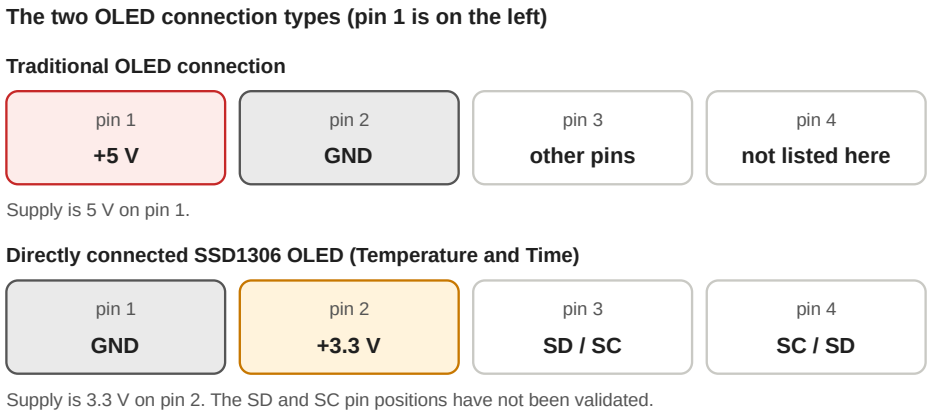


Figure 7 — The two OLED connection types

The OLED connectors are **not the same** on every display:

- **Traditional OLED connection:** +5 V on pin 1 and GND on pin 2.
- **Directly connected SSD1306 OLEDs** (the Temperature and Time displays): GND on pin 1 and +3.3 V on pin 2.

The SD and SC (data and clock) pin positions of the directly connected displays have not been validated. Do not plug one type of cable into the other type of connector: pins 1 and 2 are reversed and the supply voltages differ.

4 Lowrance AirMap GPS

`LowranceAirMapGps` makes the PC the GPS for a **Lowrance AirMap 2000C**. It receives the aircraft position from whichever bridge is running (Prepar3D / FSX, MSFS, DCS World or X-Plane) and sends it to the GPS over a serial cable as NMEA sentences twice a second.

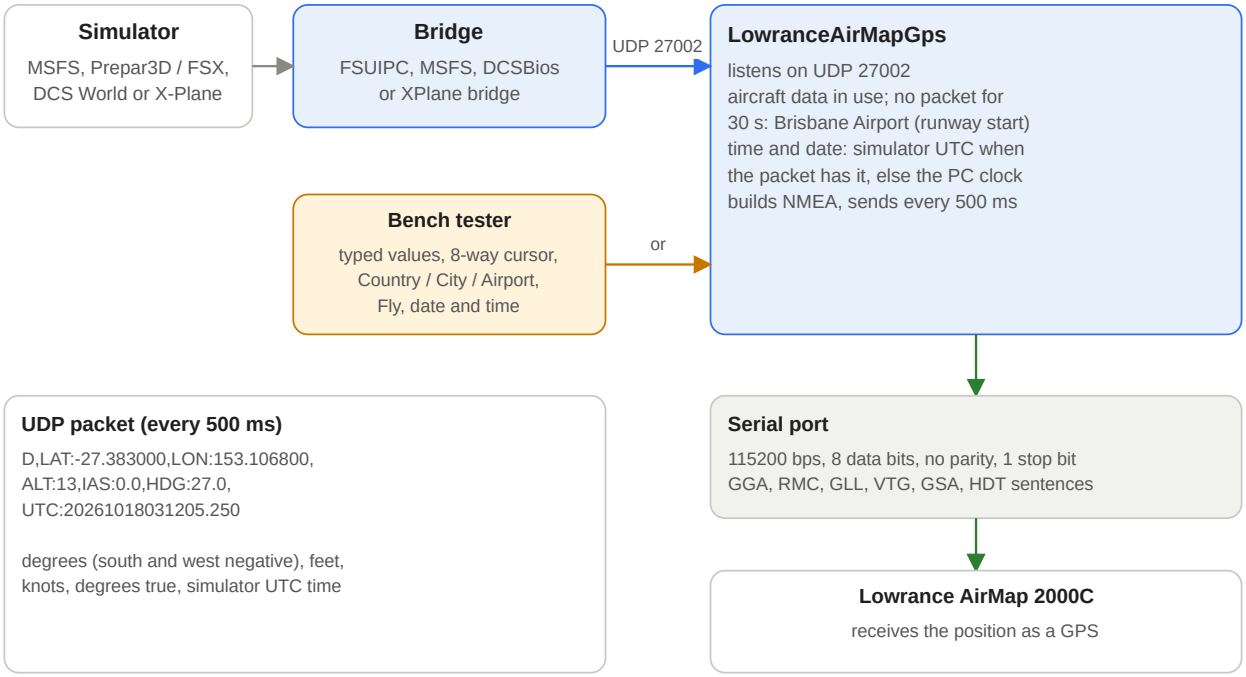


Figure 8 — From the simulator to the GPS

4.1 Setting it up

1. Connect the GPS to the PC with its serial cable (a USB serial adapter works).
2. Run `LowranceAirMapGps.exe`. The first time it asks which serial port the GPS is on; it remembers the answer (*Change serial port...* changes it later). The port runs at **115200 bps, 8 data bits, no parity, 1 stop bit**.
3. Start the simulator (MSFS, Prepar3D / FSX, DCS World or X-Plane) and its bridge on the same PC. All four bridges send the aircraft information to UDP port 27002 without any setting; if the GPS interface runs on a different PC, change `GpsHost` in the bridge's `Form1.cs`. With DCS World set the bridge's **Mission time zone** so the GPS shows the right UTC time.
4. The window shows the serial port state, the number of packets received, where the position comes from, the values being sent and the time source. The sentences being sent are listed at the bottom.

4.2 What the GPS is told

- **Position, altitude, airspeed and heading** from the simulator (any aircraft, in every simulator: the bridges read the standard position, altitude, speed and heading values, not a particular aircraft's gauges), sent as GGA, RMC, GLL, VTG, GSA and HDT sentences. NMEA has no airspeed sentence a GPS accepts, so the airspeed is sent as the GPS speed.
- **Date and time** are the simulator's UTC, ticking in real time between packets (with DCS the mission clock less the **Mission time zone** set in the DCS bridge). If the packets have no time, or nothing has arrived for 30 seconds, the PC clock is used. The window says which.
- **No data for 30 seconds** (the simulator is paused, closed or not running): the GPS is told it is at Brisbane Airport, at the start of the runway, speed 0. It returns to the live position as soon as packets arrive again.

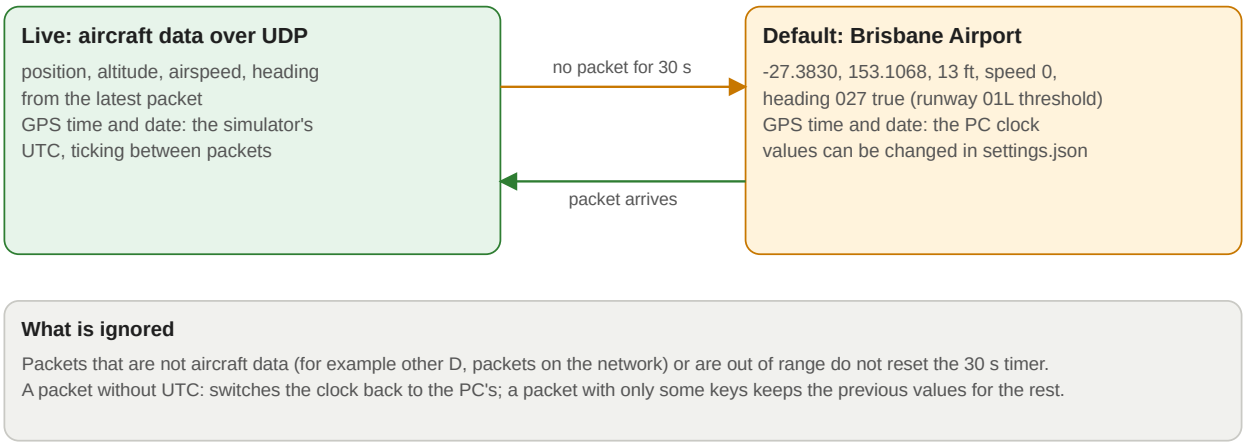


Figure 9 — Live data and the Brisbane default

The default position and the UDP port can be changed in `%AppData%\LowranceAirMapGps\settings.json`. The Brisbane coordinates came from SkyVector and were not checked against the AIP.

4.3 Testing without the simulator

`LowranceAirMapGpsBenchTester` sends the same packets from a window:

- Choose a **Country, City and Airport** to put the aircraft on the ground at that airport.
- Type the **latitude, longitude, altitude, speed and heading**, and press Enter after each.
- Use the **8-way cursor** (the buttons or the arrow keys) to move the position in steps (default 100 m); hold to repeat.
- Tick **Fly** to move along the heading at the current speed; **Time x** speeds it up (1 to 100).
- Set the **date and time (UTC)** the GPS should show, or press *PC clock*.

Use one sender at a time (a bridge or the tester): a second one just overwrites the position. The GPS interface has not yet been tested with the real AirMap 2000C, so check that the GPS shows the position and that it accepts the 115200 bps rate.

5 Program windows

This chapter shows every window of the Windows programs in `Jet Ranger/ C# Code/`. The screenshots were taken from the current builds. The two bridges were captured with no simulator running, so they show their disconnected state; with a simulator connected the left box fills with the values being sent and the list shows the lamps lighting.

Program	Used for	Section
FSUIPCWinformsAutoCS	Bridge for Prepar3D and FSX	Bridges
MSFSWinformsAutoCS	Bridge for MSFS 2020 and 2024	Bridges
DCSBiosBridge	Bridge for DCS World (through DCS-BIOS)	Bridges
XPlaneBridge	Bridge for X-Plane (through UDP datarefs)	Bridges
JetRangerHealthMonitor	Shows which boards are alive and the last voltage reading from each	Health Monitor
SwingTester	Swings every gauge, lamps, OLEDs, tests the Auto Pilot, Audio and Huey panel LEDs, resets boards, sets brightness	SwingTester
StepperVSITester	Sends real and raw values to the stepper boards and the servo	StepperVSITester
JetRangerRadioControllerEmulator, JetRangerUpperControllerEmulator	Stand in for the radio and upper boards	Emulators
LowranceAirMapGps	Feeds the Lowrance AirMap 2000C	GPS interface
LowranceAirMapGpsBenchTester	Sends aircraft data to the GPS interface for testing	GPS bench tester

> **Careful:** the SwingTester, StepperVSITester and the two emulators send packets to the real boards (the addresses are in the boxes) as soon as you use their controls. Close them, or point them at the right addresses, before you press anything while the cockpit is powered.

5.1 Bridges

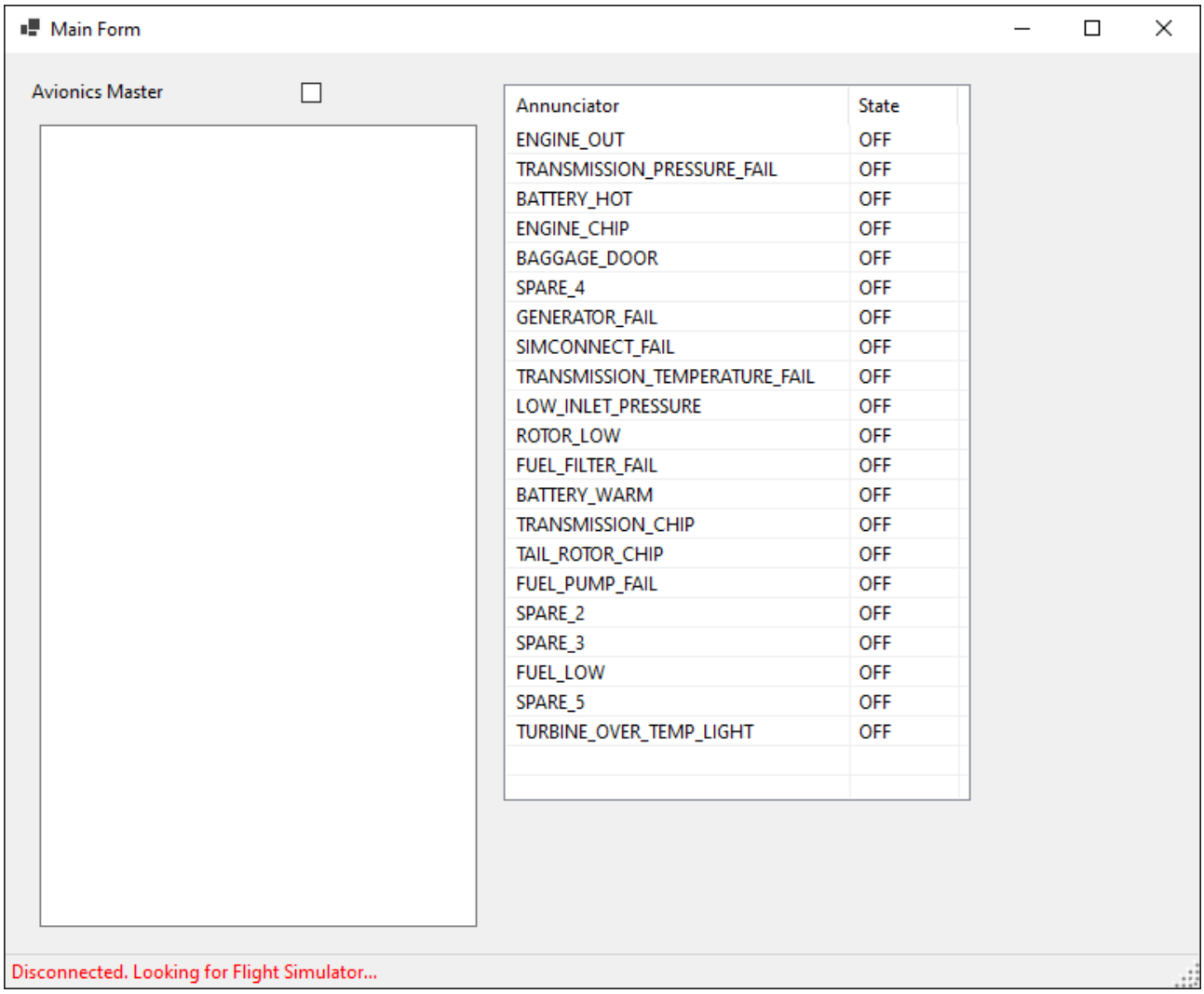


Figure 10 — FSUIPCWinformsAutoCS (Prepar3D and FSX)

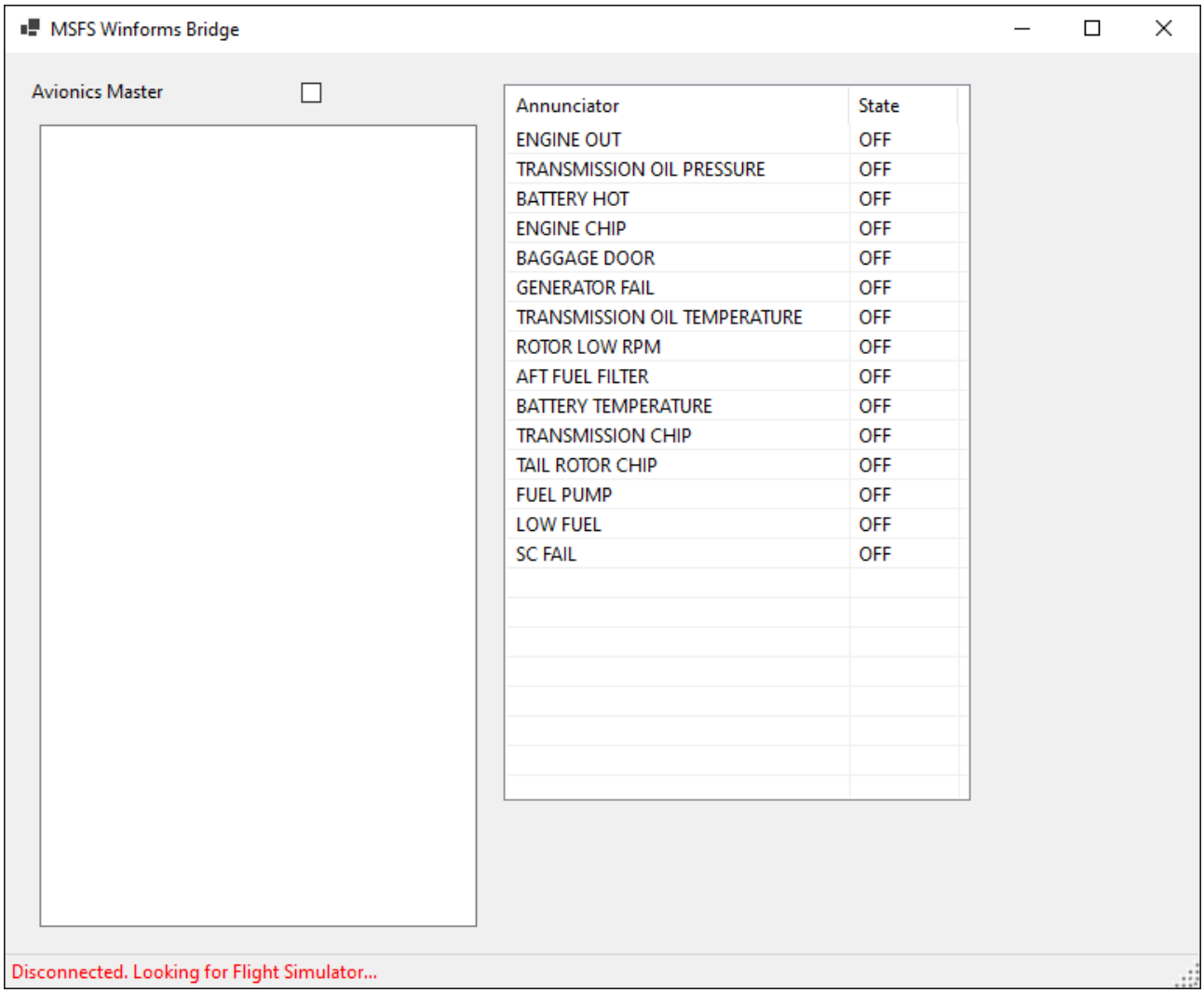


Figure 11 — MSFSWinformsAutoCS (MSFS 2020 and 2024)

Both bridges have one window with the same layout.

- **Avionics Master** check box: follows the simulator's avionics master switch and can also be used to switch it.
- **Readout box** (left): the values being sent to the boards, one per line (airspeed, altitude, bus voltages, engine and transmission values, radio frequencies and, in the MSFS bridge, the GPS position and heading). It is empty until the simulator is connected.
- **Annunciator list** (right): every warning lamp with its state, ON or OFF. The FSUIPC bridge lists the 21 lamp bits of the DodoSim aircraft; the MSFS bridge lists the 15 lamps of the panel.
- **Status bar** (bottom): *Disconnected. Looking for Flight Simulator...* in red until the simulator is found, then the connection and (MSFS) the aircraft name in green, or orange when the simulator is paused or loading and no data arrives.

Run only one bridge at a time.

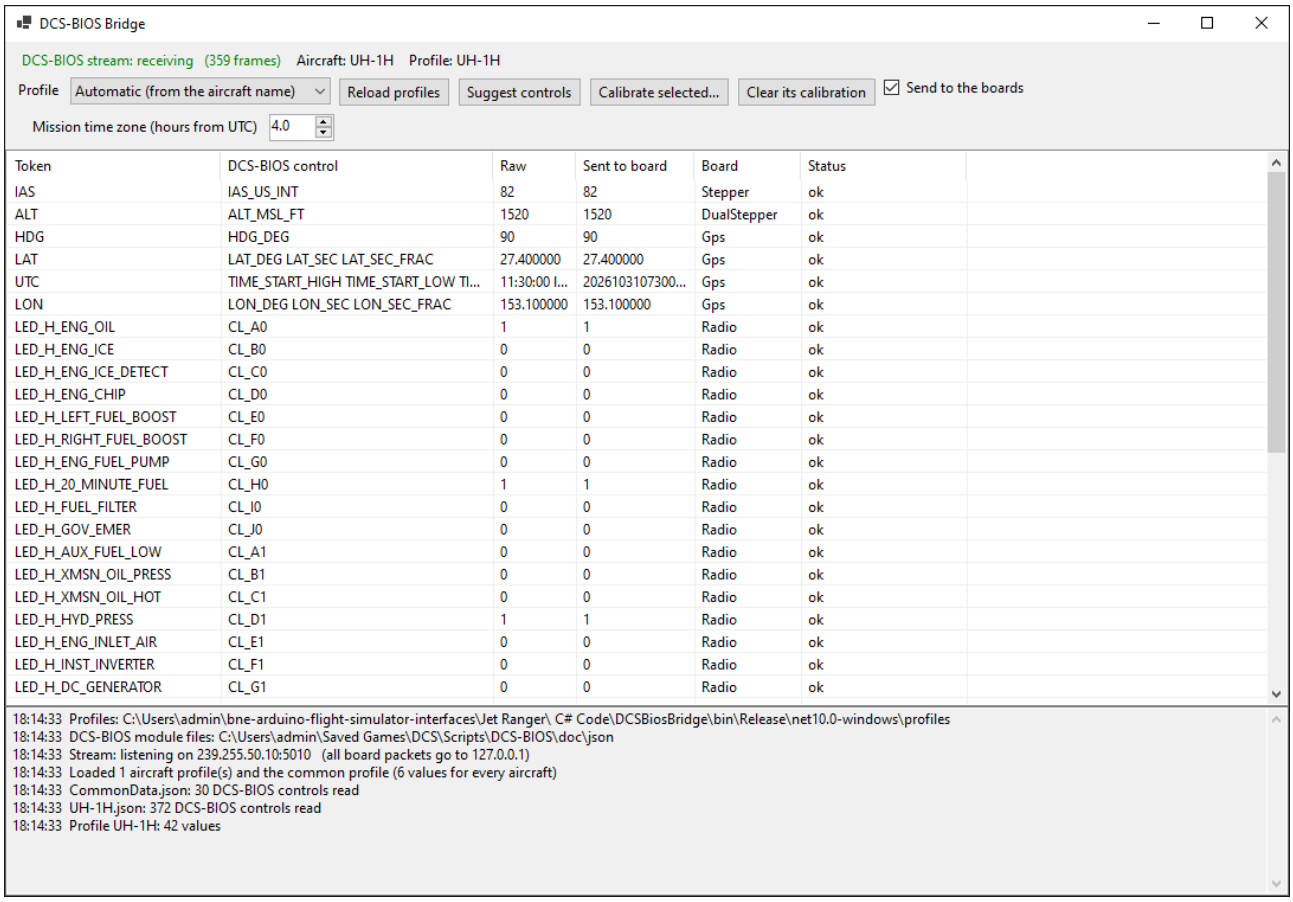


Figure 12 — DCSBiosBridge (DCS World) with a synthetic UH-1H stream: three caution lights on, the common values, the mission clock with its time zone box and the Huey LED rows (scroll for the gauges)

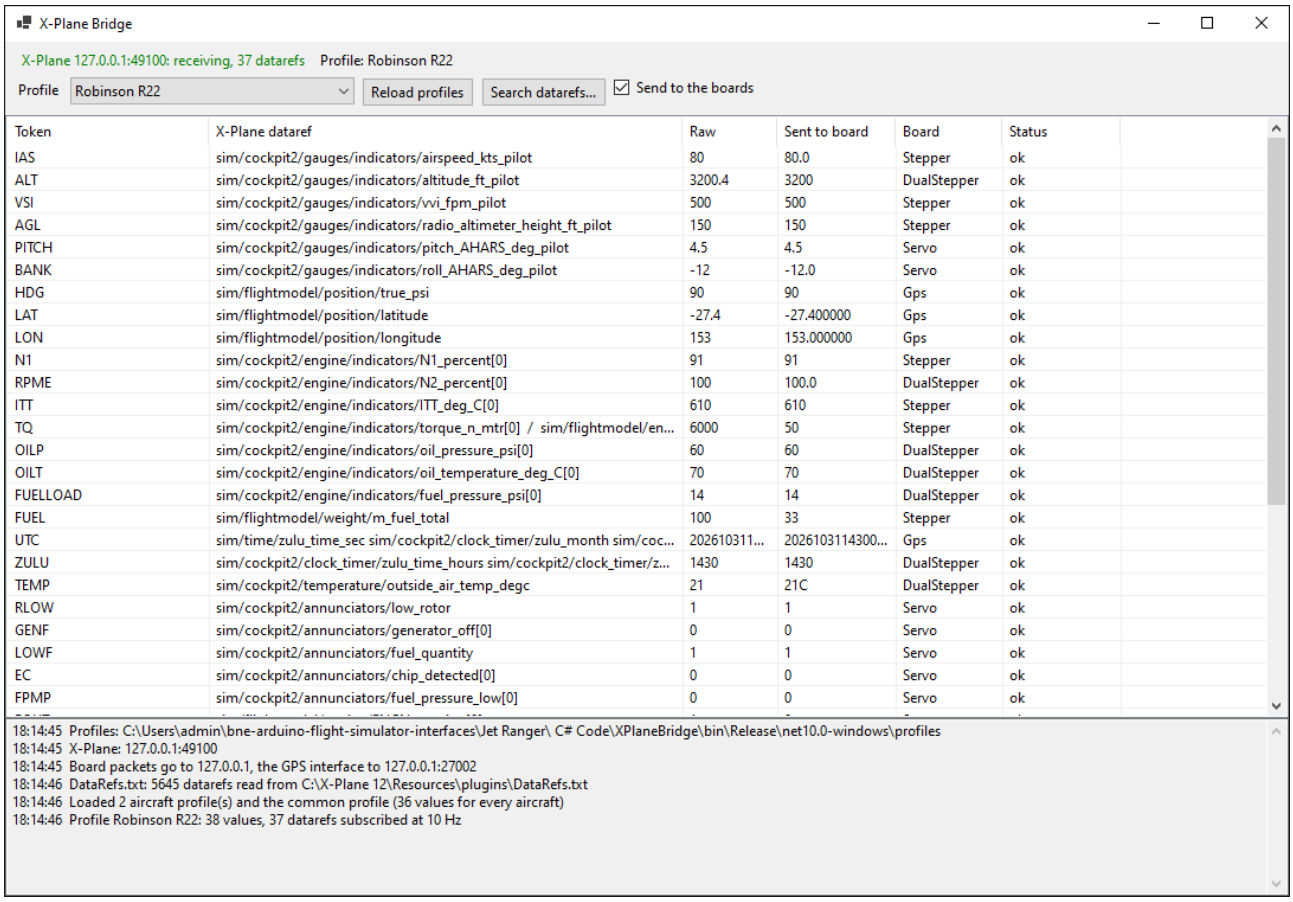


Figure 13 — XPlaneBridge (X-Plane) against a fake X-Plane with the Robinson R22 profile (scroll for the Huey LEDs, rotor speed and bus voltage)

DCSBiosBridge and **XPlaneBridge** have a different window: a status line (the DCS-BIOS stream and aircraft, or the X-Plane address and whether data is arriving), a **Profile** box (Automatic, or an aircraft chosen by hand), **Reload profiles**, a **Send to the boards** check box, a table of every token with its source control or dataref, the raw value, the value sent and the board, and a log at the bottom. The DCS bridge also has **Suggest controls** (finds controls for a new profile), **Calibrate selected...** and **Clear its calibration** for the Huey needle gauges; the X-Plane bridge has **Search datarefs...**. See Running the DCS bridge and Running the X-Plane bridge.

5.2 Health Monitor

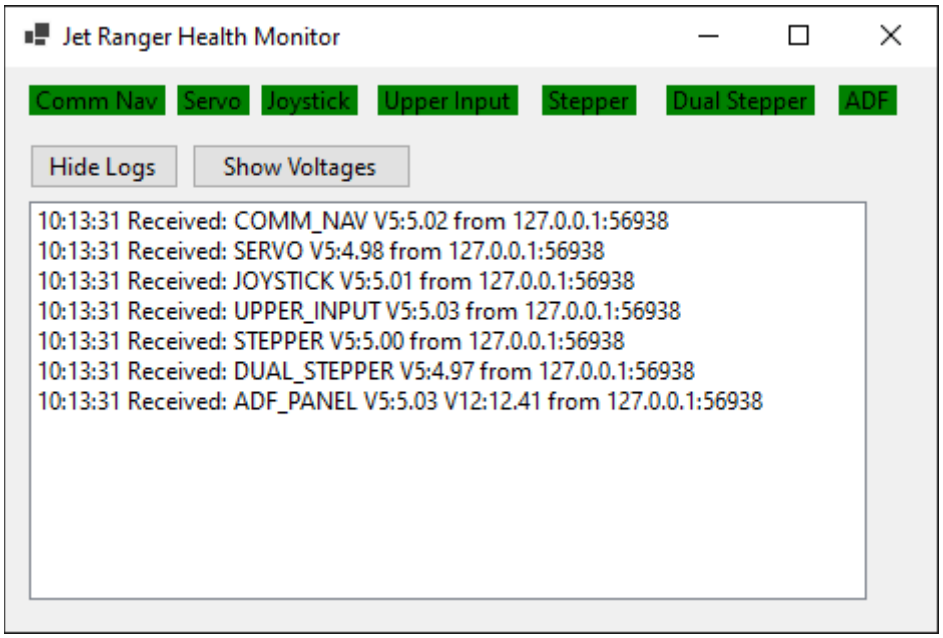


Figure 14 — JetRangerHealthMonitor: all seven boards heard

One coloured label per board: Comm Nav, Servo, Joystick, Upper Input, Stepper, Dual Stepper and ADF (the combined ADF and Upper board lights both the Upper Input and ADF labels). Each turns green when that board's 10 second heartbeat arrives on UDP 13137. The **Hide Logs** button hides the list of received heartbeats (the sender address and time of each one) and shrinks the window; it then reads **Show Logs**.

The **Show Voltages** button opens a table with one row per board:

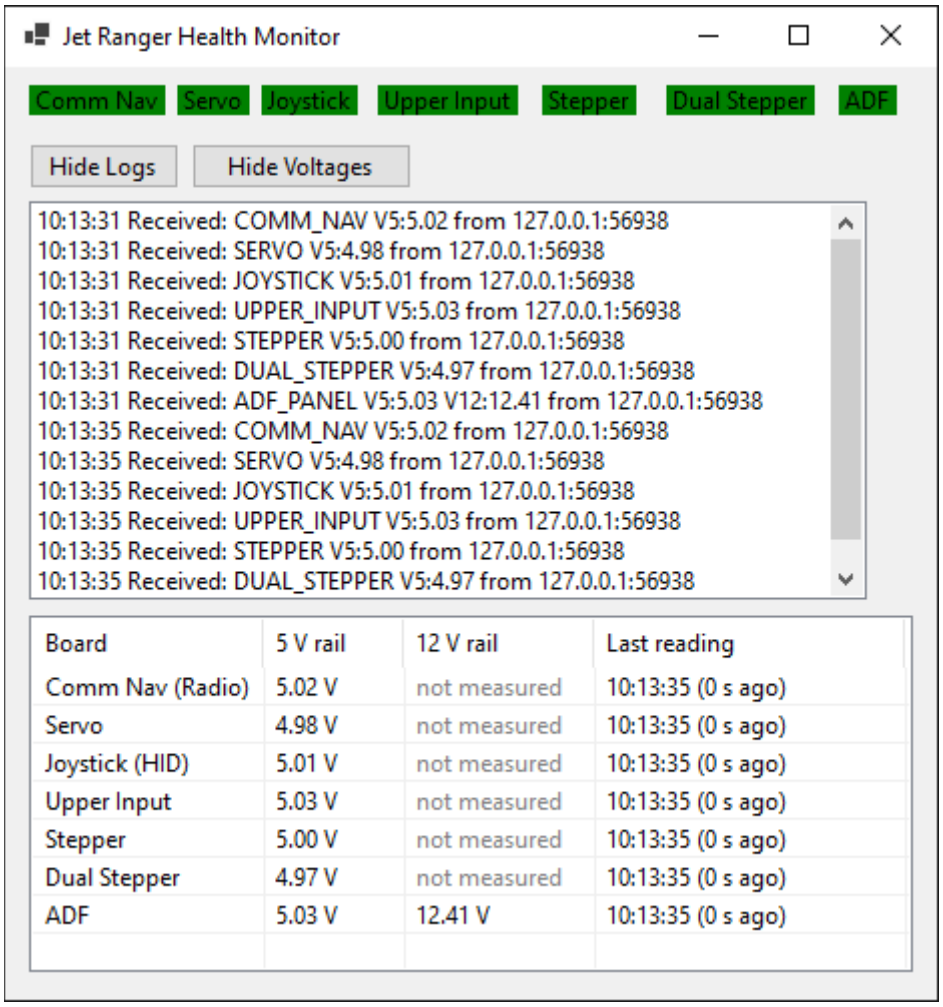


Figure 15 — Voltage table

Each board adds its 5 V rail to its heartbeat, and the ADF also adds its 12 V rail, so the table shows the last 5 V reading from every board, the **12 V** reading for the ADF only (the other rows say "not measured") and when the reading arrived. A reading outside its limits (5 V outside 4.75 to 5.25, 12 V outside 10.8 to 15.0) is orange, and a reading that is more than 35 seconds old, or that has not arrived yet, is grey. A board that has not been reflashed since the voltages were added shows "no reading" but still lights green. *Hide Voltages* closes the table.

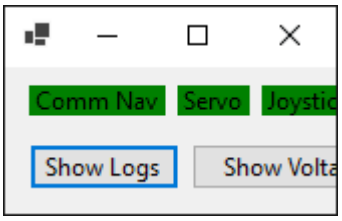


Figure 16 — Logs hidden

A board that goes quiet turns orange after 15 seconds and red after 30 seconds.

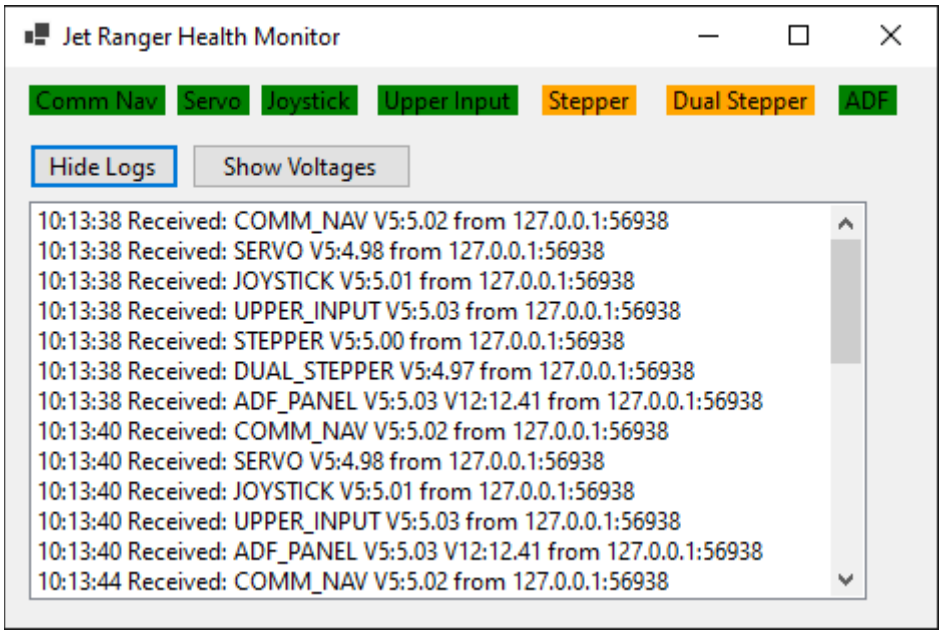


Figure 17 — Stepper and Dual Stepper quiet for 15 seconds: orange

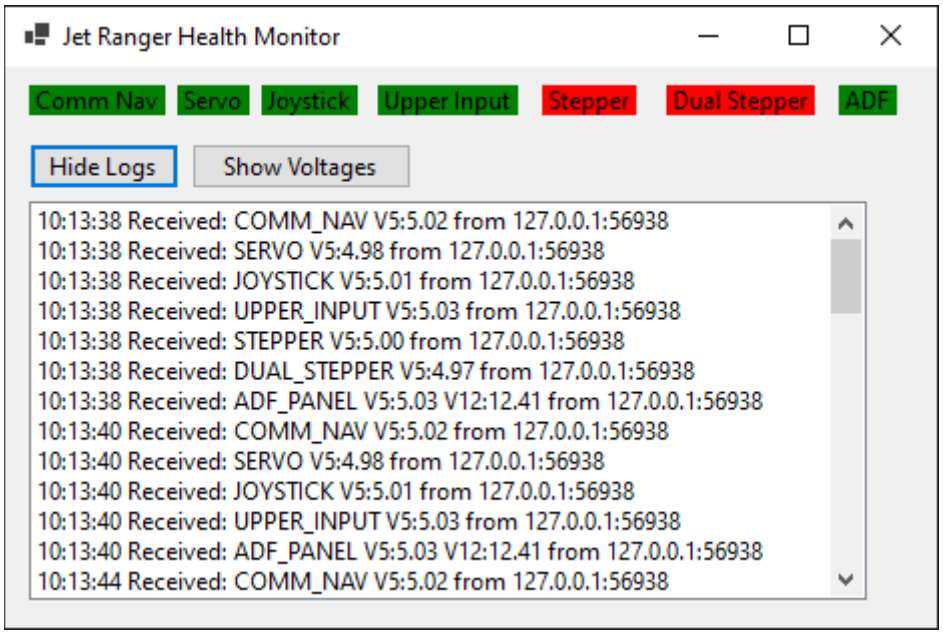


Figure 18 — Stepper and Dual Stepper quiet for 30 seconds: red

5.3 SwingTester

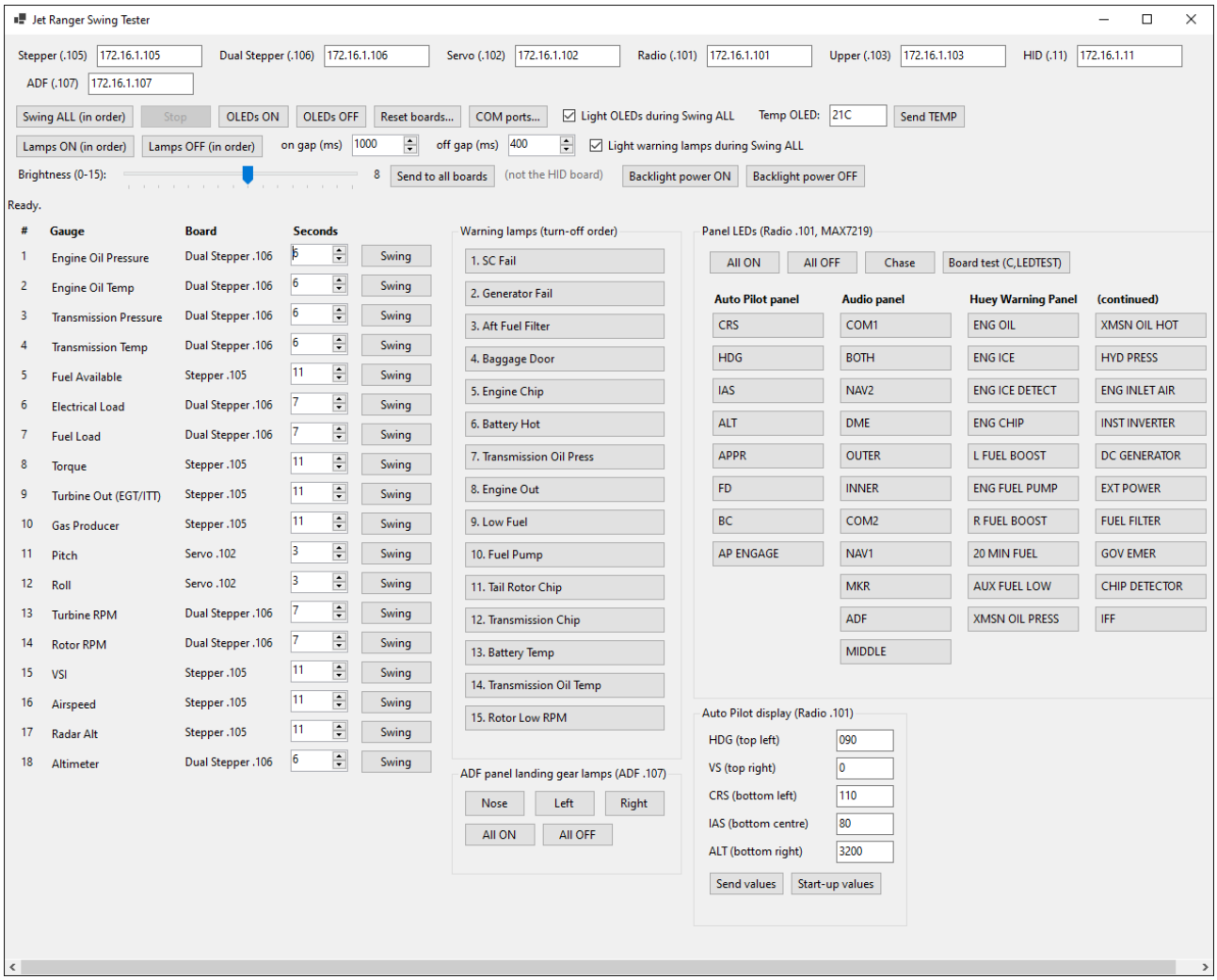


Figure 19 — SwingTester main window

- **Top row:** the IP addresses of the boards (Stepper, Dual Stepper, Servo, Radio, Upper, ADF and HID; the Upper address .103 is the retired board, the ADF and Upper board is .107).
- **Swing ALL (in order)** runs the 18 gauge swings in the fixed order, **Stop** cancels the remaining waits, **OLEDs ON / OFF** fill or restore the Dual Stepper's four OLEDs, **Reset boards...** opens the reset dialog, and **Light OLEDs during Swing ALL** controls the OLEDs. **Temp OLED** with **Send TEMP** sends text to the Dual Stepper's temperature display.
- **Lamps ON (in order)** and **Lamps OFF (in order)** with the **on gap** and **off gap** times, and **Light warning lamps during Swing ALL**.
- **Brightness (0-15)** slider with **Send to all boards** sends the backlight brightness to every board except the USB HID board (the Radio, Servo, Stepper, Dual Stepper and ADF and Upper, and the retired Upper address); **Backlight power ON / OFF** send `BPWR` to those boards to test the main bus rule.
- **Status line:** what was last sent and to which boards.
- **Panel LEDs group** (far right): one toggle for each of the 39 Auto Pilot, Audio and Huey Warning Panel LEDs (gold when on; the Huey LEDs are in two columns, *Huey Warning Panel* and *(continued)*), which lights that LED on its own through the Radio controller (`D,NAVCOM1:1,LED_xxx:1/0` or `LED_H_xxx`). **All ON** and **All OFF** do all 39, **Chase** lights them one at a time for the *on gap* time (click again to stop; a full chase of 39 takes about 39 seconds at the default), and **Board test (C,LEDTEST)** lights all 39 with the Radio's own test. The Radio on COM20 was flashed with the Huey build on 31 October 2026; the LEDs have not been checked on the panel yet.

- **Auto Pilot display group** (far right, below the panel LEDs): five boxes for HDG, VS, CRS, IAS and ALT (start-up values 090, 0, 110, 80 and 3200). **Send values** sends them to the Radio for the Auto Pilot panel's OLED and **Start-up values** puts the start-up values back.
- **ADF panel landing gear lamps** (under the warning lamps): **Nose**, **Left** and **Right** toggles (gold when on) light the three lamps on the ADF panel through the ADF controller (172.16.1.107, set in the `ADF (.107)` box at the top), with **All ON** and **All OFF**.
- **COM ports...** opens a window that lists the PC's COM ports with the board type, the USB serial number and whether another program has the port open, and can restart a USB device (see COM ports window below).
- **Gauge rows** (left): the 18 gauges with the board each is on, the wait in seconds and a **Swing** button for each. The row being swung is highlighted gold.
- **Warning lamps** (right): one button for each of the 15 lamps, in turn-off order. A lamp that is on is gold.

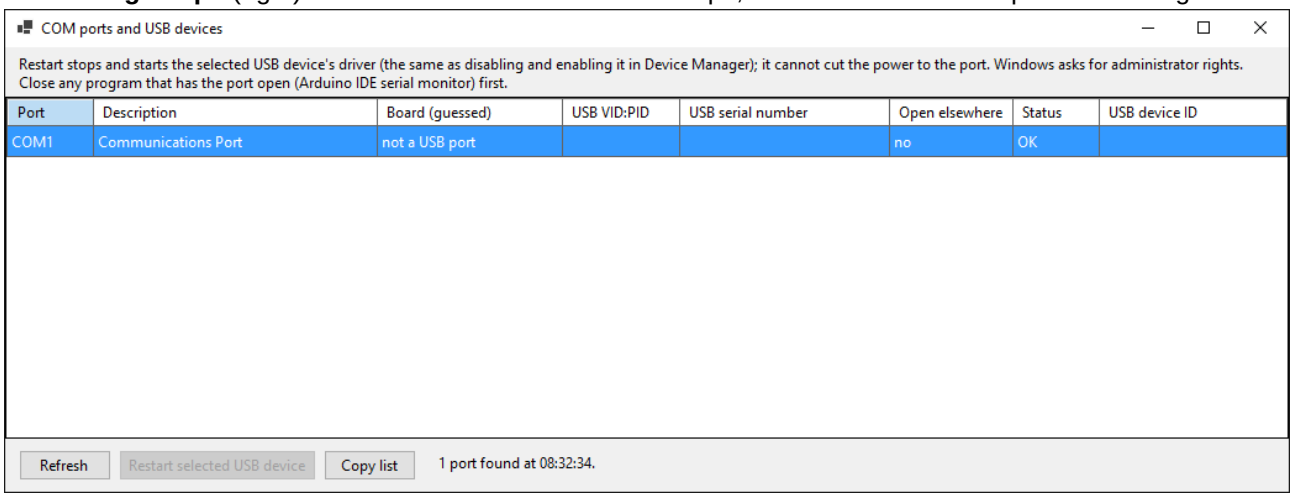


Figure 20 — The COM ports window

COM ports... lists every COM port Windows knows: the port, Windows' description, the board type (guessed from the USB vendor and product ids), the USB serial number (genuine Arduino boards have one; most clones do not), whether another program has the port open (for example the Arduino IDE serial monitor, which blocks flashing), the status and the USB device id. It refreshes by itself when a device is plugged in or removed. **Copy list** puts the table on the clipboard. **Restart selected USB device** stops and starts the selected board's USB device with `pnputil` (the same as disabling and enabling it in Device Manager, so the port reappears; it cannot cut the power): Windows asks for administrator rights, and the port must not be open in another program.

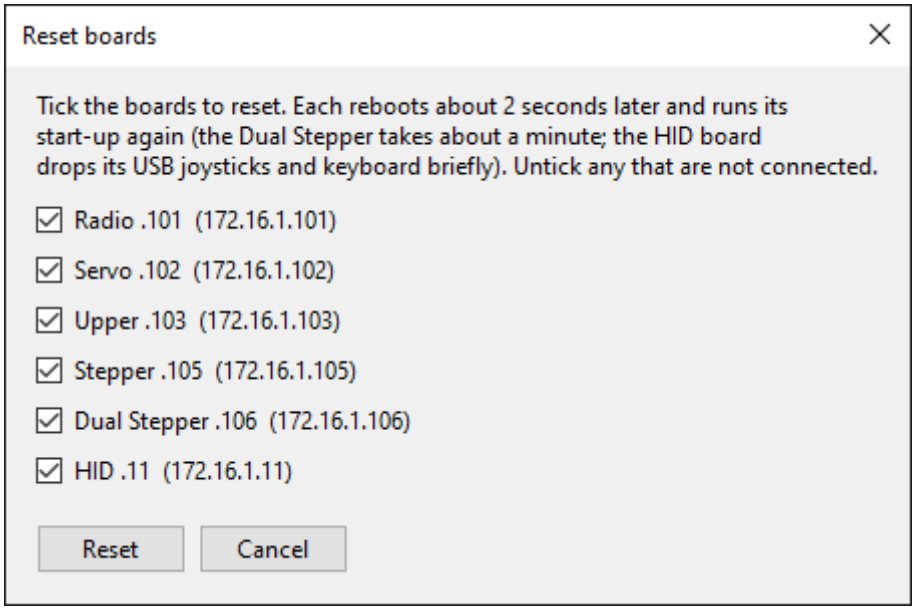


Figure 21 — The Reset boards dialog

The **Reset boards...** dialog has a tick box for each board (all ticked; untick any that are not connected). **Reset** sends `C,RESET:1` to each ticked board and **Cancel** closes the dialog without sending anything.

5.4 StepperVSITester

The window scrolls; both halves are shown.

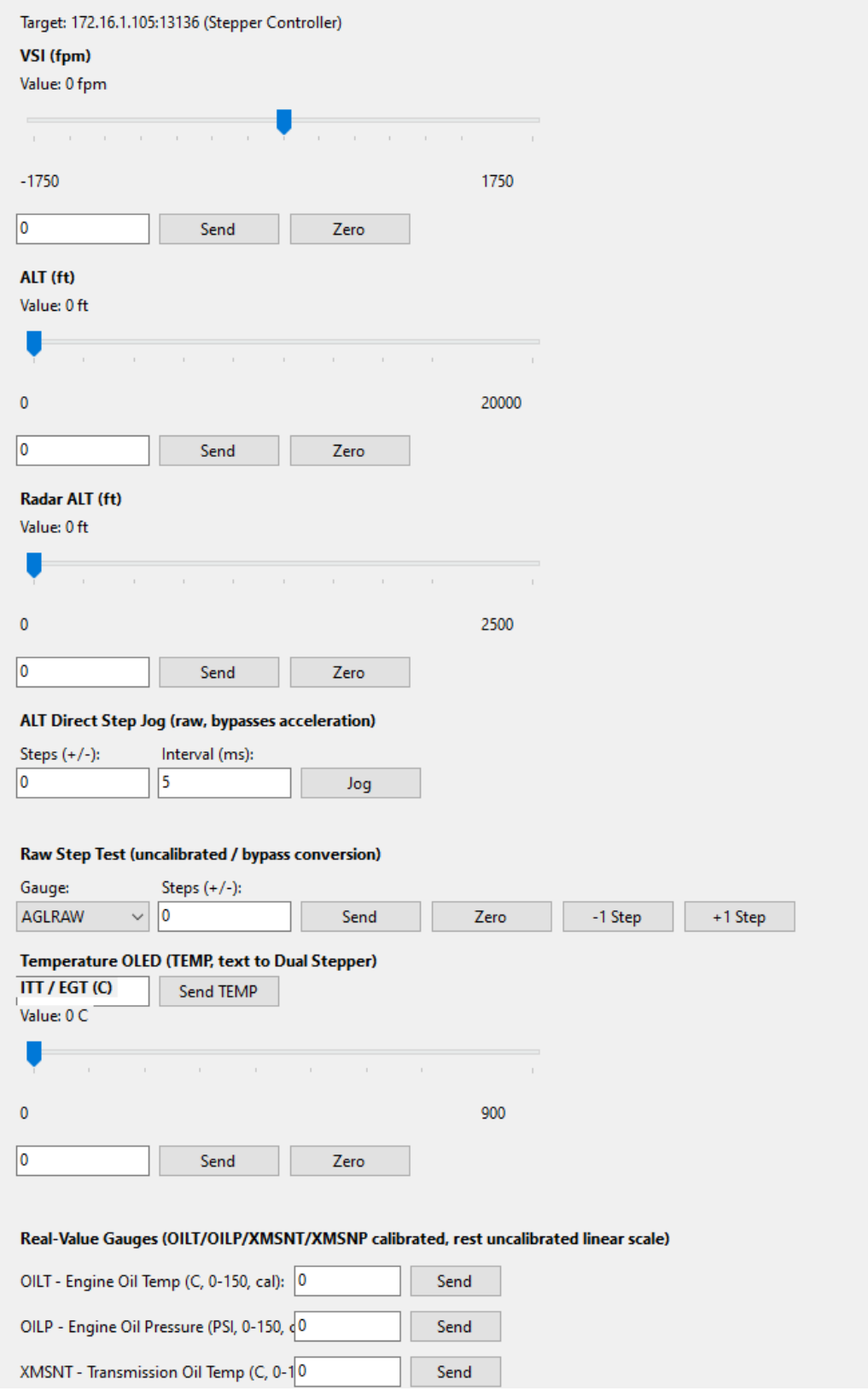


Figure 22 — StepperVSITester, top half

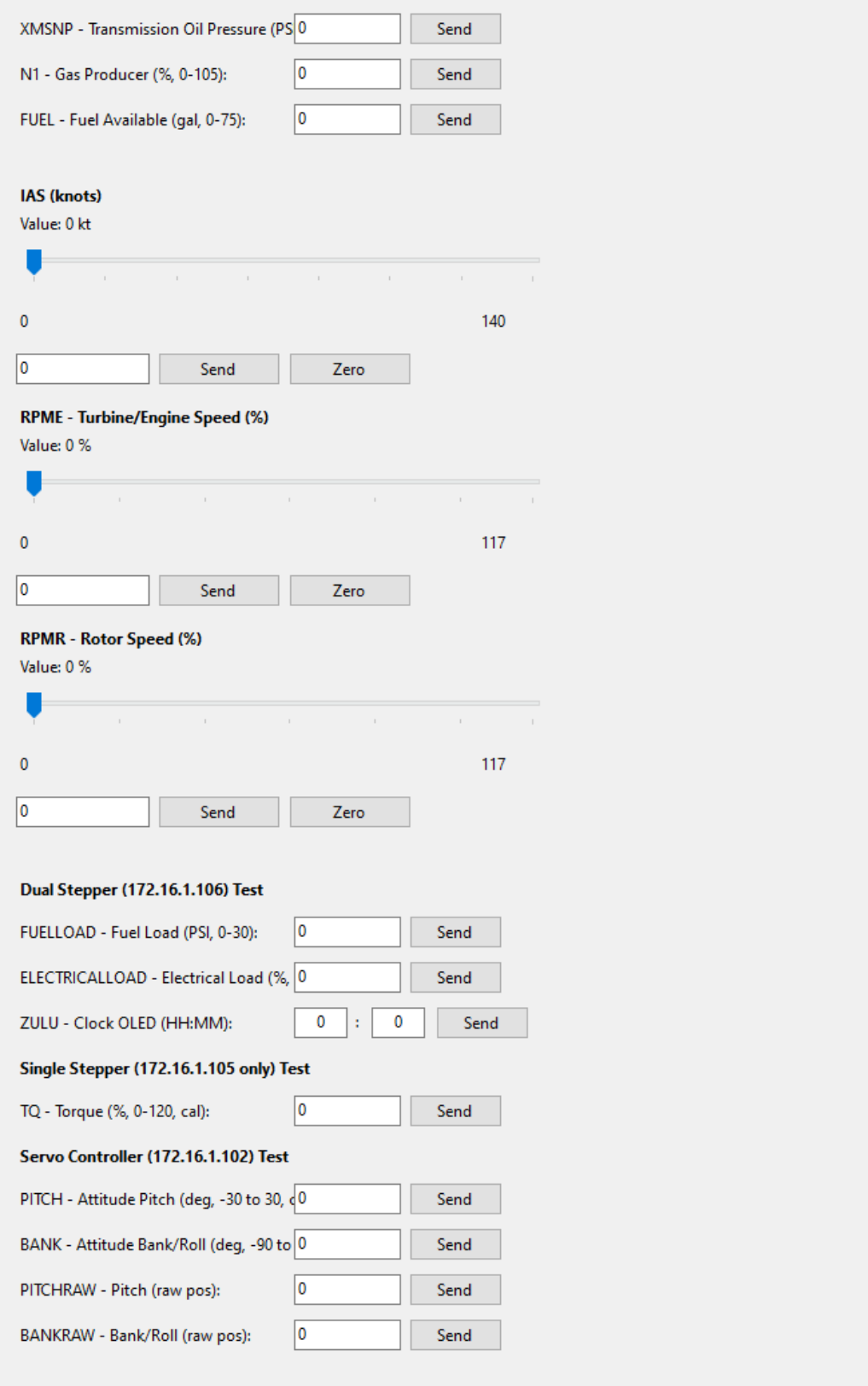


Figure 23 — StepperVSITester, bottom half

The target is shown at the top (the Stepper board, 172.16.1.105 port 13136). Each gauge has a slider, a text box, **Send** and **Zero**:

- **VSI (fpm)**, **ALT (ft)**, **Radar ALT (ft)**: real values for the vertical speed, altimeter and radar altimeter.
- **ALT Direct Step Jog** sends a signed step count at a fixed interval, bypassing the acceleration ramp.

- **Raw Step Test** sends raw steps to the chosen gauge code (with **-1 Step** and **+1 Step** nudge buttons).
- **Temperature OLED** sends text to the Dual Stepper's temperature display, and the **ITT / EGT** slider below it sends turbine out temperature.
- **Real-Value Gauges** rows for engine oil temperature and pressure, transmission oil temperature and pressure, gas producer and fuel.
- **IAS**, **RPME** and **RPMR** sliders for airspeed, engine speed and rotor speed.
- **Dual Stepper test** rows (fuel load, electrical load, the clock), **Single Stepper test** (torque) and **Servo Controller test** (pitch and bank, in degrees or as raw servo positions).

5.5 Emulators

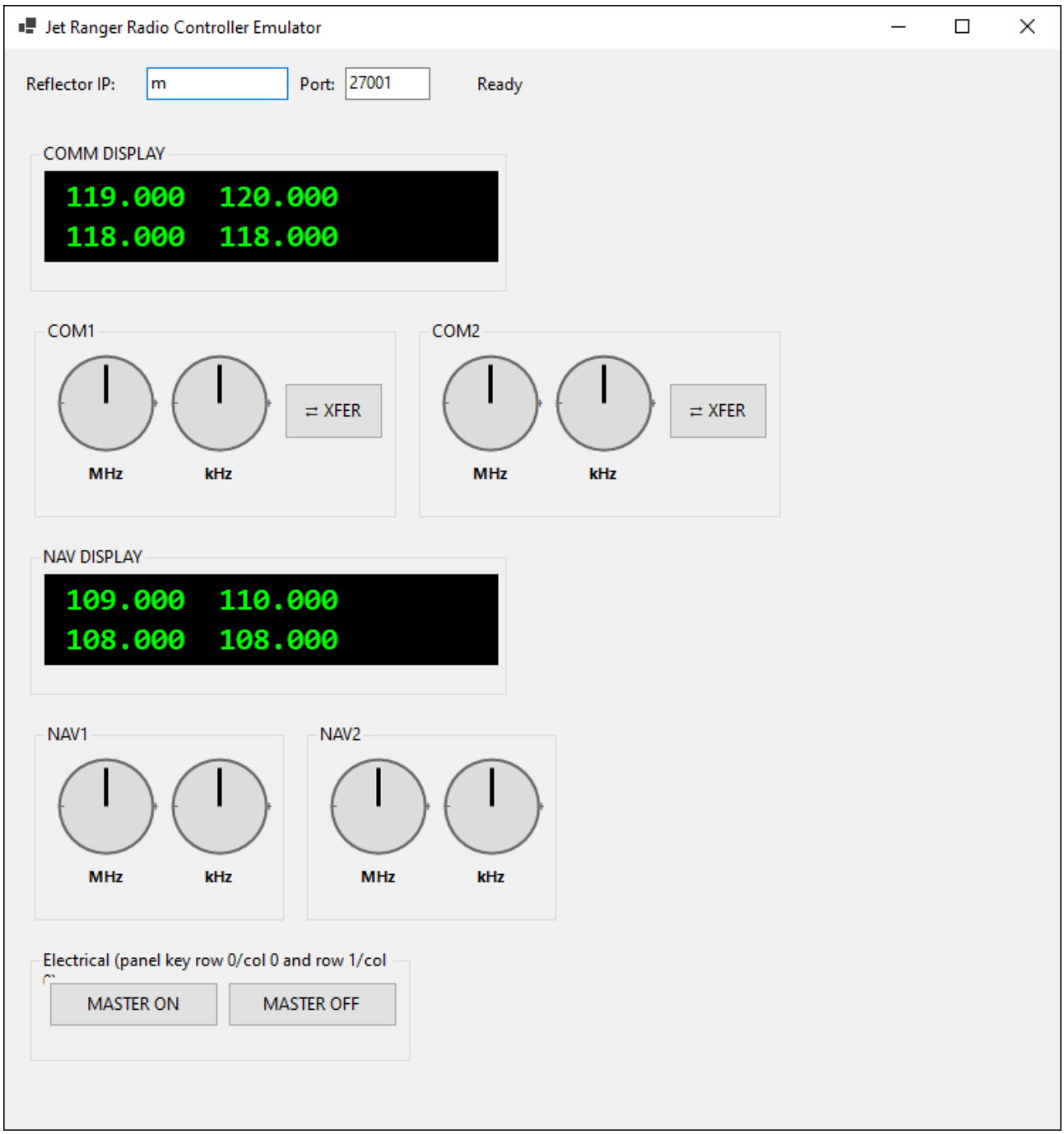


Figure 24 — JetRangerRadioControllerEmulator

The **Radio controller emulator** stands in for the radio board. Set the reflector (bridge) address and port at the top. The COMM and NAV displays show the active and standby frequencies; each radio has a **MHz** and a **kHz** knob (click the right half to turn it up one detent, the left half to turn it down) and the COM radios have an **XFER** swap button. **MASTER ON** and **MASTER OFF** send the battery and alternator commands. Every control sends its packet to the bridge on UDP 27001 at once.

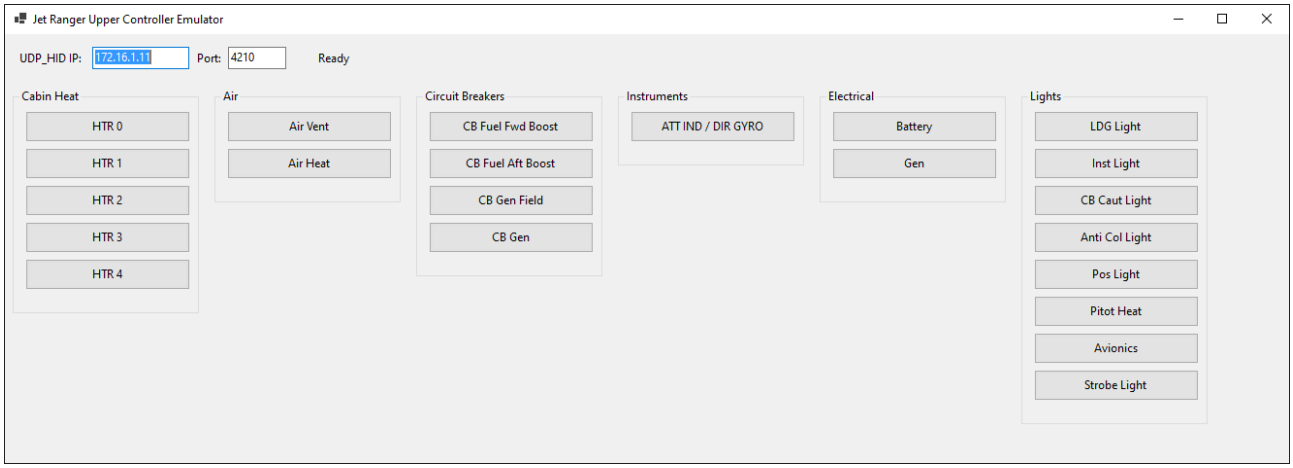


Figure 25 — JetRangerUpperControllerEmulator

The **Upper controller emulator** stands in for the upper panel (now on the combined ADF and Upper board; the switch numbers are unchanged). Set the USB HID board's address and port at the top (172.16.1.11, 4210). There is one toggle button per switch the firmware uses, in groups: Cabin Heat, Air, Circuit Breakers, Instruments, Electrical and Lights. A button turns green when its switch is on, and each press sends the switch change to the USB HID board.

5.6 GPS interface

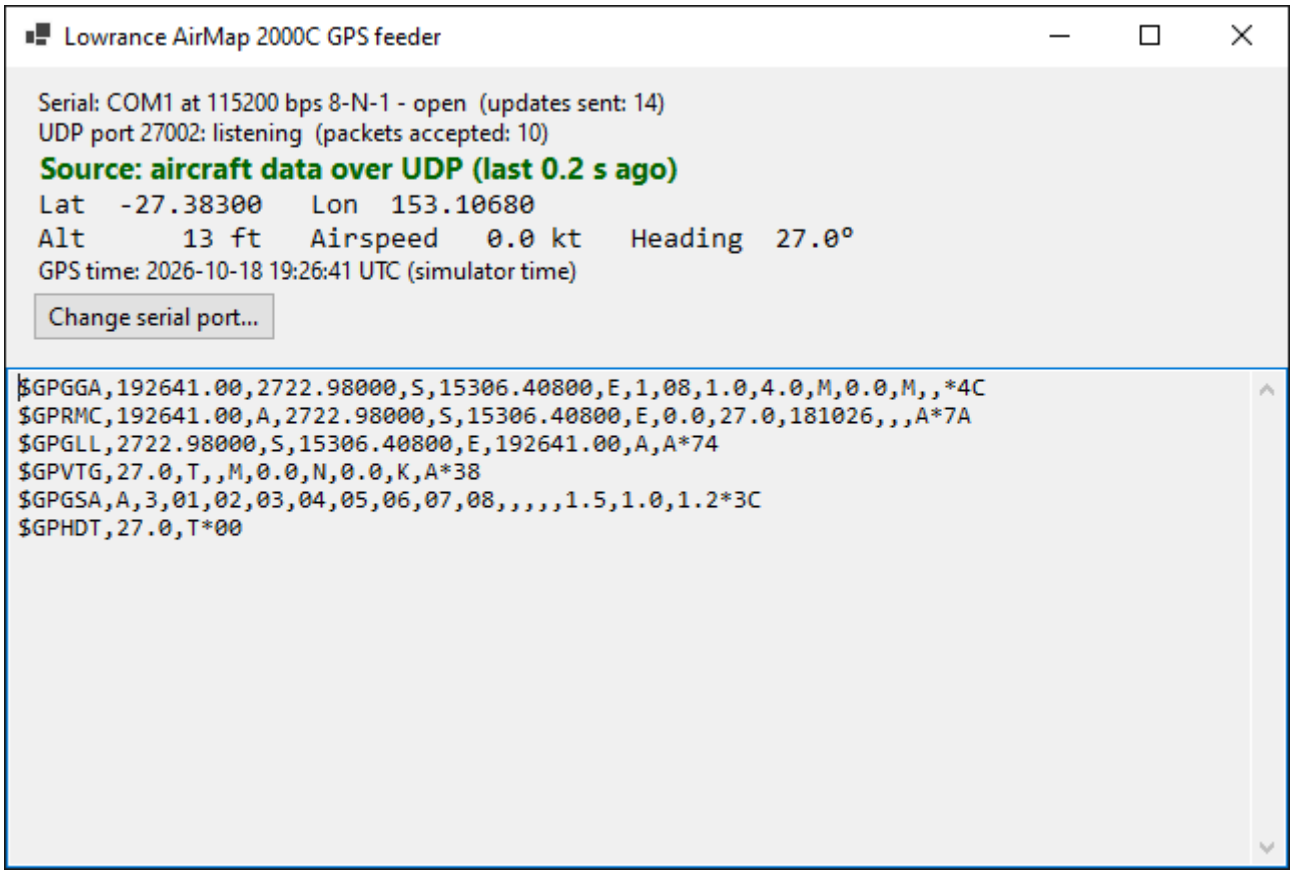


Figure 26 — LowranceAirMapGps

The window shows, from the top: the serial port, its speed and state and how many updates have been sent; the UDP port and the number of packets accepted; the **source** of the position (green *aircraft data over UDP*, orange *default - Brisbane Airport*); the latitude, longitude, altitude, airspeed and heading being sent; the **GPS time** and whether it comes from the simulator or the PC clock; the **Change serial port...** button; and the sentences being sent to the GPS.

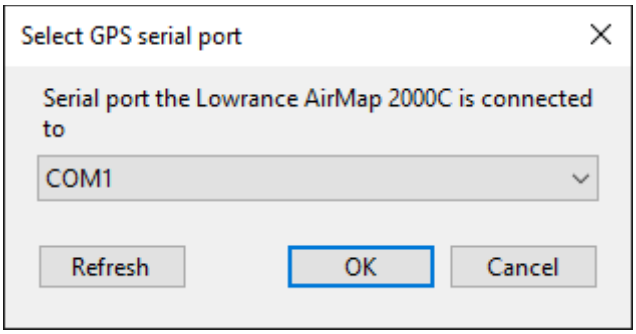


Figure 27 — The serial port dialog

The first time the program runs, or when the saved port is no longer present, this dialog asks which port the GPS is connected to. **Refresh** rereads the list of ports, **OK** saves the choice and **Cancel** leaves the port unselected (the program still runs, without sending to a GPS). **Change serial port...** opens the same dialog later.

5.7 GPS bench tester

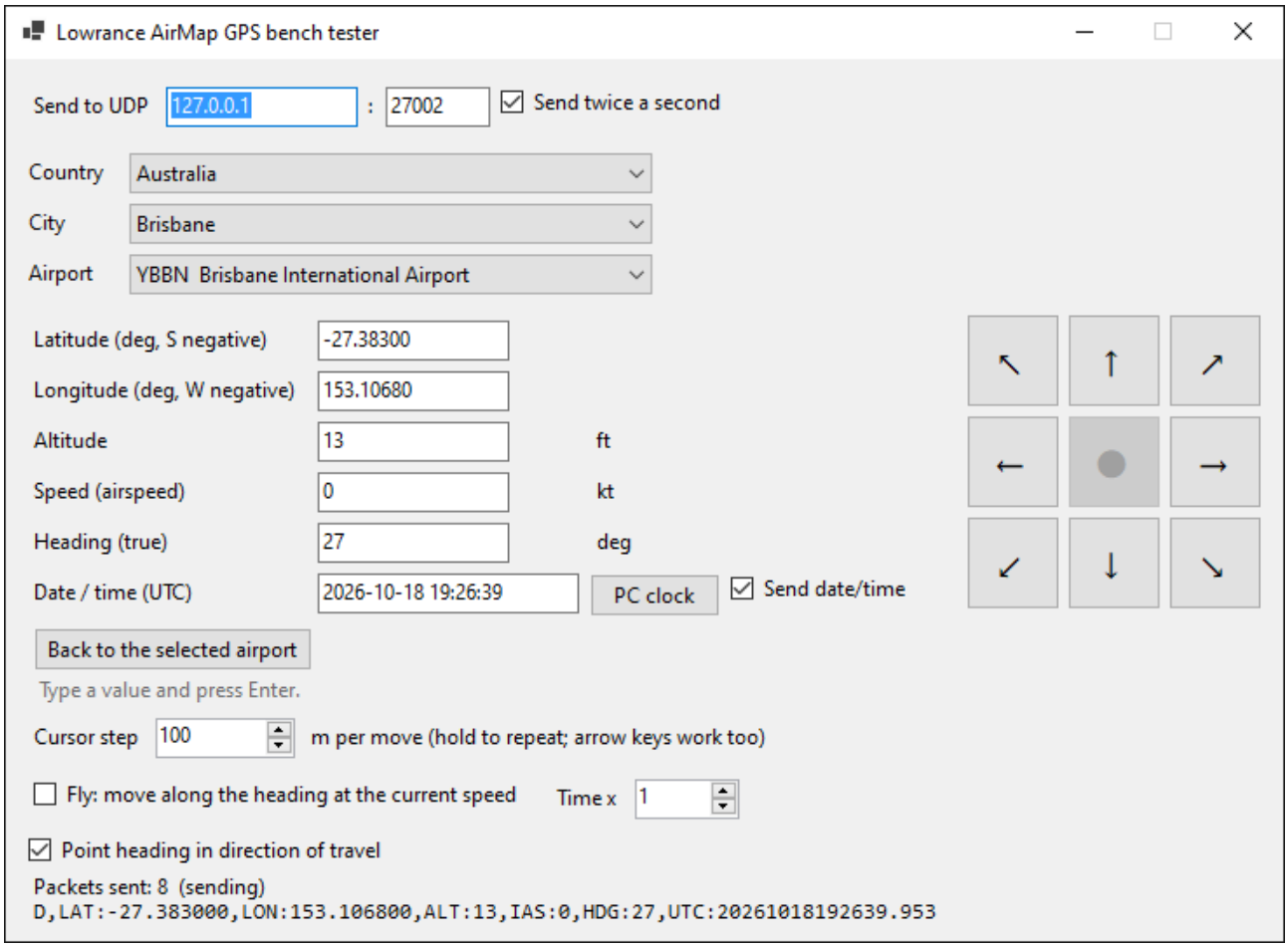


Figure 28 — LowranceAirMapGpsBenchTester

- **Send to UDP:** the address and port of the GPS interface (the same PC and port 27002 by default) and a check box to turn the sending on and off.
- **Country, City and Airport:** choose an airport to put the aircraft on the ground there.
- **Latitude, Longitude, Altitude, Speed and Heading:** type a value and press Enter (a value that cannot be read turns the box pink).
- **Date / time (UTC):** the date and time the GPS should show, **PC clock** to use the PC's, and **Send date/time** to include it.
- **Back to the selected airport** returns to the airport chosen above.
- **Eight-way cursor** (right): moves the position by the **Cursor step** in metres; the arrow keys work too. **Point heading in direction of travel** sets the heading to the direction moved.
- **Fly** moves the aircraft along its heading at its speed; **Time x** speeds it up.
- **Bottom:** the number of packets sent and the packet being sent.

6 Starting each simulator

This chapter is the start-up procedure for each simulator: **MSFS**, **Prepar3D / FSX**, **DCS World** and **X-Plane**. Every procedure includes the **Lowrance GPS interface**. You can follow it by hand, or let the start-up script do the PC side for you.

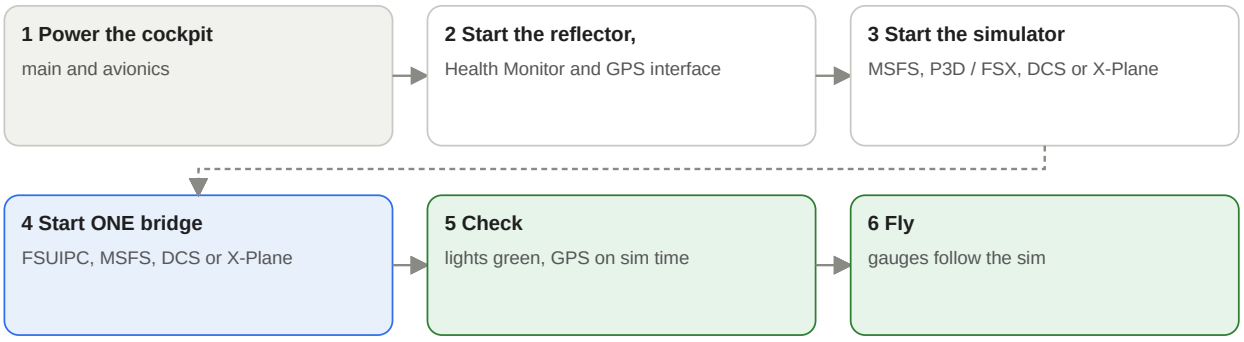


Figure 29 — Start-up order

6.1 The start-up script

`Start Jet Ranger.ps1` (in the `Jet Ranger` folder, with a `Start Jet Ranger.bat` that can be double-clicked) starts the PC programs for the simulator you choose, in the right order, and checks them. It never starts a program that is already running, so it can be run again safely.

Step	What the script does
1	Checks that your **UDP debug reflector** is listening on port 27000. It does not start a second listener; if you set `Reflector` in `Start Jet Ranger.json` it starts that command when nothing is listening
2	Stops any **other bridge** that is running (only the four Jet Ranger bridge programs), because only one bridge may run at a time
3	Starts the **Health Monitor**
4	Starts the **Lowrance GPS interface**
5	Starts the **simulator** only when you give `-StartSimulator` **and** have set a program for it in `Start Jet Ranger.json` (a program path, or an address such as `steam://...` for a Steam game); otherwise it just reminds you to start it. It does not guess where a simulator is installed
6	Starts the **bridge** for the simulator (the DCS bridge gets the time zone)
7	Waits five seconds and checks that the bridge, the Health Monitor and the GPS interface are running, and that the GPS interface owns UDP port 27002

Windows may refuse to run `.ps1` scripts directly (*running scripts is disabled on this system*; this PC does). The `.bat` gets round that by starting PowerShell with the policy bypassed for that one run, so **use the `.bat`**: double-click it (it asks which simulator) or run it from a command window with the same options. Calling the script yourself needs `powershell -ExecutionPolicy Bypass -File "Start Jet Ranger.ps1" ...`.

```
"Start Jet Ranger.bat"
"Start Jet Ranger.bat" -Simulator MSFS
"Start Jet Ranger.bat" -Simulator P3D
"Start Jet Ranger.bat" -Simulator DCS -DcsTimeZone 4
"Start Jet Ranger.bat" -Simulator XPlane -XPlaneHost 192.168.1.20
"Start Jet Ranger.bat" -Simulator MSFS -StartSimulator
"Start Jet Ranger.bat" -Simulator DCS -DryRun
"Start Jet Ranger.bat" -Stop
```

Option	Meaning
`-Simulator`	`MSFS`, `P3D`, `FSX`, `DCS` or `XPlane`. Left out, the script shows a menu
`-DcsTimeZone`	DCS only: the mission's time zone in hours from UTC (4 for the Caucasus

Option	Meaning
	and the Persian Gulf, 3 for Syria, -8 for Nevada), so the GPS shows the right UTC time
`-XPlaneHost`	X-Plane only: the PC X-Plane runs on, when it is not this one
`-StartSimulator`	Also starts the simulator, from the path (or `steam://` address) set in `Start Jet Ranger.json`; nothing is started for a simulator with no path set
`-NoGps`, `-NoHealthMonitor`, `-NoReflector`	Leave that part out
`-DryRun`	Shows what would be done without starting or stopping anything
`-Stop`	Stops the bridges, the GPS interface and the Health Monitor (the reflector and the simulator are left running)

The optional settings file `Start Jet Ranger.json` (copy `Start Jet Ranger.sample.json` and fill in what you need) holds the `Reflector` command, the `Simulators` program paths, a default `DcsTimeZone` and a default `XPlaneHost`. A program that has not been built yet is reported by name: build it with `dotnet build -c Release` in its folder under `C# Code`. The script uses the newest build (Release or Debug) of each program.

The first time the GPS interface runs, and whenever its saved serial port is not on the PC, it shows a **Select GPS serial port** window and waits: pick the port the GPS cable is on (or press Cancel to run without a GPS). It does not listen for the simulator's data until you answer, so the script's check says it is waiting.

6.2 What is the same for every simulator

1. Power the cockpit (main and avionics). The boards start and swing their gauges.
2. Make sure the **debug reflector** is running (UDP 27000).
3. Start the **Health Monitor**. Its lights turn green as the boards report in (within about 10 seconds).
4. Start the **GPS interface** (`LowranceAirMapGps.exe`) with the GPS cable on the saved serial port. Until a bridge is sending it reports the Brisbane Airport default and the PC clock.
5. Start the **simulator** and load an aircraft.
6. Start **one** bridge, and only one.
7. Check three things: the bridge says it is connected and receiving; the Health Monitor lights are green and the gauges follow the simulator; and the **GPS window says the time source is the simulator** (*simulator time*) and shows the aircraft's position, not Brisbane Airport.

If you change simulator, close the bridge first (or let the script stop it) and start the other one: the two simulators' bridges send to the same boards.

6.3 MSFS 2020 and 2024

1. Reflector, Health Monitor and GPS interface as above.
2. Start MSFS and load a Jet Ranger.
3. Run `MSFSWinformsAutoCS.exe`. The status bar turns from red *Disconnected* to green *Connected to MSFS* with the aircraft name. Orange means the simulator is paused or loading.
4. The window lists the values being sent and the lamps. With the FlyInside B206 or the CowaSim Jet Ranger the bridge uses its estimated dials (see Aircraft with estimated dials).
5. The panel's switches reach MSFS as joystick buttons: bind them in the MSFS controls menu (the DS206 events are not available in MSFS).
6. The GPS interface takes the position, altitude, airspeed and heading from SimConnect and the simulator's zulu date and time.

Script: `"Start Jet Ranger.bat" -Simulator MSFS`.

6.4 Prepar3D and FSX

1. Reflector, Health Monitor and GPS interface as above.
2. Start Prepar3D (or FSX) with FSUIPC installed and the DodoSim 206 loaded. FSUIPC must be running.

3. Run `FSUIPCWinformsAutoCS.exe`. The status bar shows when it is connected. The window lists the values being sent and the 21 annunciator bits.
4. The panel's joystick buttons must be bound to the DS206 events in FSUIPC (see Installing the DodoSim event file); without that nothing happens in the simulator when you press a switch.
5. The GPS interface takes the position, altitude, airspeed and heading from FSUIPC offsets and the simulator's UTC time from FSUIPC.

Script: `"Start Jet Ranger.bat" -Simulator P3D` (or `FSX`).

6.5 DCS World

1. Reflector, Health Monitor and GPS interface as above.
2. DCS-BIOS must be installed once (see Running the DCS bridge): `Saved Games\DCS\Scripts\DCS-BIOS` and the `Export.lua` line. It is already installed on the cockpit PC.
3. Start DCS and **load a mission and sit in the cockpit** of the aircraft (the UH-1H Huey has a profile; any aircraft drives the airspeed, altitude, heading and GPS). DCS-BIOS only sends data while a mission is running.
4. Run `DCSBiosBridge.exe`. The stream line turns green (*receiving*) and shows the aircraft name and the profile it chose.
5. Set **Mission time zone (hours from UTC)** to the theatre's offset (or give `-DcsTimeZone` to the script), so the GPS time is right. The date is the PC's.
6. Fly the Huey and calibrate the needle gauges if they are marked *uncalibrated* (see Running the DCS bridge).
7. The panel's switches reach DCS as joystick buttons: bind them in the DCS controls.
8. The GPS interface follows the aircraft's position, altitude, airspeed and heading, and the mission clock.

Script: `"Start Jet Ranger.bat" -Simulator DCS -DcsTimeZone 4`.

6.6 X-Plane

1. Reflector, Health Monitor and GPS interface as above.
2. In X-Plane open **Settings, Network** and allow incoming UDP connections (the receive port is 49000 by default).
3. Start X-Plane and load an aircraft.
4. Run `XPlaneBridge.exe`. It finds X-Plane from its beacon (or tries `127.0.0.1:49000`; use `-XPlaneHost` or `--xplane-host` if X-Plane is on another PC). The status line turns green and shows how many datarefs are arriving.
5. In the **Profile** box choose the aircraft's profile (*Robinson R22*, *Sikorsky S-76*) or *Generic (common values, any aircraft)*. X-Plane does not tell the bridge which aircraft is loaded.
6. If a gauge stays at zero, use **Search datarefs...** to find the right dataref and add it to the profile.
7. The panel's switches reach X-Plane as joystick buttons: bind them in the X-Plane joystick settings.
8. The GPS interface follows the aircraft's position, altitude, airspeed and heading, and X-Plane's zulu clock and date (the PC's year).

Script: `"Start Jet Ranger.bat" -Simulator XPlane`.

6.7 If something does not come up

Symptom	What to check
The script says a program is not built	Build it with `dotnet build -c Release` in its folder under `C# Code`
The script says nothing is listening on UDP 27002	The GPS interface is waiting at its *Select GPS serial port* window: choose the port
The GPS still shows Brisbane Airport	No bridge is sending (check the bridge status), or the simulator is paused (no data for 3 seconds stops the bridge sending)
The GPS time is hours out with DCS	Set the bridge's Mission time zone

Symptom	What to check
The bridge never connects	MSFS: aircraft loaded and not paused. Prepar3D / FSX: FSUIPC running. DCS: in the cockpit with DCS-BIOS installed. X-Plane: incoming UDP allowed
Gauges move but nothing is on the GPS	Another program owns UDP 27002 (close it), or two bridges are running

7 Network

All devices sit on one wired `172.16.1.x` subnet with fixed addresses; there is no DHCP on the boards.

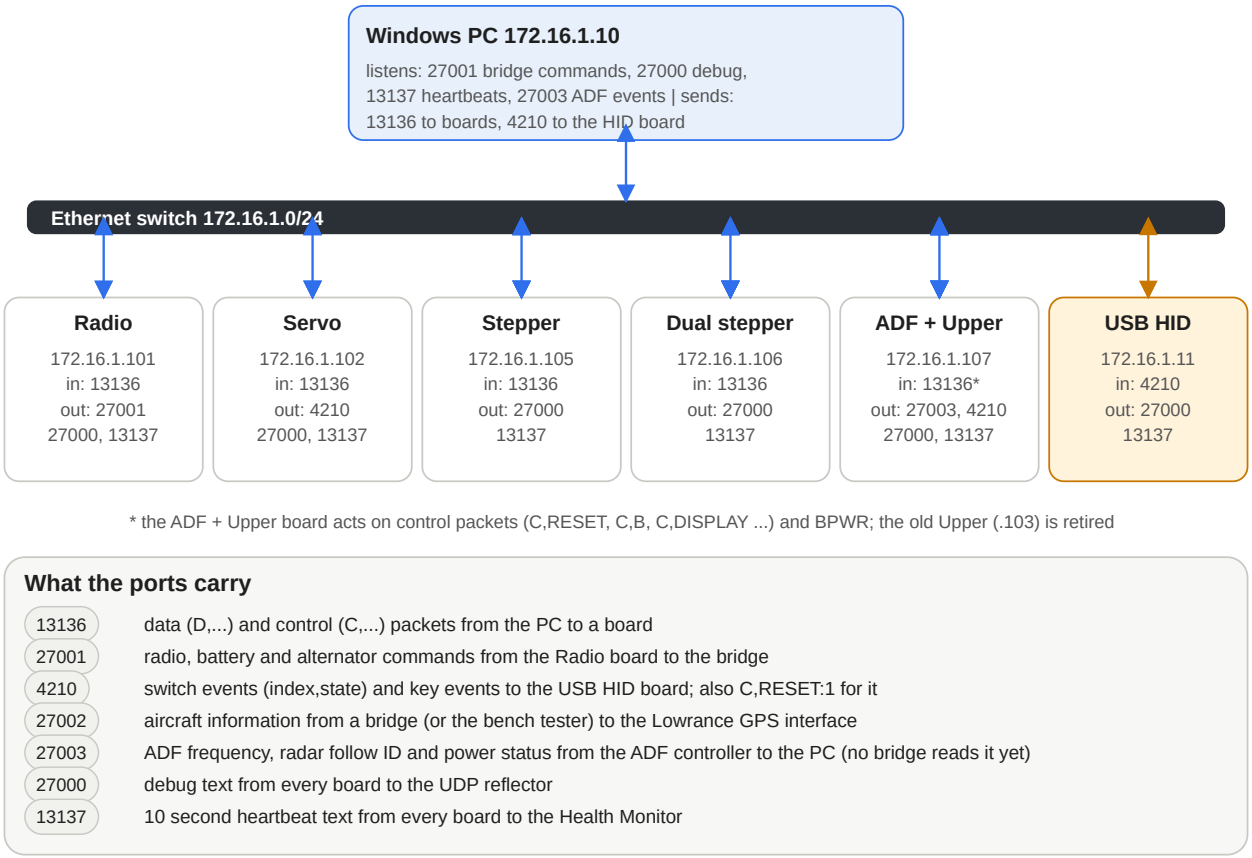


Figure 30 — Who talks to whom

Device	IP address	MAC	Keep-alive text
Primary PC (simulator, bridge, debug listener, Health Monitor)	172.16.1.10		
USB HID board	172.16.1.11	DE:AD:BE:EF:FE:ED	`JOYSTICK`
Radio controller	172.16.1.101	A8:61:0A:9E:83:01	`COMM_NAV`
Servo board	172.16.1.102	A8:61:0A:9E:83:02	`SERVO`
Stepper board	172.16.1.105	A8:61:0A:67:83:69	`STEPPER`
Dual stepper board	172.16.1.106	A8:61:0A:67:83:6A	`DUAL_STEPPER`
ADF and Upper controller	172.16.1.107	A8:61:0A:67:83:6B	`ADF_PANEL` and `UPPER_INPUT`

Every live board has its own IP and MAC address. The old Upper controller (172.16.1.103) was merged into the ADF controller on 31 October 2026 and no longer exists; the merged board sends both heartbeat names so the Health Monitor still shows an Upper light. The bridges and the SwingTester still send some packets to .103, which go nowhere. The firmware sends its debug and keep-alive traffic to `172.16.1.10` (`reflectorIP`), so the primary PC must hold that address, or `reflectorIP` must be changed and the sketch uploaded again.

UDP port	Direction	Purpose
13136	PC to boards	Gauge, lamp and radio data (`D,`) and control (`C,`) packets
27001	Radio and ADF controllers to PC bridge (FSUIPC and MSFS bridges only)	Standby frequencies, radio swaps, and the Auto Pilot, Audio and ADF panel controls (for example `AP_MASTER`, `GEAR_DOWN`)
27002	Bridge to the GPS interface (same PC)	Aircraft position, altitude, airspeed, heading and simulator time, every 500 ms

UDP port	Direction	Purpose
27003	ADF controller to the PC	ADF frequency, radar follow ID and power status (no bridge reads it yet)
5010	DCS to the DCS bridge	The DCS-BIOS export stream (multicast 239.255.50.10, on the PC)
49000	X-Plane and the X-Plane bridge	Dataref requests to X-Plane and its answers (X-Plane's beacon is multicast 239.255.1.1 port 49707)
4210	Servo, Radio and ADF and Upper controllers to USB HID board	Switch changes (`index,state`) and keypad keys (`KEY,key,state`)
13137	Boards to Health Monitor	Keep-alive text every 10 seconds
27000	Boards to debug listener	Debug messages, each prefixed with the board name

7.1 Packet format

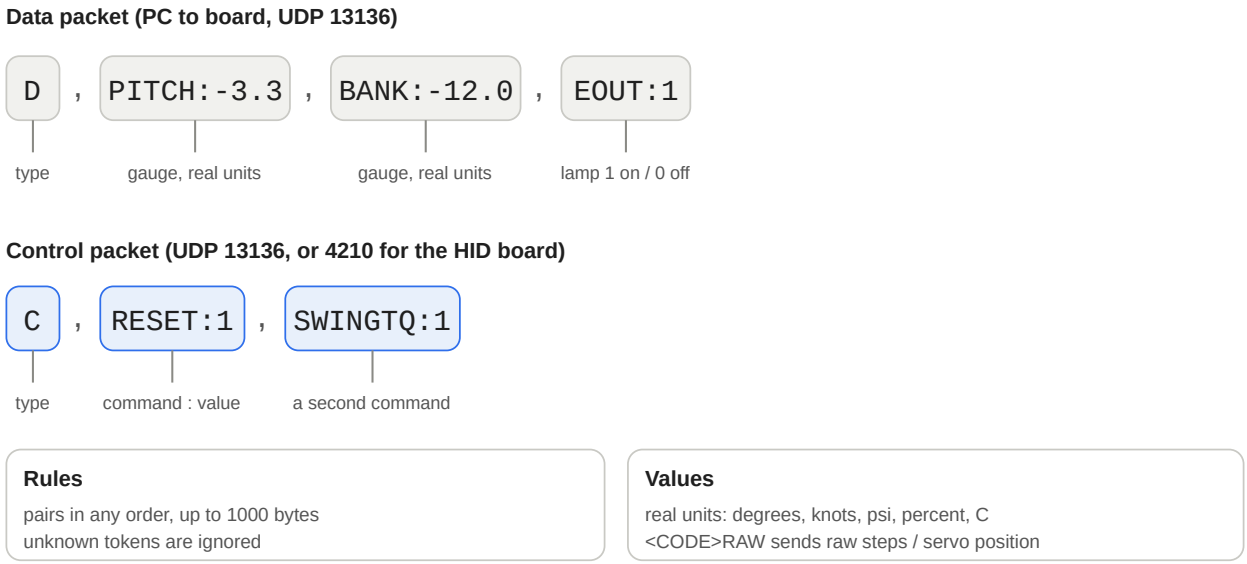


Figure 31 — Data and control packets

A packet can carry any number of pairs, in any order, up to 1000 bytes. Unknown tokens are ignored, so the PC can send only the values that changed. For example `D,PITCH:5,BANK:-10,LOWF:0` sent to the servo board sets 5 degrees pitch, minus 10 degrees bank and turns off Low Fuel.

To test without the simulator, send a packet from a PC on the subnet, for example with PowerShell:

```
$u = New-Object System.Net.Sockets.UdpClient
$b = [Text.Encoding]::ASCII.GetBytes('D,PITCH:5,RLOW:1')
$u.Send($b, $b.Length, '172.16.1.102', 13136)
```

7.2 What a board does

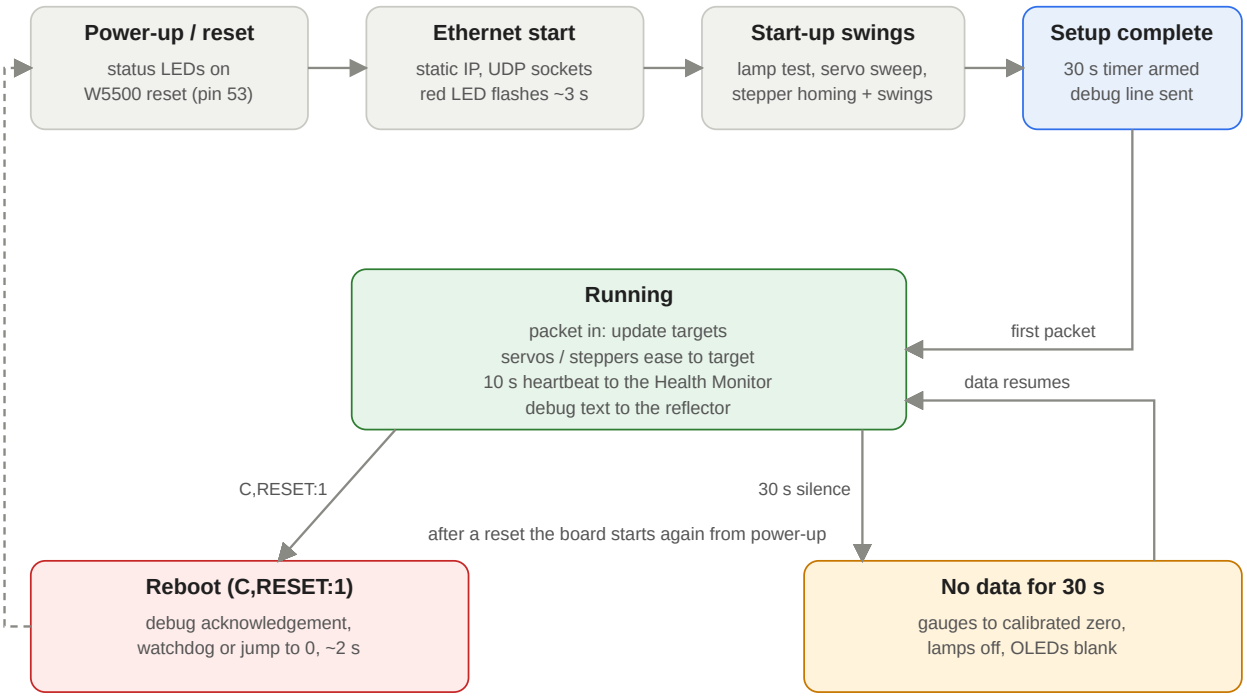


Figure 32 — Board life cycle

After start-up and its self-test a gauge board checks UDP 13136 every 5 ms and moves each needle smoothly towards its target. It sends its keep-alive every 10 seconds, with its 5 V rail after the board name (and the ADF controller its 12 V rail as well) for the Health Monitor. If no packet arrives for **30 seconds** every needle returns to zero (the Servo board also switches its lamps off, the Dual Stepper blanks its OLEDs and the Radio blanks its displays); the next packet brings everything back.

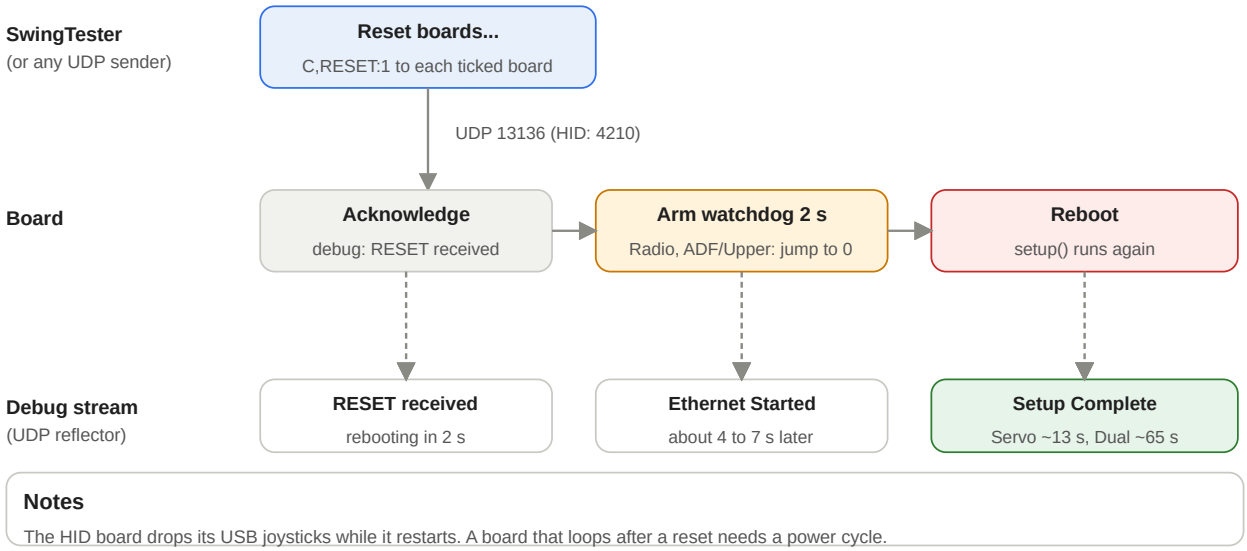


Figure 33 — Remote reset

The control packet `C,RESET:1` reboots a board about 2 seconds later. The SwingTester's *Reset boards...* button sends it to the boards you tick.

7.3 Unreachable hosts and the fast Ethernet timeout

Every sketch (Radio, Servo, Stepper, Dual Stepper, ADF and Upper and the USB HID board) sends UDP packets to other devices: debug text and keep-alives to the PC, commands from the Radio to the bridge, switch changes from the ADF and Upper controller to the USB HID board, and brightness from the ADF controller to the other panels. Before the Ethernet module can send to an address on the network it has to learn that address's MAC address by an ARP request. If the destination is switched off, unplugged or not yet running (the PC, or another board that is still starting), nobody answers.

The W5500 Ethernet chip then waits and asks again. Its default is a 200 ms timeout with 8 attempts, so one send to a missing device can hold the sketch up for about **1.6 seconds**. The sketch cannot do anything else while it waits: encoders and switches are not scanned, needles do not move, and every debug line sent to a PC that is off would cost another 1.6 seconds.

To stop that, each sketch calls `Ethernet.setRetransmissionTimeout(10)` straight after `Ethernet.begin()`. The timeout becomes **10 ms** and the number of attempts stays at 8, so the worst case for one send to a missing device is about **80 ms**. The panel keeps responding when the PC is off, a bridge is not running, or a board is unplugged, and the boards that are present are not slowed in any way you would notice.

- A missing device is not free: each send to it still costs up to 80 ms, so a sketch that sends many packets to a dead address (for example heavy debug traffic with the PC off) will run a little slower than normal. Switch the PC on, or turn debug traffic down, if a board feels sluggish with no PC.
- 10 ms is far longer than a real device on the cockpit network needs to answer an ARP request, so working devices are not affected. If you ever add a device on a slow or long network link, and packets to it are lost at first, raise the value in that sketch.
- The ADF controller sends its brightness to one board per pass through its loop for the same reason: a switched-off board never holds up the knobs and keypad for more than that one send.

7.4 When the panel loses power

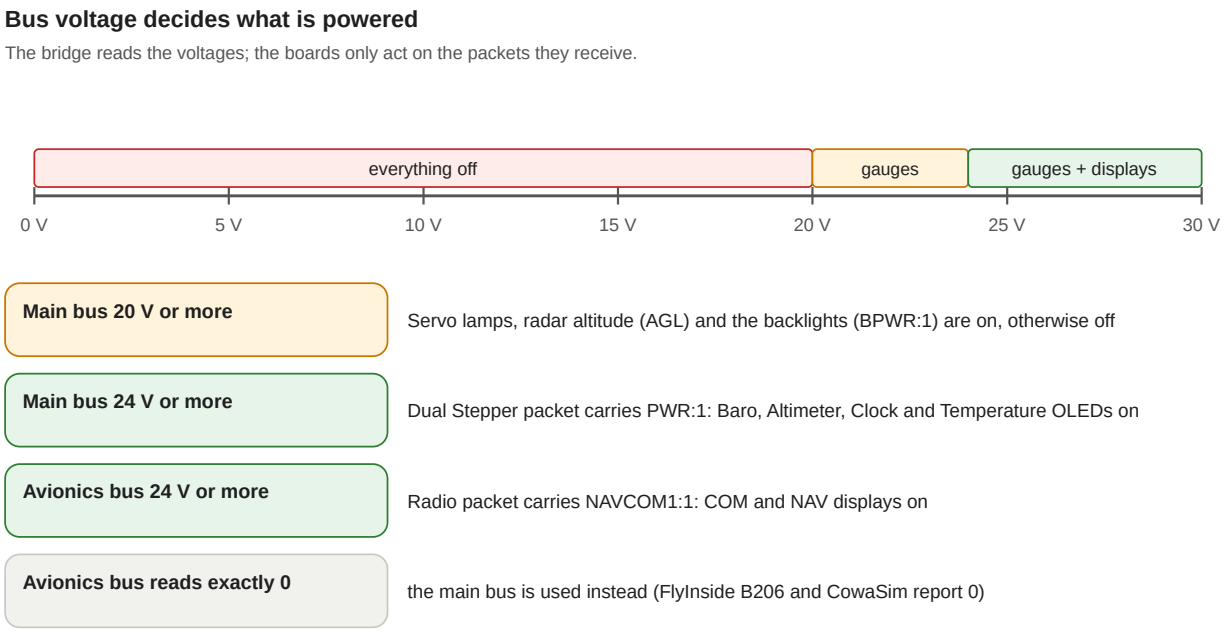


Figure 34 — Bus voltage rules

The bridge turns parts of the panel off when the simulated electrics drop: lamps, radar altitude and the backlights below 20 V on the main bus, the Dual Stepper's four OLEDs below 24 V on the main bus, and the Radio's COM and NAV displays below 24 V on the avionics bus (the main bus is used if the aircraft reports the avionics bus as 0).

7.5 Backlight

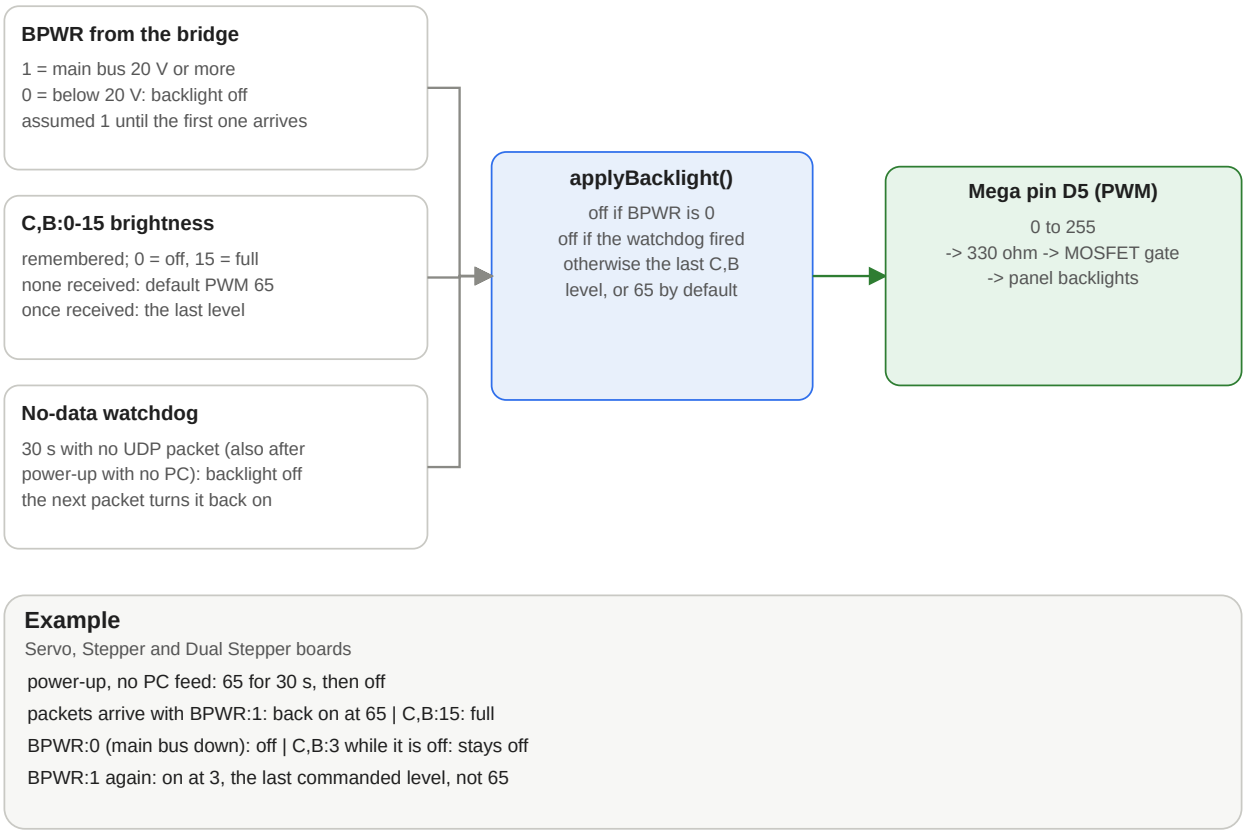


Figure 35 — How the backlight level is decided

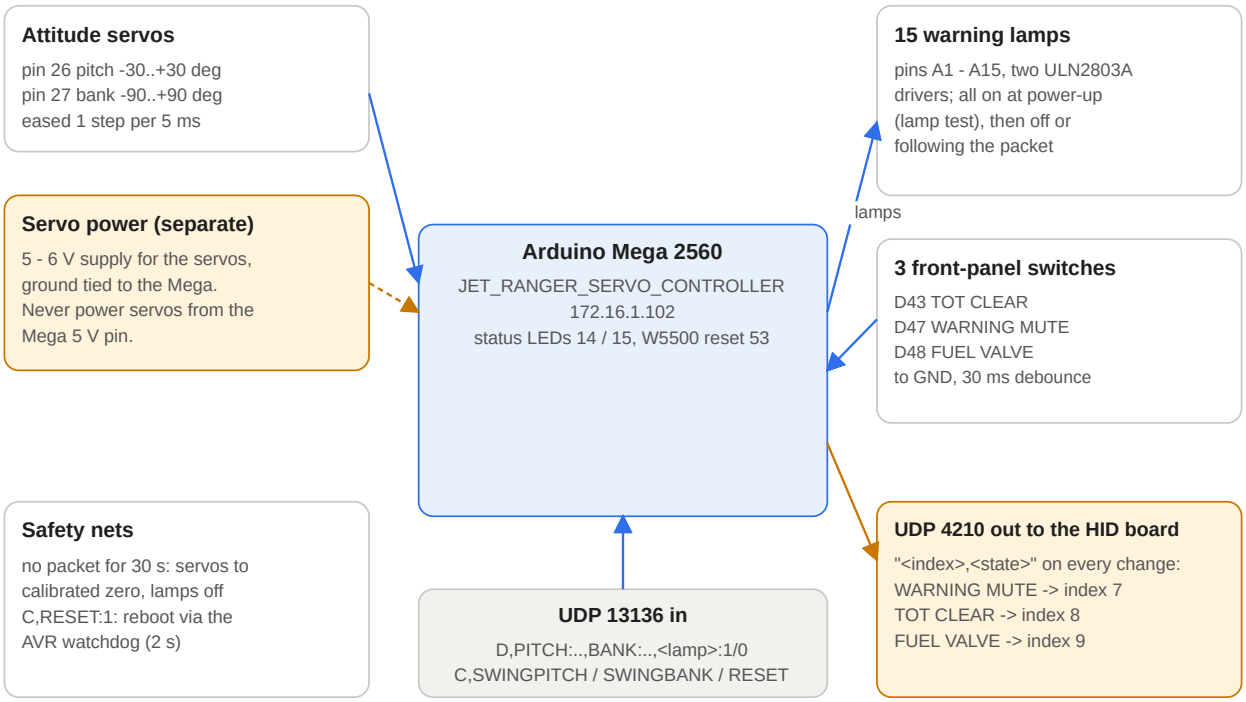
The Radio, Servo, Stepper, Dual Stepper and ADF and Upper boards drive their panel backlights from Mega pin D5 through a MOSFET on the PCB (on the Radio these are the Audio, Auto Pilot and COM and NAV panels; its "no data" rule is the same 30 seconds that blanks its displays, restarted by any knob or key; on the ADF and Upper board a switch, key or knob restarts it). Three things decide the level:

- **Main bus power (`BPWR`).** The bridge sends `BPWR:1` while the main bus is 20 V or more and `BPWR:0` below that, which switches the backlight off. Until the first `BPWR` arrives the board leaves the backlight on.
- **Brightness (`C,B:0` to `C,B:15`).** 0 is off and 15 is full. The board remembers the last level. If no brightness command has been sent the backlight uses a default level (PWM 65); once one has been sent, every time power returns the backlight comes back at the **last level you set**, not the default. A level sent while the bus is down is applied when power returns.
- **No data.** After 30 seconds with no UDP packet, including after the board is powered up with no PC sending, the backlight goes off along with the gauges. The next packet turns it back on.

The **SwingTester** has a brightness slider that sends the level to every board except the USB HID board, and *Backlight power ON / OFF* buttons for testing the main bus rule (including the ADF and Upper board). The **ADF controller** can send the level from its brightness pot to the other boards, but the pot is switched off for the moment. The Radio also uses the level for its panel LEDs. The ADF controller also blanks its display and lamps with the Radio's displays.

8 Arduino servo controller

The servo board is an Arduino Mega 2560 with an Ethernet module, on the Jet Ranger Front Panel Servo PCB at 172.16.1.102. It drives the attitude indicator's two servos, switches the 15 caution and warning lamps and reads three front-panel switches. It runs `JET_RANGER_SERVO_CONTROLLER.ino`. The other gauges that were once servo-driven now run on the two stepper boards.



Joystick 1 buttons 8, 9 and 10 in Windows (the firmware counts from 0). FSUIPC needs them bound to DS206 events.

Calibration: PITCH_DEG_TABLE -30 deg 179, 0 deg 113, +30 deg 70; BANK_DEG_TABLE -90 deg 5, 0 deg 93, +90 deg 179 (servo positions).

Figure 36 — Servo board

8.1 Uploading the firmware

1. Open the sketch in the Arduino IDE and select *Board: Arduino Mega or Mega 2560* (or use `arduino-cli`, see Flashing a board).
2. Install the libraries it uses: Ethernet (with SPI), Servo, Ewma (jonnieZG) and DCS-BIOS.
3. Upload, then watch the start-up self-test: both servos sweep and all 15 lamps light.

8.2 Servos

Instrument	Mega pin	UDP token	Range
Attitude bank (roll)	27	`BANK`	-90 to +90 degrees
Attitude pitch	26	`PITCH`	-30 to +30 degrees

Both tokens take attitude in degrees, with one decimal place allowed (`D,PITCH:4.5,BANK:-12`). The board turns degrees into a servo position with the tables in Calibration. `BANKRAW` and `PITCHRAW` take a raw servo position (0 to 180) and skip the tables. The PCB still has 16 servo headers from the earlier design; only pins 26 and 27 are used.

8.3 Other pins

Function	Pin	Behaviour
Green status LED	14	Flashes alternately with red every 200 ms while running
Red status LED	15	Fast flash during the 3 second Ethernet start-up, then alternates with green
Ethernet module reset	53	Pulsed low for 2 ms at power-up
Backlight	5	PWM to the backlight MOSFET (`BACKLIGHT_PWM`); see Backlight
Lamps	A1 to A15	Digital outputs, HIGH = lit (see Reference)
TOT clear switch	43	Switch to GND, debounced 30 ms; each change goes to the USB HID board as Joystick 1 button 9 (index 8)
Warning mute switch	47	The same, Joystick 1 button 8 (index 7)
Fuel valve switch	48	The same, Joystick 1 button 10 (index 9)

The 15 warning lamps on the Servo board

Green: the bridge drives it from the simulator. Grey: needs aircraft-specific variables, so it stays off.

ENGINE OUT pin A1 token EOUT engine speed < 60 %	**ROTOR RPM LOW** pin A2 token RLOW rotor <= 95 %	**XMSN OIL PRESS** pin A3 token TOPW pressure < 40 psi
XMSN OIL TEMP pin A4 token TOTW always off	**BATTERY HOT** pin A5 token BHOT always off	**BATTERY TEMP** pin A6 token BTMP always off
ENGINE CHIP pin A7 token EC always off	**TAIL ROTOR CHIP** pin A8 token TRC always off	**XMSN CHIP** pin A9 token TC always off
BAGGAGE DOOR pin A10 token BD always off	**AFT FUEL FILTER** pin A11 token FFLTR always off	**FUEL PUMP** pin A12 token FPMP always off
GENERATOR FAIL pin A13 token GENF engine speed < 10 %	**LOW FUEL** pin A14 token LOWF fuel <= 17 gal	**SC FAIL** pin A15 token SCF always off

All lamps go out when the main bus is below 20 V. Sent as <token>:1 / <token>:0 in every Servo packet.

The lamp test at power-up lights all 15; a no-data timeout turns them all off.

Figure 37 — The 15 lamps and their pins

8.4 What happens at power-up

1. All 15 lamps switch on as a lamp test.
2. The Ethernet module is reset and starts at 172.16.1.102; the red LED flashes for 3 seconds.
3. *Ethernet Started* goes to the debug listener.
4. Pitch, then bank, sweeps from minimum to maximum and back (about 2 seconds each) and parks at its calibrated level position.
5. All lamps switch off.
6. The starting state of the three switches is sent to the USB HID board, *Setup Complete* is logged and the 30 second watchdog starts.

8.5 Normal running

- Every 5 ms the board checks UDP 13136 and stores each value as a new target; each servo steps one position towards its target, so the attitude indicator moves smoothly.
- A servo that has not moved for 1 second is detached to stop buzz; the next movement re-attaches it.

- Lamps change as soon as a 0 or 1 arrives.
- `C,SWINGPITCH:1` or `C,SWINGBANK:1` re-runs one sweep on demand (the board pauses for about 2 seconds).

9 Stepper controller (13 Stepper board)

A second Arduino Mega 2560, on the Jet Ranger 13 Stepper PCB at 172.16.1.105, drives X27-style stepper gauges directly from its pins. Unlike the servo board it takes real units and converts them to needle steps with calibration tables in `JET_RANGER_STEPPER_CONTROLLER.ino`.

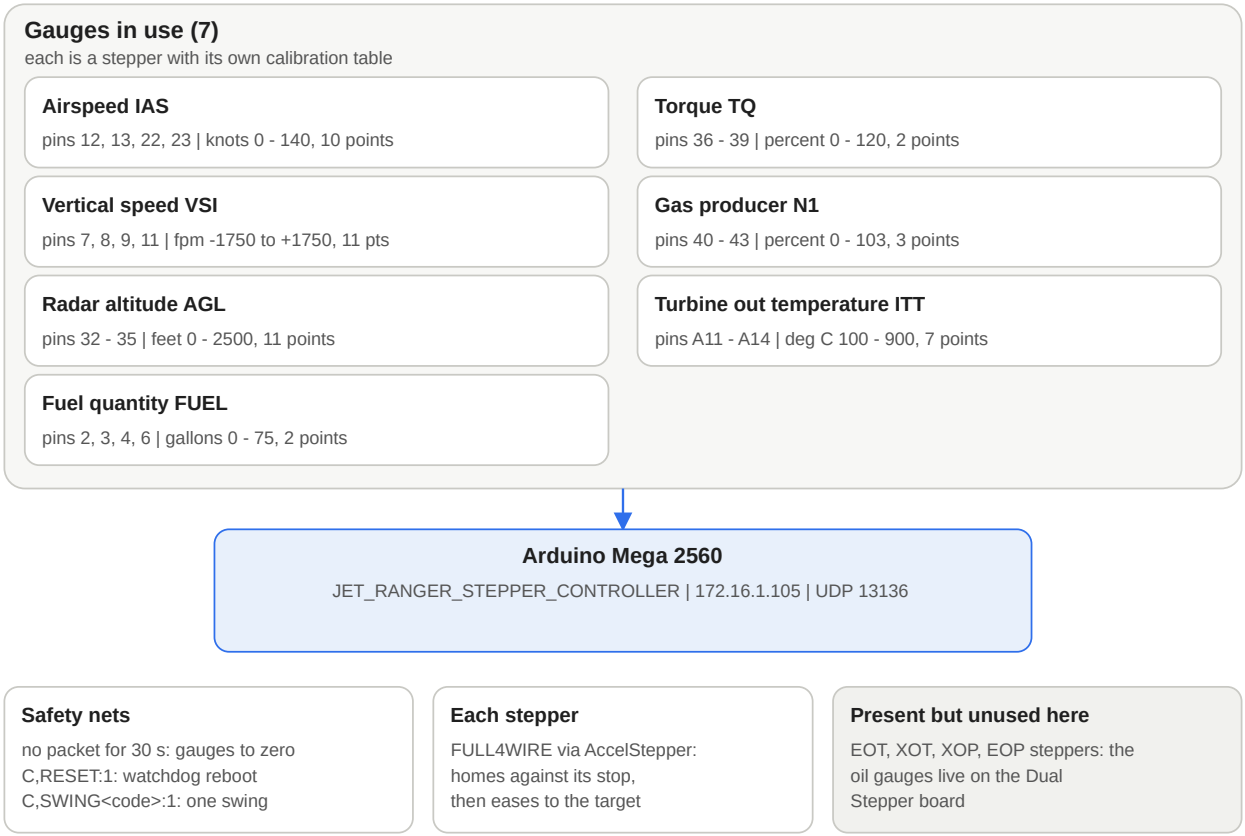


Figure 38 — Stepper board

Gauge	Token	Mega pins	Units	Calibration
Airspeed	`IAS`	12, 13, 22, 23	knots 0 to 140	10-point table
Vertical speed	`VSI`	7, 8, 9, 11	ft/min, plus and minus 1750	11-point table
Radar altitude	`AGL`	32 to 35	feet 0 to 2500	11-point table
Torque	`TQ`	36 to 39	percent 0 to 120	2-point table
Gas producer	`N1`	40 to 43	percent 0 to 103	3-point table
Turbine out temperature	`ITT`	A11 to A14	degrees C 100 to 900	7-point table
Fuel quantity	`FUEL`	2, 3, 4, 6	US gallons 0 to 75	2-point table

Every token has a raw twin with `RAW` on the end (for example `IASRAW:300`). It moves the needle by raw steps and skips the table, which is how new calibration points are measured on the bench. The board also has four oil and transmission steppers with placeholder scales; those gauges are driven from the Dual Stepper, so wire each physical gauge to one board only. Pins 24 to 31 are free.

9.1 Behaviour

- **Power-up:** status LEDs flash, Ethernet starts, backlighting ramps up, then each gauge does a self-test sweep in turn: fuel, torque, turbine out, gas producer, VSI, airspeed, radar altitude. Each sweep has its own on/off switch (`SwingFA`, `SwingTQ` and so on) near the top of the sketch.
- **Running:** every 5 ms the board checks UDP 13136 and moves each needle towards its target with smooth acceleration.
- **No-data watchdog:** after 30 seconds without a packet every needle returns to its zero.

- **Zero trims:** each gauge has a `...ZeroOffset` value to nudge its true zero without touching the table.
- **Keep-alive:** `STEPPER` to the Health Monitor every 10 seconds; debug text goes to UDP 27000.

9.2 Testing a gauge on demand

Send a control packet to re-run one gauge's sweep without rebooting, for example `C,SWINGTQ:1`.

Command	Gauge
`SWINGFUEL`	Fuel quantity
`SWINGTQ`	Torque
`SWINGITT`	Turbine out temperature
`SWINGN1`	Gas producer
`SWINGVSI`	Vertical speed
`SWINGIAS`	Airspeed
`SWINGAGL`	Radar altitude

The **StepperVSITester** app can send every token and raw token to this board without the simulator. The **SwingTester** app sends these commands, and the Dual Stepper and servo sweeps, one at a time or all in a fixed order:

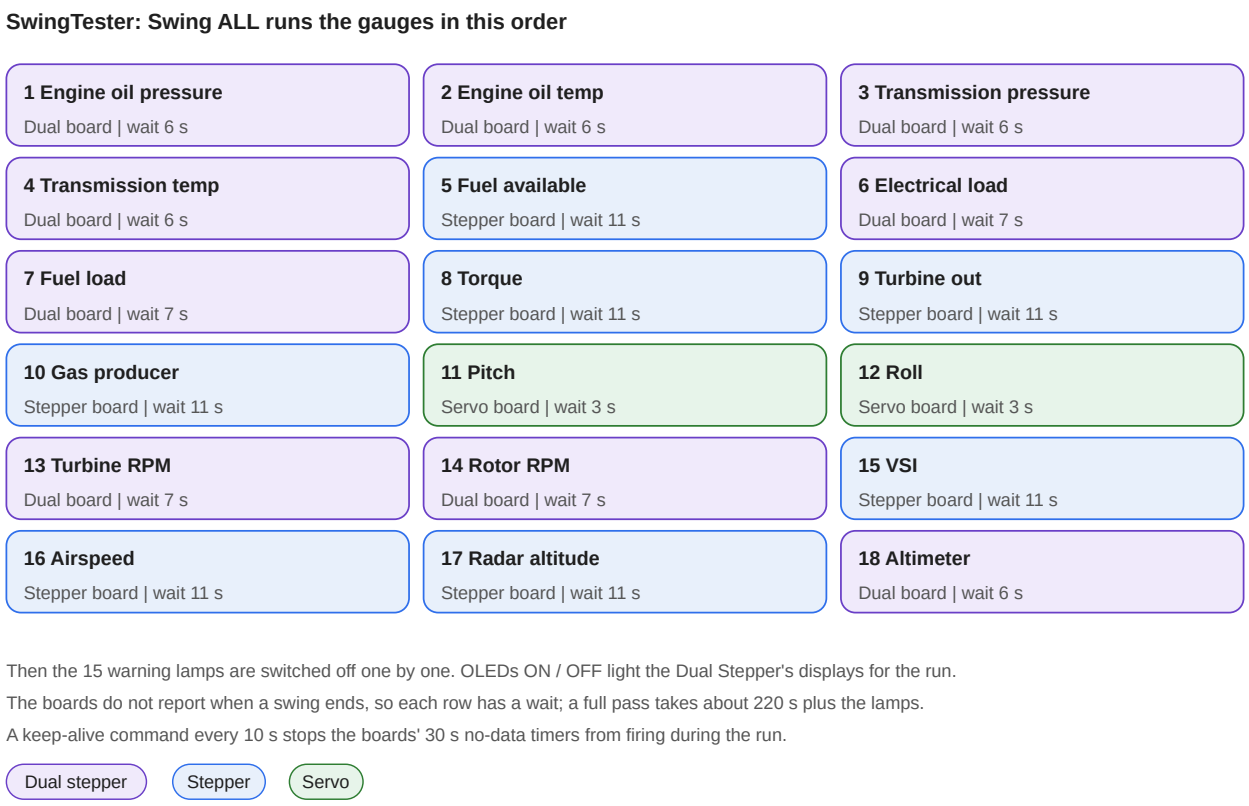


Figure 39 — Swing ALL sequence

It lights the OLEDs and the lamps during the run and sends a keep-alive so the boards' 30 second watchdogs do not fire. *Reset boards...* reboots the boards you tick (Radio, Servo, Upper (the retired .103 address), Stepper, Dual Stepper, USB HID) by sending each one `C,RESET:1` (the ADF and Upper board .107 is not in the list yet); each restarts about 2 seconds later and repeats its power-up test.

9.3 Known issues

- The backlight PWM was once on pin 8, which the PCB routes to VSI coil B. It is now pin 5, matching the PCB's `BACKLIGHT_PWM` net, in both stepper sketches, and it follows `BPWR`, `C,B:n` and the 30 second no-data watchdog (see Backlight).
- Pin 32 is also claimed by an unused roll stepper in the code.

- The oil, transmission and placeholder gauges use a shared linear scale, not bench-measured tables.

10 OLED dual stepper controller

A third Mega, on the Jet Ranger OLED Dual Stepper PCB at 172.16.1.106, drives stepper gauges plus four small OLED displays. It runs `JET_RANGER_OLED_DUAL_STEPPER_CONTROLLER.ino` and, like the 13 Stepper board, takes real units.

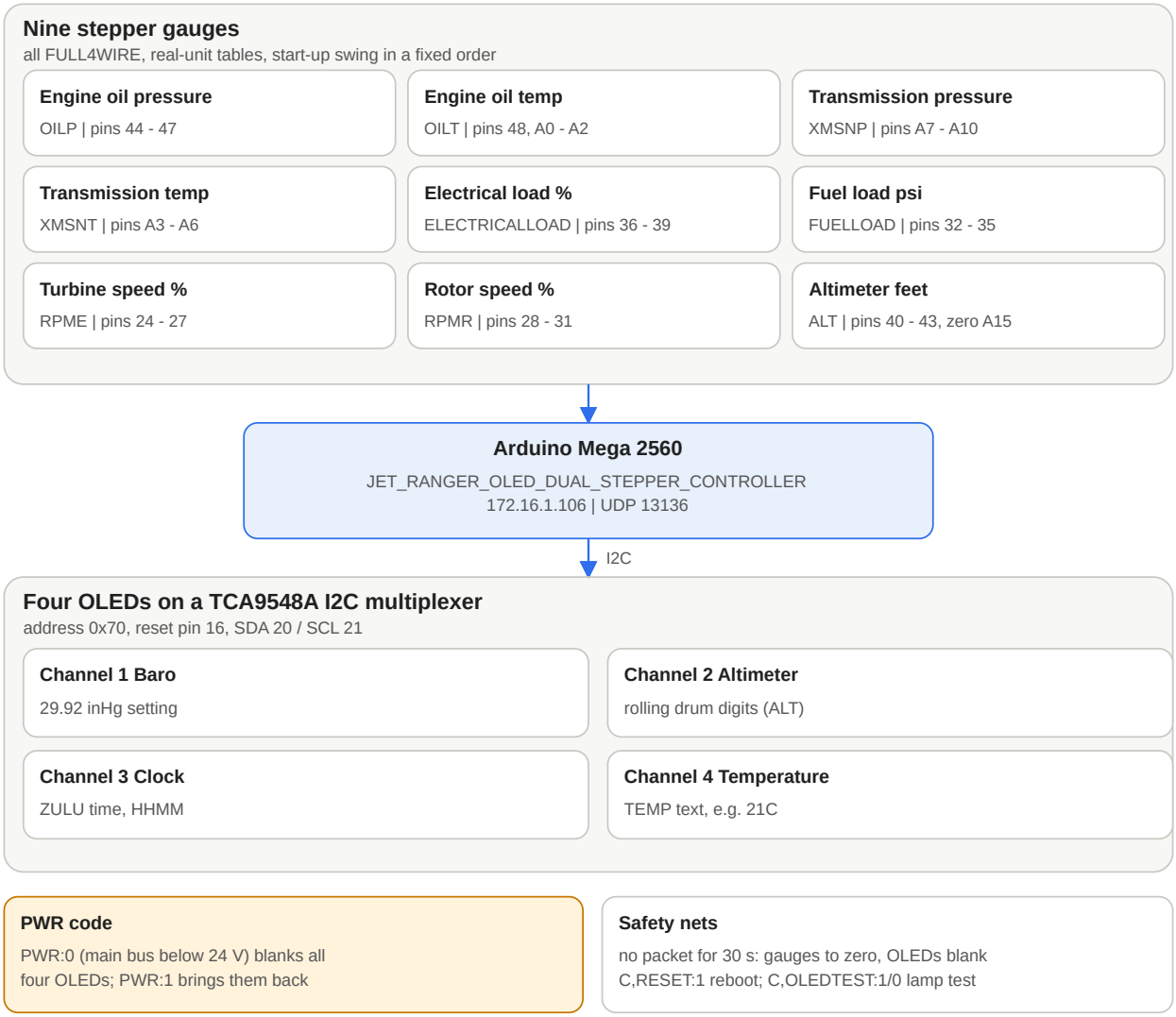


Figure 40 — Dual stepper board

Gauge	Token	Mega pins	Units	Calibration
Engine speed (N2)	`RPME`	24 to 27	percent, one decimal	4-point table
Rotor speed	`RPMR`	28 to 31	percent, one decimal	4-point table
Altimeter	`ALT`	40 to 43, zero sensor A15	feet	720 steps per 1000 ft
Fuel load	`FUELLOAD`	32 to 35	psi	2-point table (0 to 30 psi)
Electrical load	`ELECTRICALLOAD`	36 to 39	percent	2-point table
Engine oil temperature	`OILT`	48, A0 to A2	degrees C	4-point table
Engine oil pressure	`OILP`	44 to 47	psi	4-point table
Transmission oil temperature	`XMSNT`	A3 to A6	degrees C	4-point table
Transmission oil pressure	`XMSNP`	A7 to A10	psi	5-point table

Turbine out temperature (`ITT`) and fuel quantity (`FUEL`) are also accepted, as on the 13 Stepper board. Wire each physical gauge to one board only; the matching channel on the other board then simply has nothing connected. Each token also has a `RAW` twin for bench work. `IAS` is not handled here; the FSUIPC and MSFS bridges send airspeed to the 13 Stepper board with one decimal place so the needle moves smoothly.

10.1 OLED displays

The four 128 by 32 SSD1306 OLEDs share the Mega's I2C pins (20 SDA, 21 SCL) through a TCA9548A multiplexer at address 0x70. The firmware pulses the mux reset on pin 16 at start-up.

Display	Mux channel	UDP token	Shows
Barometric setting	1	none yet	`2992` at boot; not updated
Altimeter digits	2	`ALT`	Rolling-drum altitude, raw MSL (barometric correction off)
Clock	3	`ZULU`	Zulu time from HHMM, for example `ZULU:1430`
Temperature	4	`TEMP`	Text as sent, for example `TEMP:25C`

Send `C,OLEDTEST:1` to light all four displays solid and `C,OLEDTEST:0` to restore them. When the main bus is below 24 V the bridge sends `PWR:0` and all four displays go blank; `PWR:1` turns them back on and they repaint within about a second. The altimeter needle keeps following altitude. The lamp test still lights all four displays whatever `PWR` says.

10.2 Behaviour

- **Power-up:** OLEDs start first, then each gauge sweeps in turn: engine oil pressure, engine oil temperature, transmission pressure, transmission temperature, electrical load, fuel load, engine RPM, rotor RPM, altimeter. The altimeter seeks its zero sensor before sweeping. Start-up takes about 65 seconds.
- **No-data watchdog:** after 30 seconds without a packet the needles return to zero and all OLEDs blank.
- **On-demand sweeps:** `C,SWINGOILP:1`, `SWINGOILT`, `SWINGXMSNP`, `SWINGXMSNT`, `SWINGELECTRICALLOAD`, `SWINGFUELLOAD`, `SWINGRPME`, `SWINGRPMR`, `SWINGALT`. Each blocks the board for 5 to 7 seconds.
- **Keep-alive:** `DUAL_STEPPER` to the Health Monitor every 10 seconds, so it shows separately from the 13 Stepper board.

10.3 Known issues

- RAM use is 75 percent or more; avoid adding more OLEDs or string-heavy code.
- The latest bench I2C scan found OLEDs on mux channels 1 to 4; check by eye that each shows its text.
- Engine and rotor RPM tables assume 0 percent is 0 steps; `RSoffset` is not yet measured.
- The altimeter zero seek travels up to 10,080 steps (about 14 turns) before giving up.

11 Radio controller (COM and NAV)

The radio panel is another Mega at 172.16.1.101, running `JET_RANGER_RADIO_CONTROLLER.ino`. It shows the COM and NAV frequencies on two OLEDs, reads eight tuning knobs and a 4 by 5 button matrix, serves the Auto Pilot and Audio panels (switches, knobs and 19 LEDs), and sends knob and button actions back to the PC.

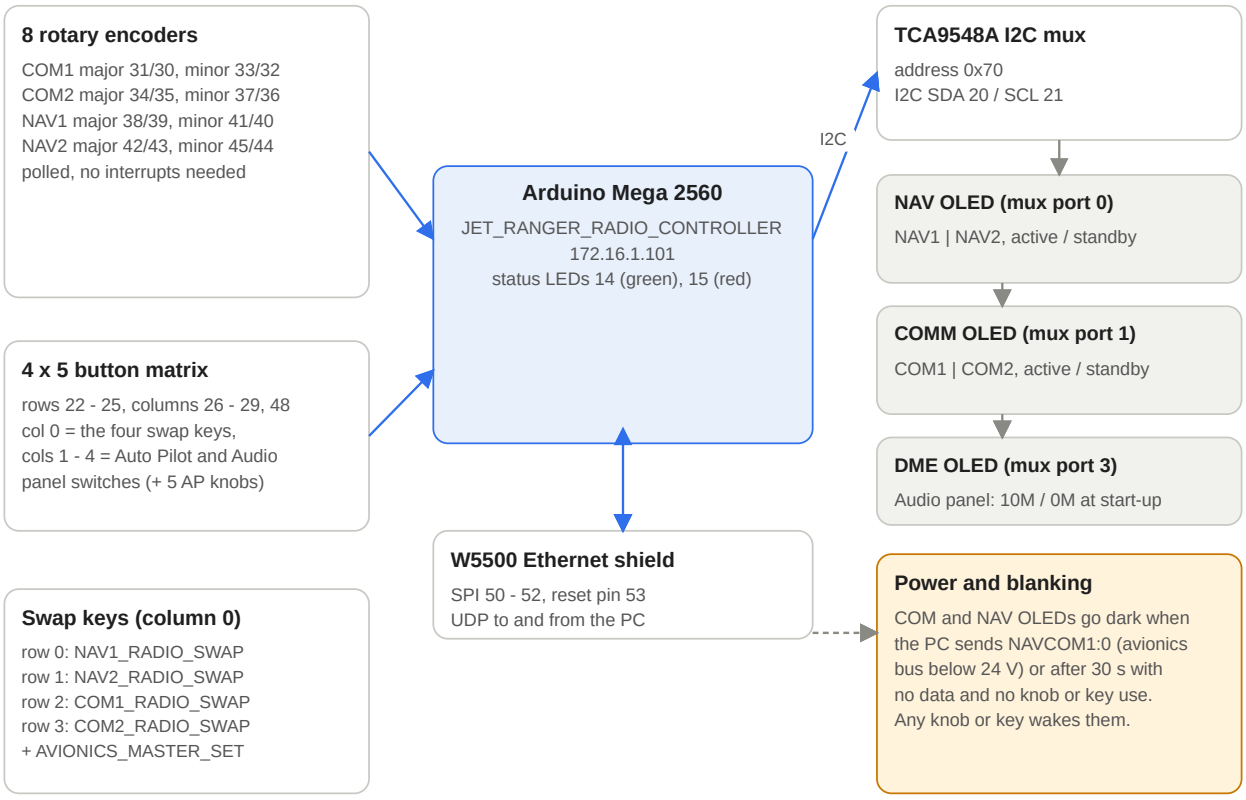


Figure 41 — Radio panel

11.1 Boards

PCB	Role	Main parts
Jet Ranger Radio Auto Pilot	Radio controller board	Mega 2560 shield, Ethernet module, TCA9548A I2C mux
Jet Ranger COM and NAV	Front panel board behind the radio faceplate	8 rotary encoders, push buttons, four frequency-swap switches

11.2 Inputs and outputs

Function	Mega pins	Notes
COM1 major and minor knob	31, 30 and 33, 32	Encoder A, B
COM2 major and minor knob	34, 35 and 37, 36	Encoder A, B
NAV1 major and minor knob	38, 39 and 41, 40	Encoder A, B
NAV2 major and minor knob	42, 43 and 45, 44	Encoder A, B
Button matrix	rows 22 to 25, columns 26, 27, 28, 29 and 48	4 by 5: column 0 the COM and NAV swap switches, columns 1 and 2 the Auto Pilot panel,

Function	Mega pins	Notes
		columns 3 and 4 the Audio panel
Auto Pilot knobs	CRS 11, 12; ALT A4, A5; VS 46, 47; IAS 7, 6; HDG 8, 9	Encoder A, B
Panel LEDs (MAX7219)	DIN A15, CLK A14, LOAD A13	19 LEDs on the Auto Pilot and Audio panels and the 20 Huey Warning Panel LEDs
Huey Warning Panel switches	A9 TEST, A10 RESET, A8 BRT / DIM	Switches to ground; the panel is hand wired at the moment (see Huey Warning Panel)
Backlight	D5	IRF540N MOSFET for the Audio, Auto Pilot and COM and NAV panel backlights
OLEDs	I2C through the TCA9548A (0x70)	Port 0 = NAV OLED, port 1 = COM OLED, both at 0x3C
Status LEDs	14 green, 15 red	Red flashes while Ethernet starts
Ethernet module	SPI 50 to 52, reset 53	

11.3 Data from the PC (UDP 13136)

Token	Meaning	Example
`C1A`, `C1S`	COM1 active and standby frequency	`C1A:119.500`
`C2A`, `C2S`	COM2 active and standby frequency	`C2S:121.900`
`NAVCOM1`	Radio power (0 = off): the COM and NAV displays and the panel LEDs go off	`NAVCOM1:1`
`MAINBUS`	Main bus voltage	
`LED_xxx`	One of the 19 Auto Pilot and Audio panel LEDs (1 = on)	`LED_HDG:1`
`LED_H_xxx`	One of the 20 Huey Warning Panel LEDs (1 = on); the DCS bridge sends all 20 for the Huey, the X-Plane bridge ten approximations, the FSUIPC and MSFS bridges none	`LED_H_ENG_OIL:1`
`APHDG`, `APCRS`, `APIAS`, `APVS`, `APALT`	The five values on the Auto Pilot panel's OLED	`APHDG:090`
`BPWR`	Backlight power: 1 = main bus 20 V or more, 0 = backlight off	`BPWR:1`
`C,B:n`	Brightness 0 to 15: the panel backlight level (0 off) and the panel LEDs' intensity	`C,B:5`
`C,LEDTEST:n`	1 lights all 39 panel LEDs, 0 returns them to the simulator's state	`C,LEDTEST:1`

11.4 Commands to the PC (UDP 27001)

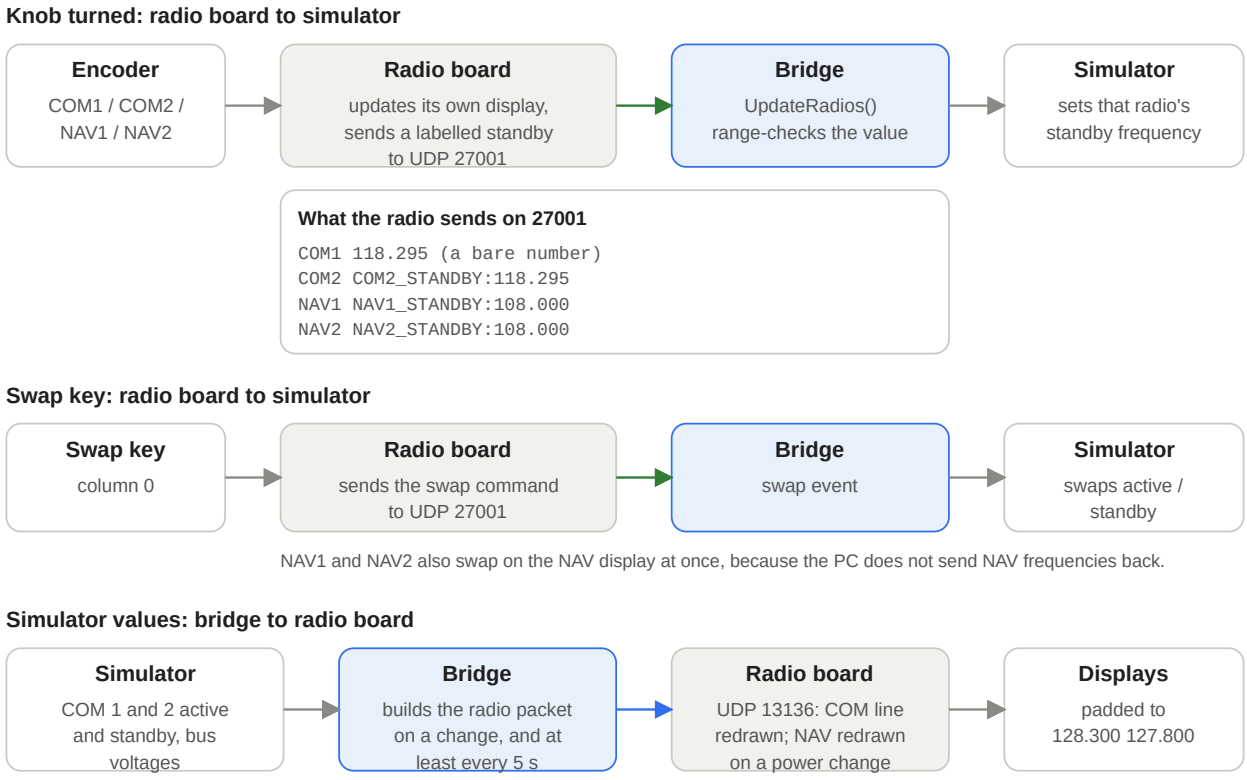


Figure 42 — Where each input goes

Turning a knob changes the standby frequency on the OLED and sends the new value to the PC. The four swap switches share matrix column 0:

Switch (row, column 0)	Command sent
Row 0: NAV1 FREQ	`NAV1_RADIO_SWAP`
Row 1: NAV2 FREQ	`NAV2_RADIO_SWAP`
Row 2: COMM1	`COM1_RADIO_SWAP`
Row 3: COMM2	`COM2_RADIO_SWAP`, then `AVIONICS_MASTER_SET`

The bridge listens on 27001 and turns these into simulator events. NAV1 and NAV2 also swap their active and standby frequencies on the NAV OLED straight away, because the PC does not send NAV frequencies back; the COM swaps change the display only when the simulator returns the swapped frequencies. Other buttons are only logged to the debug listener.

Each knob sends its standby frequency with the radio's name: COM1 as the bare number (for example `118.295`), the others as `COM2_STANDBY:118.295`, `NAV1_STANDBY:108.000` or `NAV2_STANDBY:108.000`. Nothing is sent during the first 1.5 seconds after power-up, so the knobs' start positions do not change the simulator. Each OLED line always shows two frequencies in a fixed 16-column layout.

Tuning steps follow the standard channel spacing. The COM knobs run 118.000 to 136.975 MHz in 25 kHz channels (the major knob picks 118 to 136, the minor knob 000, 025, 050 and 075). The NAV knobs run 108.00 to 117.95 MHz in 50 kHz steps (major 108 to 117, minor 000, 050, 100 and so on). Each minor knob wraps round at the ends. 8.33 kHz channels are not offered. FSUIPC holds COM frequencies to 10 kHz, so in that simulator a channel such as 118.025 may appear as 118.02 or 118.03.

11.5 Auto Pilot and Audio panels

The Auto Pilot and Audio panels plug into the radio controller. Each panel has eight switches, and the Auto Pilot panel also has five knobs (CRS, ALT, VS, IAS and HDG). Every switch and knob is sent **two ways**:

1. as a **joystick button** to the USB HID board (**Joystick 3** for the Auto Pilot panel, **Joystick 4** for the Audio and ADF panels), which you can bind in FSUIPC or MSFS like any other button. A switch holds its button down while pressed; a knob click is a short press, clockwise on one button and anticlockwise on the next;
2. as a **command to the simulator bridge** on UDP 27001, which the FSUIPC and MSFS bridges turn straight into the simulator control of that name. A switch sends it when pressed; a knob sends one for each click (with a repeat count when you turn it quickly). Nothing needs binding for these.

Nothing is sent for the first 1.5 seconds after power-up. The full list, with the ADF panel's switches, is below. The joystick buttons used (Joystick 3 and 4) are not used by any other switch.

Panel	Control	Matrix or pin	Joystick button (index)	Command to the bridge (UDP 27001)
Auto Pilot	AP ENGAGE	row 0, col 1	Joystick 3, button 1 (64)	`AP_MASTER`
Auto Pilot	HDG	row 1, col 1	Joystick 3, button 2 (65)	`AP_HDG_HOLD`
Auto Pilot	CRS	row 2, col 1	Joystick 3, button 3 (66)	`AP_NAV1_HOLD`
Auto Pilot	ALT	row 3, col 1	Joystick 3, button 4 (67)	`AP_ALT_HOLD`
Auto Pilot	APPR	row 0, col 2	Joystick 3, button 5 (68)	`AP_APR_HOLD`
Auto Pilot	FD	row 1, col 2	Joystick 3, button 6 (69)	`TOGGLE_FLIGHT_DIRECTOR`
Auto Pilot	BC	row 2, col 2	Joystick 3, button 7 (70)	`AP_BC_HOLD`
Auto Pilot	IAS	row 3, col 2	Joystick 3, button 8 (71)	`AP_PANEL_SPEED_HOLD`
Auto Pilot	CRS knob clockwise, anticlockwise	D11, D12	Joystick 3, buttons 9, 10 (72, 73)	`VOR1_OBI_INC`, `VOR1_OBI_DEC`
Auto Pilot	ALT knob	A4, A5	Joystick 3, buttons 11, 12 (74, 75)	`AP_ALT_VAR_INC`, `AP_ALT_VAR_DEC`
Auto Pilot	VS knob	D46, D47	Joystick 3, buttons 13, 14 (76, 77)	`AP_VS_VAR_INC`, `AP_VS_VAR_DEC`
Auto Pilot	IAS knob	D7, D6	Joystick 3, buttons 15, 16 (78, 79)	`AP_SPD_VAR_INC`, `AP_SPD_VAR_DEC`
Auto Pilot	HDG knob	D8, D9	Joystick 3, buttons 17, 18 (80, 81)	`HEADING_BUG_INC`, `HEADING_BUG_DEC`
Audio	COM1	row 0, col 3	Joystick 4, button 1 (96)	`COM1_TRANSMIT_SELECT`
Audio	BOTH	row 1, col 3	Joystick 4, button 2 (97)	`COM_RECEIVE_ALL_TOGGLE`
Audio	NAV2	row 2, col 3	Joystick 4, button 3 (98)	`RADIO_VOR2_IDENT_TOGGLE`
Audio	DME	row 3, col 3	Joystick 4, button 4 (99)	`RADIO_DME1_IDENT_TOGGLE`
Audio	COM2	row 0, col 4	Joystick 4, button 5 (100)	`COM2_TRANSMIT_SELECT`
Audio	NAV1	row 1, col 4	Joystick 4, button 6 (101)	`RADIO_VOR1_IDENT_TOGGLE`
Audio	MKR	row 2, col 4	Joystick 4, button 7 (102)	`MARKER_SOUND_TOGGLE`
Audio	ADF	row 3, col 4	Joystick 4, button 8 (103)	`RADIO_ADF_IDENT_TOGGLE`
ADF	Landing gear switch	ADF controller D31	Joystick 4, button 9 (104)	`GEAR_SET:1` when closed, `GEAR_SET:0` when open
ADF	Flaps switch	ADF controller D32	Joystick 4, button 10 (105)	`FLAPS_SET:1` when closed, `FLAPS_SET:0` when open
ADF	Speed brakes switch	ADF controller D33	Joystick 4, button 11 (106)	`SPOILERS_SET:1` when closed, `SPOILERS_SET:0` when open
ADF	Brakes button	ADF controller D34	Joystick 4, button 12 (107)	`PARKING_BRAKES` when pressed
ADF	ADF major knob clockwise, anticlockwise	ADF controller D6, D7	Joystick 4, buttons 13, 14 (108, 109)	none (the knob tunes the ADF frequency, `ADF_FREQ`)

Panel	Control	Matrix or pin	Joystick button (index)	Command to the bridge (UDP 27001)
ADF	ADF minor knob clockwise, anticlockwise	ADF controller D8, D9	Joystick 4, buttons 15, 16 (110, 111)	none
ADF	Anti-ice switch	ADF controller D36	Joystick 4, button 17 (112)	`ANTI_ICE_SET:1` when closed, `ANTI_ICE_SET:0` when open
ADF	Hyd boost switch	ADF controller D35	Joystick 4, button 18 (113)	none (bind the joystick button in FSUIPC for the Dodo)
ADF	Caution dim switch	ADF controller D37	Joystick 4, button 19 (114)	none (bind the joystick button in FSUIPC for the Dodo)
Huey	TEST switch	Radio controller A9	Joystick 4, button 20 (115)	none (bind the joystick button in the simulator)
Huey	RESET switch	Radio controller A10	Joystick 4, button 21 (116)	none
Huey	BRT / DIM switch	Radio controller A8	Joystick 4, button 22 (117)	none

If a knob turns the wrong way in the simulator, swap its two encoder pins in the sketch. The rows and columns come from the panel schematics; check each switch on the bench with the debug listener, which prints a line for every press (for example `AP HDG pressed (joystick button index 65)`), and `C,MATRIX:1` for the raw matrix.

11.6 Auto Pilot and Audio panel LEDs

One MAX7219 chip on the Radio Auto Pilot PCB lights the 19 LEDs on the two panels (and, since the Huey Warning Panel was added to the sketch, 20 more: see Huey Warning Panel below). Each LED sits between one segment and one digit of the chip:

LED	Token	MAX7219 digit	Segment		LED	Token	MAX7219 digit	Segment
CRS	`LED_CRS`	0	A		COM1	`LED_COM1`	2	A
HDG	`LED_HDG`	0	B		BOTH	`LED_BOTH`	2	B
IAS	`LED_IAS`	0	C		NAV2	`LED_NAV2`	2	C
ALT	`LED_ALT`	0	D		DME	`LED_DME`	2	D
APPR	`LED_APPR`	1	A		OUTER	`LED_OUTER`	2	E
FD	`LED_FD`	1	B		INNER	`LED_INNER`	2	F
BC	`LED_BC`	1	C		COM2	`LED_COM2`	3	A
AP ENGAGE	`LED_APENG`	1	D		NAV1	`LED_NAV1`	3	B
					MKR	`LED_MKR`	3	C
					ADF	`LED_ADF`	3	D
					MIDDLE	`LED_MID`	3	E

The bridges send each LED as a `LED_xxx:0/1` pair in the radio packet and the radio lights it, so the panel follows the simulator: the Auto Pilot LEDs show the autopilot and flight director modes (CRS is NAV1 lock), the COM1, COM2, BOTH, NAV1, NAV2, DME, ADF and MKR LEDs show the audio panel selections, and OUTER, MIDDLE and INNER light as you fly over a marker beacon. The LEDs go off with the radio power (avionics bus below 24 V, or no data and no use for 30 seconds). `C,LEDTEST:1` lights all 39 (including the Huey panel's) for checking the wiring and `C,LEDTEST:0` returns them to the simulator's state. The brightness command `C,B:0-15` (the SwingTester slider, or the ADF panel's pot) sets their brightness, and 0 switches them off. The **SwingTester** can light each LED on its own (see Program windows). The FSUIPC bridge reads standard FSUIPC offsets that the DodoSim aircraft may not all drive, and the MSFS bridge has not been checked against the simulator, so some LEDs may stay off until that is confirmed.

11.7 Huey Warning Panel

The Huey Warning Panel has 20 caution LEDs and three switches. **It is wired by hand to the Radio controller at the moment** (there is no PCB for it in this guide), and the Radio sketch supports it. The LEDs are on the same MAX7219 as the other two panels, digits 4 to 7, with the tokens `LED_H_ENG_OIL`, `LED_H_ENG_ICE`, `LED_H_ENG_ICE_DETECT`, `LED_H_ENG_CHIP`, `LED_H_LEFT_FUEL_BOOST`, `LED_H_ENG_FUEL_PUMP`, `LED_H_RIGHT_FUEL_BOOST`, `LED_H_20_MINUTE_FUEL`, `LED_H_AUX_FUEL_LOW`,

`LED_H_XMSN_OIL_PRESS`, `LED_H_XMSN_OIL_HOT`, `LED_H_HYD_PRESS`, `LED_H_ENG_INLET_AIR`, `LED_H_INST_INVERTER`, `LED_H_DC_GENERATOR`, `LED_H_EXT_POWER`, `LED_H_FUEL_FILTER`, `LED_H_GOV_EMER`, `LED_H_CHIP_DETECTOR` and `LED_H_IFF`. The three switches (TEST on A9, RESET on A10, BRT / DIM on A8; TEST and RESET are the two positions of one switch) are joystick buttons 20 to 22 on Joystick 4 (indexes 115 to 117). **While TEST is closed all 20 LEDs light** (only while the radio has power). The bridges do not send the `LED_H_xxx` tokens yet, so for now the LEDs light only from the SwingTester (the *Panel LEDs* group has a toggle for each), the lamp test or a test packet. Scanning eight digits instead of four makes the Auto Pilot and Audio LEDs dimmer than before. The Radio on COM20 was flashed with this code on 31 October 2026; the switches and LEDs have not been checked on the panel yet.

11.8 Auto Pilot display

The Auto Pilot panel's OLED sits behind a mask, so each value has its own window on the two 16-character lines: **HDG** at the left hand end of the top row and **VS** at the right hand end, **CRS** at the left hand end of the bottom row, **IAS** in the centre of the bottom row and **ALT** at its right hand end. At start-up it shows heading 090, course 110, airspeed 80, vertical speed 0 and altitude 3200:

```
090           0
110    80   3200
```

The bridges then keep the five values up to date from the simulator's autopilot: the heading bug, the NAV1 course (OBS), the autopilot airspeed, vertical speed and altitude settings (heading and course as three digits, north is 360). The display goes blank with the radio power and comes back with it, like the COM and NAV displays. The **SwingTester** has an *Auto Pilot display* group to send test values.

11.9 When the USB HID board is not connected

Every Audio and Auto Pilot switch and knob click is also sent to the USB HID board. If that board is switched off or unplugged, the radio now notices when a send fails, stops sending joystick buttons for 30 seconds, and says so in the monitor (`USB HID board 172.16.1.11 not answering`), so the panel stays responsive. When the board returns the buttons that are held down are sent again.

11.10 Backlight

The radio controller switches the backlights of the Audio, Auto Pilot and COM and NAV panels from Mega pin D5, using the same rules as the other panels (see Network): on only while the main bus is 20 V or more (`BPWR`), at the last brightness sent (`C,B:0-15`) or a default until one arrives, and off when the radio has had no data and no knob or key use for 30 seconds (a knob or key brings it back). The SwingTester's brightness slider and *Backlight power ON / OFF* buttons test it, and the ADF panel's brightness pot sets it. The brightness also sets the intensity of the Auto Pilot and Audio panel LEDs.

11.11 Display behaviour

- If no data arrives from the PC for 30 seconds and no knob or button is used, the COM, NAV, Auto Pilot and DME displays blank (as if the radio had no power) and the radio tells the ADF controller to blank its display and lamps too. Turning a knob, pressing a button or new data switches them back on.
- When the avionics bus is below 24 V the bridge tells the radio the power is off (`NAVCOM1:0`) and the COM, NAV, Auto Pilot and DME displays (and the ADF panel's display) go blank together; they light together again when the bus rises.
- At power-up the matrix is ignored for the first 1.5 seconds so a floating input cannot send false button presses. Each switch is checked every millisecond and reported about 12 ms after it changes, and a noisy reading cannot hold a release back.
- At start-up the radio scans the I2C bus and sends what it finds, behind each multiplexer channel, to the debug monitor (lines starting `I2C`); send `C,I2CSCAN:1` to scan again. The Audio panel's OLED is the **DME display**: it shows `10M` on the top line and `0M` on the bottom line after start-up.
- The radio sends `COMM_NAV` to the Health Monitor every 10 seconds.

11.12 Known issues

- Only COM1 and COM2 frequencies arrive from the PC; there are no NAV frequency tokens yet, although the NAV knobs are read.
- The Radio Auto Pilot PCB has volume pots that the radio sketch does not read yet.
- Four encoders (COM1 major and minor, COM2 minor, NAV1 minor) have A and B the other way round from the PCB labels; it only matters if one of those knobs tunes backwards.
- If every button in the matrix's first column stops working, send `C,MATRIX:1` to port 13136: the radio prints the raw H or L state of each column to the debug listener. A column that reads L with nothing pressed is a hardware fault on that Mega pin or board (a faulty or badly seated Mega did exactly this on the bench).

12 Joystick (HID) interface

Overhead and console switches reach the simulator as ordinary USB joystick buttons. The ADF and Upper controller scans the overhead switches (the old Upper controller's job) and sends each change over UDP to a small USB board, `UDP_TO_HID`, which Windows sees as four joysticks and a keyboard. The joysticks have buttons only: no axes and no hat switches. Bind them like any other controller, in FSUIPC for Prepar3D and FSX or in the MSFS controls menu.

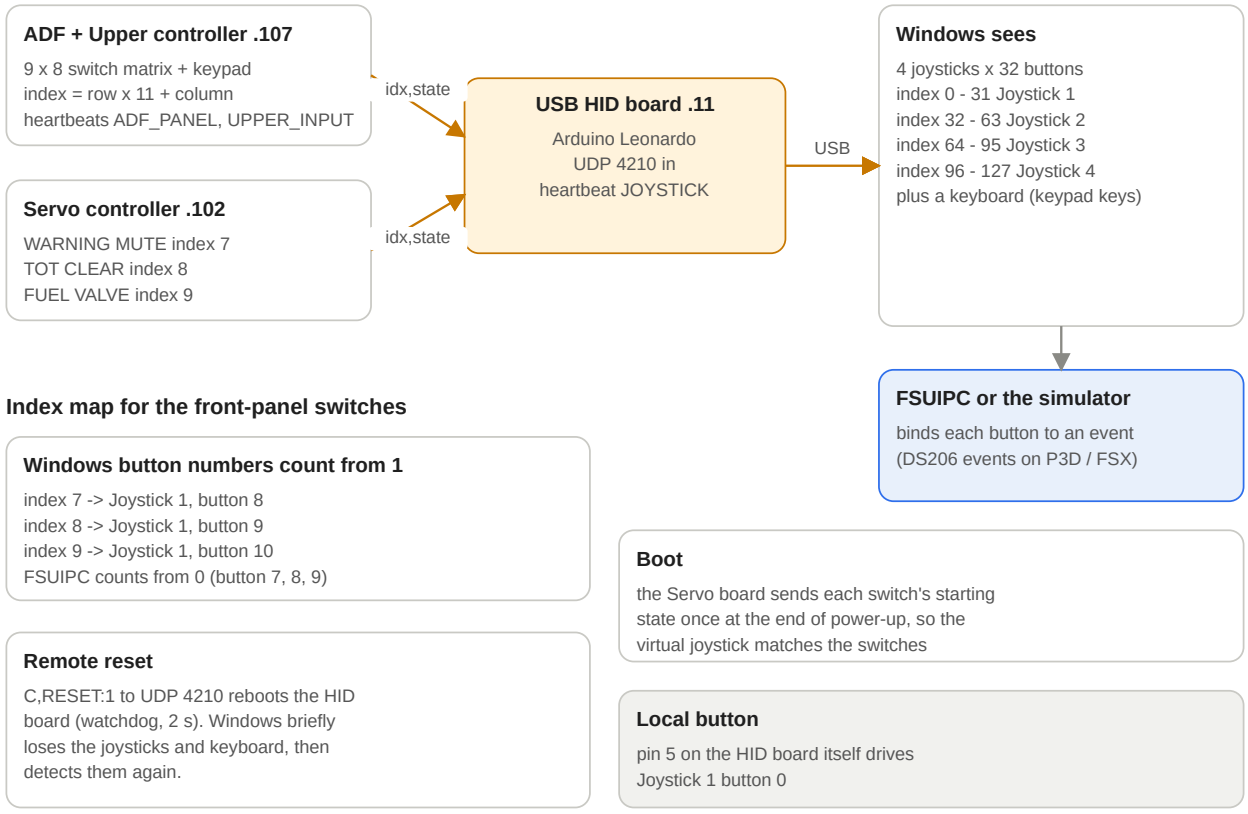


Figure 43 — From switch to joystick button

Board	Sketch	Address	Role
ADF and Upper controller (Mega 2560)	`JET_RANGER_ADF_CONTROLLER.ino`	172.16.1.107	Scans a 9 by 8 switch matrix (rows on pins 22 to 30, columns on 38 to 45) and a 4 by 3 keypad (A9 to A15), as well as the ADF panel. It replaces the old Upper controller (`JET_RANGER_UPPER_CONTROLLER.ino`, 172.16.1.103)
USB HID board (Leonardo class)	`UDP_TO_HID.ino`	172.16.1.11	Listens on UDP 4210 and presents 4 joysticks of 32 buttons plus a keyboard
Servo board (Mega 2560)	`JET_RANGER_SERVO_CONTROLLER.ino`	172.16.1.102	Also sends three front-panel switches (warning mute, TOT clear, fuel valve) as index 7, 8 and 9

12.1 How a switch becomes a joystick button

1. The ADF and Upper controller numbers each matrix position as `index = row x 11 + column` (0 to 95). The numbering is the old Upper board's, so existing bindings still apply; the matrix is now 9 rows by 8 columns instead of 16 by 11.
2. When a switch changes it sends `index,state` to `172.16.1.11:4210`, with state 1 = switch on and 0 = off.

3. The HID board picks joystick = index div 32 and button = index mod 32, then holds that button down while the switch is on.
4. Keypad keys are sent as `KEY,<key>,<state>` and typed as keyboard keys 0 to 9, `*` and `#`.
5. The servo board sends the same format for its three switches: index 7 (warning mute, pin 47), 8 (TOT clear, pin 43) and 9 (fuel valve, pin 48), with 1 = closed.

Index range	Windows device
0 to 31	Joystick 1
32 to 63	Joystick 2
64 to 95	Joystick 3
96 to 127	Joystick 4
128 and above	Not sent to Windows (out of range); nothing sends these any more

12.2 Auto Pilot, Audio and ADF panel buttons

The radio controller (Auto Pilot, Audio and Huey panels) and the ADF controller send their switches and knob clicks in **Joystick 3** (indexes 64 to 95, Auto Pilot panel) and **Joystick 4** (96 to 127, Audio, ADF and Huey panels), buttons nothing else uses (the Upper matrix's rows 6 to 8, indexes 66 to 95, are wired but have no switches; do not fit any at 66 to 73 or 77 to 81). Windows numbers them from 1:

Index	Windows button	Source
64 to 71	Joystick 3, buttons 1 to 8	Auto Pilot switches: ENGAGE, HDG, CRS, ALT, APPR, FD, BC, IAS
72 to 81	Joystick 3, buttons 9 to 18	Auto Pilot knobs (clockwise, anticlockwise): CRS, ALT, VS, IAS, HDG
96 to 103	Joystick 4, buttons 1 to 8	Audio switches: COM1, BOTH, NAV2, DME, COM2, NAV1, MKR, ADF
104 to 107	Joystick 4, buttons 9 to 12	ADF panel: landing gear, flaps, speed brakes, brakes
108 to 111	Joystick 4, buttons 13 to 16	ADF knobs (clockwise, anticlockwise): major, minor
112 to 114	Joystick 4, buttons 17 to 19	ADF panel: anti-ice, hyd boost, caution dim
115 to 117	Joystick 4, buttons 20 to 22	Huey Warning Panel: TEST, RESET, BRT / DIM

Every one of these is also sent to the simulator bridge as a control (see the radio controller chapter), so you only need to bind the joystick buttons if you want to use them differently.

12.3 Switch mapping in use

From `UpperSwitchMapping.ods`. Button numbers are as Windows and the simulators show them (1 to 32); the firmware counts from 0. A latching switch holds its button down while on, so bind both press and release: in FSUIPC use the `_SET` event with parameter 1 on press and 0 on release.

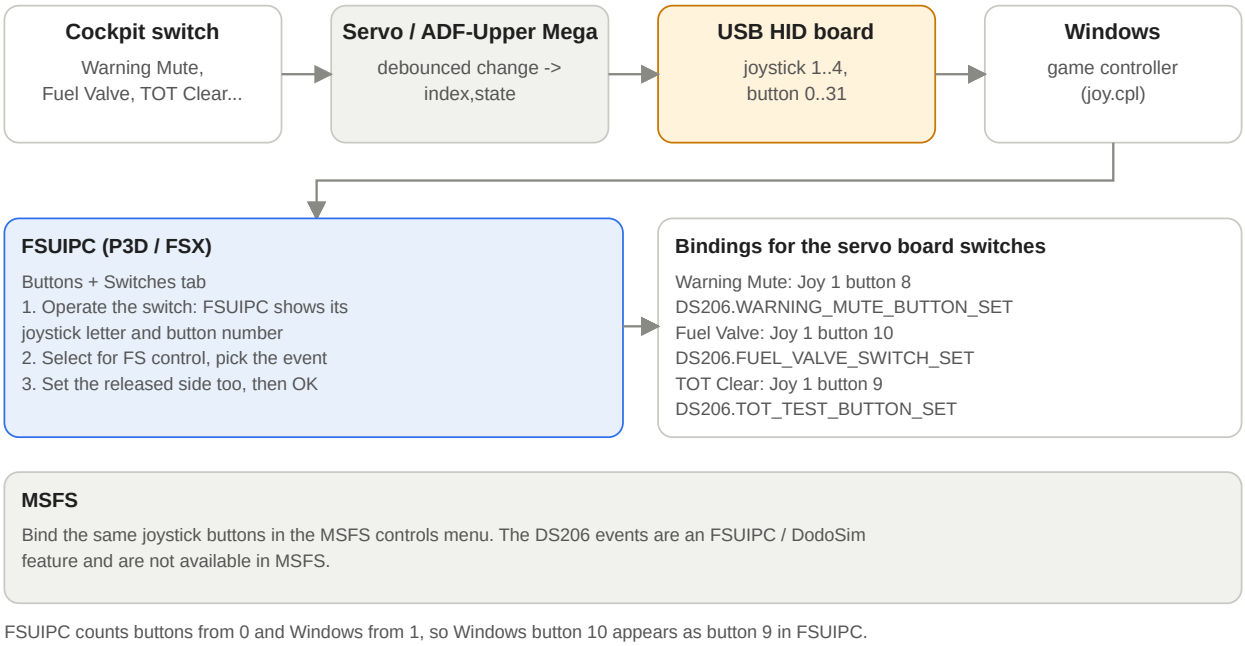
Index	Source	Switch	Joystick, button	Suggested DodoSim event
0 to 4	Upper 0, 0-4	Heater rotary HTR 0 to 4	J1, 1 to 5	
5	Upper 0, 5	Air vent	J1, 6	`HEAT_VENT_SWITCH_SET`
6	Upper 0, 6	Air heat	J1, 7	`HEAT_VENT_SWITCH_SET`
7	Servo D47	Warning mute	J1, 8	`DS206.WARNING_MUTE_BUTTON_SET`
8	Servo D43	TOT clear (TOT light test)	J1, 9	`DS206.TOT_TEST_BUTTON_SET`
9	Servo D48	Fuel valve	J1, 10	`DS206.FUEL_VALVE_SWITCH_SET`
11	Upper 1, 0	Forward fuel boost breaker	J1, 12	`FORE_FUEL_PUMP_CB_SET`
12	Upper 1, 1	Aft fuel boost breaker	J1, 13	`AFT_FUEL_PUMP_CB_SET`
13	Upper 1, 2	Generator field breaker	J1, 14	`GENERATOR_FIELD_CB_SET`
14	Upper 1, 3	Generator breaker	J1, 15	`GENERATOR_RESET_CB_SET`
15	Upper 1, 4	Directional gyro / attitude indicator	J1, 16	`DIR_GYRO_SWITCH_SET`
16	Upper 1, 5	Battery	J1, 17	`BATTERY_SWITCH_SET`
17	Upper 1, 6	Generator (on / start)	J1, 18	`GENERATOR_SWITCH_SET`
22	Upper 2, 0	Landing light	J1, 23	`LANDING_LIGHT_SWITCH_SET`
23	Upper 2, 1	Instrument lights	J1, 24	`PANEL_LIGHTS_SWITCH_SET`
24	Upper 2, 2	Caution light breaker	J1, 25	`CAUTION_LT_CB_SET`
25	Upper 2, 3	Anti-collision light	J1, 26	`COLLISION_LIGHTS_SET`

Index	Source	Switch	Joystick, button	Suggested DodoSim event
26	Upper 2, 4	Position lights	J1, 27	`POSITION_LIGHTS_SET`
27	Upper 2, 5	Pitot heat	J1, 28	`PITOT_HEAT_SWITCH_SET`
28	Upper 2, 6	Avionics	J1, 29	`AVIONICS_SWITCH_SET`
39	Upper 3, 6	Strobe light	J2, 8	none yet

The suggested events are the closest match in `DODSIM206V2.EVT`; the bindings actually saved in FSUIPC or MSFS are not in the repository. All other matrix positions are wired but unassigned.

12.4 Setting FSUIPC events from the joystick buttons

A joystick button does nothing in the simulator until it is bound to an event in FSUIPC. The **Warning Mute** and **Fuel Valve** switches on the servo board must be set this way before they work; the **TOT clear** switch beside them is bound the same way. This applies to Prepar3D and FSX with FSUIPC and `DODSIM206V2.EVT` installed. In MSFS, bind the buttons in the controls menu instead; the DS206 events are not available there.



FSUIPC counts buttons from 0 and Windows from 1, so Windows button 10 appears as button 9 in FSUIPC.

Figure 44 — Binding a switch

1. Start the simulator with the USB HID board connected. Windows should list the four joysticks (Game Controllers, `joy.cpl`); pressing a switch lights the matching button there.
2. In the simulator open FSUIPC from the Add-ons menu and choose the **Buttons + Switches** tab.
3. Operate the switch you want to bind (for example flip the Fuel Valve switch). FSUIPC shows the joystick letter and button number it saw. FSUIPC counts buttons from 0 and Windows from 1, so Windows button 10 shows as button 9 in FSUIPC.
4. Under the pressed (or closed) side, tick *Select for FS control*, choose the event from the list and type the parameter in the box beside it. The DS206 events are listed alphabetically.
5. A switch that stays on needs the released side set as well: tick *Select for FS control* there, choose the same event and enter the opposite parameter, so the cockpit follows the switch in both directions.
6. Click OK, flip the switch and check the result in the cockpit. After an event file update FSUIPC re-indexes the events alphabetically, so check these bindings again.

- **Warning Mute:** Joystick 1 button 8 (FSUIPC button 7). Event `DS206.WARNING_MUTE_BUTTON_SET`, parameter 1 when pressed and 0 when released. Alternatively bind `DS206.WARNING_MUTE_BUTTON_TRIGGER` (no parameter) to the pressed side only.

- **Fuel Valve:** Joystick 1 button 10 (FSUIPC button 9). Event `DS206.FUEL_VALVE_SWITCH_SET`, parameter 1 when closed and 0 when released. The valve in the simulator follows the switch; if it ends up the wrong way round, swap the two parameters.
- **TOT clear (TOT light test):** Joystick 1 button 9 (FSUIPC button 8). Event `DS206.TOT_TEST_BUTTON_SET`, parameter 1 when pressed and 0 when released.

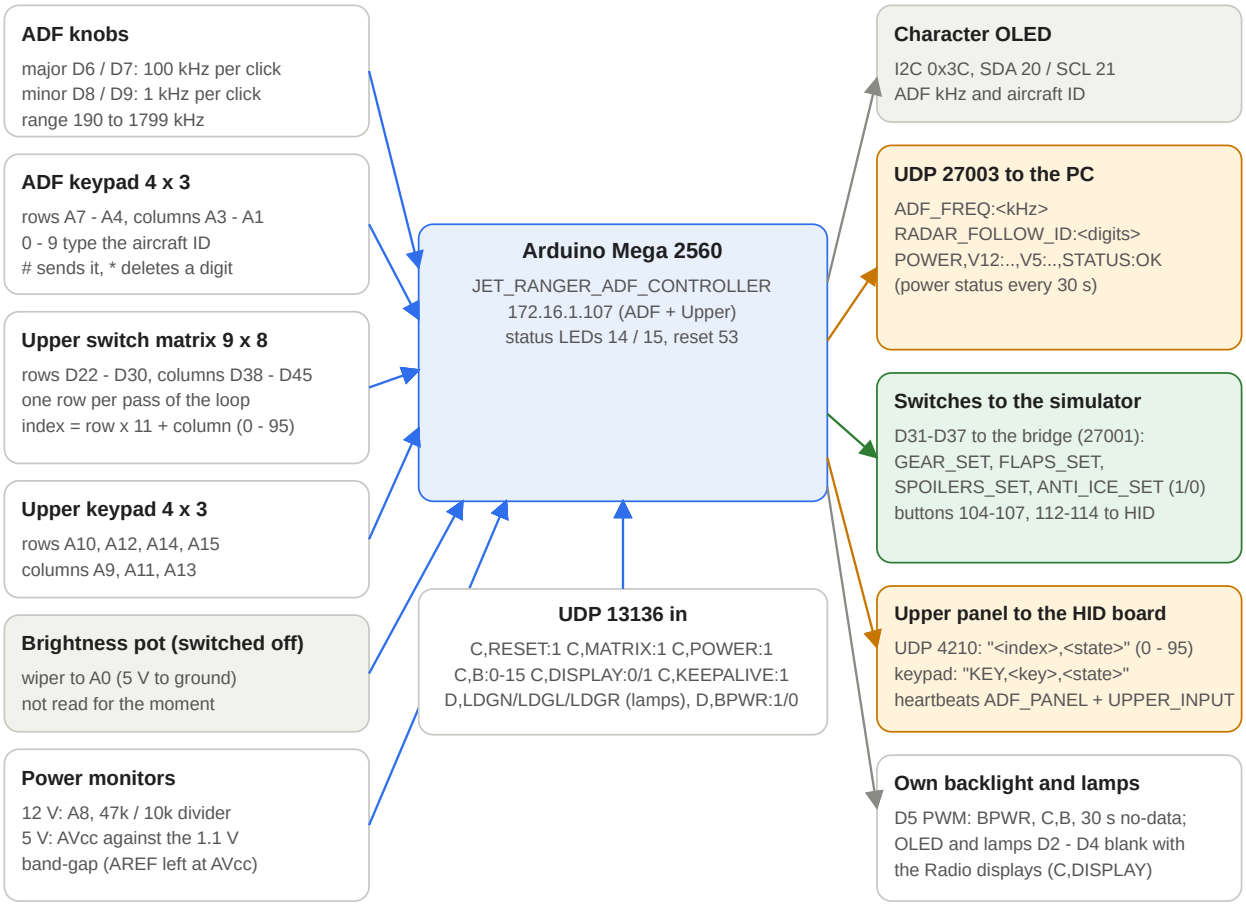
If FSUIPC does not see a switch, check that Windows shows the button moving in `joy.cpl` first. If it does not, the problem is on the board side: the servo board's debug messages report each change and the HID board logs the button it sets.

12.5 Known issues

- The HID board's own test button on pin 5 also drives Joystick 1 button 1, the same button as HTR 0.
- The old Upper sketch's DCS-BIOS stubs were not carried into the merged ADF and Upper sketch, so nothing is sent to DCS; only the joystick path works.
- The HID board reboots when it receives `C,RESET:1` (or `RESET`) on UDP 4210. Its USB connection drops while it restarts, so Windows briefly loses the four joysticks and the keyboard; re-check any game that has them open.
- The servo board's three switches use indexes 7, 8 and 9, which Windows shows as Joystick 1 buttons 8, 9 and 10. If they were meant to be Windows buttons 7, 8 and 9, button 7 is already the Air heat switch (index 6), so the indexes would have to move.

13 ADF and Upper controller

The ADF panel is another Mega, at 172.16.1.107, running `JET_RANGER_ADF_CONTROLLER.ino` on the Jet Ranger Upper ADF Controller PCB (revision 1.0), the combined ADF and Upper board. Since 31 October 2026 it also does the job of the old Upper controller (172.16.1.103): it scans the **overhead switch matrix** and the **Upper keypad** and sends them to the USB HID board (see Upper panel switches and keypad below), so there is no separate Upper board any more. It tunes the ADF with two knobs, takes the ID of an aircraft to follow on the radar from a keypad, sends its landing gear, flaps, speed brakes, brakes, anti-ice, hyd boost and caution dim switches to the simulator, lights the three landing gear lamps, blanks its display, lamps and backlight with the other panels, and watches the 12 V and 5 V supplies. The PCB has a brightness pot, but it is **switched off for the moment**. The sketch is flashed (COM26); the 12 V reading, the landing gear lamps and the ADF keypad have been checked on the bench, but the Upper matrix and keypad, the switches at their new pins and the blanking and backlight rules have not.



The toggle switches send their position as a value (every change and every 5 s); the brakes push button sends PARKING_BRAKES.
The Upper panel (switch matrix and keypad) was the separate Upper controller, 172.16.1.103, until it was merged into this sketch.

Figure 45 — ADF panel

13.1 Using the panel

Control	What it does
Major knob	Changes the ADF frequency by 100 kHz per click
Minor knob	Changes it by 1 kHz per click (faster when you turn it quickly). The range is 190 to 1799 kHz
Keypad 0 to 9	Types the ID of the aircraft to follow (up to 8 digits)
Keypad #	Sends the ID

Control	What it does
Keypad *	Deletes the last digit; on an empty entry it stops following
Landing gear, flaps and speed brakes switches	Set the simulator to match the switch position (see Panel switches)
Brakes button	Toggles the parking brake
Anti-ice, hyd boost and caution dim switches	Joystick 4 buttons 17 to 19; anti-ice also sets the simulator's anti-ice (see Panel switches)
Brightness pot	Not used for the moment: the panel backlight stays at its default and no brightness is sent to the other panels

The OLED shows the frequency on the first line (`ADF  340 kHz`) and the ID on the second: `ID: 1234_` while you type, `FOLLOW 1234` once it is sent, or `FOLLOW OFF`. An entry you leave unfinished for 20 seconds is dropped.

Brightness pot. The pot on the PCB (A0) can set the backlight of this panel and of the Radio, Servo, Stepper and Dual Stepper panels, but `USE_BRIGHTNESS_POT` in the sketch is 0, so it does nothing: this panel's backlight stays at its default (level 4) until a `C,B` command sets it, and the other panels follow the SwingTester slider or their own defaults. Set it to 1 and reflash to bring the pot back.

13.2 Inputs and outputs

Function	Mega pins	Notes
ADF major knob	D6, D7	Encoder A, B
ADF minor knob	D8, D9	Encoder A, B
ADF keypad	scan rows A7, A6, A5, A4, scan columns A3, A2, A1	4 by 3; the lines arrive in reverse order on this PCB; see the key table below
Upper switch matrix	rows D22 to D30, columns D38 to D45	9 by 8, 72 switches (see Upper panel switches and keypad)
Upper keypad	rows A10, A12, A14, A15, columns A9, A11, A13	4 by 3
Panel brightness pot	A0	Wiper, 5 V to ground; not read while the pot is switched off
12 V monitor	A8	47 k and 10 k divider with 0.1 uF
5 V monitor	AREF / AVcc	Measured against the internal 1.1 V reference; the AREF pin is left at its default
Backlight	D5	IRF540N MOSFET: level 4 by default, then `C,B`; off with `BPWR:0`, the no-data timeout or no data (see Panel blanking and backlight)
OLED	I2C D20, D21 (0x3C)	2 lines of 16 characters
Status LEDs	D14 green, D15 red	Red flashes while Ethernet starts
Landing gear lamps	D2, D3, D4	Nose, left, right: lit by data packets (see below)
Landing gear, flaps, speed brakes and brakes switches	D31 to D34	Switches to ground; see Panel switches
Hyd boost, anti-ice and caution dim switches	D35, D36, D37	Switches to ground (connector J20); see Panel switches
Ethernet module	SPI 50 to 52, reset 53	

Keypad keys. The ADF keypad's seven lines come out of pins 18 to 24 of the ADF panel connector (J5) to A1 to A7 (connector J41). On this PCB they arrive in reverse order, so the sketch scans A7, A6, A5 and A4 as the rows and A3, A2 and A1 as the columns (the old ADF board used D22 to D28). With the first guess at the pins keys 1, 2 and 3 did nothing and most others reported as the wrong key; after the swap every key reports correctly. The key labels in the sketch were set by pressing every key and moving each label to where it was reported:

Scan row	Column 0 (A3)	Column 1 (A2)	Column 2 (A1)
0 (A7)	6	4	5
1 (A6)	9	7	8
2 (A5)	#	*	0
3 (A4)	3	1	2

If the keys ever need checking again, send `C,MATRIX:1` while holding a key: the debug listener shows which row and column changes.

13.3 Panel switches and joystick buttons

The seven switches are read when they change (30 ms debounce, to ground when on). Each change is sent two ways, like the Auto Pilot and Audio panels:

Switch	Joystick button (index)	What goes to the bridge (UDP 27001)
Landing gear (toggle)	Joystick 4, button 9 (104)	The position: `GEAR_SET:1` when closed (gear down), `GEAR_SET:0` when open
Flaps (toggle)	Joystick 4, button 10 (105)	`FLAPS_SET:1` (flaps fully down) or `FLAPS_SET:0`
Speed brakes (toggle)	Joystick 4, button 11 (106)	`SPOILERS_SET:1` (fully out) or `SPOILERS_SET:0`
Brakes (push button)	Joystick 4, button 12 (107)	`PARKING_BRAKES` when pressed
Anti-ice (toggle)	Joystick 4, button 17 (112)	`ANTI_ICE_SET:1` or `ANTI_ICE_SET:0`
Hyd boost (toggle)	Joystick 4, button 18 (113)	none
Caution dim (toggle)	Joystick 4, button 19 (114)	none
ADF major knob (clockwise, anticlockwise)	Joystick 4, buttons 13, 14 (108, 109)	none: the knob tunes the ADF frequency
ADF minor knob (clockwise, anticlockwise)	Joystick 4, buttons 15, 16 (110, 111)	none

The gear, flaps, speed brakes and anti-ice toggle switches send their **position as a value** (not an event the simulator could toggle), every time they change and again **every 5 seconds**, so the simulator always follows the switches. That includes the position they were in at power-up: about 5 seconds after the panel is switched on, the aircraft's gear, flaps and speed brakes move to match the switches. For 1.5 seconds after power-up the switches are only read. If a switch works the wrong way round in the simulator, swap its 1 and 0 in the sketch (`adfSwitches[]`). A knob click is a short joystick press.

Anti-ice, hyd boost and caution dim on the Dodo. The Dodo Jet Ranger has its own events for these switches: `ANTI_ICE_SWITCH_SET`, `HYDRAULICS_SWITCH_SET` (hyd boost) and `CAUTION_LIGHT_SWITCH_SET` (caution lights bright or dim), each taking 1 or 0. In FSUIPC's Buttons + Switches page, bind Joystick 4 buttons 17, 18 and 19 to them (the switch pins are D36 anti-ice, D35 hyd boost and D37 caution dim), with 1 on press and 0 on release (and turn on the Dodo's "Sync controls on load" option). The simulator's own controls for these are toggles, so the bridges have no reliable command for hyd boost and caution dim, and the Dodo may ignore the bridge's `ANTI_ICE_SET`: use the FSUIPC binding **or** the bridge command for anti-ice, not both.

13.4 Upper panel switches and keypad

The overhead switch matrix and the numeric keypad of the old Upper controller are scanned by this board. Each of the 72 matrix switches (9 rows by 8 columns) is sent to the USB HID board as a joystick button, numbered `row x 11 + column` (0 to 95) as before, so the FSUIPC and MSFS button assignments and the switch table in the Joystick section do not change. The matrix is smaller than the old board's 16 by 11 (176 positions). The Upper keypad types keys 0 to 9, `*` and `#` on the HID board's keyboard. The Warning Mute and TOT Test connectors on the PCB are part of the matrix (indexes 88 and 89), nothing is assigned to them yet. A switch that is closed when the board starts is reported straight away. Everything on the panel counts as input for the 30 second no-data timeout. The old Upper board's backlight is this board's backlight.

13.5 Landing gear lamps

The three lamps on the ADF panel (nose, left and right) are lit by a data packet to the ADF controller on UDP 13136: `D,LDGN:1,LDGL:1,LDGR:0` (1 = lit, 0 = off). The **SwingTester** has a toggle for each lamp and *All ON* and *All OFF* buttons (see Program windows). The simulator bridges do not send them yet, so for now they are a bench test. They go out with the rest of the panel and come back as last set.

13.6 Power status

Five seconds after start-up, and then every **30 seconds**, the board sends the two rail voltages to the PC (UDP 27003) and to the debug listener (UDP 27000):

`POWER,V12:12.41,V5:5.02,STATUS:OK`

`STATUS` is `OK`, or the rails that are out of range: `5V_LOW` (below 4.75 V), `5V_HIGH` (above 5.25 V), `12V_LOW` (below 10.8 V) or `12V_HIGH` (above 15.0 V). Send `C,POWER:1` to get a reading at once. The first time you run the board, compare the readings with a meter and trim `V12_CAL` and `V5_CAL` at the top of the sketch. The same two readings are in the heartbeat the board sends every 10 seconds (`ADF_PANEL V5:5.03 V12:12.41`, and again as `UPPER_INPUT` so the Health Monitor's Upper light stays green), and the **Health Monitor** shows them in its voltage table: this board is the only one with a 12 V reading. The 12 V divider is on A8 on this PCB (it was on A15 on the old ADF board, and a sketch reading A15 reported 5.33 V with 12.3 V on the board).

13.7 Messages to the PC (UDP 27003)

Message	Meaning
`ADF_FREQ:830`	The ADF frequency in kHz, sent 250 ms after the knobs stop
`RADAR_FOLLOW_ID:4521`	The aircraft ID typed on the keypad; `RADAR_FOLLOW_ID:0` means stop following
`POWER,V12:12.41,V5:5.02,STATUS:OK`	Power status, every 30 seconds

These use port 27003, not 27001, because the FSUIPC and MSFS bridges treat anything they do not recognise on 27001 as a COM1 standby frequency. (The panel switch commands above do go to 27001: the bridges recognise those names.) **Neither bridge reads port 27003 yet**, so the frequency and the follow ID do not reach the simulator.

13.8 Control packets

The board acts on control packets on UDP 13136: `C,RESET:1` reboots it, `C,MATRIX:1` prints the raw ADF keypad readings to the debug listener, `C,POWER:1` sends a power status, `C,B:0` to `C,B:15` sets the backlight brightness (while the pot is switched off), `C,KEEPALIVE:1` counts as data and `C,DISPLAY:0` / `C,DISPLAY:1` (sent by the Radio controller) blank and restore the panel. Of the data packets it uses the three landing gear lamp tokens and `BPWR`.

13.9 Panel blanking and backlight

The panel follows the other boards' rules:

- **With the Radio.** When the Radio controller blanks its displays (no avionics power, or no data or input for 30 seconds) it tells the ADF controller, which clears its OLED and puts the landing gear lamps out. They come back when the Radio's displays do.
- **Main bus.** The bridges send `BPWR` to the ADF controller like the other boards; `BPWR:0` switches the backlight off and `BPWR:1` brings it back.
- **Brightness.** `C,B:0` to `C,B:15` sets the backlight (the SwingTester slider sends it); the default is level 4.
- **No data.** 30 seconds with no data packet from the bridge and no knob, key or switch use blanks the OLED, the lamps and the backlight until something arrives, so the panel goes dark if the PC is not feeding it.

13.10 Checking the panel for the first time

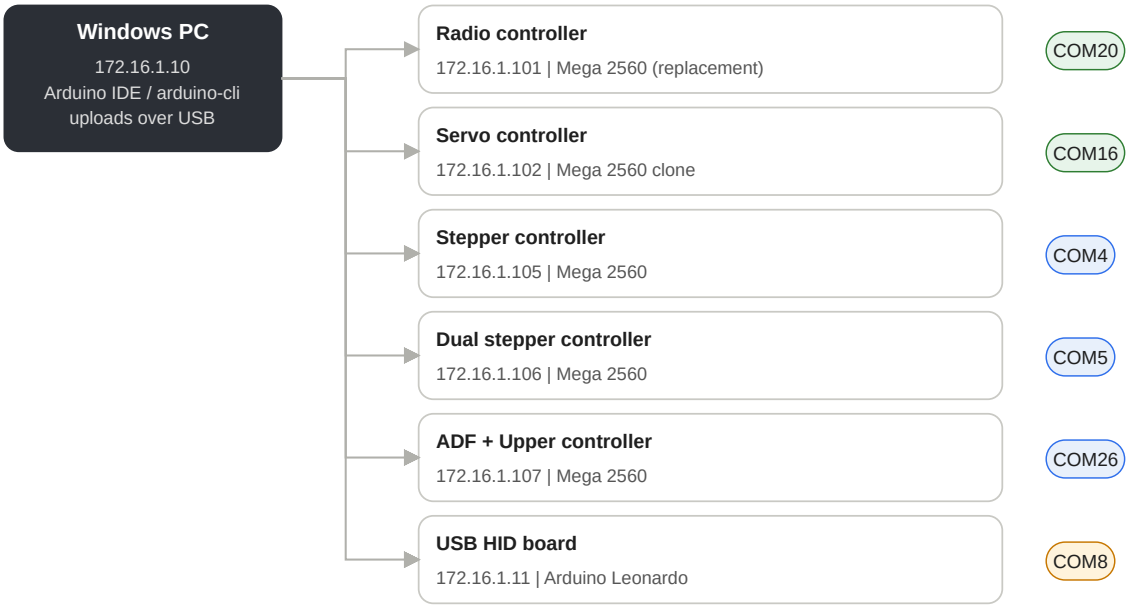
1. Flash the board (see COM ports and flashing), then watch the debug listener for `Jet Ranger ADF: Setup Complete`.
2. Press each ADF keypad key and check the OLED shows the key you pressed (`C,MATRIX:1` shows the raw readings if not). Operate a few Upper switches and keys and watch the debug listener for `Front Input - <index>:<state>` and `Upper key pressed: <key>`.
3. Turn each knob. If it tunes the wrong way, swap its two pins in the `RotaryEncoder` lines.
4. Light the three landing gear lamps from the SwingTester.

5. Check the OLED. The sketch drives the same kind of character OLED as the Radio controller; a graphics SSD1306 needs different code.
6. Compare the power status with a meter and trim `V12_CAL` and `V5_CAL`.

14 COM ports and flashing

Each Arduino is programmed over its own USB cable from the primary PC. The cockpit itself runs over Ethernet, so USB is only needed to upload a sketch.

14.1 COM port to Arduino mapping



COM numbers are the ones used at the last flash. They follow the USB socket and the board, so check them (see the guide) before an upload. Each board also has a network connection: USB is for programming only, the cockpit runs over UDP.

Figure 46 — Which board is on which COM port

Board	Address	Hardware	COM port	Sketch
Radio controller	172.16.1.101	Arduino Mega 2560 (replacement board)	**COM20**	`JET_RANGER_RADIO_CONTROLLER`
Servo controller	172.16.1.102	Mega 2560 clone	**COM16**	`JET_RANGER_SERVO_CONTROLLER`
Stepper controller	172.16.1.105	Mega 2560	**COM4**	`JET_RANGER_STEPPER_CONTROLLER`
Dual stepper controller	172.16.1.106	Mega 2560	**COM5**	`JET_RANGER_OLED_DUAL_STEPPER_CONTROLLER`
ADF and Upper controller (combined)	172.16.1.107	Mega 2560	**COM26**	`JET_RANGER_ADF_CONTROLLER` (the old `JET_RANGER_UPPER_CONTROLLER` is retired)
USB HID board	172.16.1.11	Arduino Leonardo	**COM8**	`UDP_TO_HID`

These are the ports used the last time each board was flashed. A COM number belongs to a USB socket and a board together, so it **can change**:

- The Servo controller is a clone Mega that reports no USB serial number, so Windows ties its COM number to the USB socket it is plugged into. Plug it into a different socket and it appears as a new port.
- Replacing a Mega (the Radio controller was replaced once) gives a new port number.
- The Lowrance GPS cable is separate: the GPS interface asks for its serial port the first time it runs and remembers it.

Always check the port before an upload: open Device Manager, Ports (COM and LPT), or run `arduino-cli board list`. If you are not sure which board is which, unplug one USB cable and see which port disappears.

Add the confirmed port to the table above (and to the sketch header comment) once you have uploaded to a board.

14.2 Flashing a board

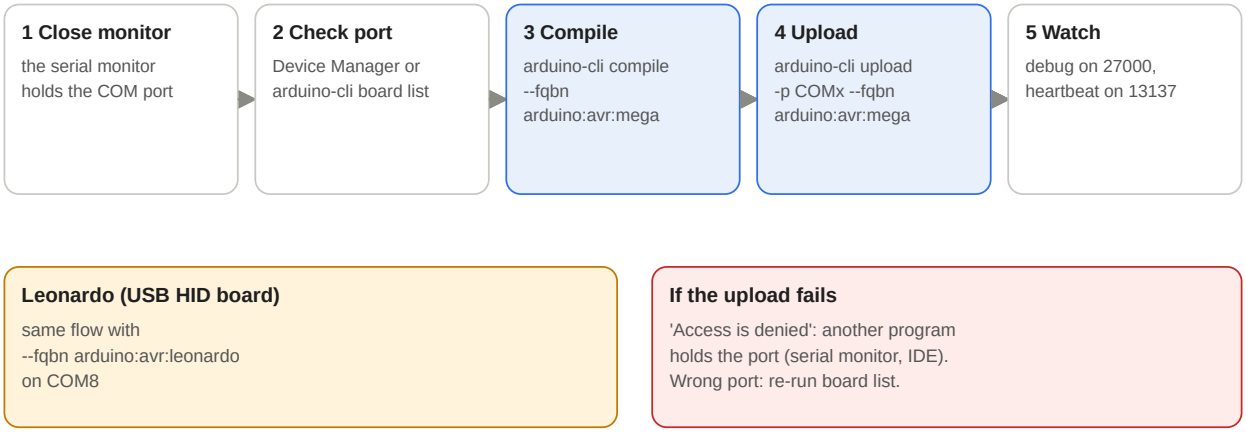


Figure 47 — Flashing a board

1. Close the Arduino IDE **serial monitor**. It holds the port, and an upload then fails with *Access is denied*.
2. Check the COM port (above).
3. Compile the sketch.
4. Upload to that port.
5. Watch the debug listener (UDP 27000) and the Health Monitor: the board prints *Ethernet Started* and *Setup Complete* and its light goes green within about 10 seconds.

```
arduino-cli compile --fqbn arduino:avr:mega "Jet Ranger Arduino
Sketches\JET_RANGER_STEPPER_CONTROLLER"
arduino-cli upload -p COM4 --fqbn arduino:avr:mega "Jet Ranger Arduino
Sketches\JET_RANGER_STEPPER_CONTROLLER"
```

For the Leonardo (USB HID board) use `--fqbn arduino:avr:leonardo` and its port, COM8. Flash only the board you are working on. If a board ever reboot-loops after a reset packet, power-cycle it and flash it again.

15 PCBs and wiring

The Mega sits on a carrier PCB that breaks its pins out to servo headers, lamp outputs, gauge sockets and the Ethernet module. KiCad projects for every board are in `Jet Ranger/PCBs` in the repository, with Gerbers in each `MANUFACTURING` folder. The list below was brought up to date on 18 October 2026.

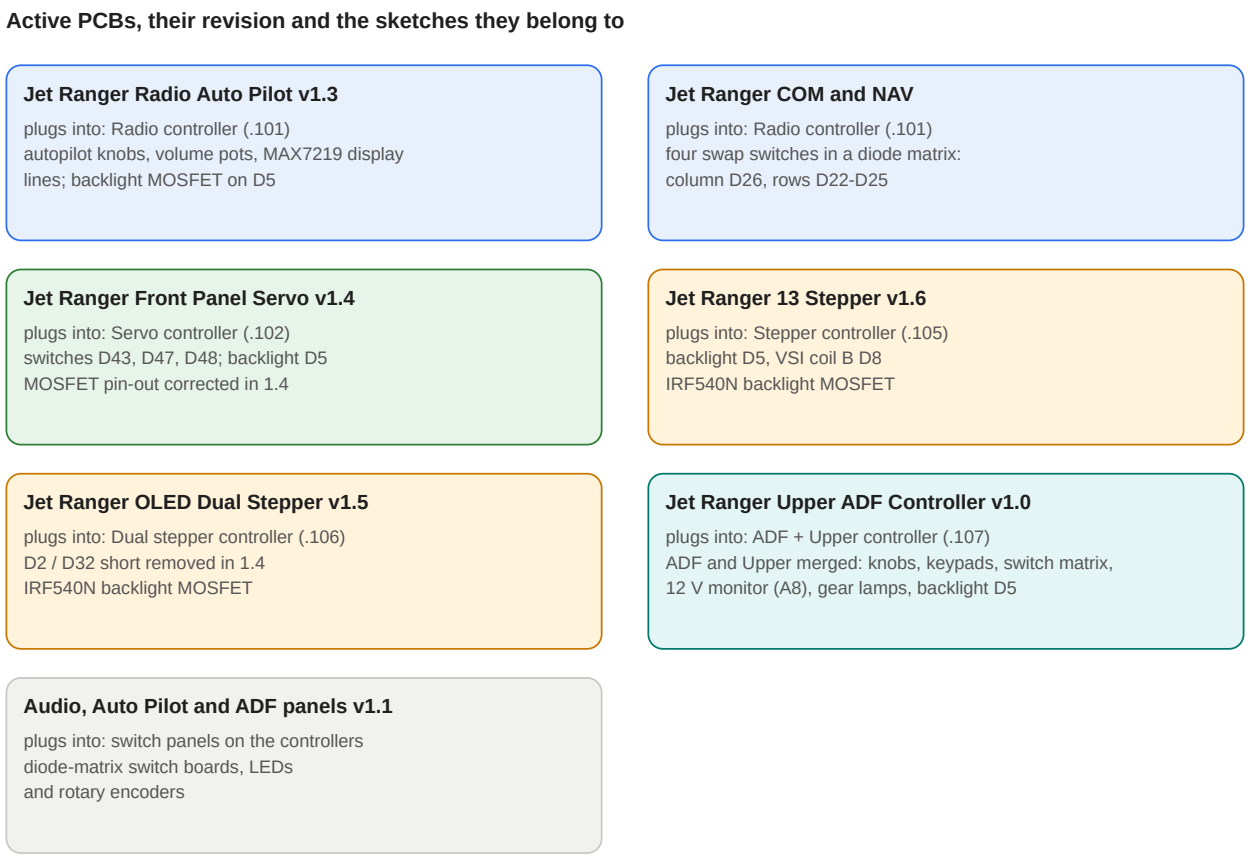


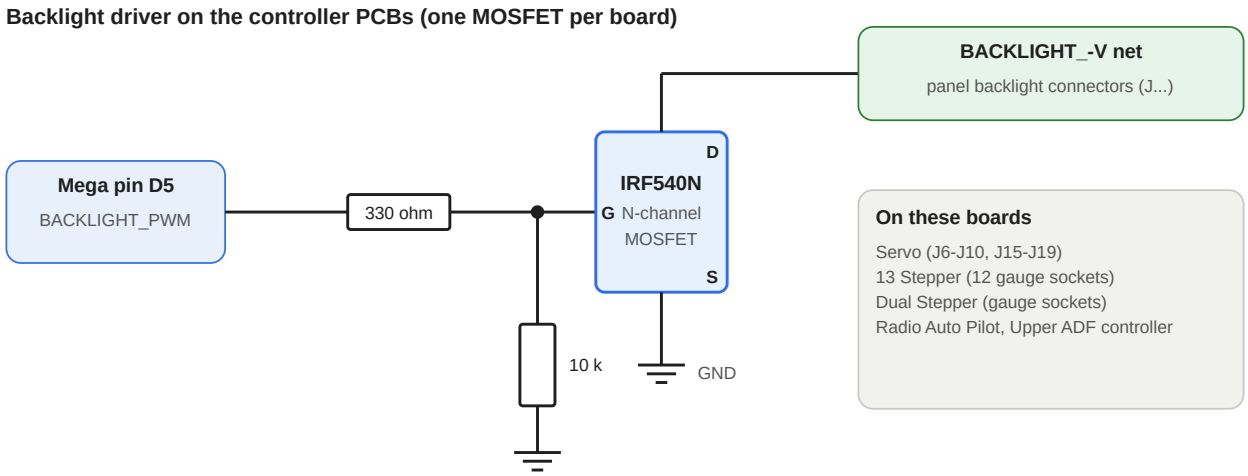
Figure 48 — Active PCBs, their revision and their sketches

PCB	Revision	Used by	Main parts
Jet Ranger Front Panel Servo	1.4	Servo board	Mega shield, ULN2803A lamp drivers, servo headers, lamp outputs, three switch inputs, backlight MOSFET
Jet Ranger 13 Stepper	1.6	Stepper board	Mega shield, USR-ES1 Ethernet, 14 six-pin gauge sockets, 12 V lamp supply with polyfuses, backlight MOSFET
Jet Ranger OLED Dual Stepper	1.5	Dual stepper board	Mega shield, USR-ES1, six-pin gauge sockets, TCA9548A sockets for the OLEDs, backlight MOSFET
Jet Ranger X27 Small Stepper	not set	Each stepper gauge	One X27 or VID29 stepper, backlight LED, six-pin connector to the controller
Jet Ranger Radio Auto Pilot	1.3	Radio controller	Radio and autopilot controller with the backlight MOSFET
Jet Ranger COM and NAV	not set	Radio panel	8 rotary encoders, push buttons, four frequency-swap switches
Jet Ranger Upper ADF Controller	1.0	ADF and Upper controller (replaces	ADF knobs, ADF keypad, overhead

PCB	Revision	Used by	Main parts
		the Upper Controller 1.4 and ADF Controller 1.6 boards)	switch matrix and Upper keypad, panel brightness pot, 12 V monitor (A8), gear lamps, anti-ice, hyd boost and caution dim switch inputs, I2C OLED, backlight MOSFET
Jet Ranger Audio Panel, Auto Pilot Panel, ADF Panel	1.1	Switch panels	Diode-matrix switch boards with LEDs and rotary encoders

The X27 steppers run straight from the Mega's pins, so the stepper boards need no driver chips; the 12 V input feeds lamps and backlights. The schematic and board layout of every PCB are in the *PCB schematics and layouts* chapter at the end of this guide.

15.1 Backlight MOSFET (revisions of 18 October 2026)



Revisions 1.3 to 1.5 (18 October 2026) replaced the generic MOSFET symbol, which had source and drain swapped, with the IRF540N.
A board built from an earlier revision has the transistor pinned wrongly: check it, or fit it the right way round.

Figure 49 — The backlight driver

Every controller board switches its panel backlights with one **IRF540N** MOSFET: Mega pin D5 (`BACKLIGHT_PWM`) goes through 330 ohm to the gate, a 10 k resistor holds the gate to ground, the drain is the shared backlight net and the source is ground. The earlier revisions used a generic MOSFET symbol with **source and drain swapped**, so the 1.4 Servo, 1.3 Radio Auto Pilot, 1.4 Upper, 1.3 ADF Controller and 1.5 Stepper and Dual Stepper boards were revised (the combined Upper ADF Controller started from the corrected ADF design) to use the IRF540N with the correct pin-out. If you have a board from an earlier revision, check the transistor before relying on the backlight.

15.1.1 Backlight resistor values

The backlight circuit uses two groups of resistors.

Where	Resistor	Value
Controller board, Mega pin D5 to the MOSFET gate	Gate resistor	330 ohm
Controller board, MOSFET gate to ground	Gate pull-down	10 kohm
Backlight LEDs wired in a chain on the 12 V backlight supply, 3 LEDs in the chain	Series resistor	150 ohm
Backlight LEDs wired in a chain on the 12 V backlight supply, 2 LEDs in the chain	Series resistor	330 ohm
Backlight LEDs wired in a chain on the 12 V backlight supply, 1 LED in the chain	Series resistor	560 ohm

The more LEDs there are in a chain, the lower the series resistor, because the LEDs use up more of the supply voltage. The Jet Ranger X27 Small Stepper board has two backlight LEDs in one chain; its schematic currently shows a **470 ohm** resistor, which is not the 330 ohm in this table, so check the value fitted.

Version 1.4 of the OLED Dual Stepper PCB (12 October 2026) separated Mega pins 2 and 32. Earlier versions joined them in copper because of a copy-and-paste net name, so the fuel quantity and fuel load gauges shared coil A. On a board older than 1.4 fit only one of those two gauges or cut the trace between the pins.

Lamp pins on the Servo PCB. Ten of the 15 lamp outputs are on different Mega pins from the sketch (see Known issues); test each lamp on its own.

Radio Auto Pilot knobs. The PCB also carries the autopilot knobs, volume pots, display drivers and a fifth matrix column that the radio sketch does not read yet, and four of its encoders have A and B swapped relative to the sketch.

15.2 Connections on the servo carrier board

Connector	Count	Pins	Wiring
Servo headers	16 on the PCB, 2 used	26 (pitch), 27 (bank)	Signal, +V, GND; +V from the servo rail
Lamp outputs	15	A1 to A15	Pin HIGH = lamp on; through a driver for lamps above 20 mA
Spare outputs	3	not assigned	Reserved for future lights
Ethernet module	1	SPI 50 to 52, chip select, reset 53	Reset is pulsed by the firmware at start-up
Status LEDs	2	14 green, 15 red	With series resistors to GND
Servo power input	1		5 to 6 V; ground joined to Mega GND
Front-panel switch inputs	3	D43, D47, D48	Each to GND through a switch (SW1 TOT clear, SW3 warning mute, SW2 fuel valve); internal pull-up, debounced in firmware

The rule that matters most: **never power servos from the Mega's 5 V pin.** Servos draw current spikes when they move, which can reset the Mega and drop the Ethernet link.

15.3 Wiring checklist

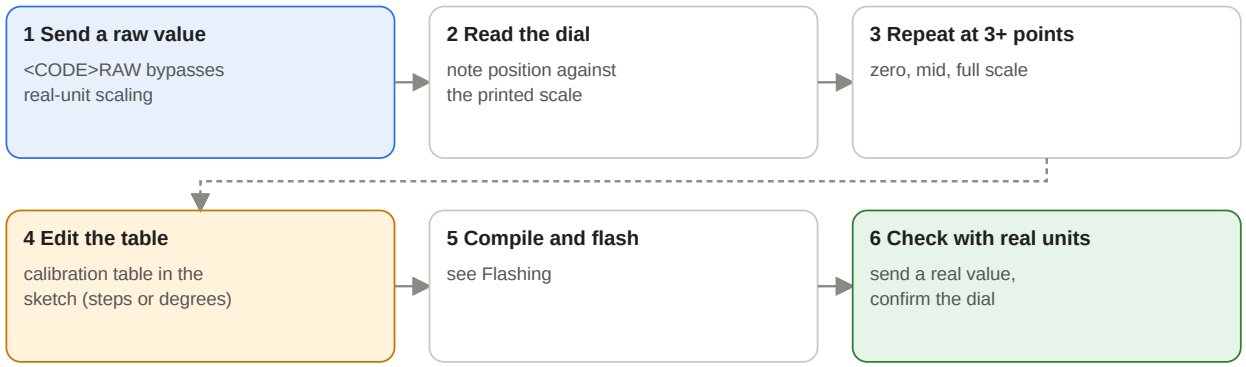
- Servo rail fed from its own 5 to 6 V supply, not the Mega.
- Supply ground joined to Mega GND (signals need a shared reference).
- Each servo plugged in signal-side matching the header silkscreen.
- Lamps checked during the power-up lamp test.
- Ethernet cable to the same switch as the primary PC.
- Board serial number noted in the sketch header.

16 Calibration

The servo board turns attitude degrees into a servo position (0 to 180) with two small tables in the sketch. The stepper boards do the same for steps. Between table points the boards interpolate in a straight line; beyond the ends they hold the end value.

Instrument	Attitude	Servo position
Pitch (`PITCH_DEG_TABLE`)	-30 degrees	179
Pitch	0 (level)	113
Pitch	+30 degrees	70
Bank (`BANK_DEG_TABLE`)	-90 degrees	5
Bank	0 (level)	93
Bank	+90 degrees	179

The self-test sweep uses `aServMinPosition[]` and `aServMaxPosition[]` (bank 5 to 179, pitch 166 to 70) and parks at `aServZeroPosition[]` (bank 93, pitch 113).



Typical tables: pitch -30 / 0 / +30 degrees, bank -90 / 0 / +90 degrees, one row per stepper or servo gauge.

Sending <CODE>RAW:n moves the motor to step n directly, so any dial can be calibrated without a simulator.

Figure 50 — The calibration loop

16.1 Calibrating a servo

1. Send raw positions with `BANKRAW` or `PITCHRAW` (for example `D,PITCHRAW:113`) until the instrument shows a known attitude.
2. Note the position for each attitude you check: at least both ends and level.
3. Enter them in `PITCH_DEG_TABLE` or `BANK_DEG_TABLE`, and set `aServZeroPosition[]` to the level position so the self-test parks there.
4. Upload, then check with real values, for example `D,PITCH:0` and `D,BANK:30`.

16.2 Calibrating a stepper gauge

1. Send raw steps with the `RAW` token (for example `D,TQRAW:300`) and read the dial.
2. Repeat at zero, mid scale and full scale (more points where the scale is not linear).
3. Edit the gauge's table in the sketch (`TQ_PCT_TABLE`, `IAS_KT_TABLE` and so on) and its `...ZeroOffset` trim if the zero is off.
4. Upload and check with real values (`D,TQ:60`).

The StepperVSITester sends every token and raw token without a simulator. Do not drive a gauge hard against its mechanical stop: a stalled servo or stepper draws high current and gets hot.

17 Troubleshooting

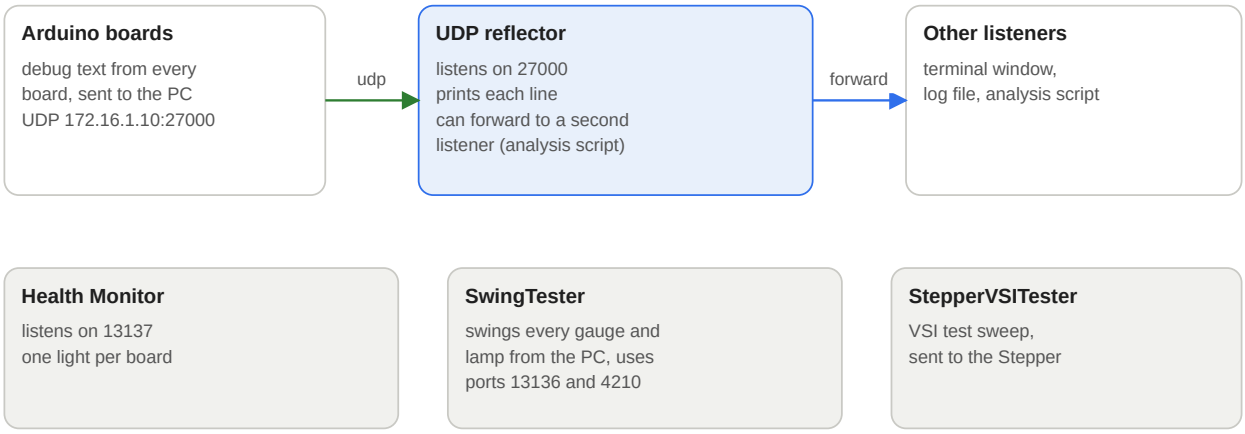


Figure 51 — Debug and test tools

Symptom	Likely cause	What to check
Red LED keeps fast-flashing, no gauge sweep	Board stuck in Ethernet start-up	Ethernet module seated; reset line on pin 53; module powered
Sweep runs but gauges never move in flight	Packets not reaching UDP 13136	PC on 172.16.1.x; Windows firewall allows outbound UDP; the bridge sends to the right boards (servo .102, steppers .105 and .106); only one bridge running
No debug messages, or the Health Monitor shows a board as down	Listener or Health Monitor not running, or PC address wrong	PC at 172.16.1.10; listener bound to UDP 27000, Health Monitor to UDP 13137; firewall allows inbound UDP on both
One gauge pinned at an end stop	Value outside the calibration table, or table points wrong	Send the `RAW` token to find the true limits, then re-measure the table (see Calibration)
The DCS bridge says no stream	DCS not running, or DCS-BIOS not installed or its `Export.lua` line missing	Start DCS with an aircraft loaded; check `Saved Games\DCS\Scripts\Export.lua` has the DCS-BIOS line; no other program should block multicast 239.255.50.10 port 5010
A Huey gauge reads wrong or sits against a stop	The needle-position scale is a first guess (*uncalibrated*)	Calibrate it in the DCS bridge (select the row, Calibrate selected...) at two or more readings
The X-Plane bridge says no data yet	X-Plane not running, UDP receive off, or the wrong host or port	Allow incoming UDP in X-Plane's network settings; use `--xplane-host` and `--xplane-port`; check the Windows firewall
One gauge stays at zero with the X-Plane bridge	The aircraft does not fill in that dataref	Use Search datarefs... and add the right one to a profile file
All gauges drop to zero after 30 seconds	No packets arriving (watchdog)	Bridge running and connected; simulator not paused; network link
A board feels slower, or its knobs and needles stutter, while the PC (or another board) is off	Each packet sent to a missing device waits up to 80 ms for an ARP answer (the fast Ethernet timeout)	Switch the missing device on, or reduce debug traffic; see *Unreachable hosts and the fast Ethernet timeout* in the Network chapter
Attitude indicator lags fast changes	1 degree per 5 ms step limit	Expected; reduce `servoCheckInterval` to speed up
Pressing an Audio or Auto Pilot switch is slow, or a joystick button seems delayed	The USB HID board is switched off or unplugged: each send to it waits for an	The radio now pauses joystick sends for 30 seconds after one fails and says so in

Symptom	Likely cause	What to check
	answer that never comes	the monitor (`USB HID board ... not answering`); connect the HID board and the buttons return
Servo buzzes or jitters	Weak servo supply	Separate 5 to 6 V supply for servos, common ground with the Mega
A backlight is off	Main bus below 20 V (`BPWR:0`), no packets for 30 seconds, or brightness set to 0	Check the main bus voltage in the bridge window; check the bridge is running; send a brightness with the SwingTester slider
A backlight stays on at power-up with no PC	Old firmware without the no-data rule, or the board has never been flashed with the backlight code	Flash the current Radio, Servo, Stepper and Dual Stepper sketches; the backlight should go out about 30 seconds after power-up
The backlight is always on or flickers	MOSFET pinned wrongly on a board from an earlier revision (source and drain swapped)	Check the transistor against the revision notes (PCBs and wiring)
The GPS shows Brisbane Airport	No aircraft data for 30 seconds	Check a bridge or the bench tester is running on the same PC and sending to UDP 27002; check no other program owns the port
The GPS time is hours out with DCS	The DCS mission clock is the theatre's local time and the bridge's **Mission time zone** box is still 0	Set the box to the theatre's hours from UTC (for example 4 for the Caucasus); the GPS date is the PC's
The GPS shows the PC time, not the simulator time	The packets carry no `UTC:` (for example the FSUIPC time read failed)	The window says *PC clock*; check the bridge version
The GPS interface cannot open its port	Wrong COM port, or another program holds it	Use *Change serial port...*; close other serial programs
A lamp never lights	Wiring or the lamp's driver	It should light at the power-up lamp test; if not, check the A pin and its driver
A radio knob tunes the wrong radio	Bare frequency sent instead of a labelled one	Use the real board (it labels COM2, NAV1 and NAV2); the Radio emulator sends bare numbers, so only COM1 tunes
COM and NAV displays blank	Avionics bus below 24 V, or no data and no knob use for 30 seconds	Switch avionics on; turn a knob to wake them
Dual stepper OLEDs blank	Main bus below 24 V (`PWR:0`)	Switch the battery on; check the main bus voltage in the bridge window
Radio swap switches do nothing	Column 0 of the matrix stuck	Send `C,MATRIX:1` and read the debug listener (see Radio controller)
FSUIPC binding fires the wrong event	EVT re-indexed after an update	Re-map the binding in FSUIPC
Joystick buttons vanish after a reset	USB HID board rebooted	Windows re-detects them within seconds; re-check games that had them open
Upload fails with *Access is denied*	Another program holds the COM port	Close the serial monitor and the Arduino IDE port; check the port with `arduino-cli board list`
Upload goes to the wrong board	COM port moved	Re-check Device Manager; see COM ports and flashing
Data reaches the boards only in bursts, with gaps of several seconds	The PC's Ethernet link to the board switch is dropping in and out	In Windows, check the network adapter stays connected; check the 5 V supply of the network switch stays above 4.75 V (it once dipped below and the switch dropped its ports), then the cable and switch port

17.1 Known issues in the code

Servo firmware (`JET_RANGER_SERVO_CONTROLLER.ino`)

- `powerAvailable` is set from `NAVCOM1` but nothing uses it, so the attitude indicator stays live when the simulator's power is off (the bridge sends all lamps off below 20 V).
- Several lamps log *Transmission Temperature changed* to the debug listener, whichever lamp changed.

- The PCB labels 10 of the 15 lamp outputs on different pins from the sketch (A1/A2, A5/A6, A7 to A10, A11/A12). Test each lamp on its own, for example `D,RLOW:1`, and correct whichever side is wrong.

FSUIPC event file (`DODSIM206V2.EVT`)

- The line before event 133 is missing its `//`, so FSUIPC may reject or mis-read it.
- Events 189 and 190 are named the wrong way round: 189 is called `_SET` but toggles, 190 is called `_TOGGLE` but takes 1 or 2.
- Events 69 to 71 mix up the instrument and cockpit light breakers: 69 is named `COCKPIT_LT_CB_TOGGLE`, and `INST_LT_CB_SET` appears twice.

17.2 Open questions

- Power supply rating for the servo rail and the 12 V lamp supply.
- Whether the estimates for the FlyInside and CowaSim aircraft hold at other power settings.

18 Reference: UDP tokens

From the current sketches. Gauge tokens take real units; lamp tokens take 0 (off) or 1 (on). `UDP_Interface_Values.xlsx` still lists the old 16-servo pin plan, so use these tables instead.

18.1 Gauge tokens

Token	Gauge	Board	Units
`BANK`	Attitude bank	Servo (.102)	degrees, plus and minus 90
`PITCH`	Attitude pitch	Servo (.102)	degrees, plus and minus 30
`IAS`	Airspeed	Stepper (.105)	knots, one decimal allowed
`VSI`	Vertical speed	Stepper (.105)	ft/min
`AGL`	Radar altitude	Stepper (.105)	feet
`TQ`	Engine torque	Stepper (.105)	%
`N1`	Gas producer	Stepper (.105)	%
`ITT`	Turbine out temperature	Stepper (.105) or Dual (.106)	degrees C
`FUEL`	Fuel quantity	Stepper (.105) or Dual (.106)	US gal
`OILT`	Engine oil temperature	Dual (.106) or Stepper	degrees C
`OILP`	Engine oil pressure	Dual (.106) or Stepper	psi
`XMSNT`	Transmission oil temperature	Dual (.106) or Stepper	degrees C
`XMSNP`	Transmission oil pressure	Dual (.106) or Stepper	psi
`RPME`	Engine RPM (N2)	Dual (.106)	%, one decimal
`RPMR`	Rotor RPM	Dual (.106)	%, one decimal
`ALT`	Altimeter	Dual (.106)	feet
`FUELLOAD`	Fuel load	Dual (.106)	psi
`ELECTRICALLOAD`	Electrical load	Dual (.106)	%

Every gauge token also has a `RAW` twin (for example `PITCHRAW`, `IASRAW`) that takes a raw servo position or stepper step count and skips the calibration table.

18.2 Lamp tokens (servo board)

Token	Lamp	Pin
`EOUT`	Engine out	A1
`RLOW`	Rotor RPM low	A2
`TOPW`	Transmission oil pressure	A3
`TOTW`	Transmission oil temperature	A4
`BHOT`	Battery hot	A5
`BTMP`	Battery temperature	A6
`EC`	Engine chip	A7
`TRC`	Tail rotor chip	A8
`TC`	Transmission chip	A9
`BD`	Baggage door	A10
`FFLTR`	Aft fuel filter	A12
`FPMP`	Fuel pump	A11
`GENF`	Generator fail	A13
`LOWF`	Low fuel	A14
`SCF`	SC fail (standby compass)	A15

18.3 Aircraft information to the GPS interface (UDP 27002)

`D,LAT:<deg>,LON:<deg>,ALT:<feet>,IAS:<knots>,HDG:<degrees true>,UTC:<yyyyMMddHHmmss[.fff]>`, sent every 500 ms by a bridge or the bench tester. Any subset of keys works, and the CSV form `lat,lon,altitude_ft,airspeed_kt,heading_deg[,utc]` is accepted too. South and west are negative.

18.4 Status and control tokens

Token	Packet	Meaning
NAVCOM1	D	0 = avionics bus below 24 V (radio COM and NAV displays blank, panel LEDs off), 1 = on
LED_CRS, LED_HDG, LED_IAS, LED_ALT, LED_APPR, LED_FD, LED_BC, LED_APENG	D	Auto Pilot panel LEDs on the radio controller (1 = on)
LED_COM1, LED_BOTH, LED_NAV2, LED_DME, LED_OUTER, LED_INNER, LED_COM2, LED_NAV1, LED_MKR, LED_ADF, LED_MID	D	Audio panel LEDs on the radio controller (1 = on)
APHDG, APCRS, APIAS, APVS, APALT	D	The values on the Auto Pilot panel's OLED (radio controller): heading and course as three digits (north is 360), airspeed in knots, vertical speed in feet per minute and altitude in feet
LDGN, LDGL, LDGR	D	The landing gear lamps on the ADF panel (nose, left, right): 1 = lit
BPWR	D	Backlight power for the Radio, Servo, Stepper, Dual Stepper and ADF and Upper controllers: 1 = main bus 20 V or more, 0 = backlight off. Sent first in the Servo and Stepper packets, after PWR in the Dual Stepper packet, after NAVCOM1 in the Radio packet and on its own to the ADF and Upper board
PWR	D	Dual stepper board: 1 = all four OLEDs on, 0 = blank them (main bus below 24 V). Sent first in the packet
ZULU	D	Zulu time as HHMM, for example 1430 (dual stepper clock OLED)
TEMP	D	Temperature text for the dual stepper's temperature OLED, for example 25C
B	C	Backlight brightness 0 to 15 (0 off, 15 full). Radio, Servo, Stepper, Dual Stepper and ADF and Upper (the Radio also sets the brightness of its Auto Pilot, Audio and Huey panel LEDs from it). The ADF controller can send it from its brightness pot, which is switched off for the moment
SWING<CODE>	C	Runs one gauge's start-up sweep on demand, for example C,SWINGTQ:1 (stepper boards) or C,SWINGPITCH:1 (servo board); the value is ignored
OLEDTEST	C	1 fills the four OLEDs solid white, 0 restores them (dual stepper board)
KEEPALIVE	C	Ignored command that still counts as received data and restarts the 30 second no-data timer (SwingTester sends it during a run)
RESET	C	Reboots the board about 2 seconds later (servo, stepper, dual stepper, radio and ADF and Upper controllers on port 13136; USB HID board on port 4210). The radio and ADF and Upper controllers use a software reset because a watchdog reset hung the radio's bootloader; the upper controller's reset has not been bench-tested
MATRIX	C	Radio and ADF controllers: print the raw button matrix to the debug listener
POWER	C	ADF controller: send a power status (12 V and 5 V rails) now
LEDTEST	C	Radio controller: 1 lights all 19 Auto Pilot and Audio panel LEDs, 0 returns them to the simulator's state

19 PCB schematics and layouts

The schematic (the top sheet of each project) and the board layout of every PCB project in `Jet Ranger/PCBs/`, exported from KiCad 9. The layout shows the front and back copper (red and blue), the silkscreen (yellow) and the board outline, viewed from the front. The images are bitmaps: zoom in on the PDF or the document to read the labels, or open the KiCad project for the full detail. The schematic and layout images are regenerated from the KiCad files, so they show the boards as they were on the day the document was built, including any changes not yet committed.

19.1 Jet Ranger Radio Auto Pilot

Revision 1.3 (change log; the title block shows 1.2, 2026-10-17). Mega shield for the Radio controller (172.16.1.101). Carries the USR-ES1 Ethernet module, the TCA9548A I2C multiplexer for the OLEDs, the MAX7219 that lights the 19 Auto Pilot and Audio panel LEDs, the backlight MOSFET (D5), the COM and NAV panel connector (`COMM_NAV`), the `AUTOPILOT` and `AUDIO` panel connectors, five matrix columns and the Auto Pilot knob inputs.

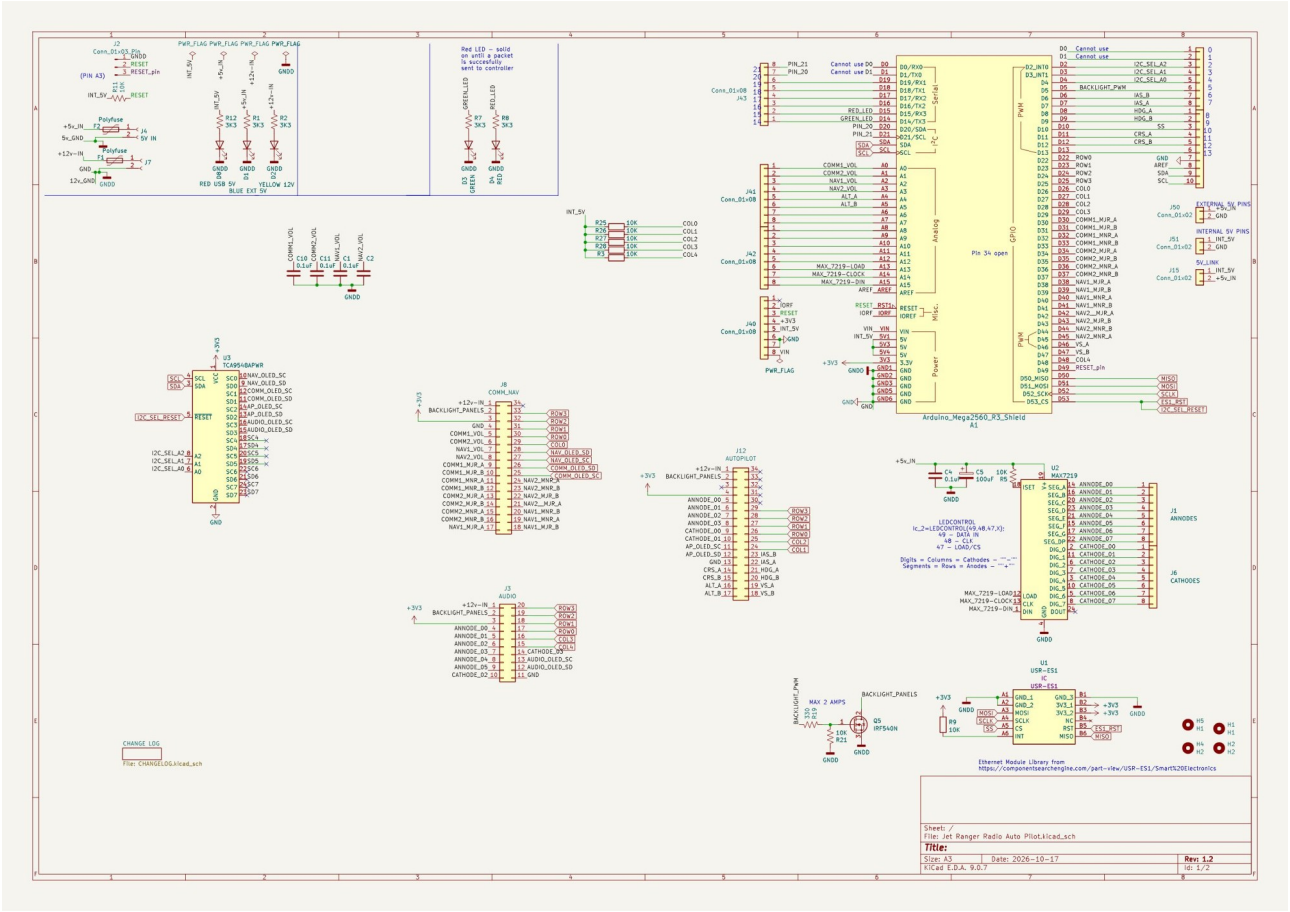


Figure 52 — Jet Ranger Radio Auto Pilot schematic

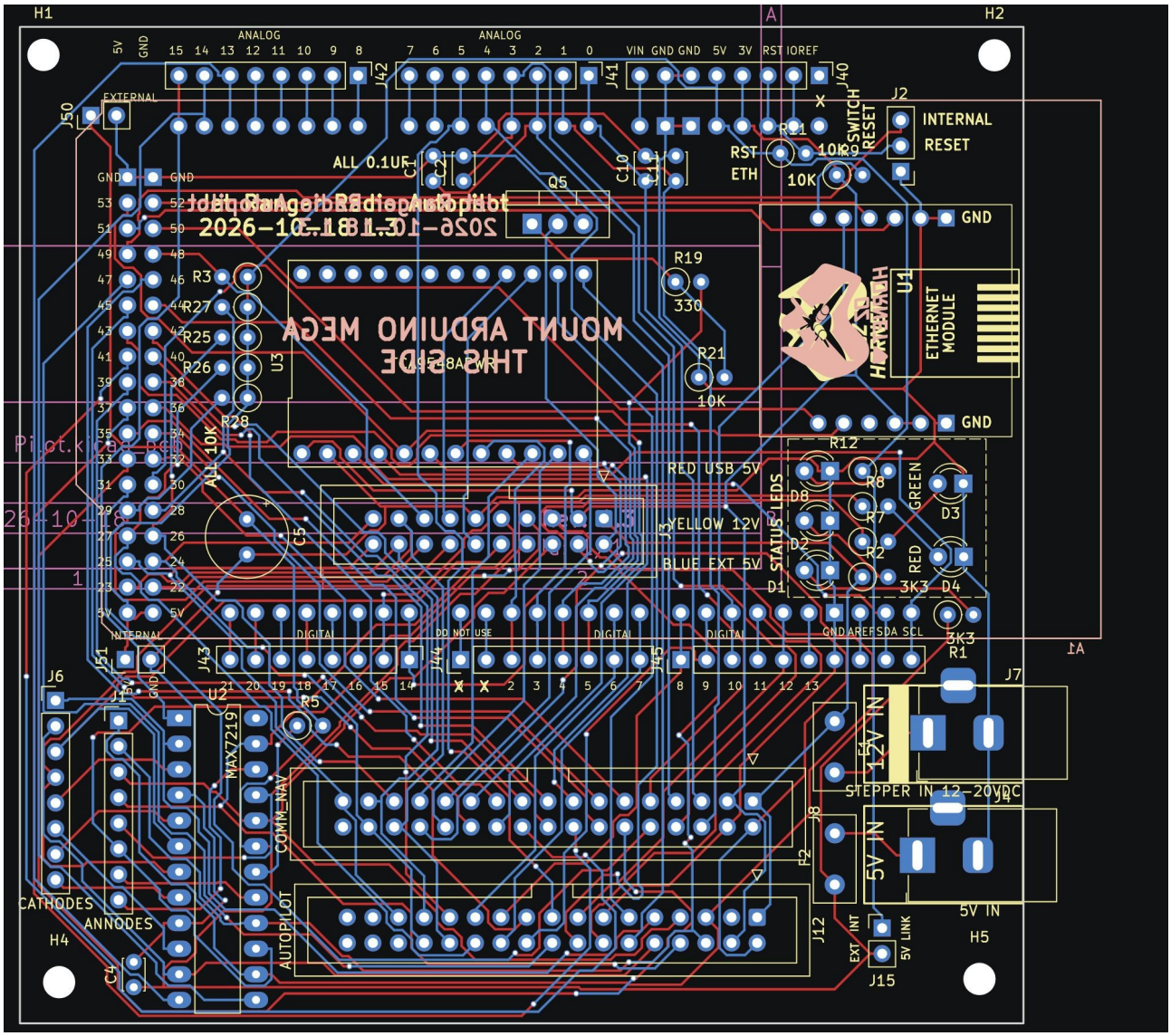


Figure 53 — Jet Ranger Radio Auto Pilot board layout

19.2 Jet Ranger COM and NAV

No revision set (title block dated 2024-08-10). Radio panel board behind the COM and NAV faceplate. Eight rotary encoders (a major and a minor knob for COM1, COM2, NAV1 and NAV2), the four frequency-swap switches in a diode matrix on column 0, four volume pots, and sockets for the two character OLEDs.

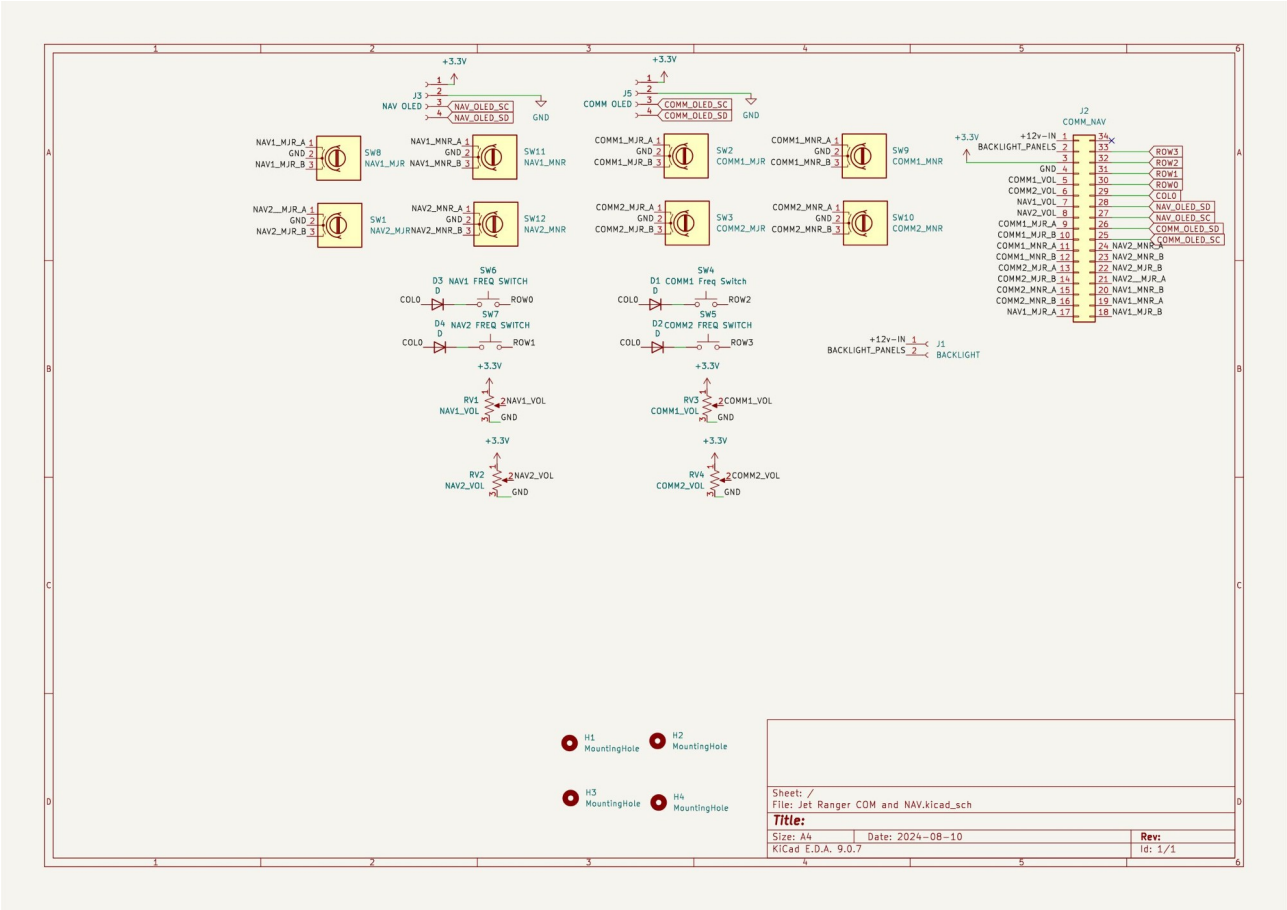


Figure 54 — Jet Ranger COM and NAV schematic

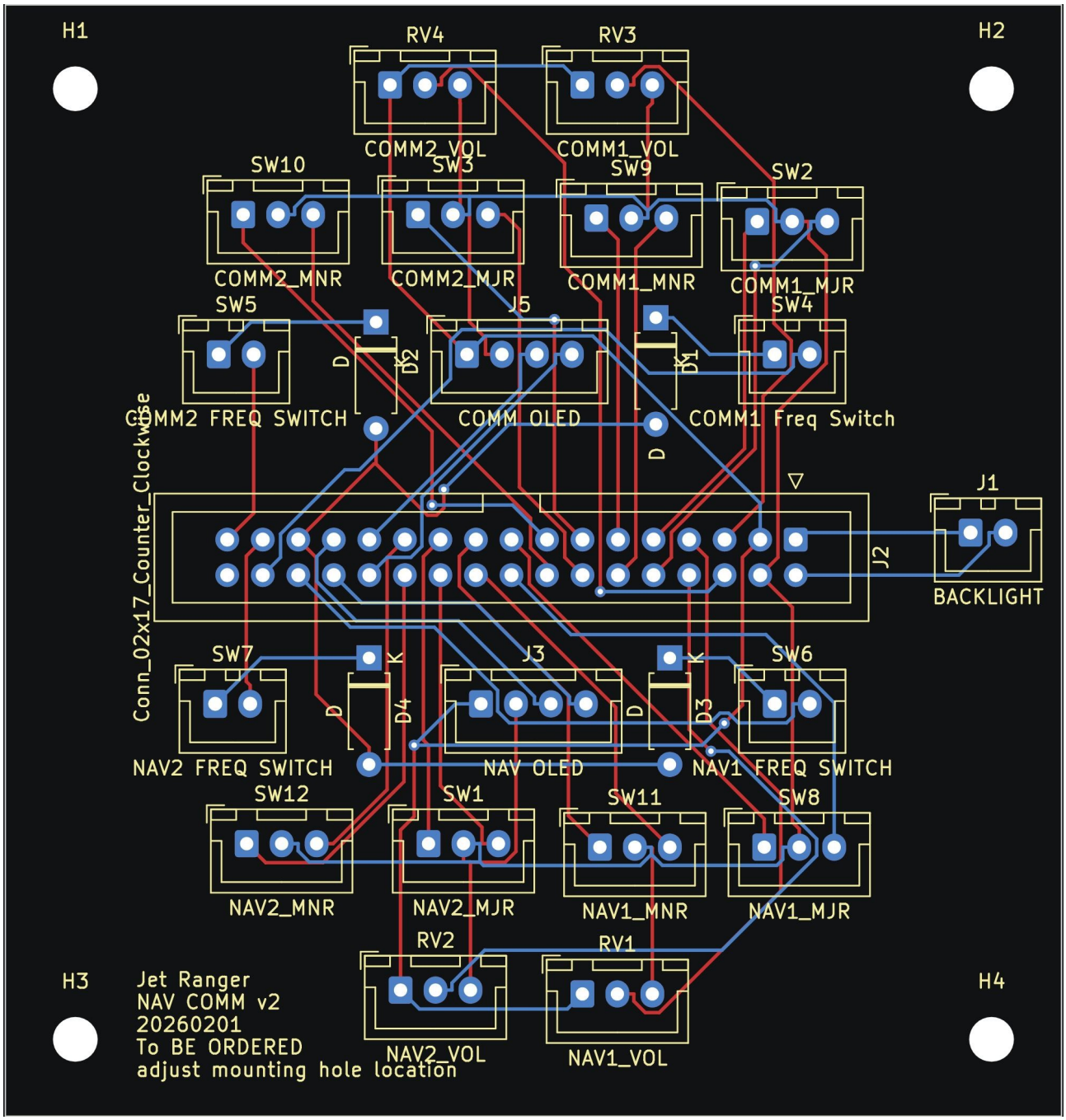


Figure 55 — Jet Ranger COM and NAV board layout

19.3 Jet Ranger Front Panel Servo

Revision 1.4 (2026-10-18). Mega shield for the Servo controller (172.16.1.102). ULN2803A lamp drivers for the 15 warning lamps, headers for the pitch and bank servos, three front-panel switch inputs, the USR-ES1 Ethernet module and the backlight MOSFET. Version 1.4 corrected the MOSFET pin-out.

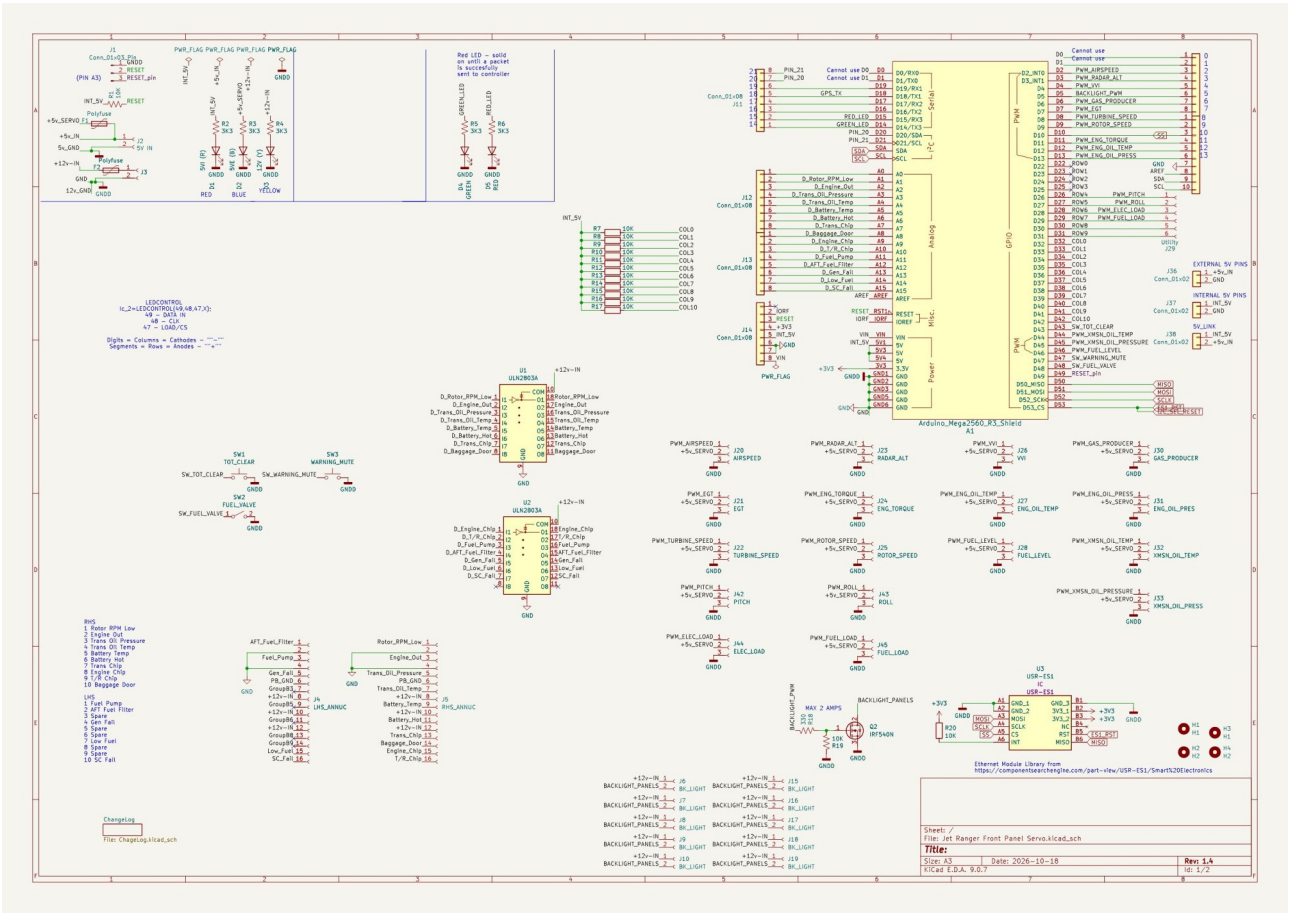


Figure 56 — Jet Ranger Front Panel Servo schematic

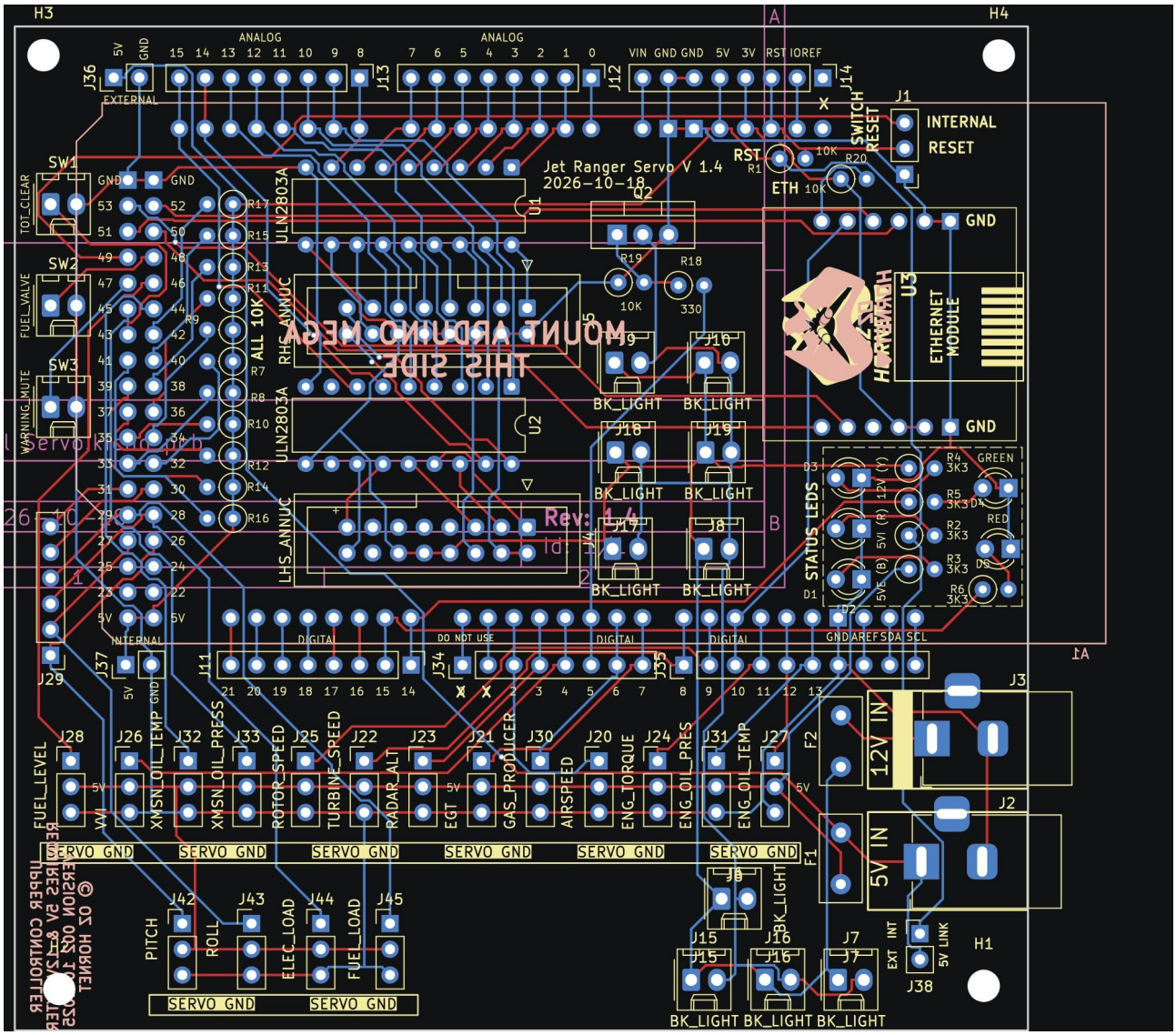


Figure 57 — Jet Ranger Front Panel Servo board layout

19.4 Jet Ranger 13 Stepper

Title block revision 1.6 (2026-10-18); the change log lists changes up to 1.5. Mega shield for the Stepper controller (172.16.1.105). Fourteen six-pin gauge sockets for the X27 steppers, a 12 V lamp supply with polyfuses, the USR-ES1 Ethernet module and the backlight MOSFET. 1.4 moved the LEDs away from the USB connectors and 1.5 selected the IRF540N.

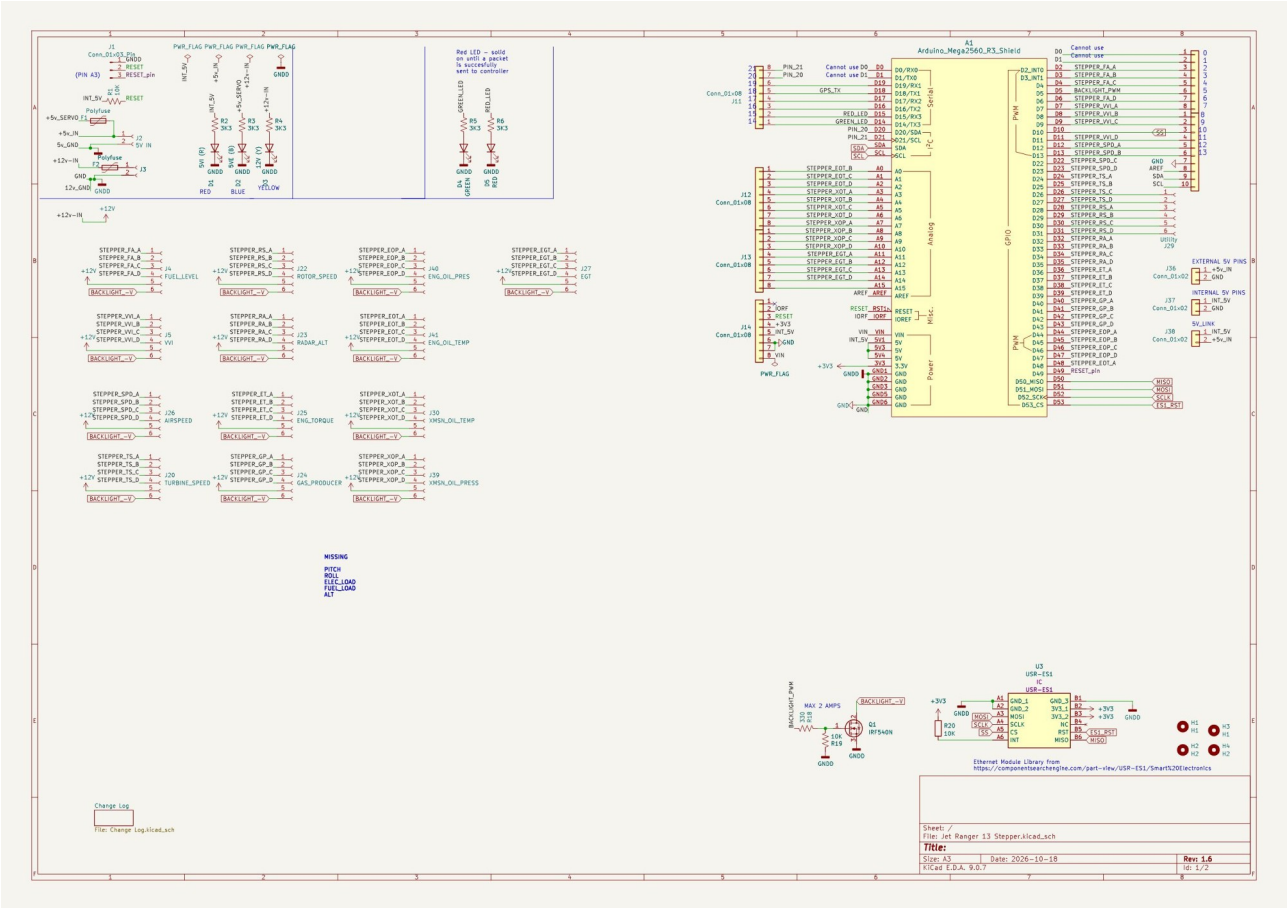


Figure 58 — Jet Ranger 13 Stepper schematic

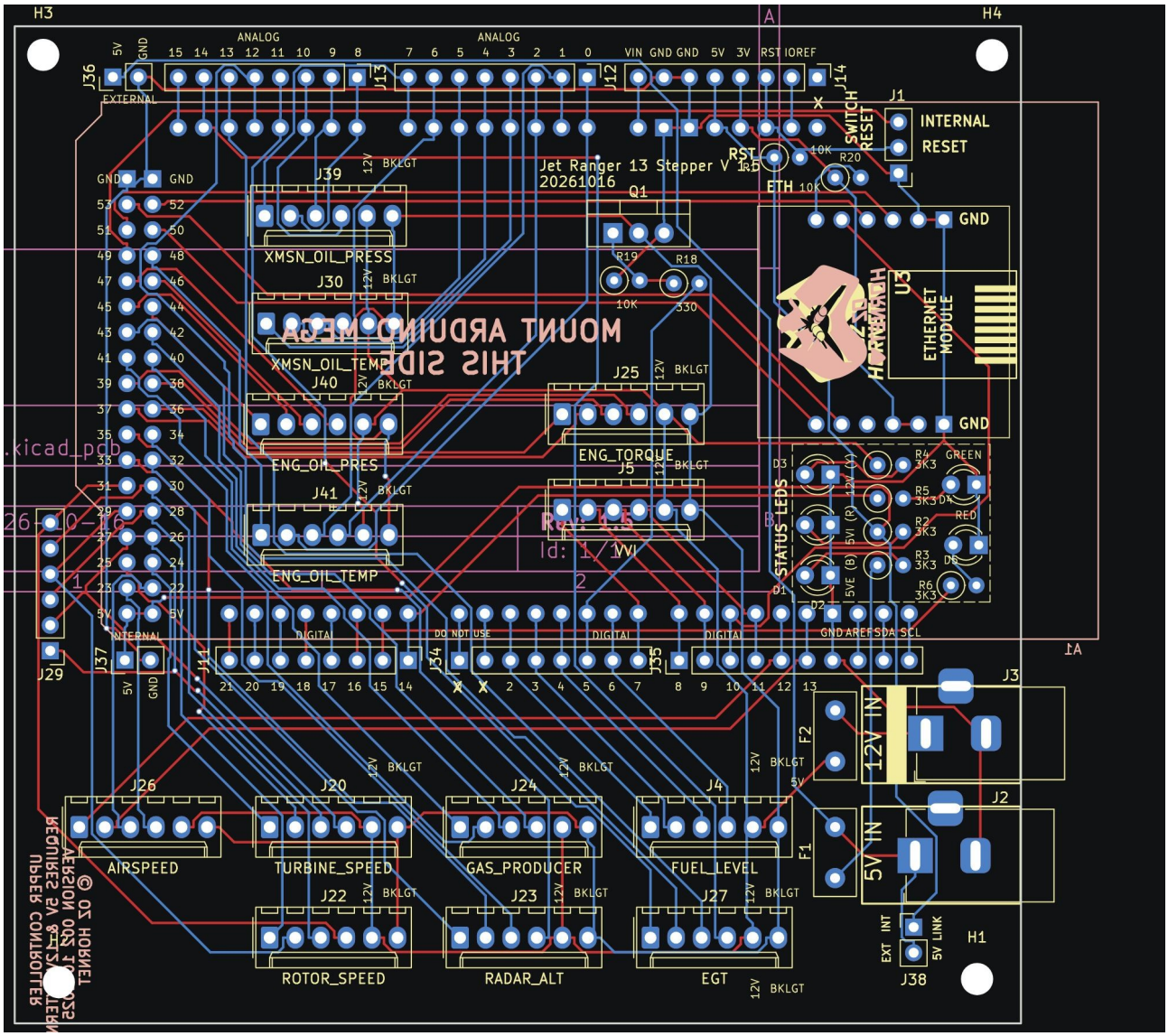


Figure 59 — Jet Ranger 13 Stepper board layout

19.5 Jet Ranger OLED Dual Stepper

Revision 1.5 (2026-10-18). Mega shield for the Dual Stepper controller (172.16.1.106). Six-pin gauge sockets for the oil, load, RPM and altimeter gauges, TCA9548A sockets for the four OLEDs (barometer, altimeter, clock, temperature), the USR-ES1 Ethernet module and the backlight MOSFET. 1.3 swapped power and ground on the four-pin OLED connections, 1.4 removed the short between Mega pins 2 and 32 and 1.5 corrected the MOSFET.

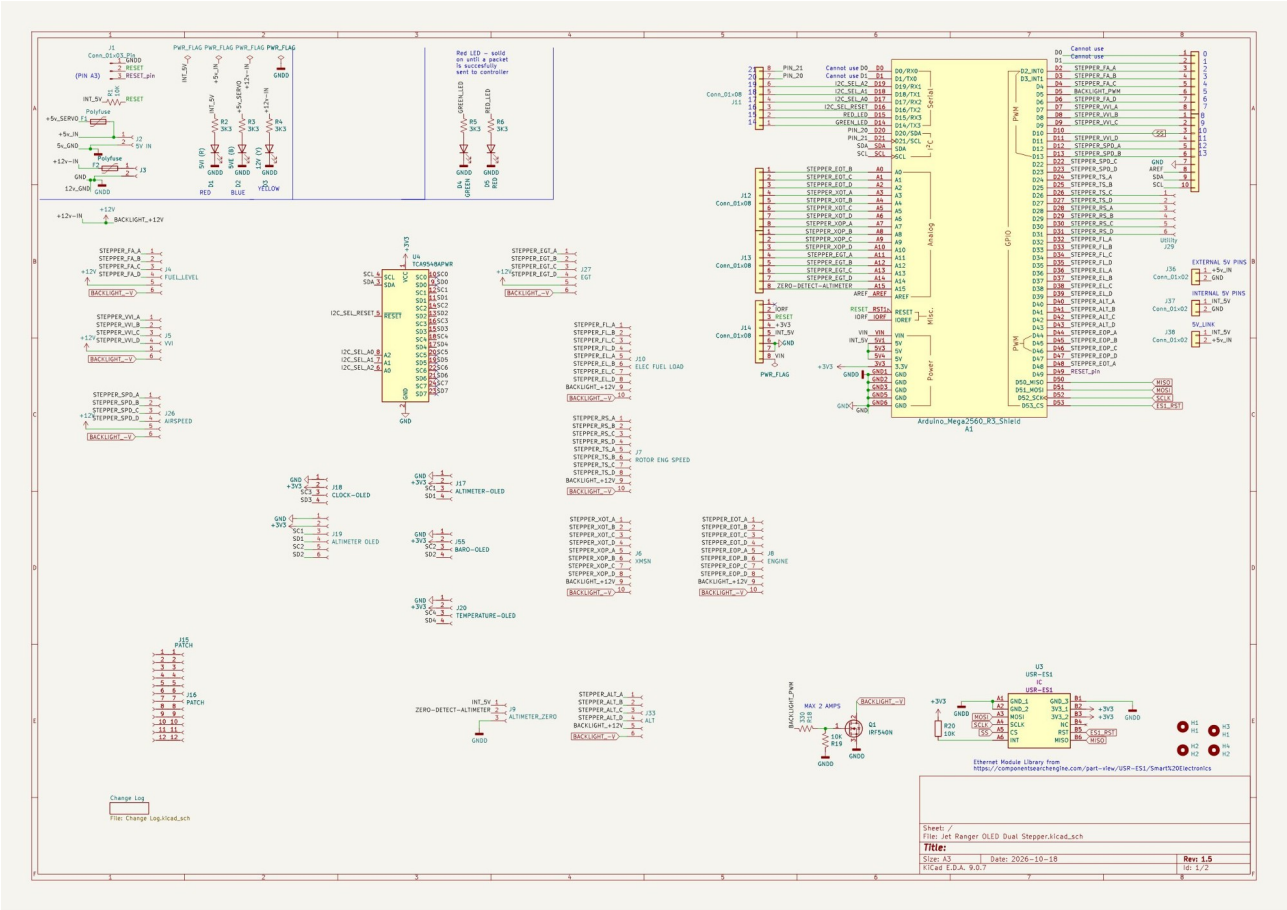


Figure 60 — Jet Ranger OLED Dual Stepper schematic

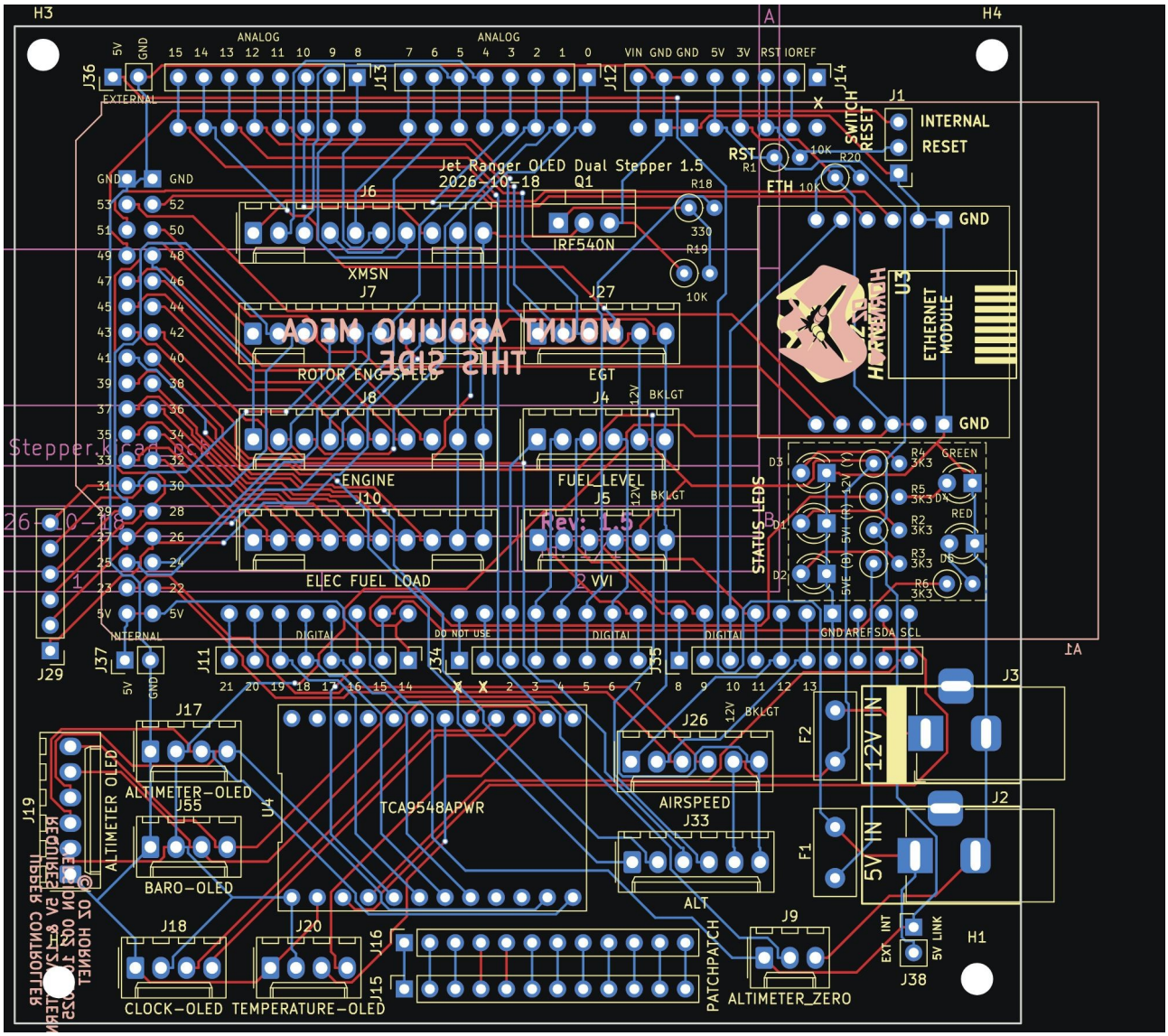


Figure 61 — Jet Ranger OLED Dual Stepper board layout

19.6 Jet Ranger X27 Small Stepper

Title block revision 2 (2022-08-04). One board for each stepper gauge. One X27 or VID29 stepper motor, two backlight LEDs with a 470 ohm resistor (the schematic value, see the backlight resistor table) and the six-pin connector to the controller.

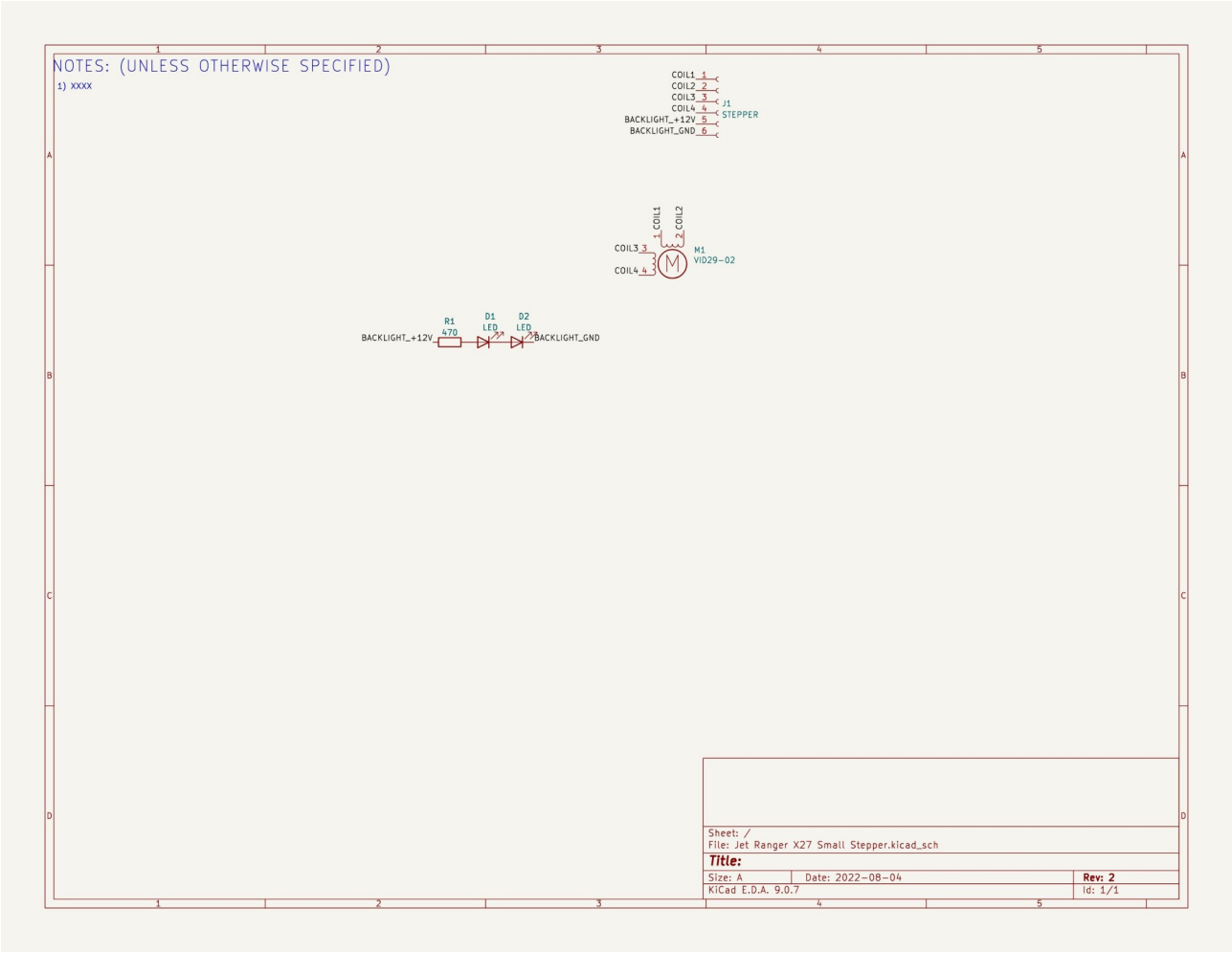


Figure 62 — Jet Ranger X27 Small Stepper schematic

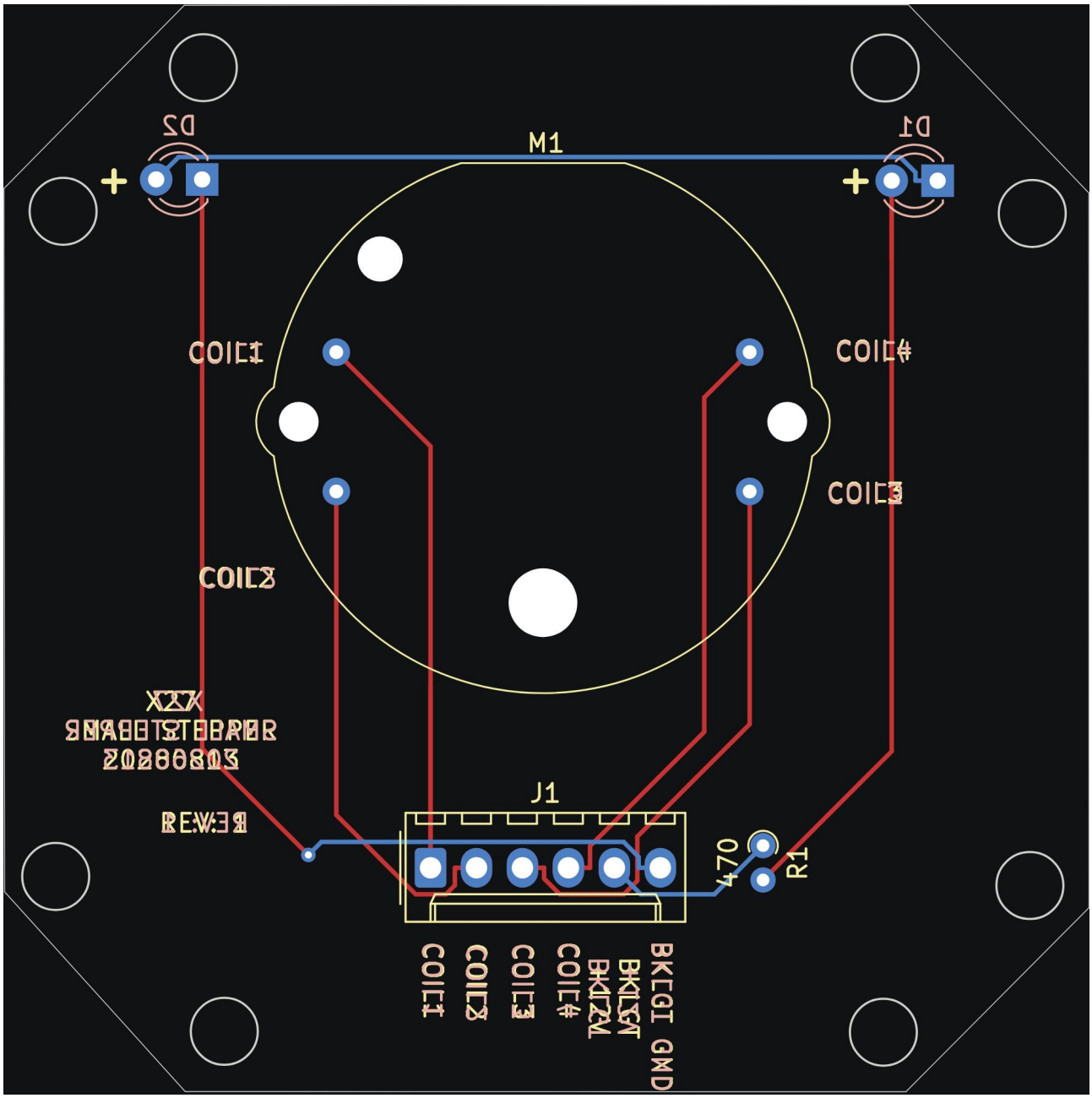


Figure 63 — Jet Ranger X27 Small Stepper board layout

19.7 Jet Ranger Upper ADF Controller

Revision 1.0 (2026-10-26). The combined ADF and Upper controller (172.16.1.107), the reference PCB for the ADF controller. It merges the Jet Ranger Upper Controller (1.4) and the Jet Ranger ADF Controller (1.6) onto one Mega, started from the ADF Controller. The ADF major and minor knobs (D6 to D9), the ADF keypad (A1 to A7, connector J41), the landing gear, flaps, speed brake and brake inputs (D31 to D34), the hyd boost, anti-ice and caution dim inputs (D35 to D37, J20), three landing gear lamps (D2 to D4), the I2C OLED, the Upper switch matrix (rows D22 to D30 on J30, columns D38 to D45 on J31 with 10 k pull-ups), the Upper keypad (A9 to A15, J32), the Warning Mute (J17) and TOT Test (J18) switches, the backlight MOSFET (D5, panel backlight on J33), a panel brightness pot on A0 and a 12 V monitor on A8 (47 k and 10 k divider). The 1x16 breakout headers J9 to J14 were removed to make room. The layout is routed (no unconnected items in KiCad's design rule check).

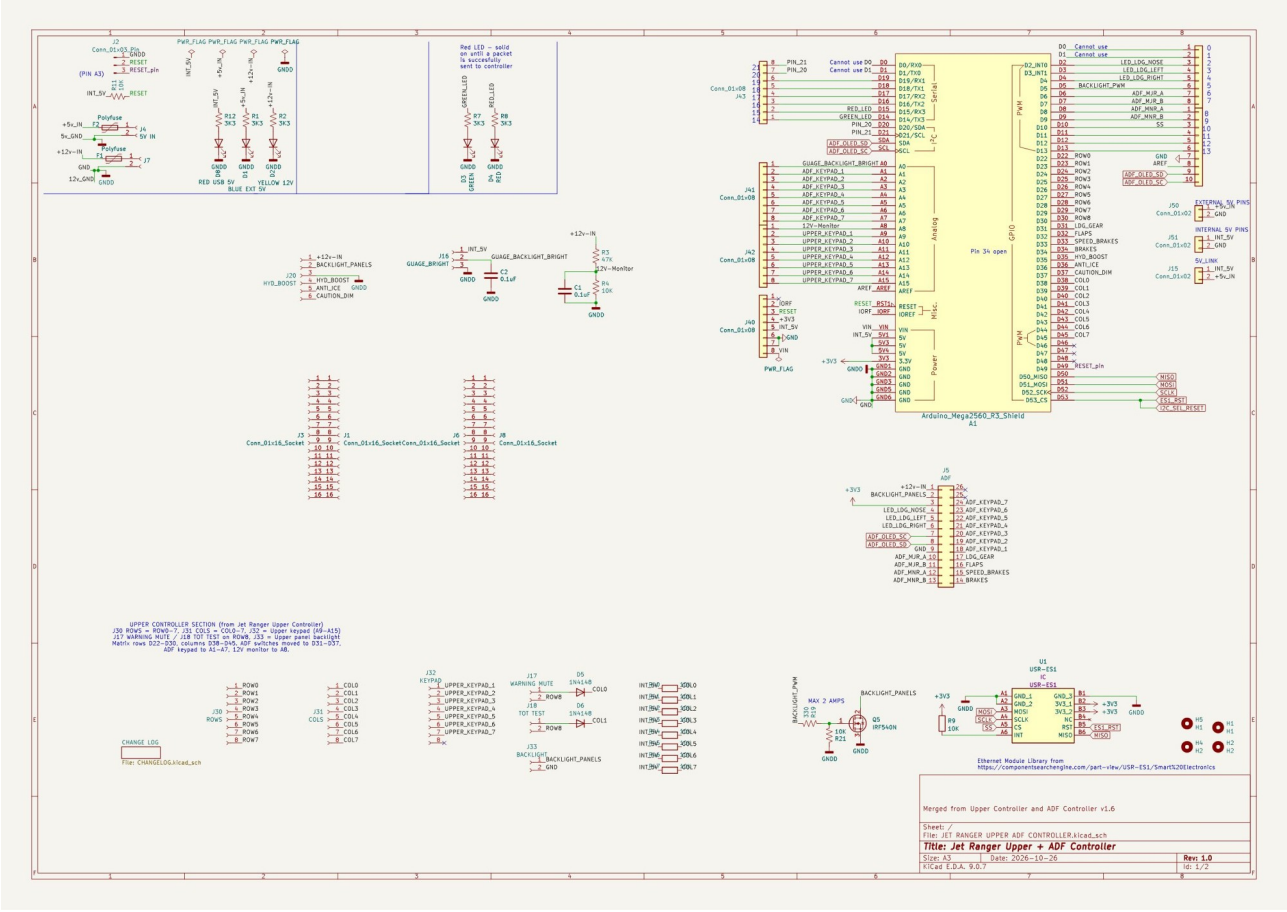


Figure 64 — Jet Ranger Upper ADF Controller schematic

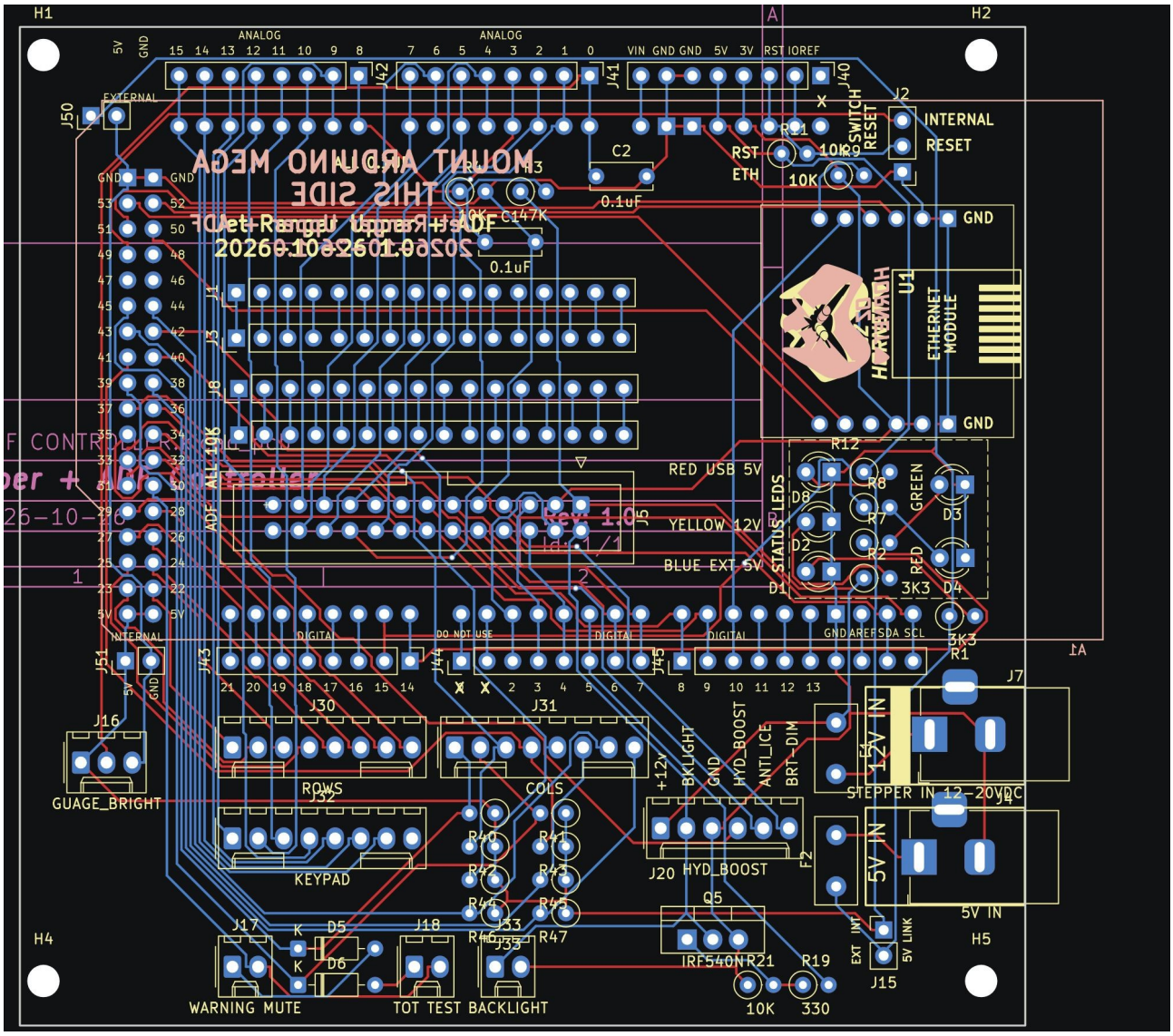


Figure 65 — Jet Ranger Upper ADF Controller board layout

19.8 Jet Ranger Audio Panel

Revision 1.1 (2026-10-17). Switch panel for the Radio Auto Pilot controller's `AUDIO` connector. Eight switches in a diode matrix on columns 3 and 4 (COM1, BOTH, NAV2, DME, COM2, NAV1, MKR, ADF), eleven LEDs on the MAX7219 digits 2 and 3, a connector for the Audio panel OLED and the backlight connector.

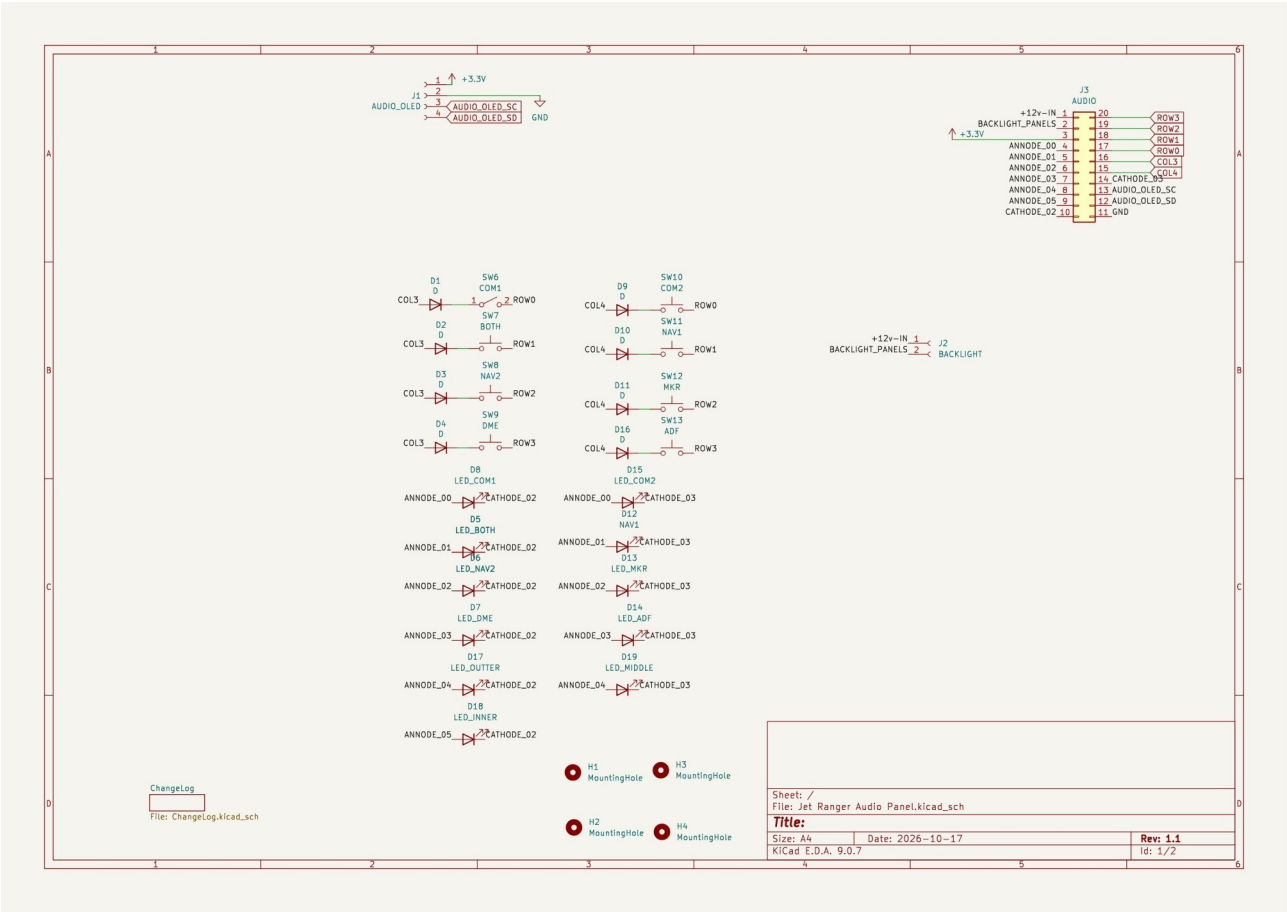


Figure 66 — Jet Ranger Audio Panel schematic

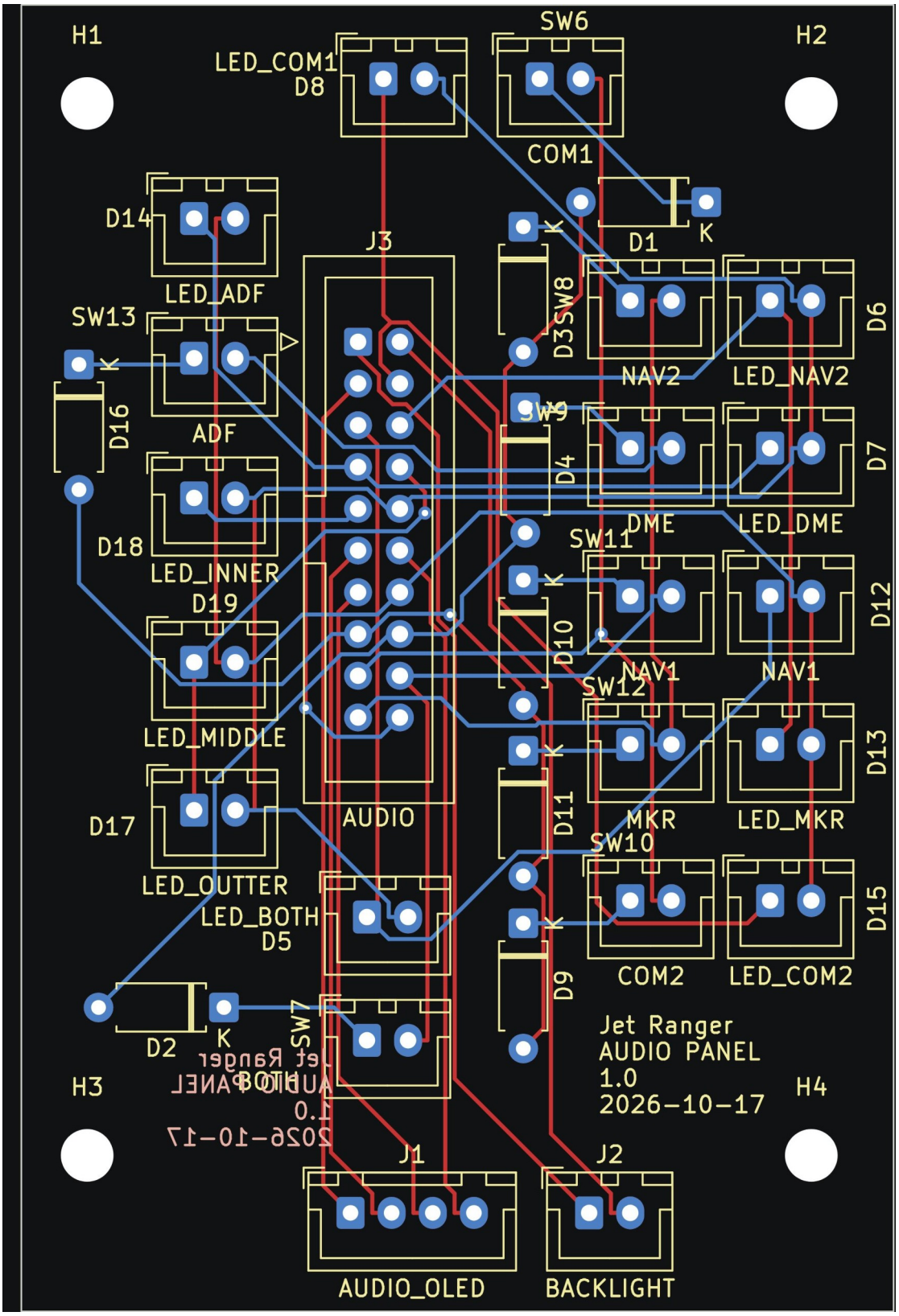


Figure 67 — Jet Ranger Audio Panel board layout

19.9 Jet Ranger Auto Pilot Panel

Revision 1.1 (2026-10-17). Switch panel for the Radio Auto Pilot controller's `AUTOPILOT` connector. Eight switches in a diode matrix on columns 1 and 2 (AP ENGAGE, HDG, CRS, ALT, APPR, FD, BC, IAS), five rotary encoders (CRS, ALT, VS, IAS, HDG), eight LEDs on the MAX7219 digits 0 and 1, the AP OLED connector, the autopilot magnet coil connection and the backlight connector.

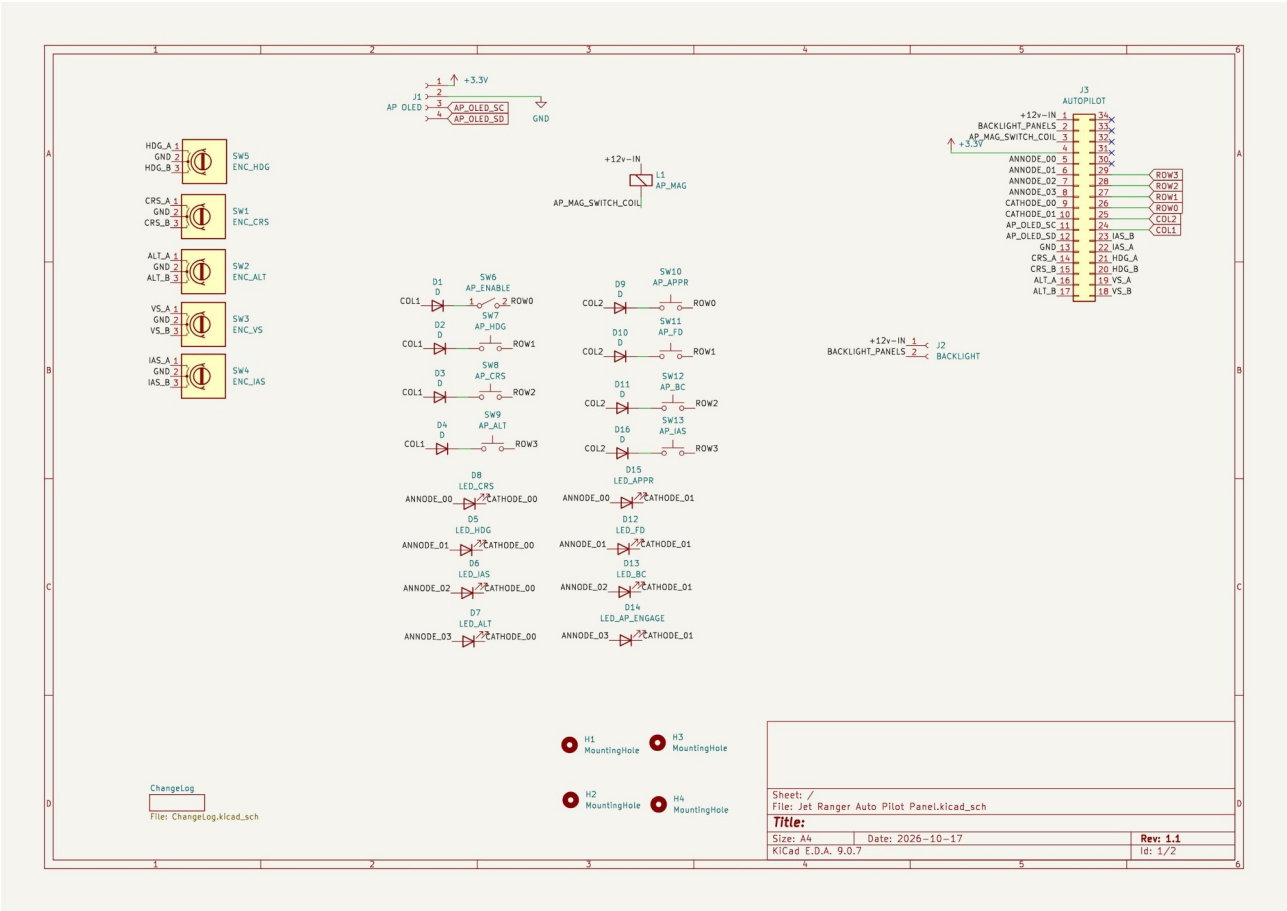


Figure 68 — Jet Ranger Auto Pilot Panel schematic

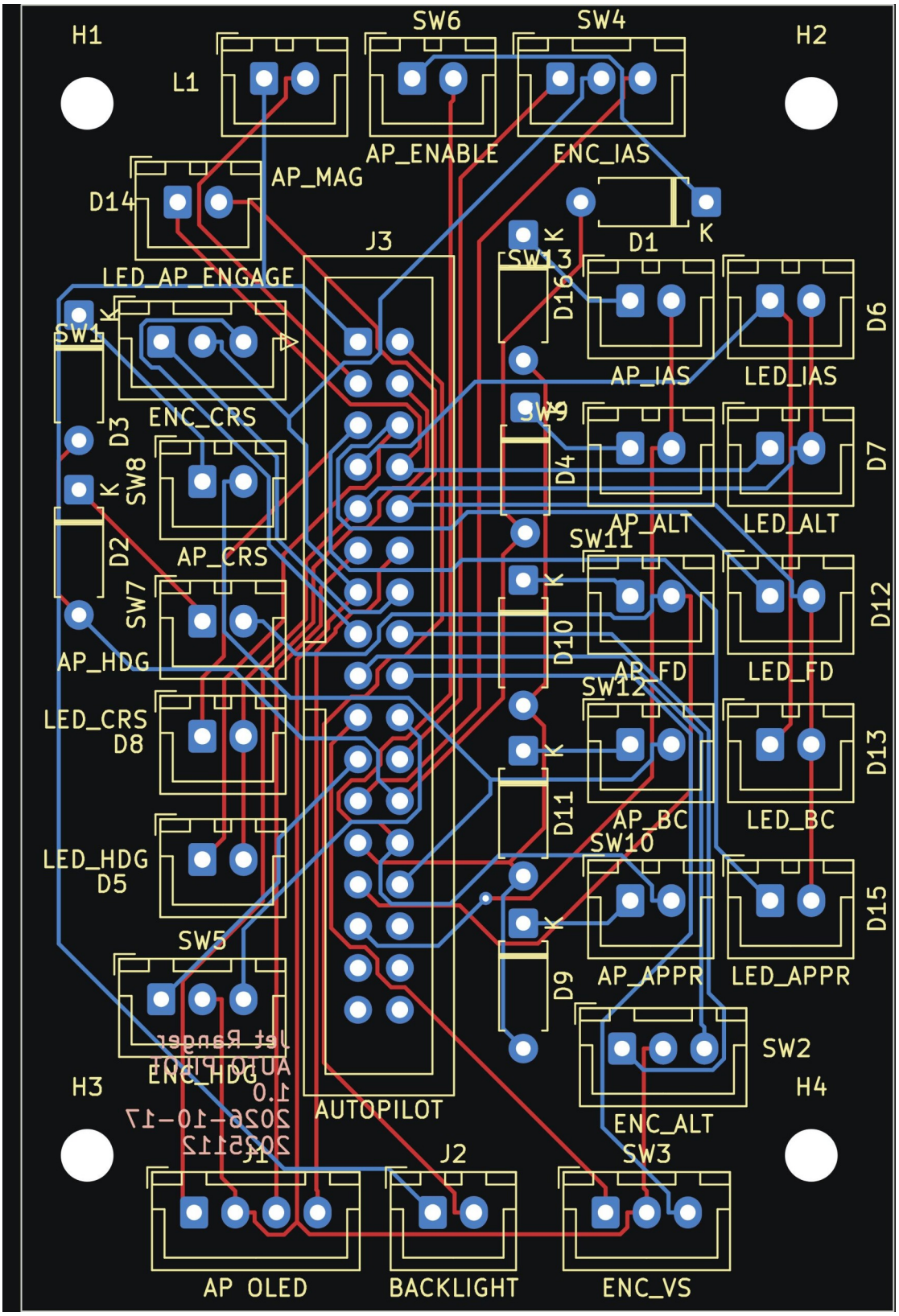


Figure 69 — Jet Ranger Auto Pilot Panel board layout

19.10 Jet Ranger ADF Panel

Revision 1.1 (2026-10-17). Panel for the ADF controller's ADF connector. The ADF major and minor encoders, the landing gear, flaps, speed brake and brakes switches, three landing gear lamps with 330 ohm resistors, the keypad connector, the ADF OLED connector and the backlight connector.

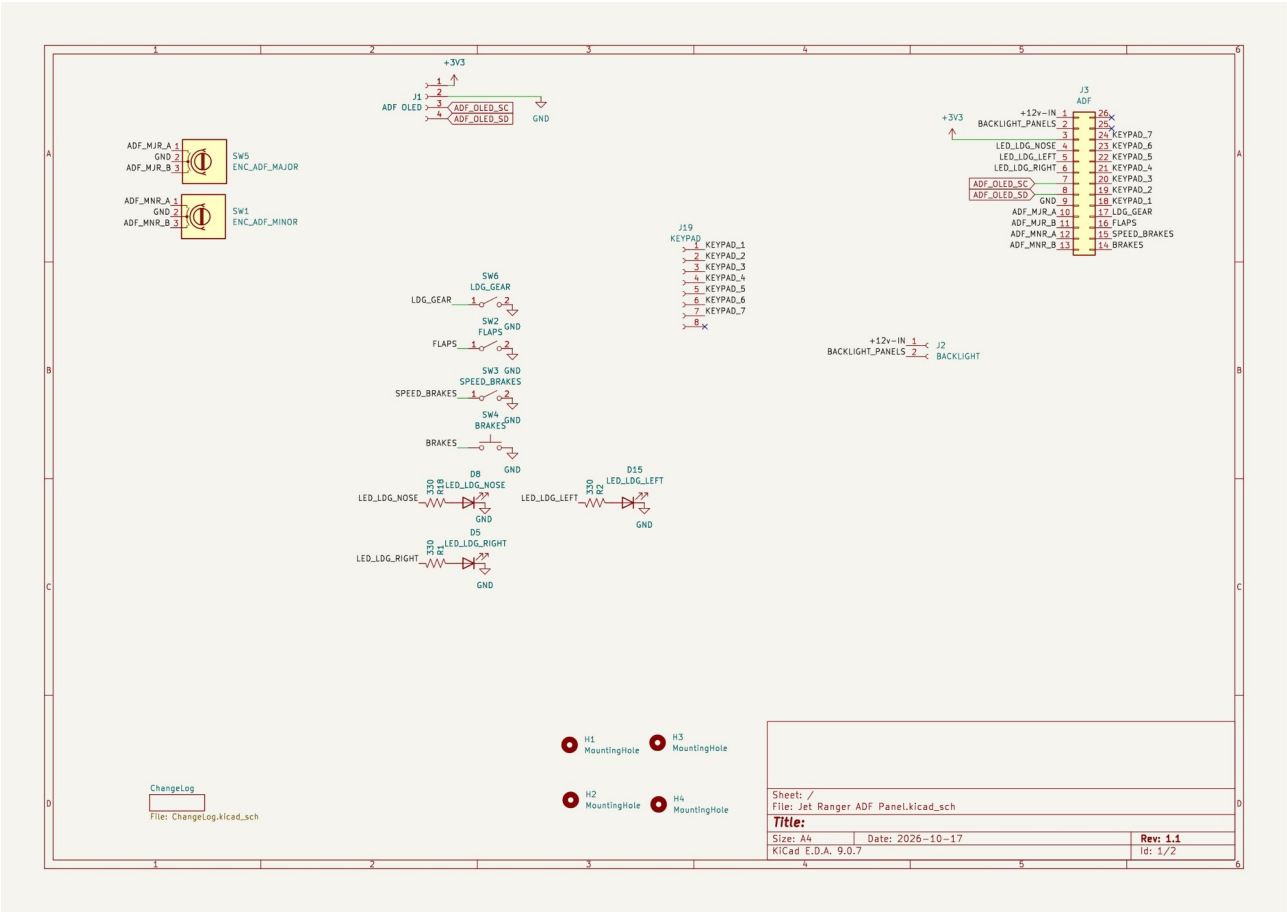


Figure 70 — Jet Ranger ADF Panel schematic

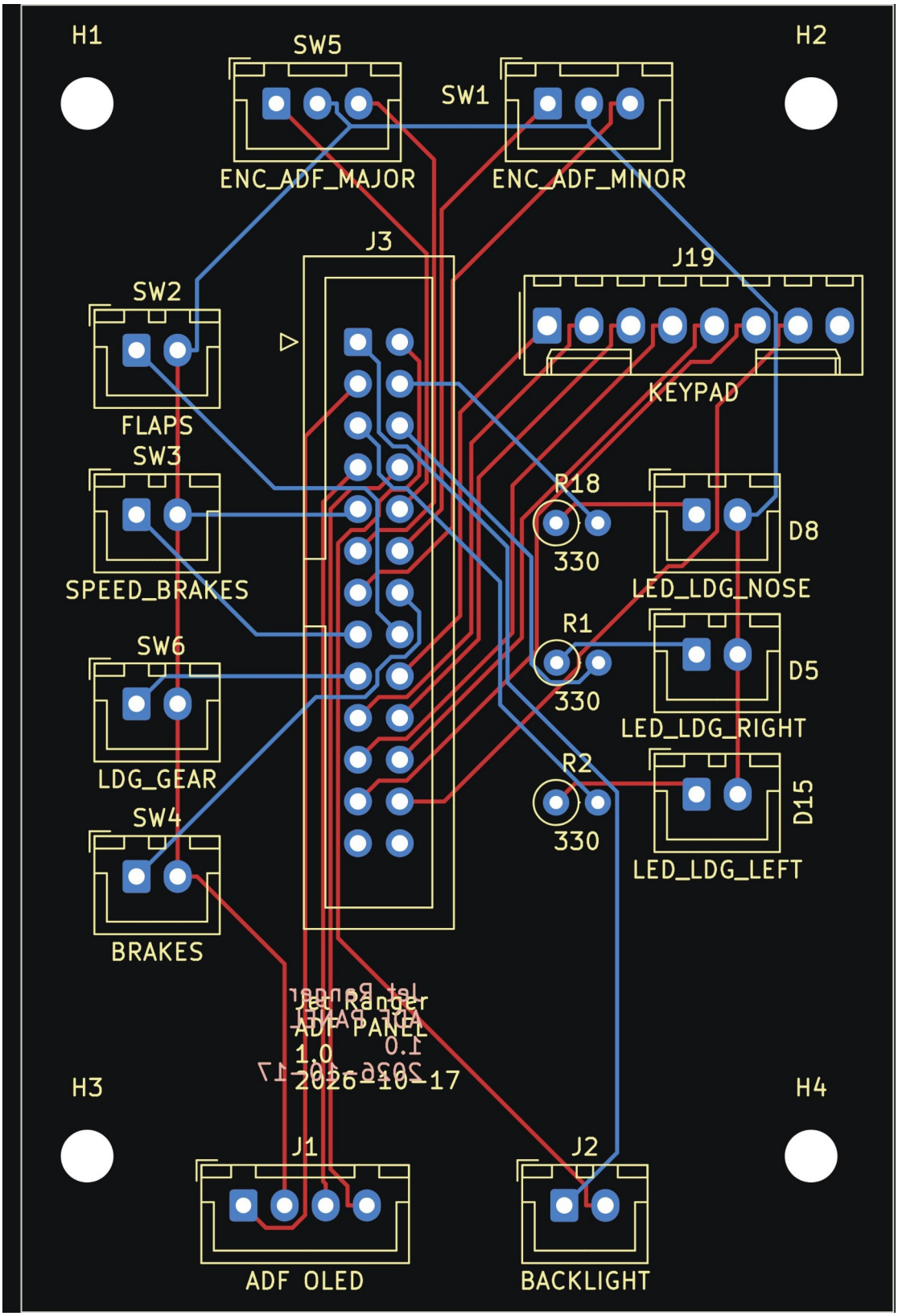


Figure 71 — Jet Ranger ADF Panel board layout

19.11 Annunciator

Schematic sheet only: the project has no board layout. The same sheet is used inside the Servo, 13 Stepper and Dual Stepper schematics. Warning lamp annunciator circuit.

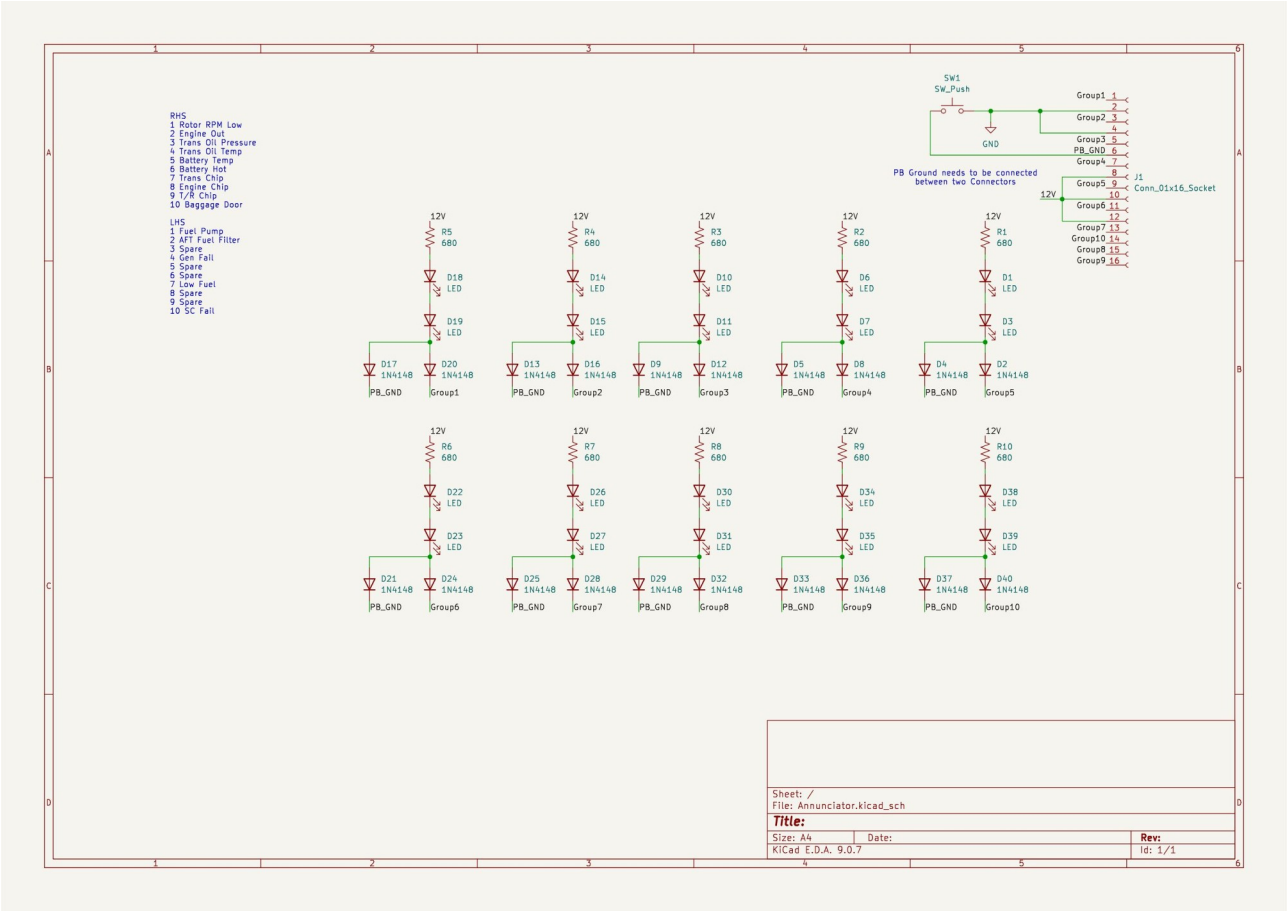


Figure 72 — Annunciator schematic

19.12 Jet Ranger Altimeter

Title block revision 2 (2022-08-04). Not covered by the sketches. Altimeter gauge board.

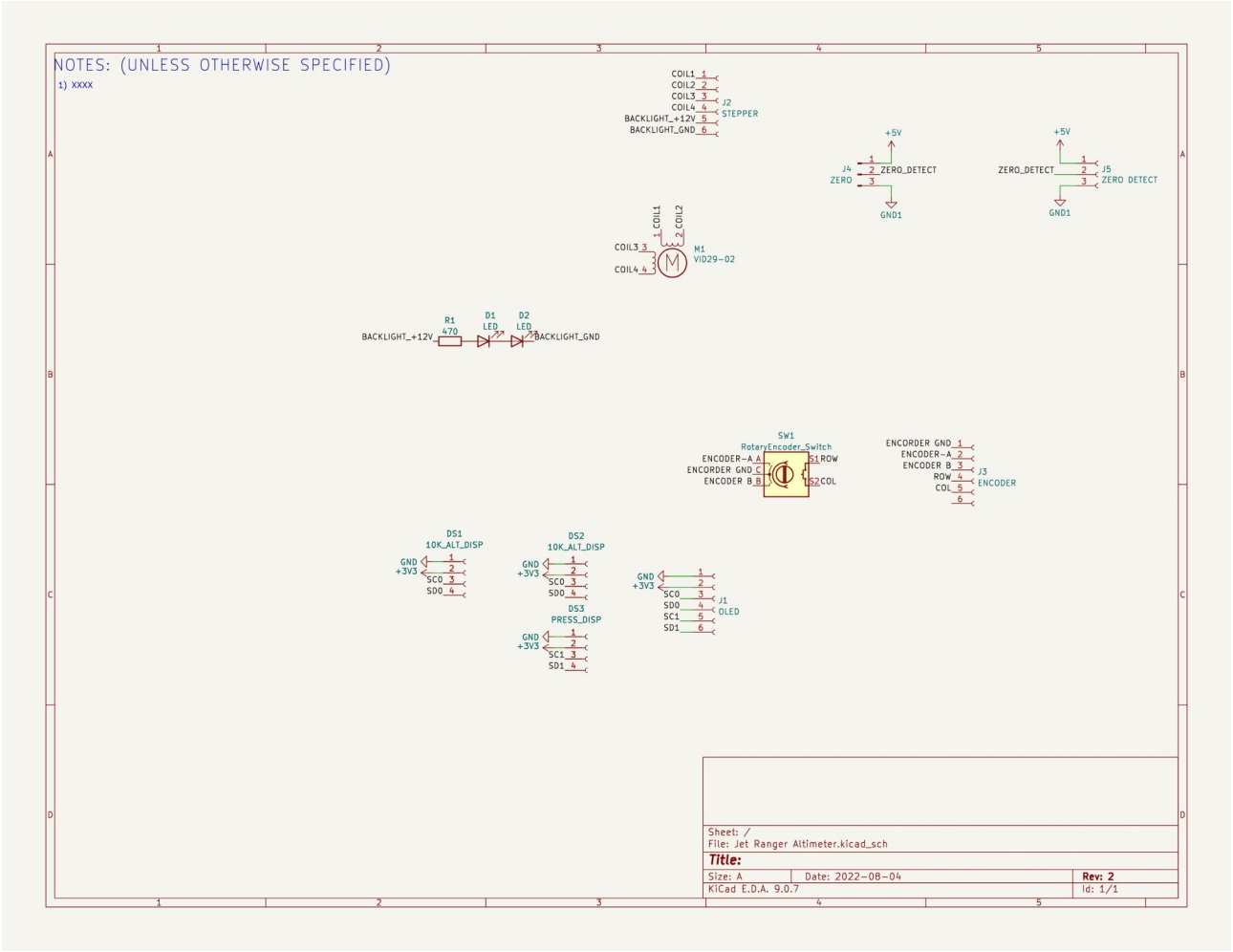


Figure 73 — Jet Ranger Altimeter schematic

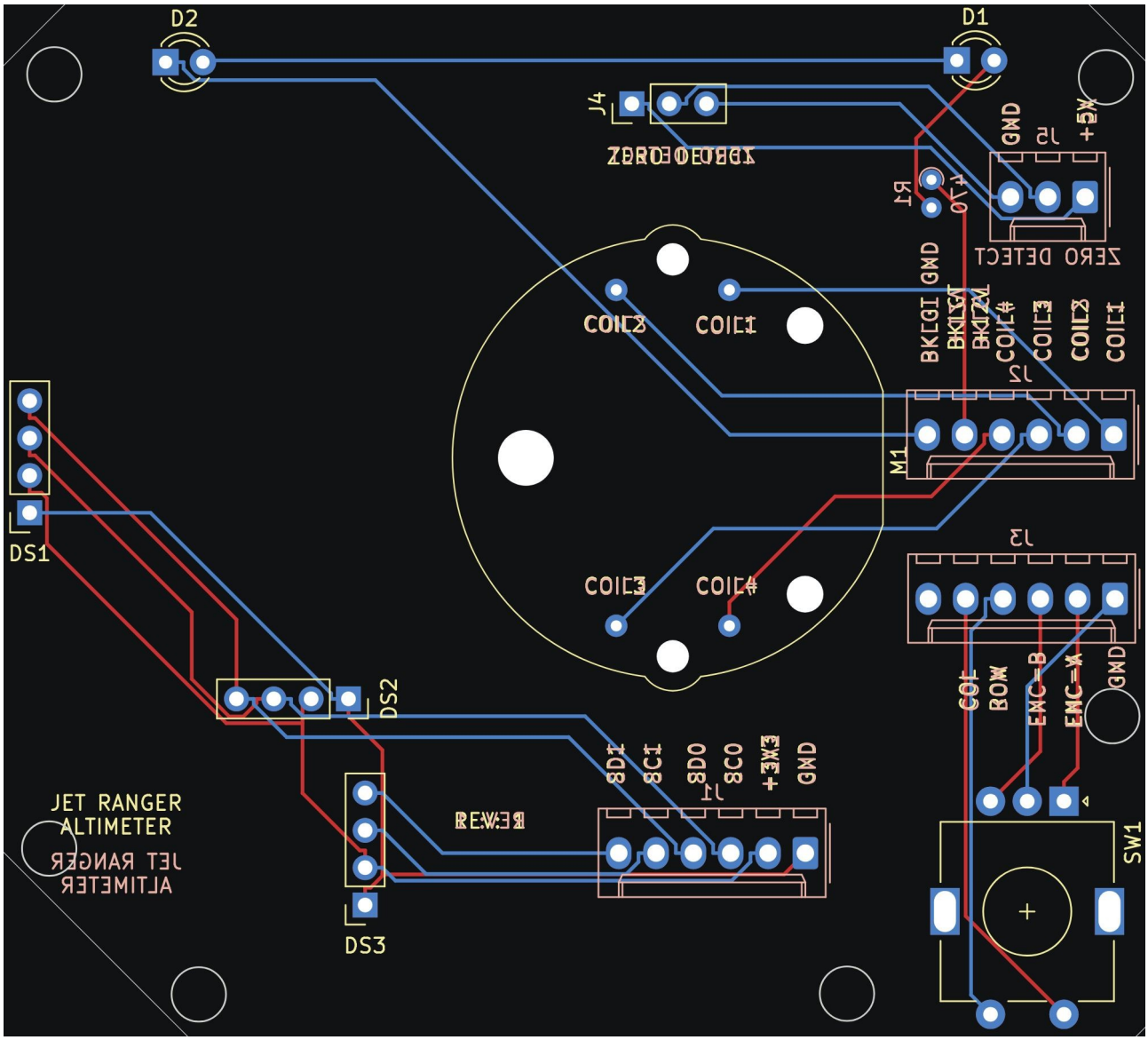


Figure 74 — Jet Ranger Altimeter board layout

19.13 Large Guage (Simmeters Large Guage Replacement)

Title block revision 2 (2022-08-04). Not covered by the sketches. Replacement board for a large Simmeters gauge.

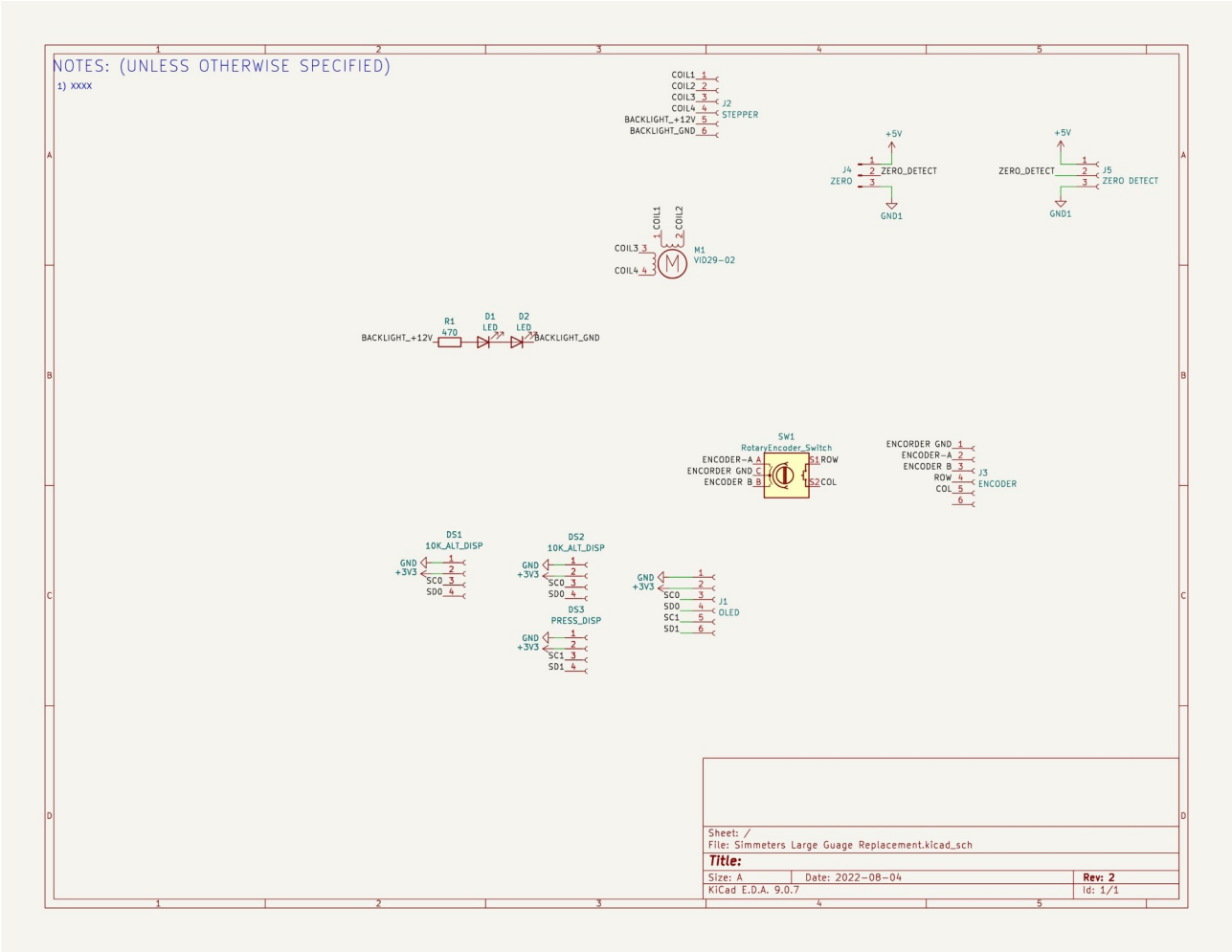


Figure 75 — Large Guage (Simmeters Large Guage Replacement) schematic

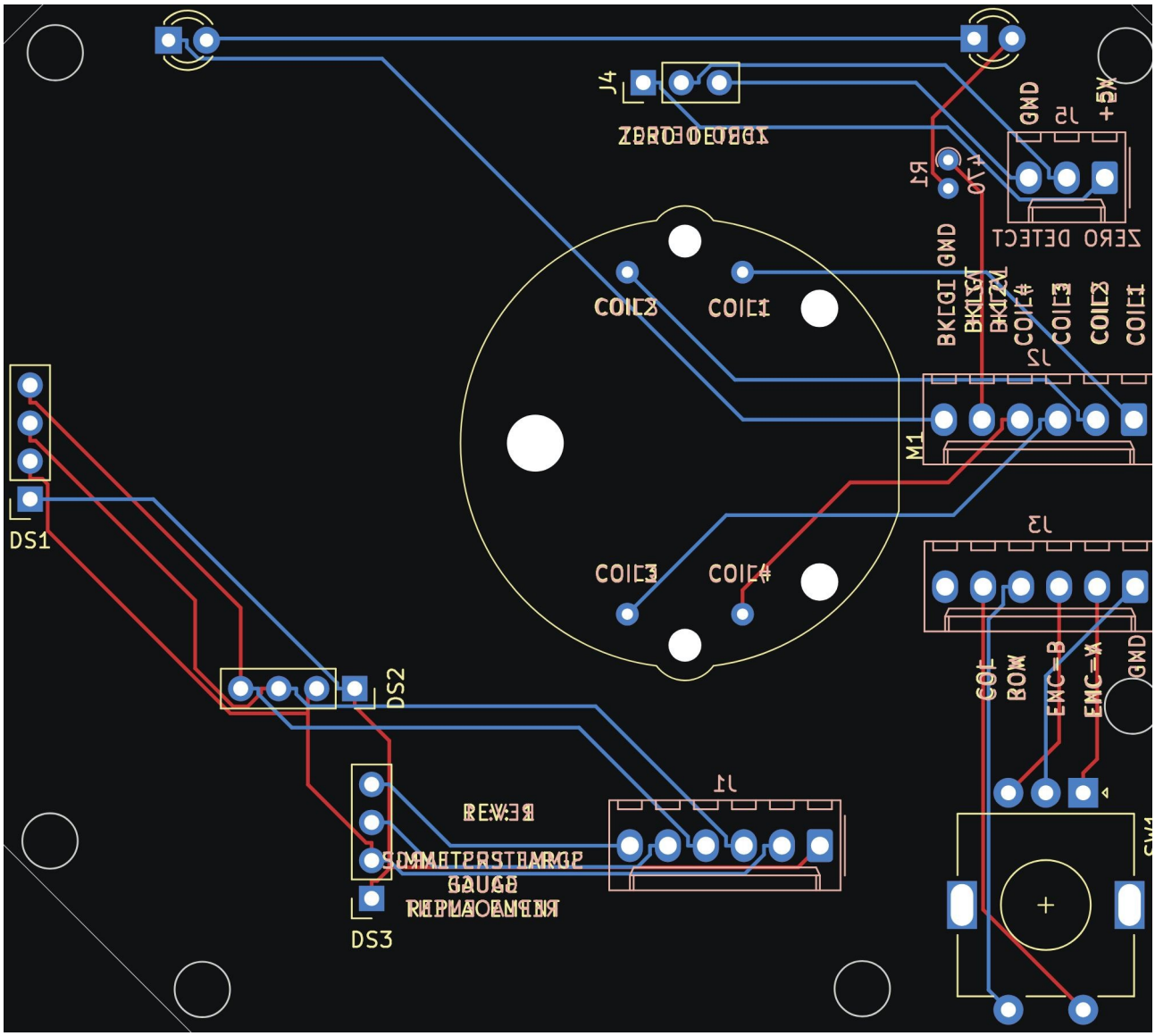


Figure 76 — Large Guage (Simmeters Large Guage Replacement) board layout

19.14 Simkits Large Dual Gauge

Title block revision 2 (2022-08-04). Not covered by the sketches. Replacement board for a large dual Simkits gauge.

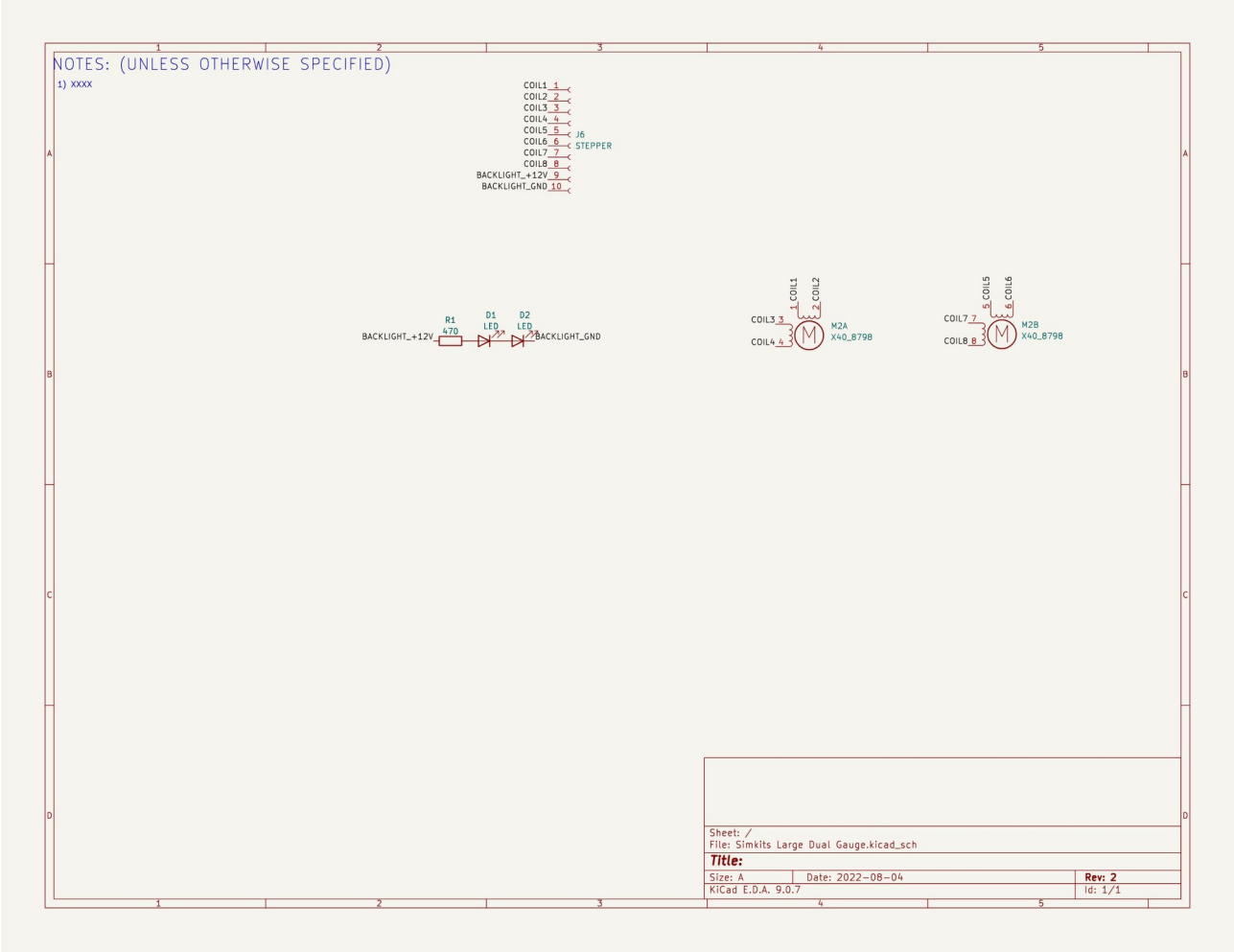


Figure 77 — Simkits Large Dual Gauge schematic

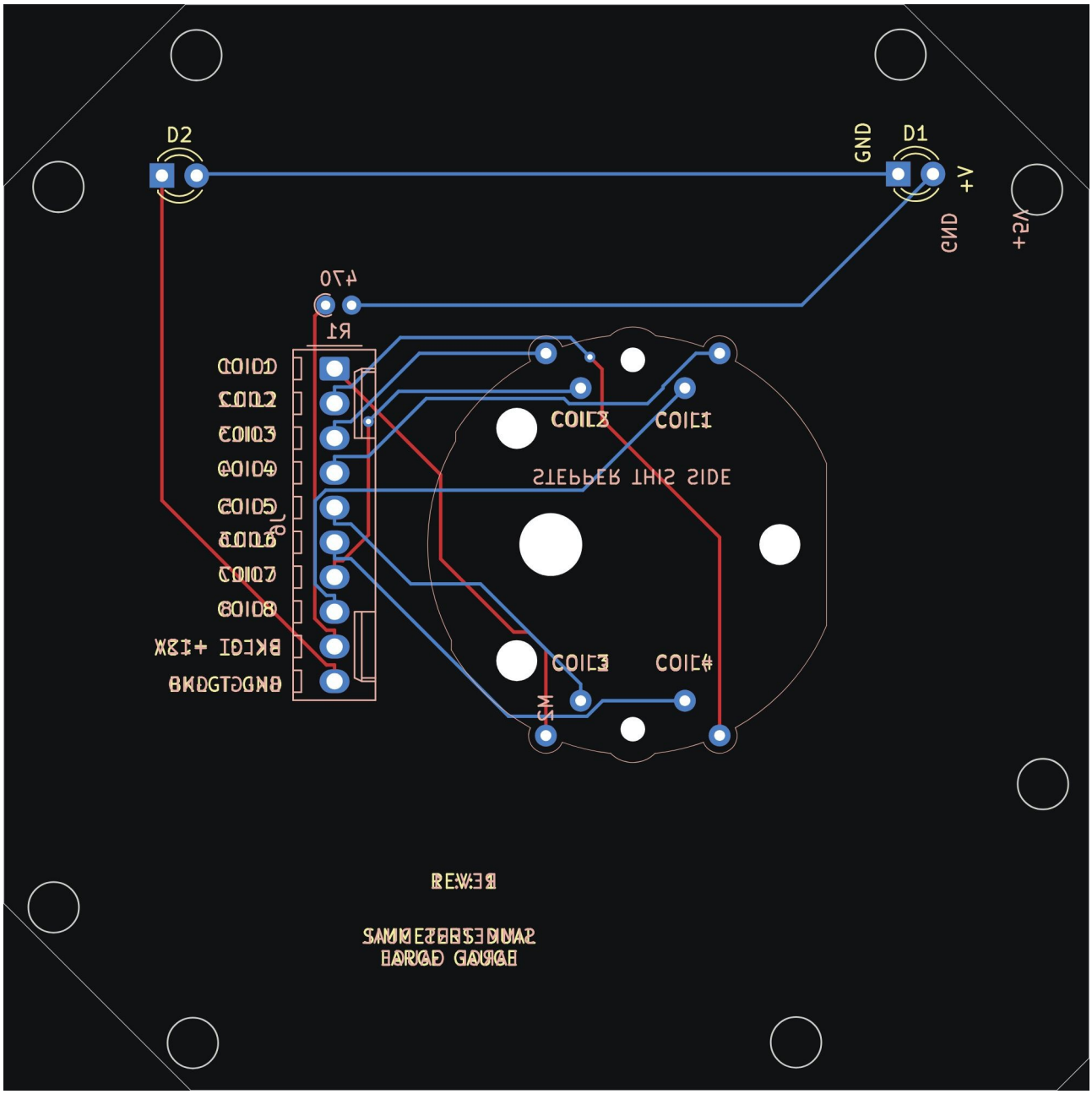


Figure 78 — Simkits Large Dual Gauge board layout

19.15 Simkits Large Guage

Title block revision 2 (2022-08-04). Not covered by the sketches. Replacement board for a large Simkits gauge.

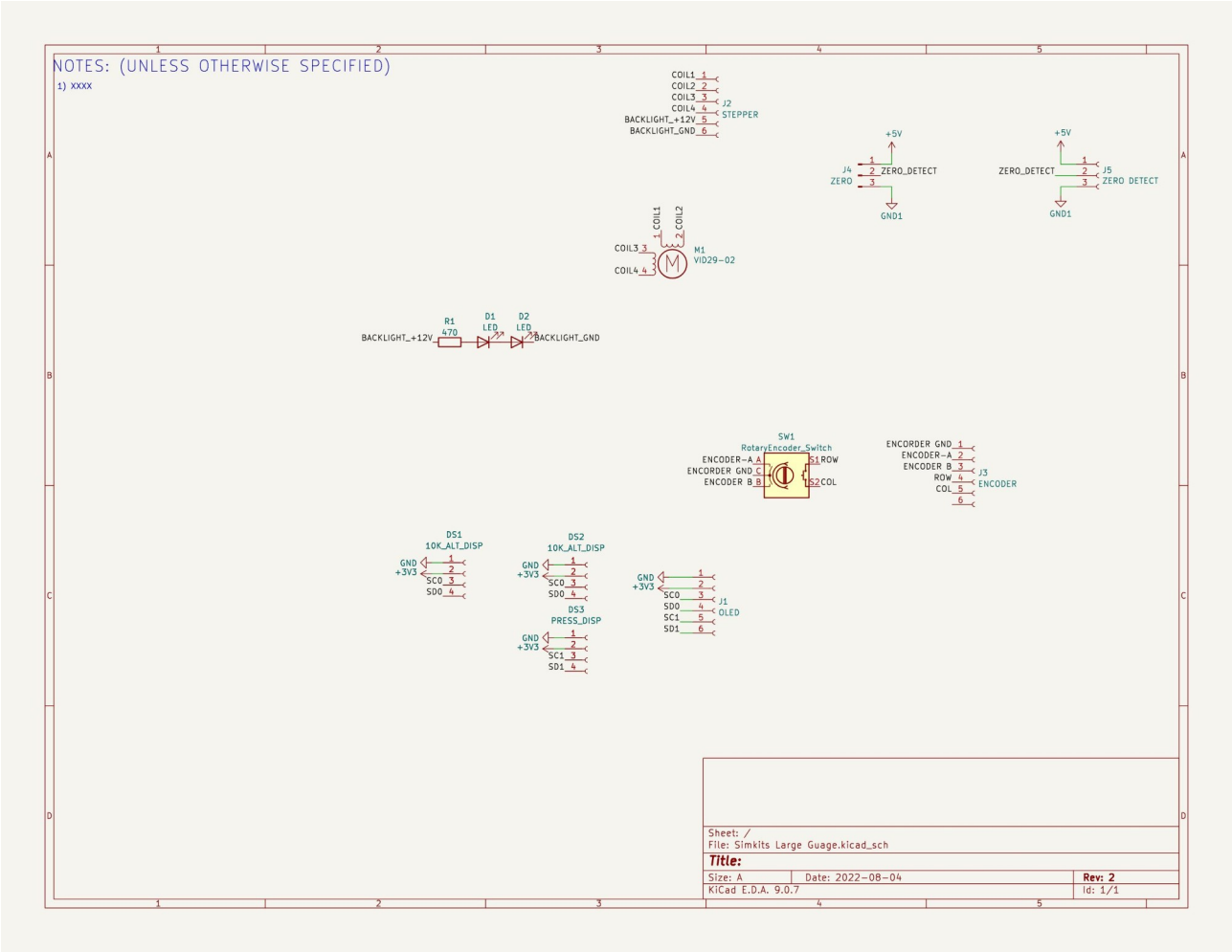


Figure 79 — Simkits Large Guage schematic

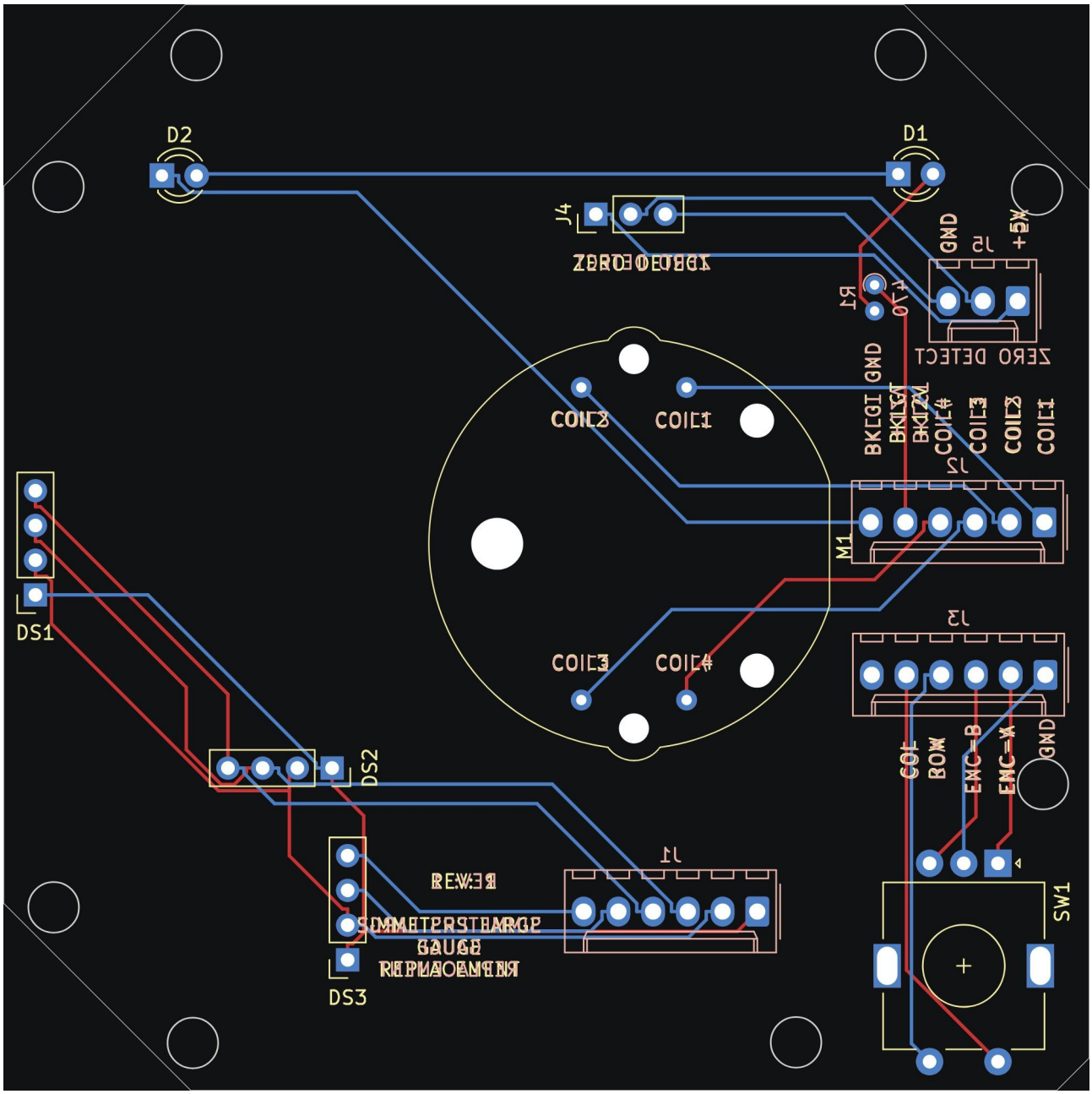


Figure 80 — Simkits Large Guage board layout

19.16 Simkits Small Dual Gauge

Title block revision 2 (2022-08-04). Not covered by the sketches. Replacement board for a small dual Simkits gauge.

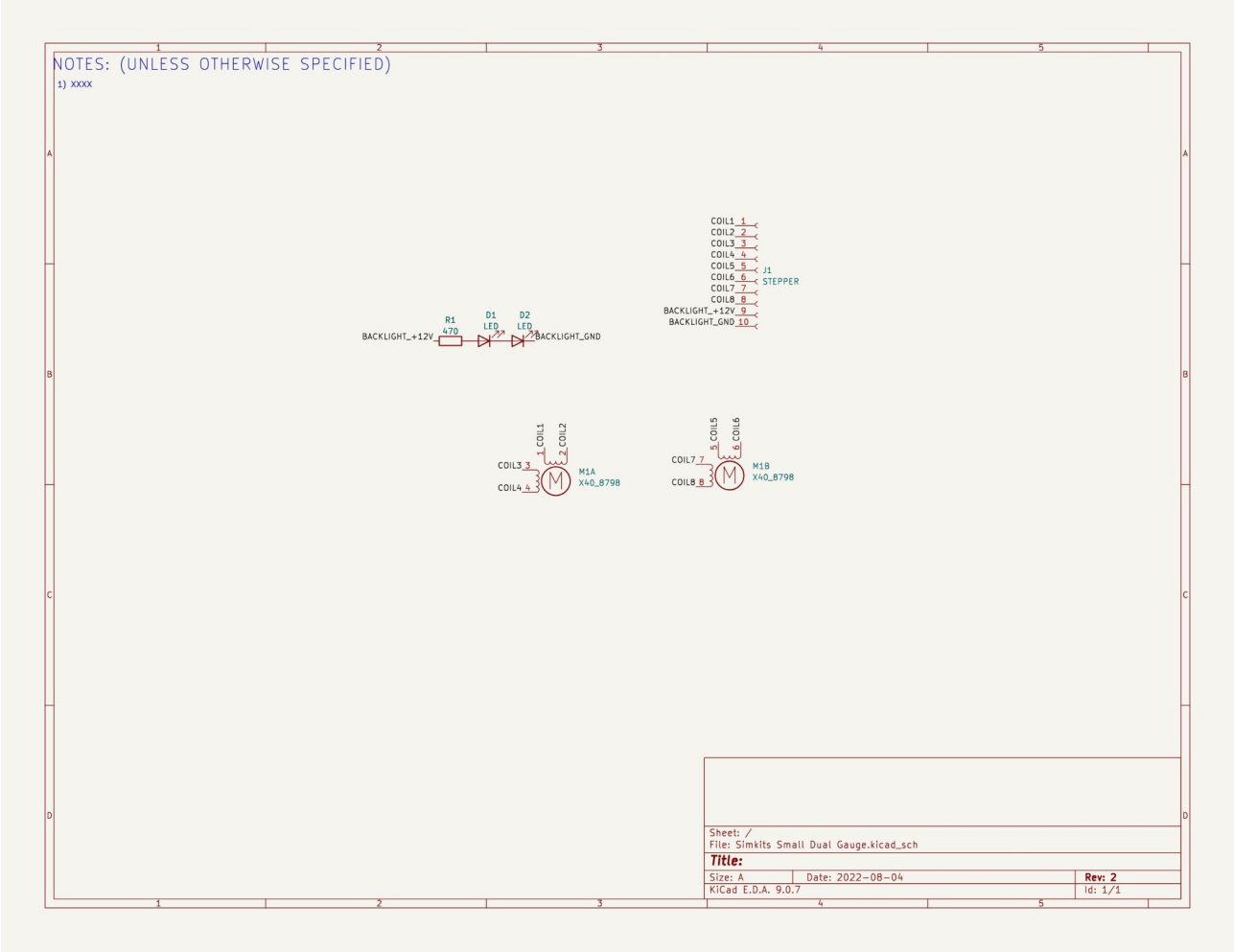


Figure 81 — Simkits Small Dual Gauge schematic

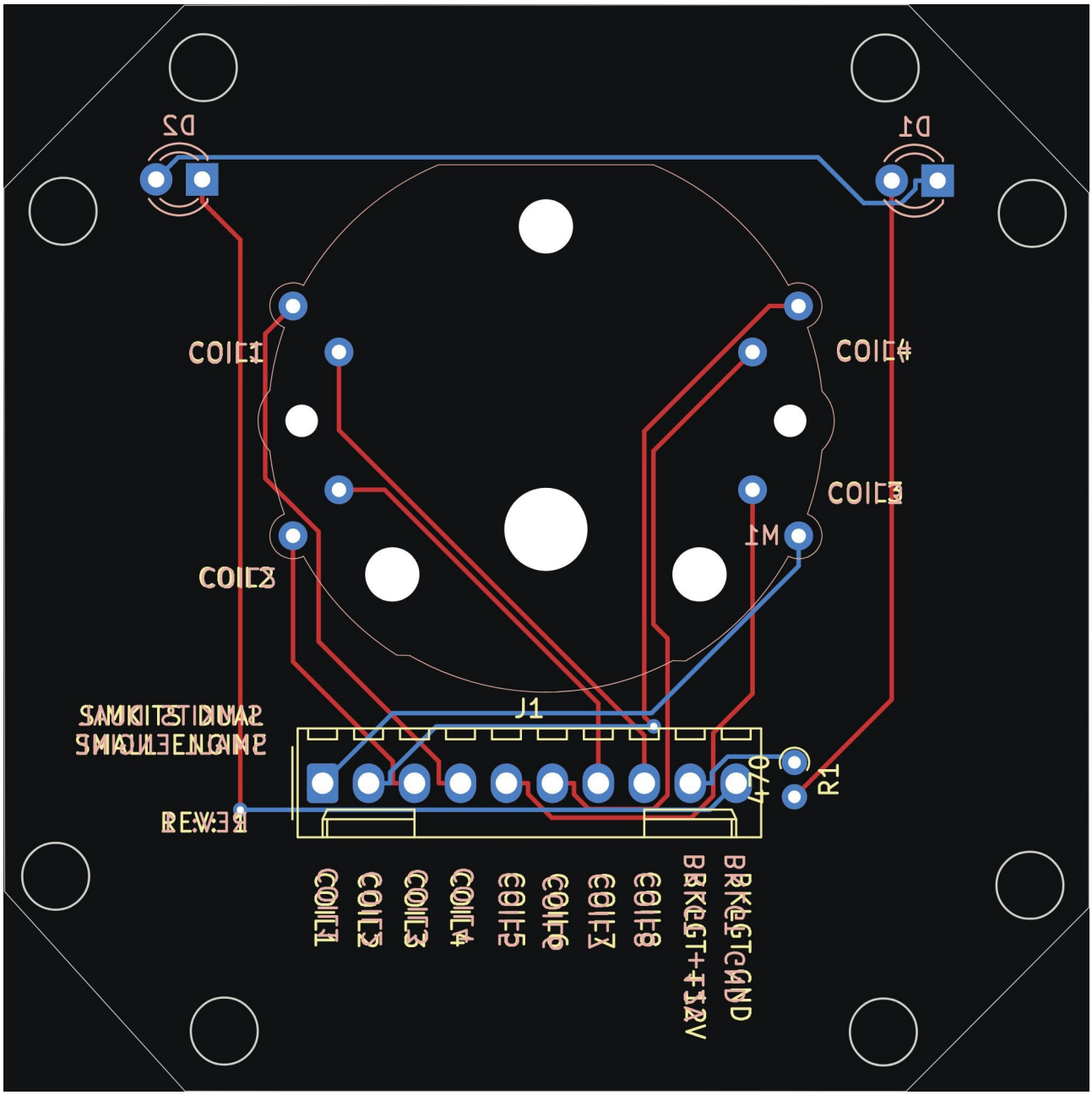


Figure 82 — Simkits Small Dual Gauge board layout

19.17 Simkits Small Guage

Title block revision 2 (2022-08-04). Not covered by the sketches. Replacement board for a small Simkits gauge.

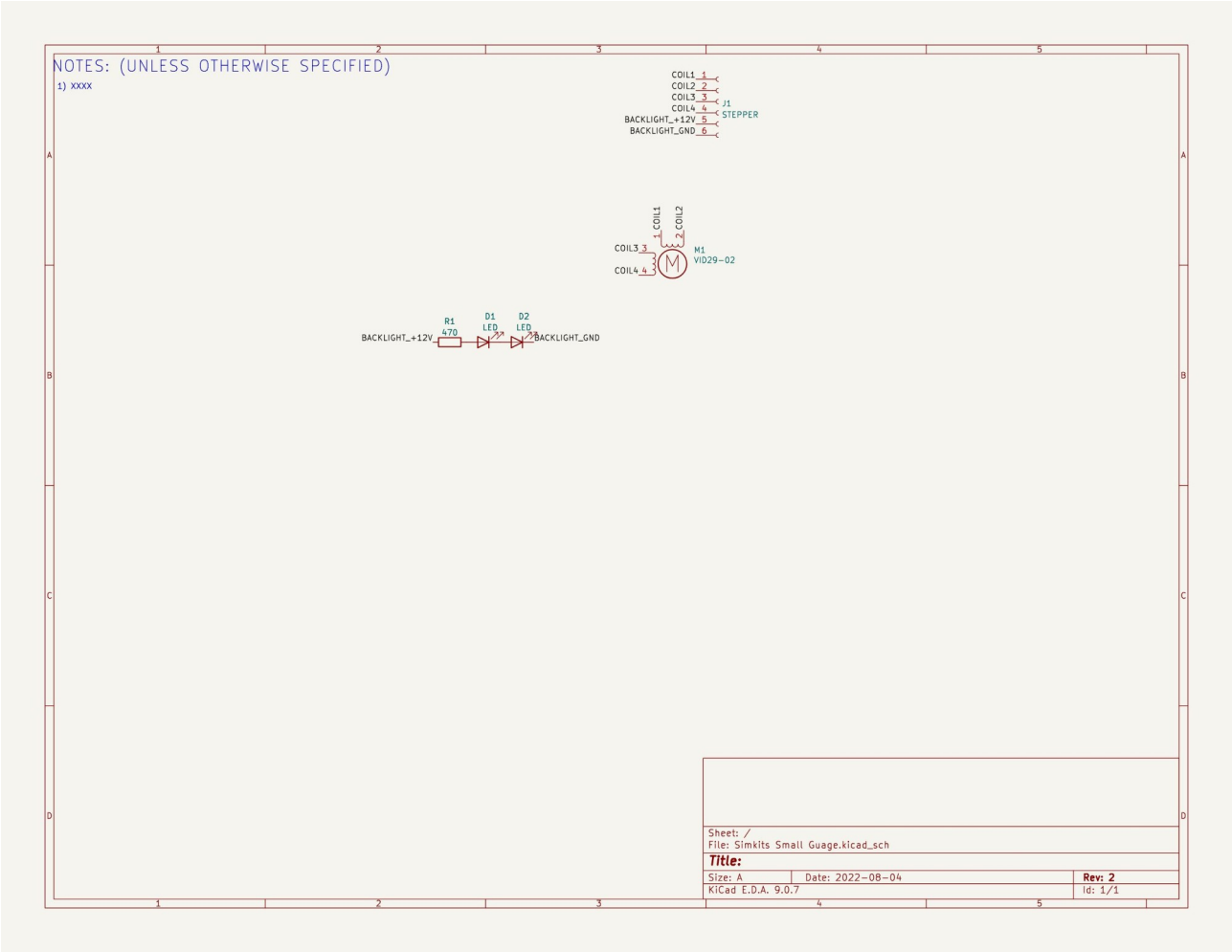


Figure 83 — Simkits Small Guage schematic

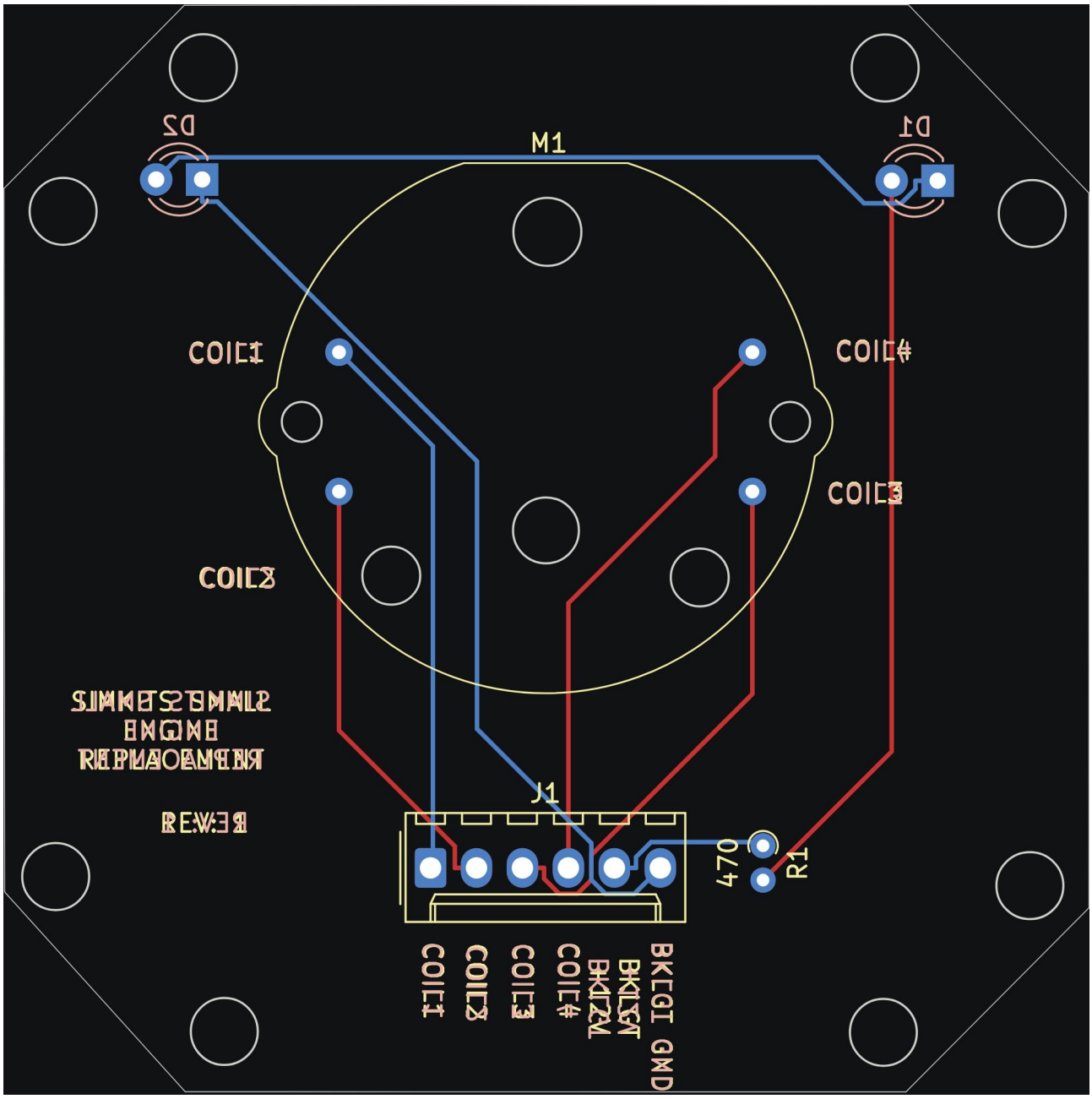


Figure 84 — Simkits Small Guage board layout

20 Installing the controller PCBs

These pictures are an aid for installing the controller boards. Each shows the **top (component) side** of a controller PCB as it is seen when the board is fitted, with only the board outline, the mounting holes, the pads and silkscreen outlines of the **external connectors**, the unlabelled round pads in rows and a double column around the edge (the Arduino Mega headers) and the outline of the Ethernet module. Every external connector is labelled with its reference (J5, SW2) and, where there is room, its name. **Pin 1 of each connector is the square pad**; on a ribbon-cable (IDC) header the small triangle also marks pin 1. Nothing else is printed, so the pictures are easy to read at the bench. They are drawn from the KiCad layout files by `export_install_views.py`, so they show the boards as they were on the day the document was built.

20.1 Jet Ranger Radio Auto Pilot

Revision 1.4 (31 October 2026, with the Huey Warning Panel connectors J20 to J23), for the Radio controller (172.16.1.101). Connectors with the name left off in the picture (a row of gauge sockets, for example) are listed in the table.

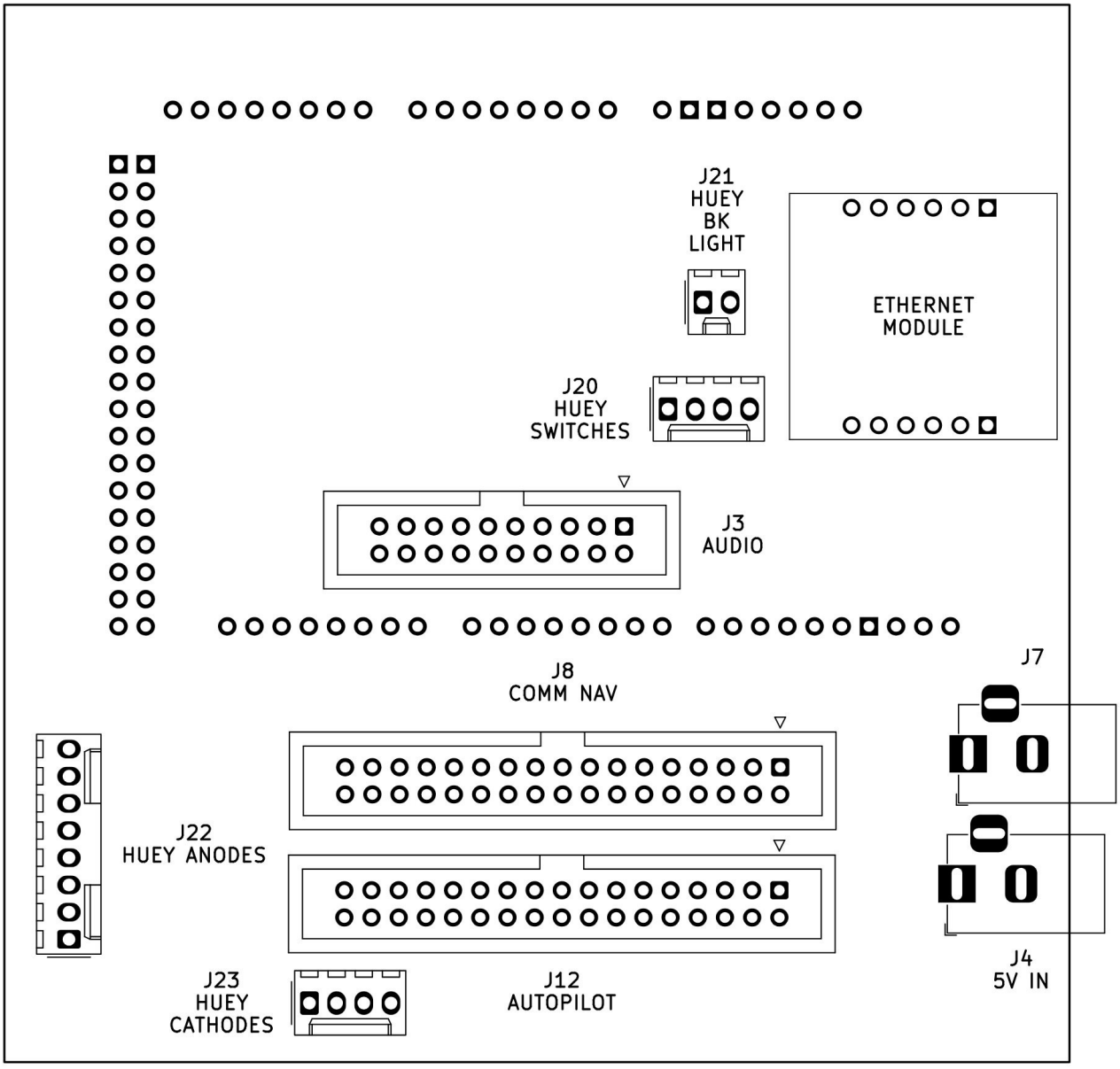


Figure 85 — Jet Ranger Radio Auto Pilot, top side with the external connectors labelled

Label	Name	Type	Pins	Notes
J3	AUDIO	IDC header	20 (2 x 10)	Audio panel
J4	5V IN	DC jack	3	5 V supply input (DC jack)
J7	STEPPER IN 12-20VDC	DC jack	3	12 to 20 V supply input (DC jack)
J8	COMM NAV	IDC header	34 (2 x 17)	COM and NAV panel
J12	AUTOPILOT	IDC header	34 (2 x 17)	Auto Pilot panel
J20	HUEY SWITCHES	Molex KK-254	4	Huey Warning Panel switches: GND, brightness / dim (A8), test (A9), reset (A10)
J21	HUEY BK LIGHT	Molex KK-254	2	Huey Warning Panel backlight: +12 V and the switched backlight
J22	HUEY ANODES	Molex KK-254	8	Huey Warning Panel LED anodes ANNODE_00 to ANNODE_07
J23	HUEY CATHODES	Molex KK-254	4	Huey Warning Panel LED cathodes CATHODE_04 to CATHODE_07 (MAX7219 digits 4 to 7)

20.2 Jet Ranger Front Panel Servo

Revision 1.4, for the Servo controller (172.16.1.102). Connectors with the name left off in the picture (a row of gauge sockets, for example) are listed in the table.

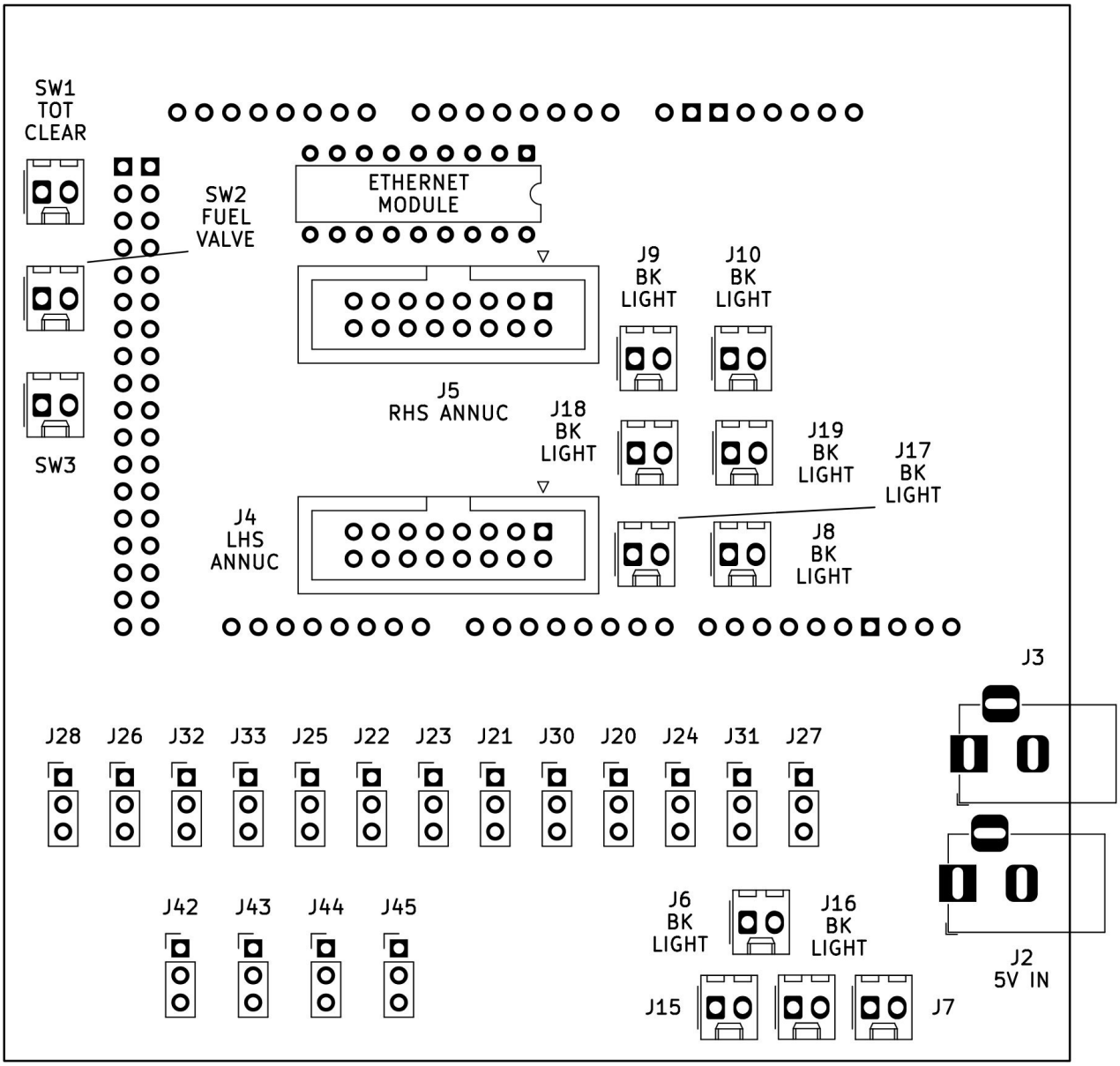


Figure 86 — Jet Ranger Front Panel Servo, top side with the external connectors labelled

Label	Name	Type	Pins	Notes
J2	5V IN	DC jack	3	5 V supply input (DC jack)
J3	STEPPER IN 12-20VDC	DC jack	3	12 to 20 V supply input (DC jack)
J4	LHS ANNUC	IDC header	16 (2 x 8)	Left hand annunciator panel
J5	RHS ANNUC	IDC header	16 (2 x 8)	Right hand annunciator panel
J6	BK LIGHT	Molex KK-254	2	
J7	BK LIGHT	Molex KK-254	2	
J8	BK LIGHT	Molex KK-254	2	
J9	BK LIGHT	Molex KK-254	2	
J10	BK LIGHT	Molex KK-254	2	

Label	Name	Type	Pins	Notes
J15	BK LIGHT	Molex KK-254	2	
J16	BK LIGHT	Molex KK-254	2	
J17	BK LIGHT	Molex KK-254	2	
J18	BK LIGHT	Molex KK-254	2	
J19	BK LIGHT	Molex KK-254	2	
J20	AIRSPEED	Pin header	3	
J21	EGT	Pin header	3	
J22	TURBINE SPEED	Pin header	3	
J23	RADAR ALT	Pin header	3	
J24	ENG TORQUE	Pin header	3	
J25	ROTOR SPEED	Pin header	3	
J26	VVI	Pin header	3	
J27	ENG OIL TEMP	Pin header	3	
J28	FUEL LEVEL	Pin header	3	
J30	GAS PRODUCER	Pin header	3	
J31	ENG OIL PRES	Pin header	3	
J32	XMSN OIL TEMP	Pin header	3	
J33	XMSN OIL PRESS	Pin header	3	
J42	PITCH	Pin header	3	Attitude indicator pitch servo
J43	ROLL	Pin header	3	Attitude indicator bank (roll) servo
J44	ELEC LOAD	Pin header	3	
J45	FUEL LOAD	Pin header	3	
SW1	TOT CLEAR	Molex KK-254	2	TOT clear switch (D43), to ground when pressed
SW2	FUEL VALVE	Molex KK-254	2	Fuel valve switch (D48), to ground when on
SW3	WARNING MUTE	Molex KK-254	2	Warning mute switch (D47), to ground when pressed

20.3 Jet Ranger 13 Stepper

Revision 1.6, for the Stepper controller (172.16.1.105). Connectors with the name left off in the picture (a row of gauge sockets, for example) are listed in the table.

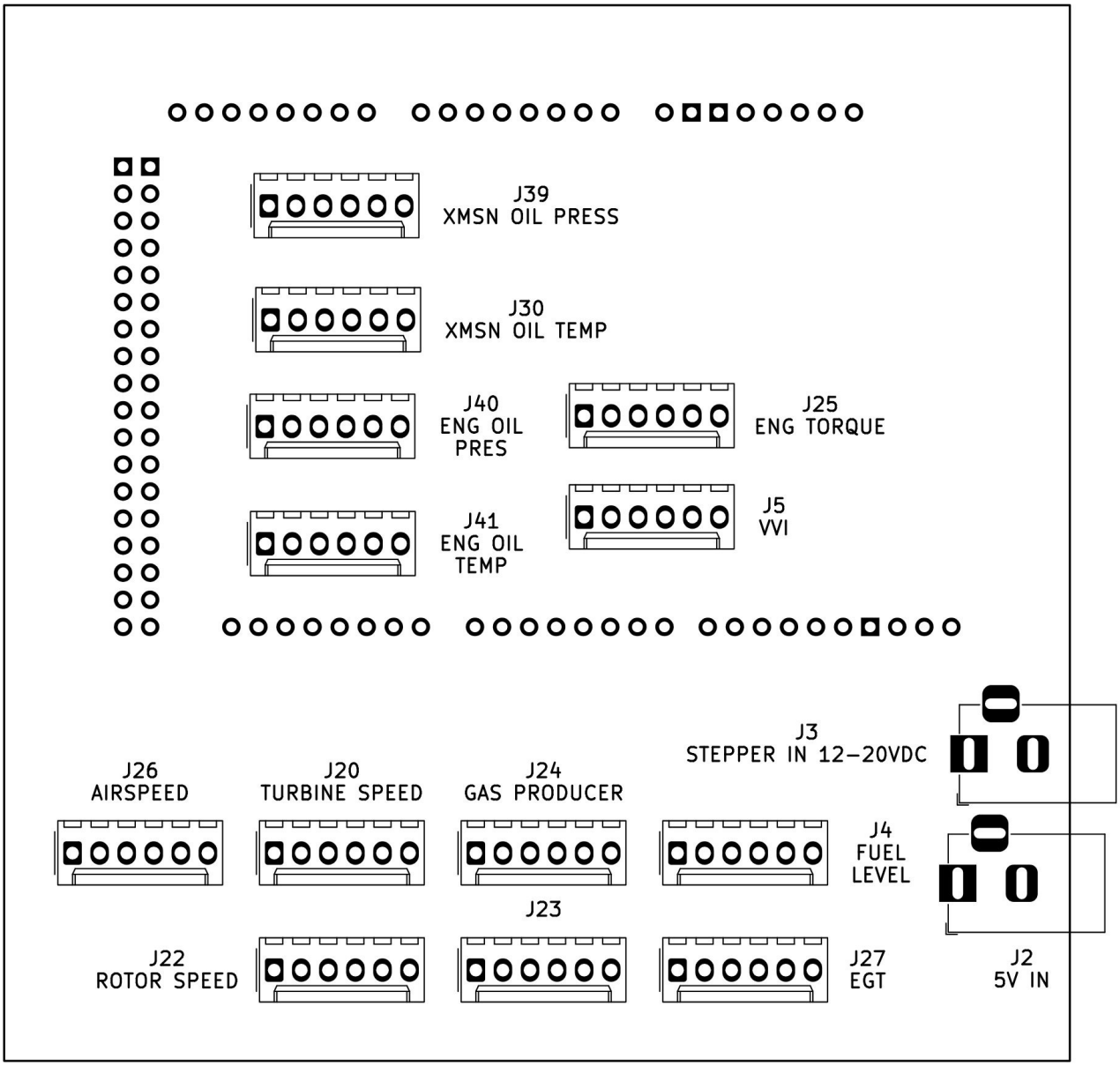


Figure 87 — Jet Ranger 13 Stepper, top side with the external connectors labelled

Label	Name	Type	Pins	Notes
J2	5V IN	DC jack	3	5 V supply input (DC jack)
J3	STEPPER IN 12-20VDC	DC jack	3	12 to 20 V supply input (DC jack)
J4	FUEL LEVEL	Molex KK-254	6	
J5	VVI	Molex KK-254	6	
J20	TURBINE SPEED	Molex KK-254	6	
J22	ROTOR SPEED	Molex KK-254	6	
J23	RADAR ALT	Molex KK-254	6	
J24	GAS PRODUCER	Molex KK-254	6	
J25	ENG TORQUE	Molex KK-254	6	
J26	AIRSPEED	Molex KK-254	6	

Label	Name	Type	Pins	Notes
J27	EGT	Molex KK-254	6	
J30	XMSN OIL TEMP	Molex KK-254	6	
J39	XMSN OIL PRESS	Molex KK-254	6	
J40	ENG OIL PRES	Molex KK-254	6	
J41	ENG OIL TEMP	Molex KK-254	6	

20.4 Jet Ranger OLED Dual Stepper

Revision 1.5, for the Dual Stepper controller (172.16.1.106). Connectors with the name left off in the picture (a row of gauge sockets, for example) are listed in the table.

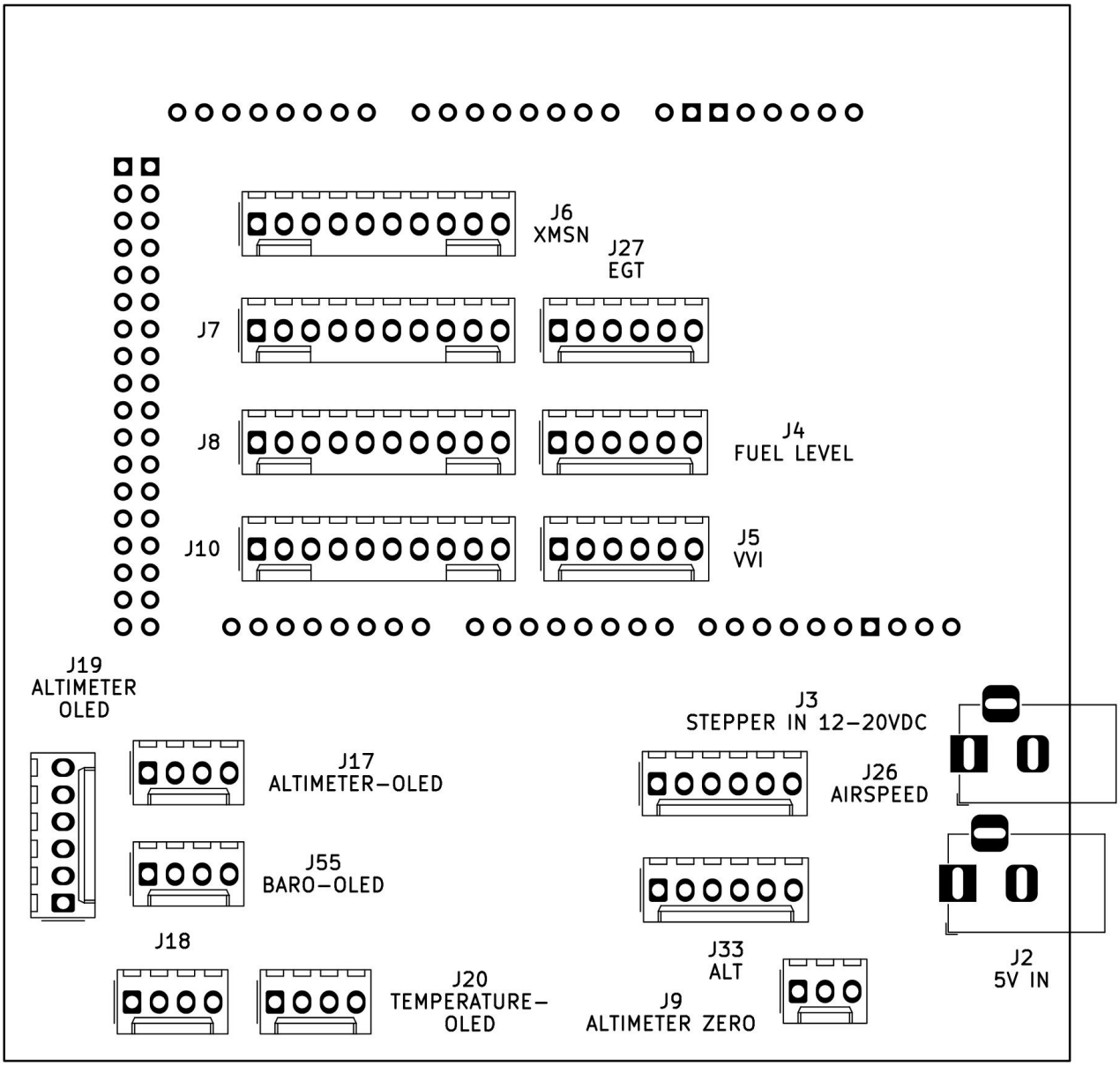


Figure 88 — Jet Ranger OLED Dual Stepper, top side with the external connectors labelled

Label	Name	Type	Pins	Notes
J2	5V IN	DC jack	3	5 V supply input (DC jack)
J3	STEPPER IN 12-20VDC	DC jack	3	12 to 20 V supply input (DC jack)
J4	FUEL LEVEL	Molex KK-254	6	
J5	VVI	Molex KK-254	6	
J6	XMSN	Molex KK-254	10	
J7	ROTOR ENG SPEED	Molex KK-254	10	
J8	ENGINE	Molex KK-254	10	
J9	ALTIMETER ZERO	Molex KK-254	3	Altimeter zero sensor
J10	ELEC FUEL LOAD	Molex KK-254	10	
J17	ALTIMETER-OLED	Molex KK-254	4	Altimeter OLED

Label	Name	Type	Pins	Notes
J18	CLOCK-OLED	Molex KK-254	4	Clock OLED
J19	ALTIMETER OLED	Molex KK-254	6	
J20	TEMPERATURE-OLED	Molex KK-254	4	Temperature OLED
J26	AIRSPEED	Molex KK-254	6	
J27	EGT	Molex KK-254	6	
J33	ALT	Molex KK-254	6	
J55	BARO-OLED	Molex KK-254	4	Barometer OLED

20.5 Jet Ranger Upper ADF Controller

Revision 1.0, for the ADF and Upper controller (172.16.1.107). Connectors with the name left off in the picture (a row of gauge sockets, for example) are listed in the table.

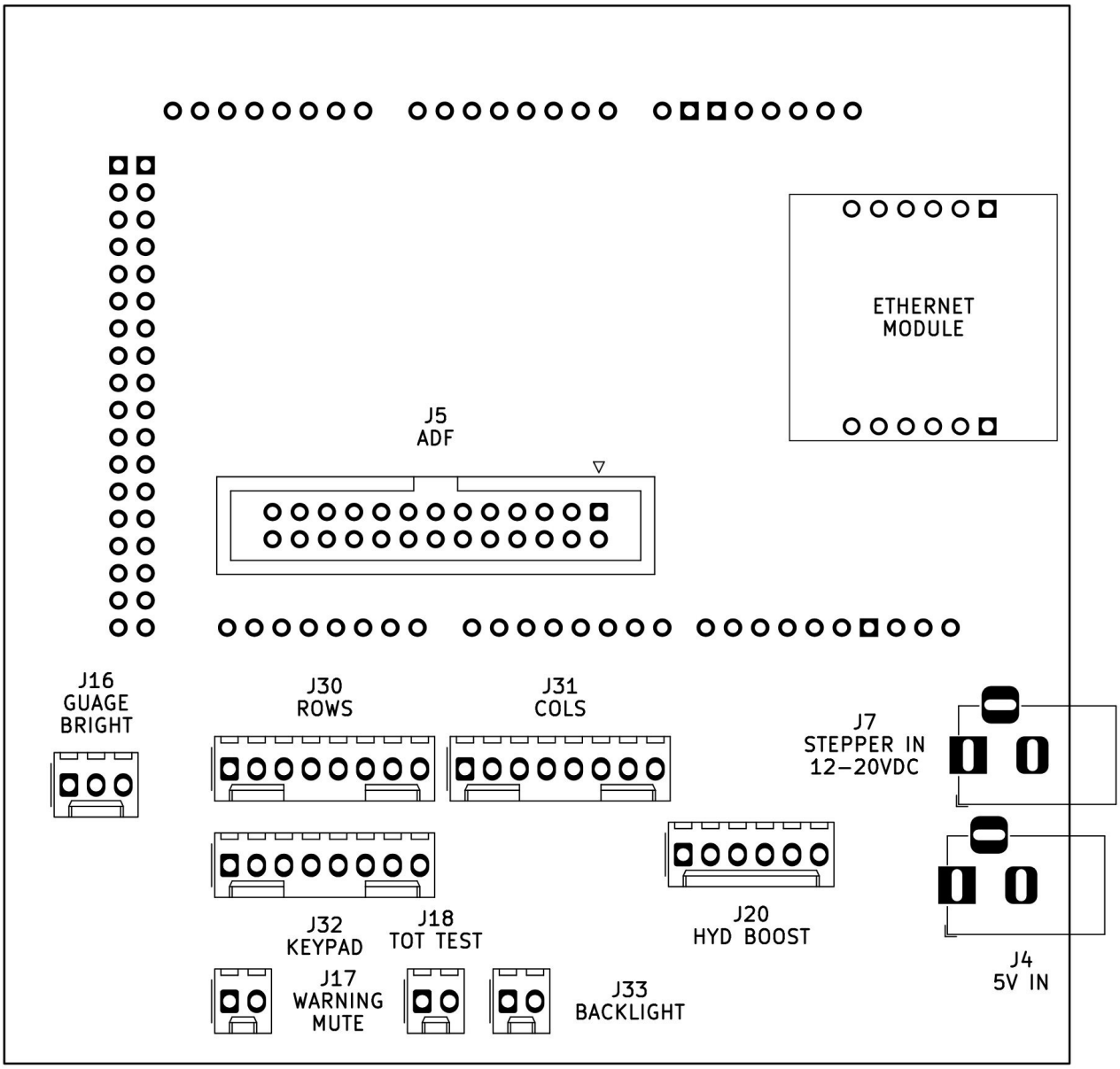


Figure 89 — Jet Ranger Upper ADF Controller, top side with the external connectors labelled

Label	Name	Type	Pins	Notes
J4	5V IN	DC jack	3	5 V supply input (DC jack)
J5	ADF	IDC header	26 (2 x 13)	ADF panel (knobs, keypad, switches, lamps)
J7	STEPPER IN 12-20VDC	DC jack	3	12 to 20 V supply input (DC jack)
J16	GUAGE BRIGHT	Molex KK-254	3	Panel brightness pot
J17	WARNING MUTE	Molex KK-254	2	Warning mute switch (matrix ROW8)
J18	TOT TEST	Molex KK-254	2	TOT test switch (matrix ROW8)
J20	HYD BOOST	Molex KK-254	6	Hyd boost, anti-ice and caution dim switches
J30	ROWS	Molex KK-254	8	Upper switch matrix rows ROW0 to ROW7
J31	COLS	Molex KK-254	8	Upper switch matrix columns COL0 to COL7
J32	KEYPAD	Molex KK-254	8	Upper numeric keypad

Label	Name	Type	Pins	Notes
J33	BACKLIGHT	Molex KK-254	2	Panel backlight

21 Instrument map

This appendix draws every gauge, lamp and display of the cockpit as the real thing (dial gauges with needles, the attitude indicator, annunciator and legend lamps, gear lights, OLED displays) and joins each one by a cable to the connector on the controller board that drives it. The connector references and names are the ones printed on the PCBs (see the installation views in Appendix F). Needles and lamps are drawn at illustrative readings and the lens colours are illustrative. Switches, knobs, keypads and the USB HID board are inputs and are not drawn. The full-size drawing is on one A3 landscape page, `Jet Ranger Instrument Map A3.pdf` (also an SVG and a PNG) in the documentation folder; it is made by `DocBuild/make_instrument_map.py`.

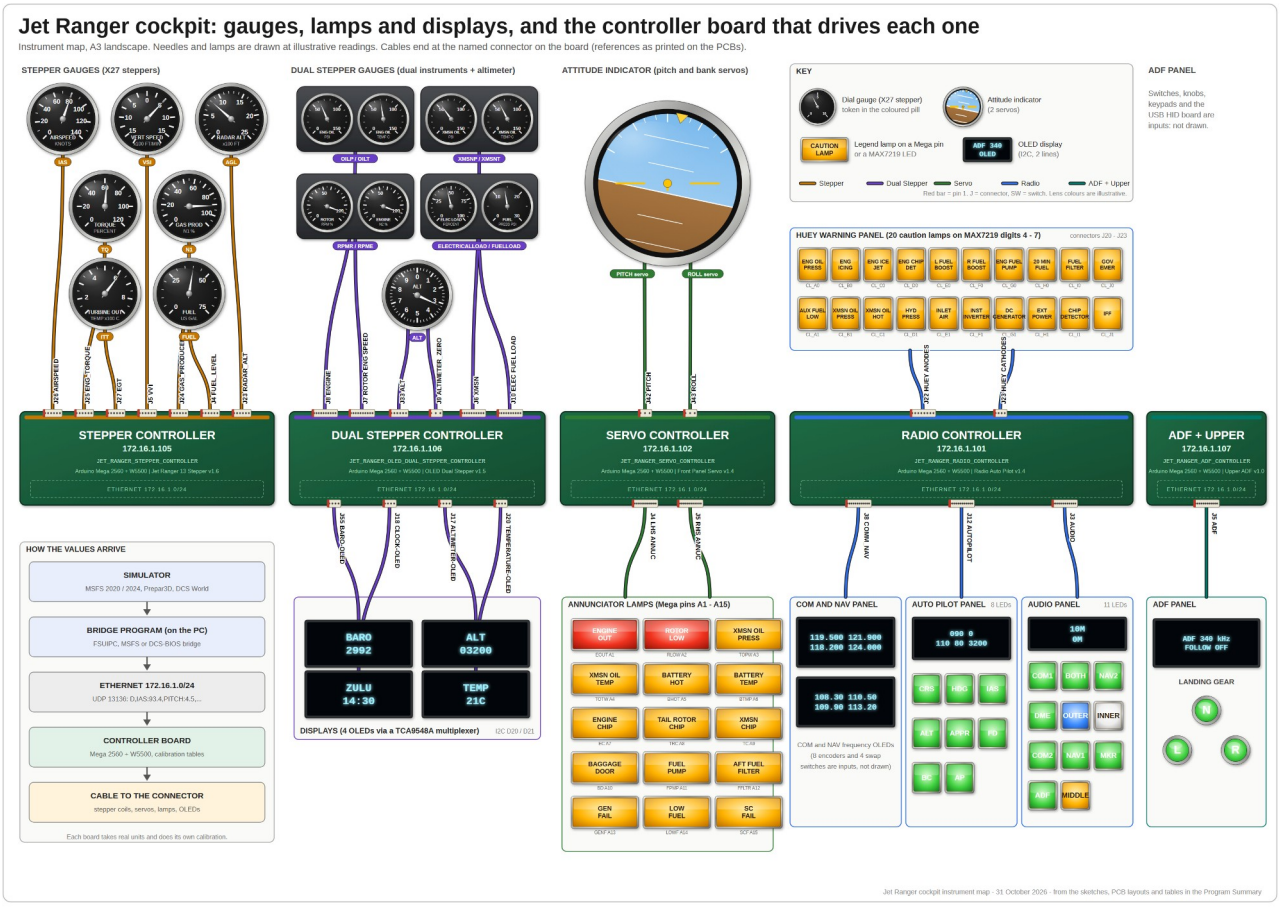


Figure 90 — The whole map (the A3 original is Jet Ranger Instrument Map A3.pdf; zoom in the PDF to read it)

21.1 Stepper gauges and the Stepper controller

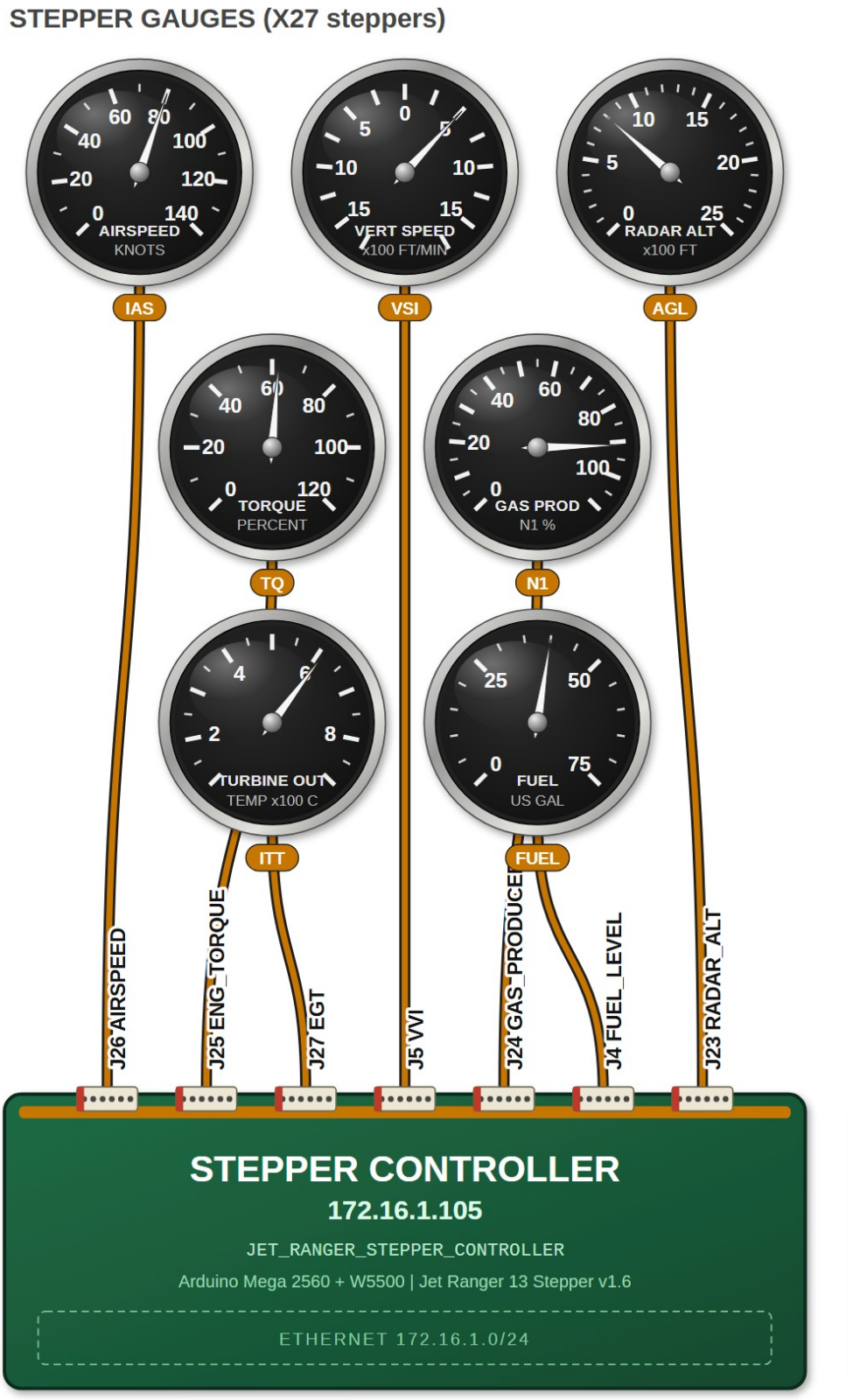


Figure 91 — Stepper gauges and the Stepper controller

21.2 Dual Stepper gauges and the Dual Stepper controller

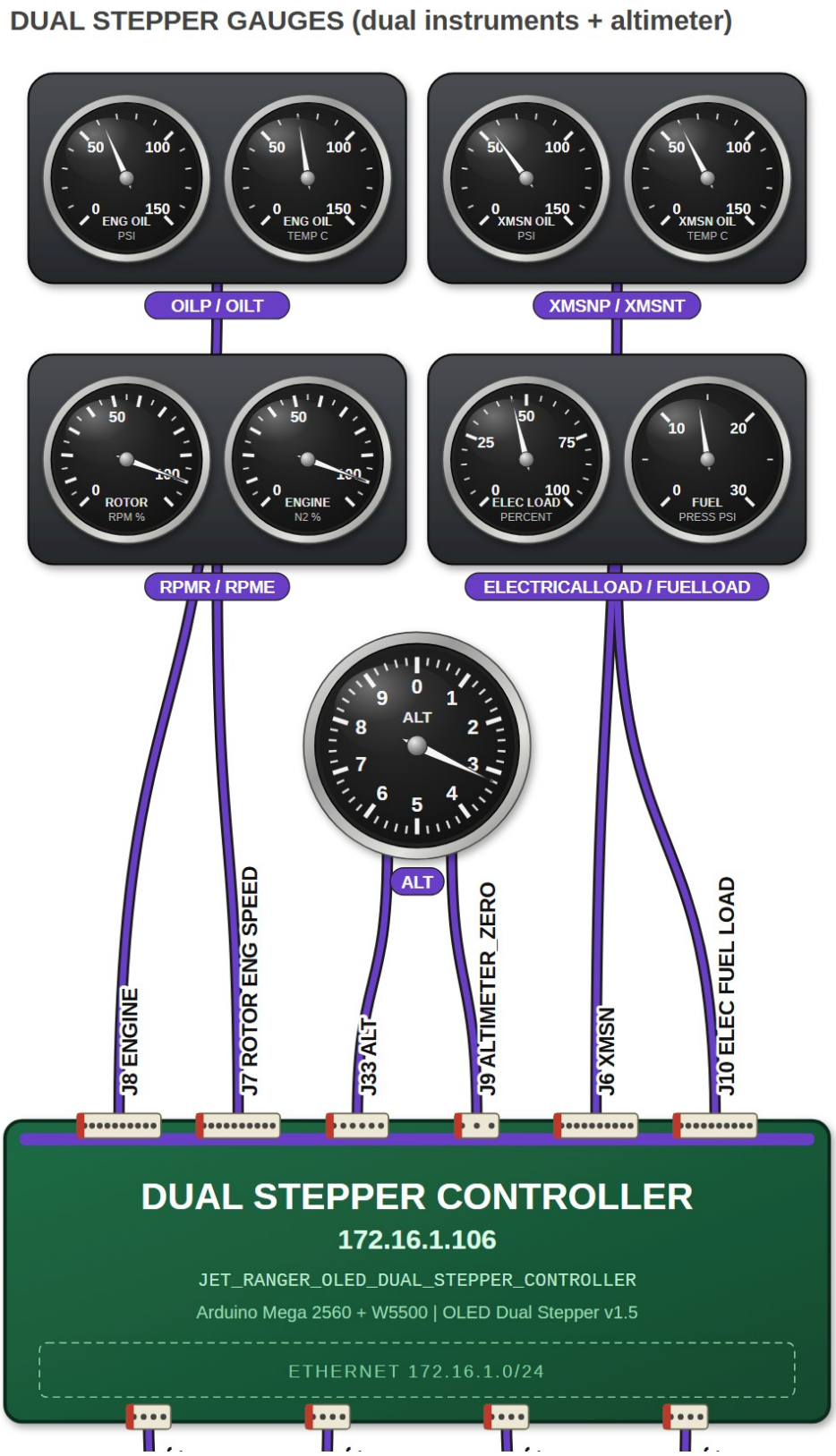


Figure 92 — Dual Stepper gauges and the Dual Stepper controller

21.3 Attitude indicator and the Servo controller

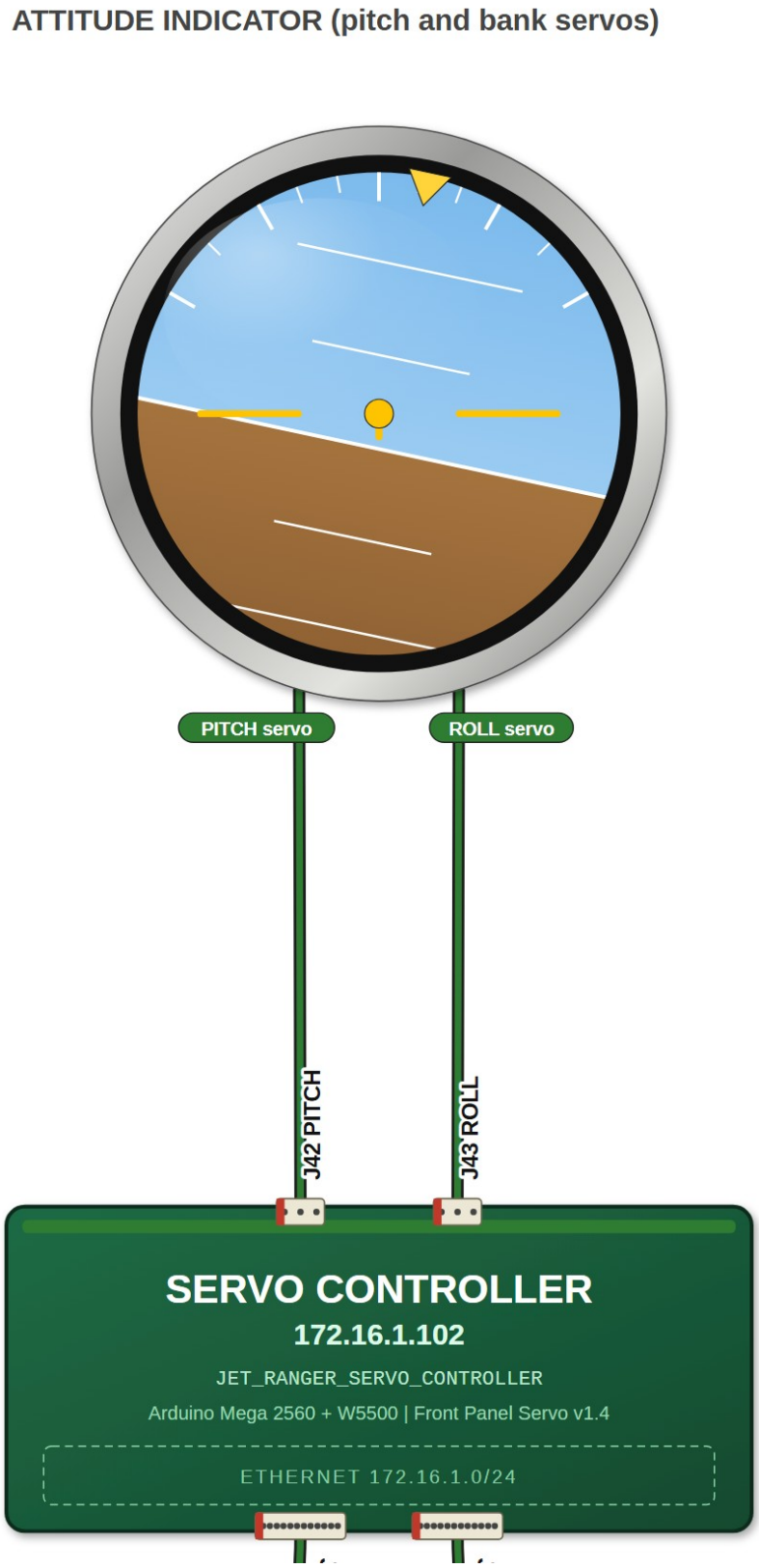


Figure 93 — Attitude indicator and the Servo controller

21.4 Huey Warning Panel, key, and the top of the Radio and ADF boards

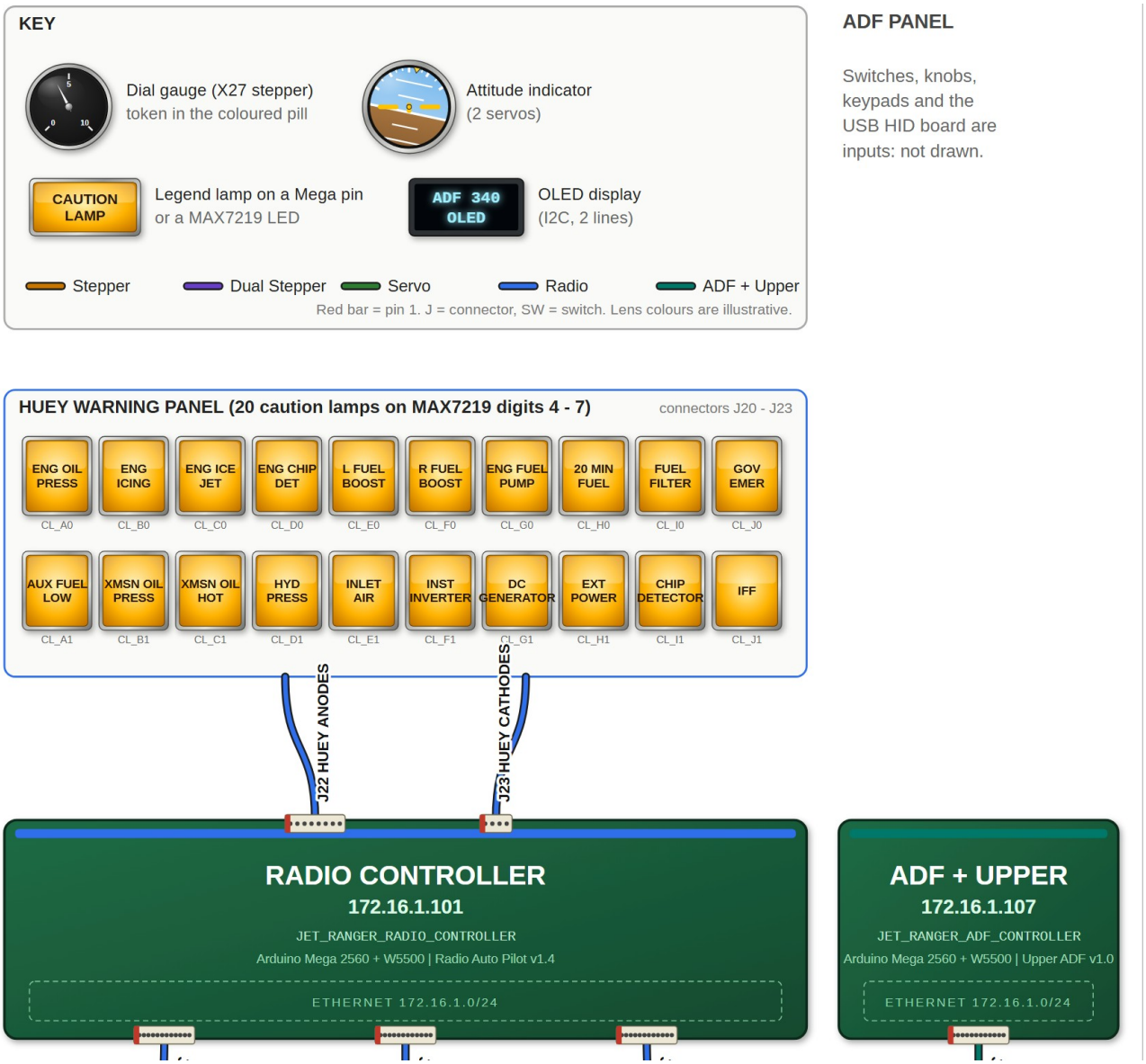


Figure 94 — Huey Warning Panel, key, and the top of the Radio and ADF boards

21.5 Dual Stepper displays and the Servo annunciator lamps

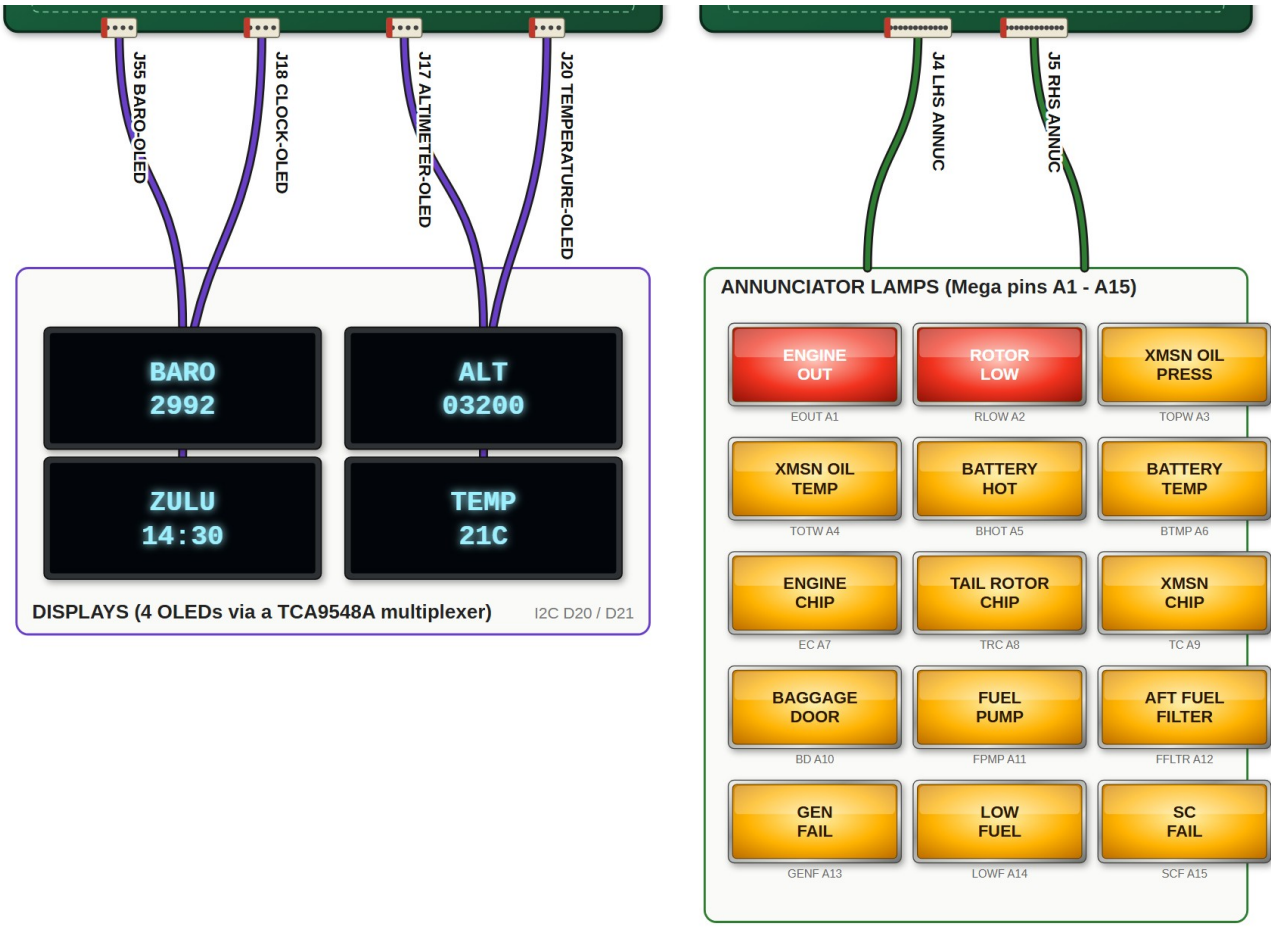


Figure 95 — Dual Stepper displays and the Servo annunciator lamps

21.6 Radio panels (COM and NAV, Auto Pilot, Audio) and the ADF panel

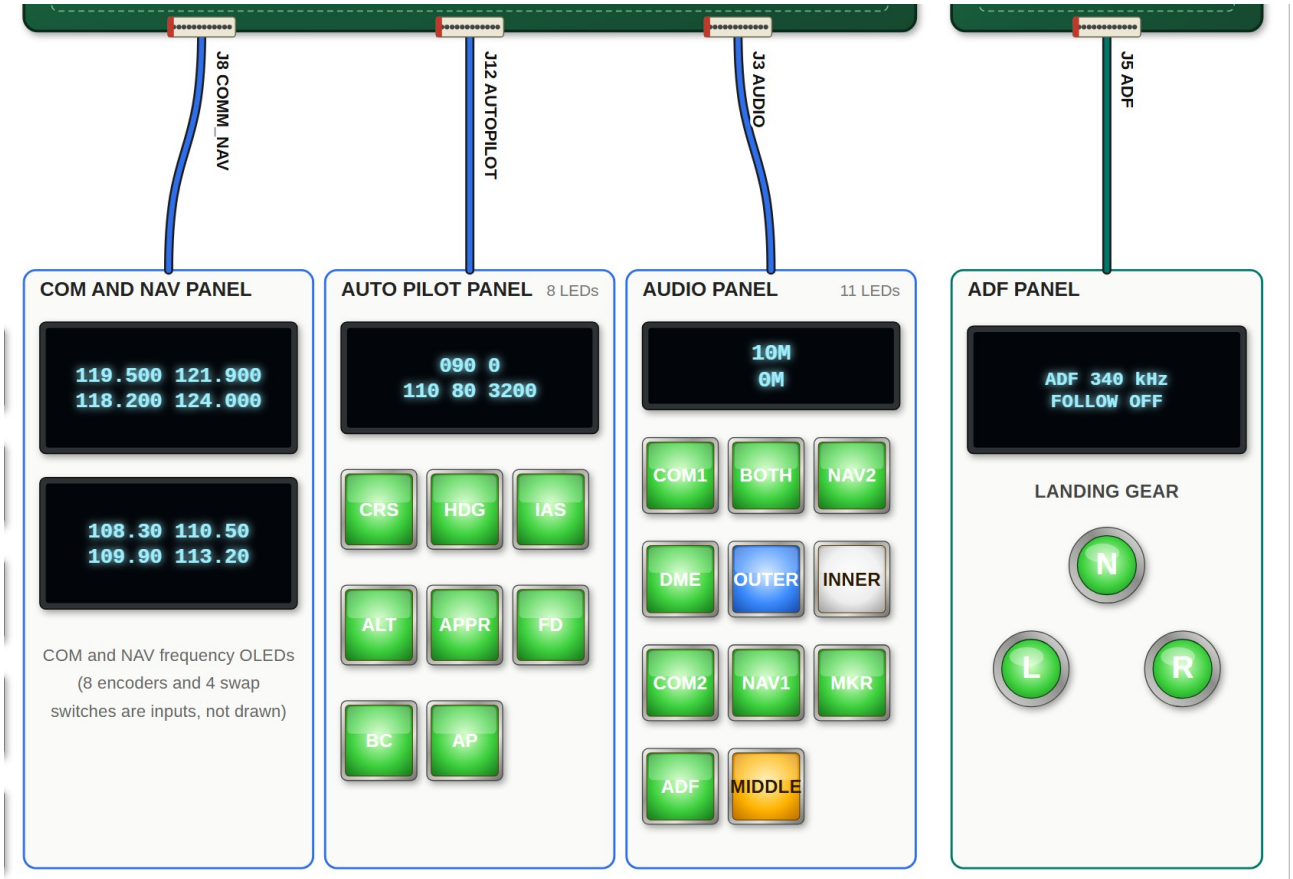


Figure 96 — Radio panels (COM and NAV, Auto Pilot, Audio) and the ADF panel

21.7 Gauges and the connectors they use

Instrument	Token	Connector on the board	Range
Airspeed	IAS	Stepper .105: J26 AIRSPEED	0 to 140 knots
Vertical speed	VSI	Stepper .105: J5 VVI	-1750 to +1750 ft/min
Radar altitude	AGL	Stepper .105: J23 RADAR_ALT	0 to 2500 ft
Torque	TQ	Stepper .105: J25 ENG_TORQUE	0 to 120 percent
Gas producer (N1)	N1	Stepper .105: J24 GAS_PRODUCER	0 to 103 percent
Turbine out temperature	ITT	Stepper .105: J27 EGT	100 to 900 C
Fuel quantity	FUEL	Stepper .105: J4 FUEL_LEVEL	0 to 75 US gallons
Engine oil pressure and temperature (dual instrument)	OILP, OILT	Dual Stepper .106: J8 ENGINE	0 to 150 psi, 0 to 150 C
Transmission oil pressure and temperature (dual instrument)	XMSNP, XMSNT	Dual Stepper .106: J6 XMSN	0 to 150 psi, 0 to 150 C
Rotor speed and engine speed N2 (dual instrument)	RPMR, RPME	Dual Stepper .106: J7 ROTOR ENG SPEED	0 to 110 percent
Electrical load and fuel load (dual instrument)	ELECTRICALLOAD, FUELLOAD	Dual Stepper .106: J10 ELEC FUEL LOAD	0 to 100 percent, 0 to 30 psi
Altimeter (and its zero sensor)	ALT	Dual Stepper .106: J33 ALT, J9 ALTIMETER_ZERO	one turn per 1000 ft
Attitude indicator pitch	PITCH	Servo .102: J42 PITCH	-30 to +30 degrees
Attitude indicator bank	BANK	Servo .102: J43 ROLL	-90 to +90 degrees

21.8 Lamps and displays and the connectors they use

Lamps and displays	Tokens	Connector on the board
15 annunciator lamps (engine out, rotor low, transmission oil pressure and temperature, battery hot and temperature, engine chip, tail rotor chip, transmission chip, baggage door, fuel pump, aft fuel filter, generator fail, low fuel, SC fail)	EOUT, RLOW, TOPW, TOTW, BHOT, BTMP, EC, TRC, TC, BD, FPMP, FFLTR, GENF, LOWF, SCF on Mega pins A1 to A15	Servo .102: J4 LHS_ANNUC and J5 RHS_ANNUC (how the lamps divide between the two is not recorded)
COM and NAV frequency OLEDs	C1A, C1S, C2A, C2S	Radio .101: J8 COMM_NAV
Auto Pilot display (OLED) and 8 LEDs	APHDG, APCRS, APIAS, APVS, APALT; LED_CRS, LED_HDG, LED_IAS, LED_ALT, LED_APPR, LED_FD, LED_BC, LED_APENG	Radio .101: J12 AUTOPILOT
DME display (OLED) and 11 audio panel LEDs	LED_COM1, LED_BOTH, LED_NAV2, LED_DME, LED_OUTER, LED_INNER, LED_COM2, LED_NAV1, LED_MKR, LED_ADF, LED_MID	Radio .101: J3 AUDIO
20 Huey Warning Panel lamps	LED_H_* (MAX7219 digits 4 to 7)	Radio .101: J22 HUEY_ANODES and J23 HUEY_CATHODES (rev 1.4 layout; the panel is hand wired today)
4 Dual Stepper OLEDs (barometer, altimeter digits, clock, temperature)	ALT, ZULU, TEMP	Dual Stepper .106: J55 BARO-OLED, J17 ALTIMETER-OLED, J18 CLOCK-OLED, J20 TEMPERATURE-OLED
ADF display (OLED) and 3 landing gear lamps	LDGN, LDGL, LDGR (Mega pins D2, D3, D4)	ADF and Upper .107: J5 ADF

Which dual instrument uses which socket is taken from the names on the PCBs (J8 ENGINE, J6 XMSN, J7 ROTOR ENG SPEED, J10 ELEC FUEL LOAD); the Stepper's turbine out temperature gauge is shown on J27 EGT; and how the 15 annunciator lamps divide between J4 and J5 is not recorded, so the drawing joins the whole lamp panel to both.